

Fundamental Data Structures, Algorithms and Complexity, and Computer Architecture

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What are CSC211 and its partner CSC212 about?

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What are CSC211 and its partner CSC212 about?

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- CSC212 has two parts— (1) It continues the themes of problem solving and programming in a high level language and understanding the time and space complexity of algorithms. Students continue to *broaden their problem solving skills*.

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- CSC212 has two parts— (1) It continues the themes of problem solving and programming in a high level language and understanding the time and space complexity of algorithms. Students continue to *broaden their problem solving skills*.
(2) Hardware is covered and we learn some assembler programming to understand clearly what the underlying hardware does when it computes a loop or executes a function recursively.

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(2) Hardware is covered and we learn some assembler programming to understand clearly what the underlying hardware does when it computes a loop or executes a function recursively. Different styles of computer architecture are compared and students gain an understanding of the interaction between high-level and low-level programming and hardware.

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- We learn data structures such as linear lists, queues, stacks, hash tables, binary search trees, priority queues.

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Fundamental Data Structures—prescribed books

Class notes for 2015

These slides form the core part of this course COS211/212.¹

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¹The class notes can be found on the Sun systems in the directory /export/second/notes/ds/ in the files ds.ps, or ds.pdf—or 4-to-a-page: 4up-ds.ps, 4up-ds.pdf. Older versions of the notes are called ds-slides.*. Read the notes.

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For *Python* there are many good, free books available on the Internet. Stick to Python 3 and learn the differences between 2 and 3.

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An introductory example—Shuffling Cards

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$\{1, 2, \dots, 52\}$, which is often written as $[1..52]$.

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This can be built in Python using a list called `deck`.

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1 deck = [0]*52
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Learn to use both abstractions.

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A shuffled deck can be built by drawing cards one-by-one randomly and stacking them onto the ‘shuffled’ deck.

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We will attempt both approaches.

Shuffling—make bad method work

- We must insert `r` into `shuffledDeck` if and only if `r` is not there.

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1 k = 0;
2 while (k < 52) {
3     r = (int)(Math.random()*52) + 1;
4     for (j=1; j<=k; j++)
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- Notice that `k < 52`. Once all the values in the array `shuffledDeck[1..51]` are filled in, getting the 52nd one by letting the loop run to `k = 52` is very time consuming.

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How long it takes on average?

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There are some other ideas to improve matters.

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- How fast is this method?

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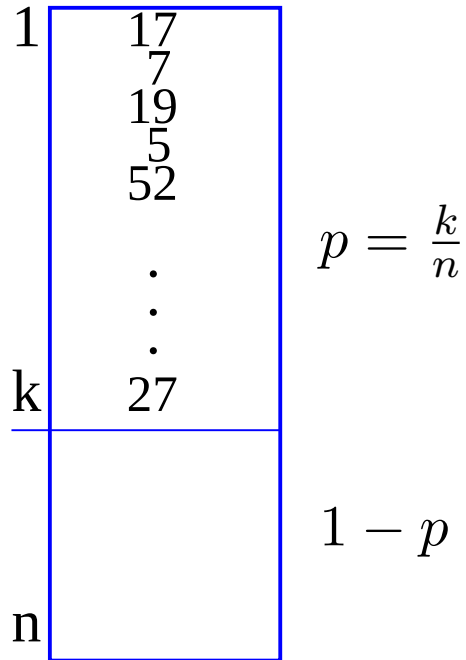
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- At first the shuffled deck starts filling rapidly, but as it gets more cards—values of r —matters slow down.
- How fast is this method?
- —It is slow.
- We will see how slow after looking at the same method in Python.

Shuffling—inserting into a shuffled deck in Python all the way

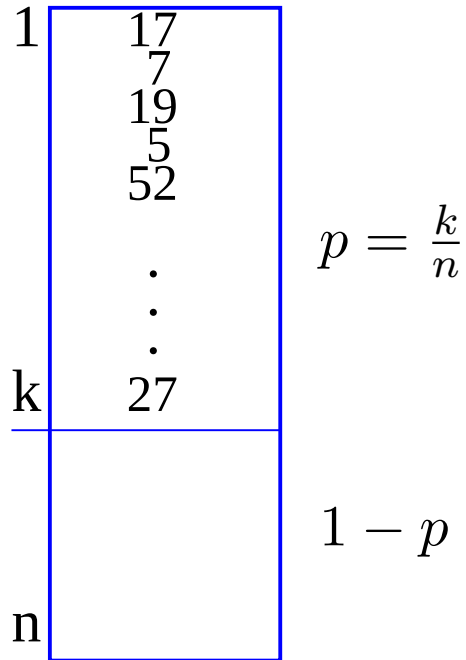
```
1 from random import random
2
3 def shuffle(N):
4     shuffledDeck = [0]*N
5     k = 0
6     while k < N:
7         r = int(random()*N)
8         for j in range (k+1):
9             if shuffledDeck[j] == r: break
10        if j == k:
11            shuffledDeck[k] = r
12            k += 1
13
14    return shuffledDeck
15
16 print (shuffle(52))
```

Shuffling—average number of probes



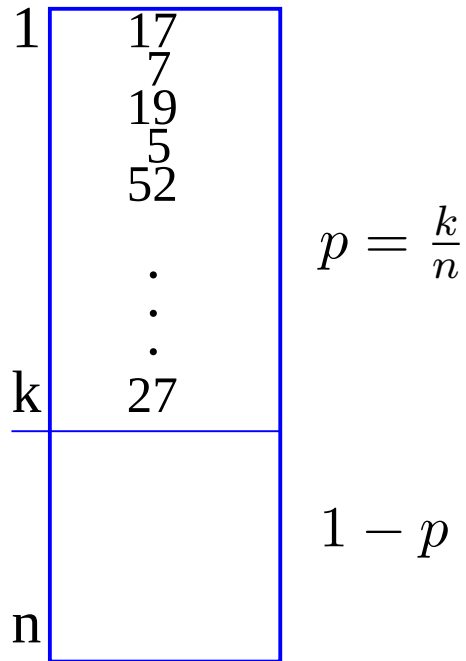
- Assume that there are already k cards in the list. The fraction $p = \frac{k}{n}$ contains the already shuffled cards in their final positions.

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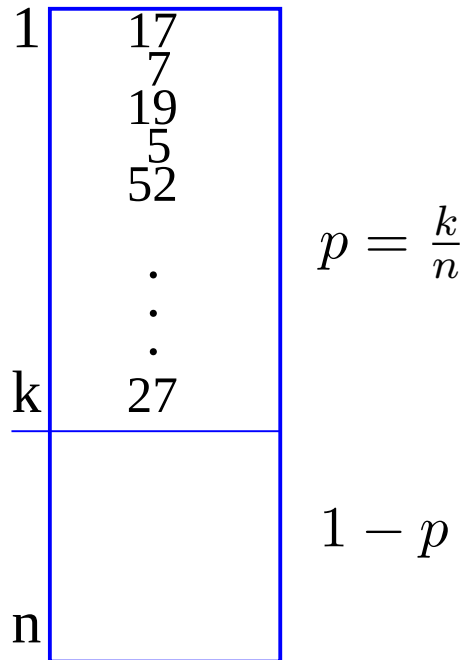
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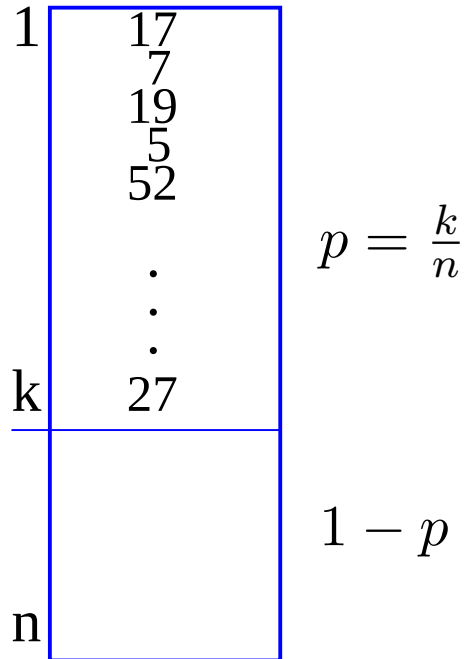
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- The $(k+1)$ th card must now be randomly drawn by generating a random number $r \in [1..52]$.

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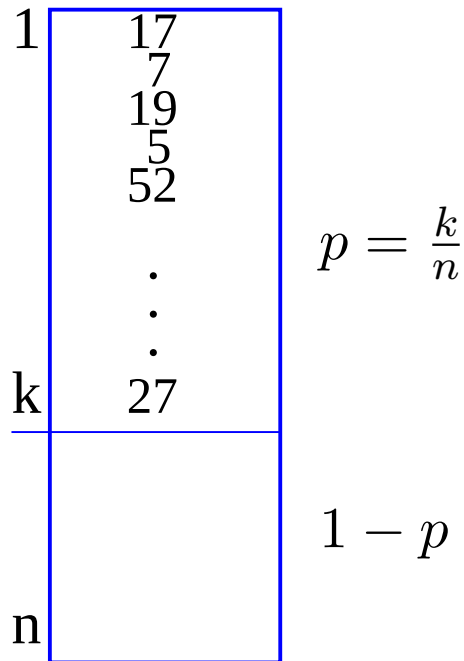
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- The rest of the list, the fraction $1 - p$ of the list is open. Here $n = 52$.
- The $(k+1)$ th card must now be randomly drawn by generating a random number $r \in [1..52]$.
- How long does it take to draw a valid card $= r$ that can be added to the list?

Shuffling—average number of probes



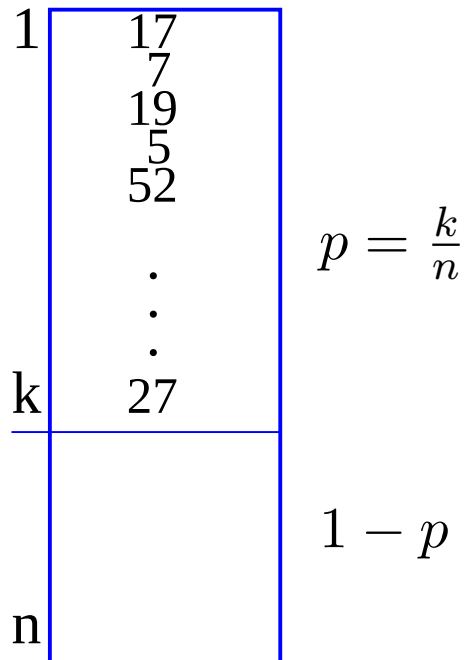
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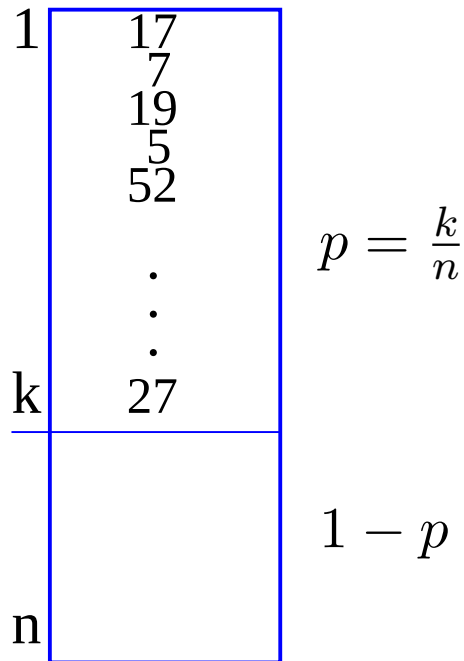
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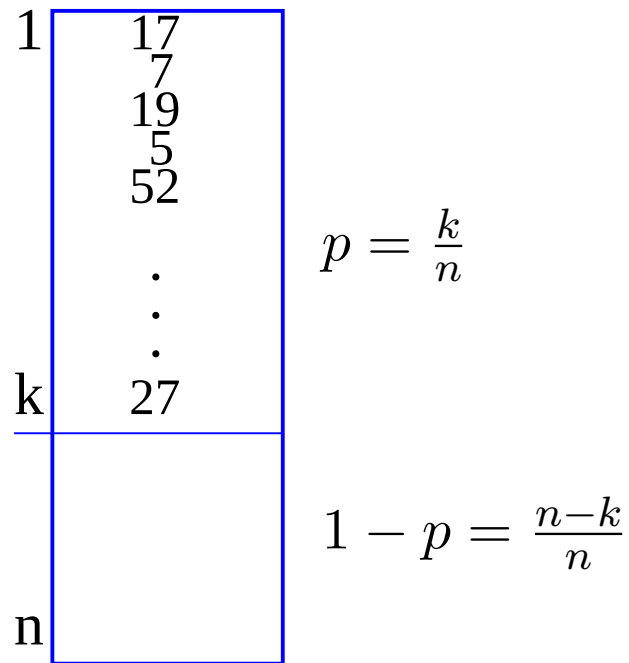
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- *How do we discover that a card has already been selected?*

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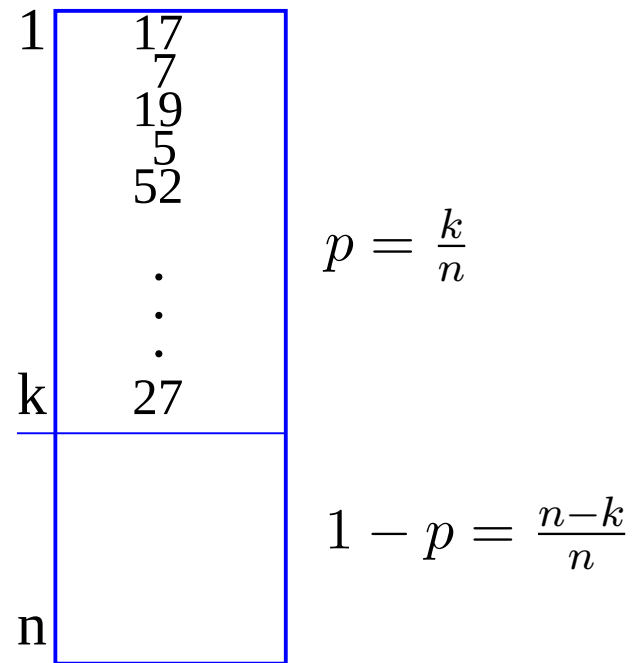
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- *How do we discover that a card has already been selected?*
- One method is to search through the list until it is found.

Shuffling—checking for the presence of $\mathbf{r}$



- Remember that we have assumed that we will find $\mathbf{r}$ in the list,

Shuffling—checking for the presence of $\mathbf{r}$



- Remember that we have assumed that we will find $\mathbf{r}$ in the list,
- In this case about half of the list of k elements must be searched to discover $\mathbf{r}$, so $\frac{k+1}{2}$ probes are done on average, for p of the time:

$$p \frac{k+1}{2}$$

Shuffling—when r is not in the list

1	17
	7
	19
	5
	52
	.
	.
	.
k	27
n	

$$p = \frac{k}{n}$$

$$1 - p = \frac{n-k}{n}$$

- For the rest of the time the whole list—now of length k —is searched, i.e. the r is compared with every one of the k items in the list, and is unequal to any of them. So k comparisons are made.

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- How often is there a new r , i.e. an r that is not in the first k elements?

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$$\left(1 - \frac{k}{52}\right)k = (1 - p)k.$$

Shuffling—when r is not in the list

- This must be summed for all 52 cards:
Number of probes =

$$\sum_{k=1}^{52} p \frac{(k+1)}{2} + (1-p)k$$

- The first term is not quite correct, since if a card is already in the list we must search *again and again* until one is found that is not in the list, and that process may have to be repeated many times.

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- Let us analyse this.

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- 3rd probe costs $(p + p^2 + p^3) \frac{k+1}{2}$

Shuffling—total number of probes

- So, how long does it take to draw the k th card?
- For $p = \frac{k}{52}$ of the time the card is already in the deck, so when a card *is not in the list*, then the entire list—currently of length k —must be searched, i.e. $(1 - p)k$.
- Suppose the generated card *is* in the list. It will be found on average after $\frac{k+1}{2}$ probes.
- But, if it *is* in the list another card must be generated, and this process must be repeated until finding a card not in the list.
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Probe time = $\left[\sum_{i=1}^{\infty} p^i \right] \frac{k+1}{2}$, but putting

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Shuffling Cards—improving the method

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- It mounts up to about 90% of the time.
- The trick is to store a bit array—incorrectly called a bit vector—that indicates which cards have been generated:

```
1 boolean[]notInList = new boolean[53];
2   for (k=1; k<=52; k++)
3       notInList[k] = true;
4   while (k < 52) {
5       r = (int)(Math.random()*52) + 1;
6       if (notInList[r]) {
7           shuffledDeck[k++] = r;
8           notInList[k] = false; } }
```

Shuffling Cards—improving the method

- The time complexity is:

$$\sum_{k=1}^{51} (1 - p)^k$$

- This is about 10% of the time taken by the original bad method.
- There is an efficient method that takes $O(n)$ time.

Shuffling Cards—much improved method

- Start with a deck of $N = 52$ cards, represented by an array of integers $\{1, 2, \dots, N\}$. Suppose the array is initialized using:

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Shuffling three cards—an experiment

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r	The deck	Effect
1	$\{1, 2, 3\}$	1 is swapped with 1
2	$\{2, 1, 3\}$	2 is swapped with 1
3	$\{3, 2, 1\}$	3 is swapped with 1

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- Next, let **card** = 2. Get a random number $r \in \{1, 2, 3\}$ and swap **deck[card]** with **deck[r]**. This yields three possibilities for each of the above configurations—a total of nine possibilities.

The deck	r	Result of swap
$\{1, 2, 3\}$	1	$\{2, 1, 3\}$
$\{1, 2, 3\}$	2	$\{1, 2, 3\}$
$\{1, 2, 3\}$	3	$\{1, 3, 2\}$
$\{2, 1, 3\}$	1	$\{1, 2, 3\}$
$\{2, 1, 3\}$	2	$\{2, 1, 3\}$
$\{2, 1, 3\}$	3	$\{2, 3, 1\}$
$\{3, 2, 1\}$	1	$\{2, 3, 1\}$
$\{3, 2, 1\}$	2	$\{3, 2, 1\}$
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- Starting with these nine orderings we next consider swapping where **card** = 3 **deck[card]** with **deck[r]** for $r \in \{1, 2, 3\}$.

Shuffling three cards—an experiment

Deck	r	After final swap
{2, 1, 3}	1	{3, 1, 2}
{2, 1, 3}	2	{2, 3, 1}
{2, 1, 3}	3	{2, 1, 3}
{1, 2, 3}	1	{3, 2, 1}
{1, 2, 3}	2	{1, 3, 2}
{1, 2, 3}	3	{1, 2, 3}
{1, 3, 2}	1	{2, 3, 1}
{1, 3, 2}	2	{1, 2, 3}
{1, 3, 2}	3	{1, 3, 2}
{1, 2, 3}	1	{3, 2, 1}
{1, 2, 3}	2	{1, 3, 2}
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{2, 3, 1}	1	{1, 3, 2}
{2, 3, 1}	2	{2, 1, 3}
{2, 3, 1}	3	{2, 3, 1}
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{2, 3, 1}	2	{2, 1, 3}
{2, 3, 1}	3	{2, 3, 1}
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Permutation	Number of occurrences	Random trial of 10^7 biased shuffles	Random trial of 10^7 <i>unbiased</i> shuffles
{1, 2, 3}	4	8890048	10001750
{1, 3, 2}	5	11116476	10003651
{2, 1, 3}	5	11104844	9997698
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{3, 1, 2}	4	8885890	10000476
{3, 2, 1}	4	8891189	10000551

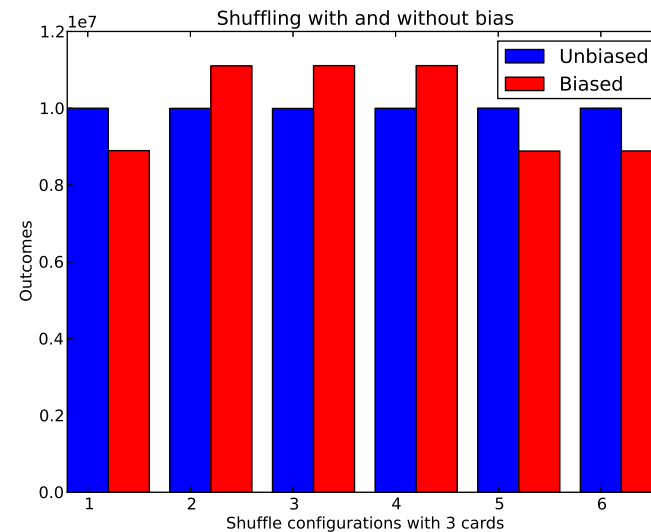
The bias is quite clear and confirms the distribution of shuffles when r is chosen randomly.

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Shuffling three cards—shuffling bias

It is not surprising that this bias occurs. We have just shown that when starting from the unshuffled deck $\{1, 2, 3\}$ it turns out that there are $3^3 = 27$ ways of reaching one of six possible outcomes. This means that each of the six possibilities can be reached in an average of 4.5 ways. so it is reached in either 4 or 5 possible ways.

If we randomly start with any other ordering a similar distribution ensues. Since the pattern of outcomes is always 4, 5, 5, 5, 4, 4 the distribution quickly evens out when the shuffle starts at the current ordering of the deck. Let us call the probability of getting the ordering s_k when starting with s_1 , $P_1(s_k)$ $P_1(s_1) = 4/27$, $P_1(s_2) = 5/27$ $P_1(s_3) = 5/27$ $P_1(s_4) = 5/27$ $P_1(s_5) = 4/27$, $P_1(s_6) = 4/27$. Starting at ordering s_2 yields the same probabilities tied to other orderings, namely $P_2(s_1) = 5/27$, $P_2(s_2) = 4/27$ $P_2(s_3) = 4/27$ $P_2(s_4) = 4/27$ $P_2(s_5) = 5/27$, $P_2(s_6) = 5/27$.

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Call the orderings $s_1 = \{1, 2, 3\}$, $s_2 = \{1, 3, 2\}$, $s_3 = \{2, 1, 3\}$, $s_4 = \{2, 3, 1\}$, $s_5 = \{3, 1, 2\}$, $s_6 = \{3, 2, 1\}$.

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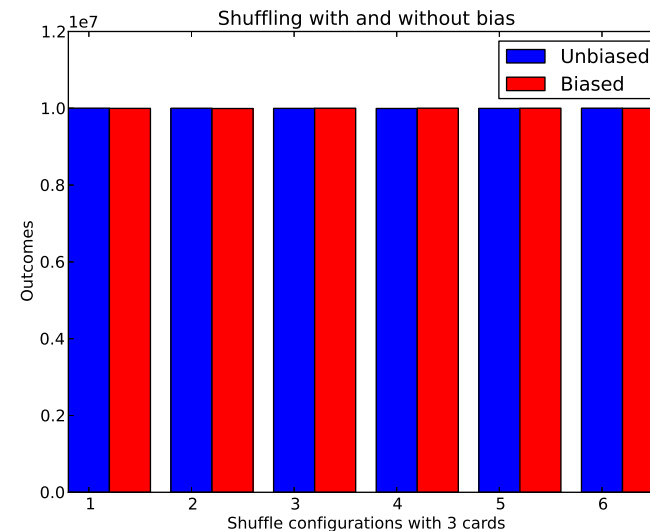
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Starting at ordering	$P(s_1)$	$P(s_2)$	$P(s_3)$	$P(s_4)$	$P(s_5)$	$P(s_6)$
s_1	4/27	5/27	5/27	5/27	4/27	4/27
s_2	5/27	4/27	4/27	4/27	5/27	5/27
s_3	5/27	5/27	4/27	5/27	4/27	4/27
s_4	4/27	4/27	5/27	4/27	5/27	5/27
s_5	5/27	5/27	4/27	4/27	4/27	5/27
s_6	4/27	4/27	5/27	5/27	5/27	4/27

It can be seen that the expected probability of getting any ordering is equal to that of any other as is evidenced by the histogram.



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- If you use it in your online poker game you are going to lose money.
- We will first show why sorting three card cards with this method yields a biased result.
- We then illustrate experimentally how badly it is biased.

Shuffling Cards—by drawing randomly from $\{1, 2, \dots, N\}$.

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Shuffling Cards—ideal shuffle

- Start with a deck cards, this time represented by an array of integers $\{0, 2, \dots, 51\}$. Initialize the array using:

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- In the following round we consider the list $\{c_1, c_2, \dots, c_{n-2}\}$ and choose an item called c_j that we swap with c_{n-2} that will never be moved again.
- We carry on like this until only one element is left,

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- We carry on like this until only one element is left, so there is no choice left, and it remains where it is.
- Next we try to program the selection and replacement.

Shuffling Cards—ideal shuffle

- Next we generalize the selection steps.
- Considering the lists $\{c_1, c_2, \dots, c_{n-i}\}$ for values of $i = 0, 1, \dots, n - 1$.
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```
1 for (i=1; i < N; i++) {  
2     j = i + (int)(Math.random()*(N-i) + 1;  
3     swap(card[N-i+1], card[j]); }
```

Shuffling Cards—another attempt

- Start by initializing the deck

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1 for (i=1; i <=n; i++)  
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- Replace the old array cards with the new array:
 $card[r + 1], \dots, card[n - 1], card[n], card[1], \dots, card[r]$

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- Does this method work?

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- Repeated this m times.
- What is the time complexity?
- Each iteration costs $O(n)$, and this must be repeated n times, so the time complexity is $mO(n) = O(mn)$.
- Does this method work? Of course it merely rearranges the position of the first element and does not shuffle the deck.

Shuffling Cards—riffing

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```
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Shuffling Cards—riffing

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```

- Generate a random number $r \in \{1, 2, \dots, n\}$.
- Then cut the cards at $card[r]$ to create two more or less half packs:
 $\{card[1], \dots, card[r]\}$ and $\{card[r + 1], \dots, card[n - 1], card[n]\}$

Shuffling Cards—riffing

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The cards also need to CUT more or less in the middle and ROTATED.

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- So, the expected number of probes is $\frac{N+1}{2}$.
- We can instead say that that average number of probes is $O(N)$.

The exact number of bits needed to store a digit

Bits in N $\lceil \log_2 N \rceil$	N	N as $2^i - 1$	digits exact $\lfloor \log_{10} N \rfloor$	$\log_{10} N$
1	1	$2^1 - 1$	0	0.000
2	3	$2^2 - 1$	0	0.301
3	7	$2^3 - 1$	0	0.845
4	15	$2^4 - 1$	1	1.176
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16	65535	$2^{16} - 1$	4	4.816
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Why is the 32-bit number interesting?

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- i.e. number of *bits* $\times 0.3$,
i.e. $n \times 0.3$ gives the accuracy in *digits*.
- To find the number of *bits* needed to store m *digits*, simply calculate $\lceil m/0.3 \rceil$, e.g. to store 7 digits, use $\lceil 7/0.3 \rceil = \lceil 23.333333 \rceil = 24$ bits.

The bits needed to store decimal number

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- A 21-bit number can represent a sign and 6 decimal digits.

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- The number of times a list of length n can be halved until its length is 0 is thus $\lceil \log_2 n \rceil$.

How long does binary search take?

- Suppose we are looking for *item* in the ordered, i.e. sorted, list $d[1], d[2], \dots, d[n]$.

```
1 def binarySearch(d, item):
2     left = 1; right = len(d) # which is n
3     while right >= left:
4         middle = (left + right)/2
5         if item == d[middle]:
6             return middle
7         else:
8             if item < d[middle]:
9                 right = middle-1
10            else:
11                left = middle+1
12    return -left # the item was not found
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- Each iteration of the **while** loop halves the length of the list that is being searched. This can only happen $\lceil \log_2 n \rceil$ times.

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- For sorted lists, binary search takes $O(\log n)$, but inserting `item` in its correct position—maintaining the sorted order—requires: (1) a look up to find where the item must be inserted in $O(\log n)$ time, and (2) on average half of the list must be moved down one place to make place for the new item, in $\frac{n+1}{2} = O(n)$ time.

Searching, Insertion and Deletion in lists

Repeating some of the previous slide:

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- So while *binary search* is excellent for looking items up, it is tardy when new items are inserted, as opposed to *linear search* which is slow for looking up, but extremely fast, i.e. $O(1)$ for appending new items at the end of the list.

$O(1)$ Searching, Insertion and Deletion— $\text{id} \in [1..m]$

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- Deletion is similarly efficient. To delete an item: (1) it must be marked as deleted, and (2) its id must be added to the list of available ids , so that this id is later available for re-use.
- The method is not very practical if the identifiers have already been allocated—we will *hash* the ids .

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- Suppose there are n persons or entries and we have a table with m positions or slots. Now n is at most m .

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- Insertion is done as follows—assuming id is not in list:

```
1  i = hash(id)
2  while d[i].isFilled:
3      i = (i + p) % m + 1
4  d[i].key = id
5  d[i].isFilled = true
```

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Insert an item into the hash table as follows:

Insert item directly when the *slot is open*

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1  i = hash(id)
2  while d[i].isFilled and d[i].key != key:
3      i = (i + p) % m + 1
4  if d[i].key == id:
5      update(d[i], item)
6  else:
7      d[i].key = id
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The $\bmod m + 1$ at the end ensures that $i_k \in [1..m]$.

A hash function with no collisions is called a *perfect hash* function.

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- Suppose k entries have been made, then the probability of a collision is α . The probability of a collision on the i^{th} probe is α^i . The probability of finding an entry in precisely i probes is $\alpha^{i-1}(1 - \alpha)$.

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$$\begin{aligned} &= \sum_{i=0}^n i\alpha^{i-1}(1 - \alpha) \\ &\leq (1 - \alpha) \sum_{i=0}^{\infty} i\alpha^{i-1} \\ &= \frac{(1 - \alpha)}{(1 - \alpha)^2} \\ &= \frac{1}{1 - \alpha} \end{aligned}$$

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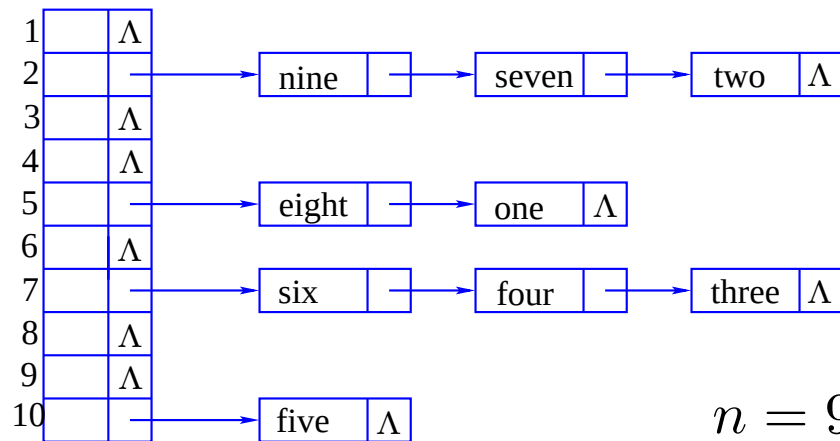
$$\leq (1 - \alpha) \sum_{i=0}^{\infty} i\alpha^{i-1}$$

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Loading as a %	α	<i>Average Search Length</i> $\frac{1}{1-\alpha}$
50%	0.50	$1/0.50 = 2.000$
75%	0.75	$1/0.25 = 4.000$
85%	0.85	$1/0.15 = 6.667$
95%	0.95	$1/0.05 = 20$
99%	0.99	$1/0.01 = 100$
99.9%	0.999	$1/0.001 = 1000$
99.99%	0.9999	$1/0.0001 = 10000$

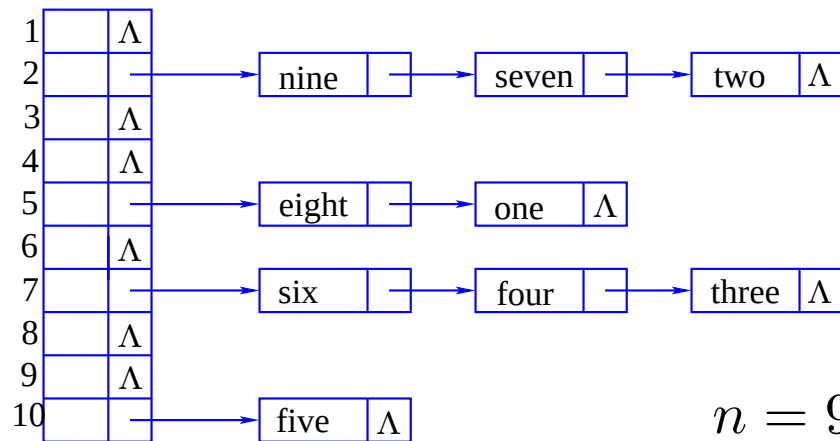
Chained hash tables



$n = 9$ and $m = 10$

- The table $d[m+1]$ has m positions and n entries.

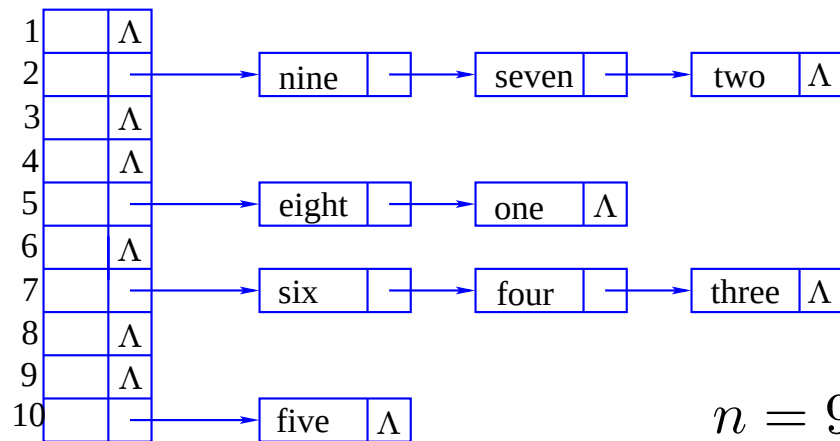
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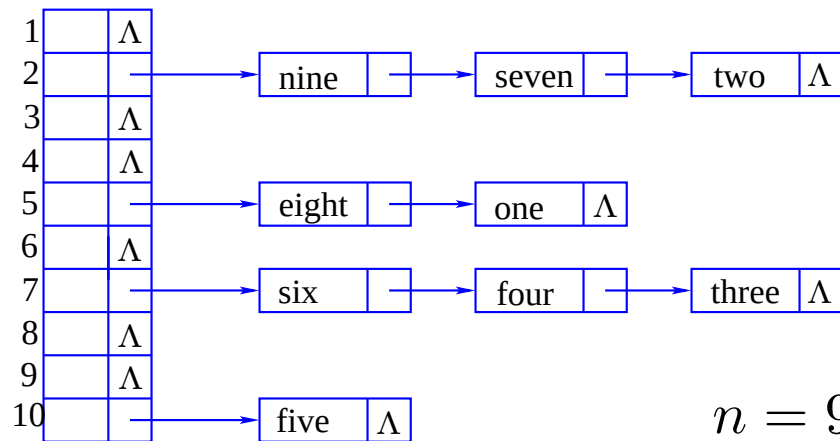
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Chained hash tables



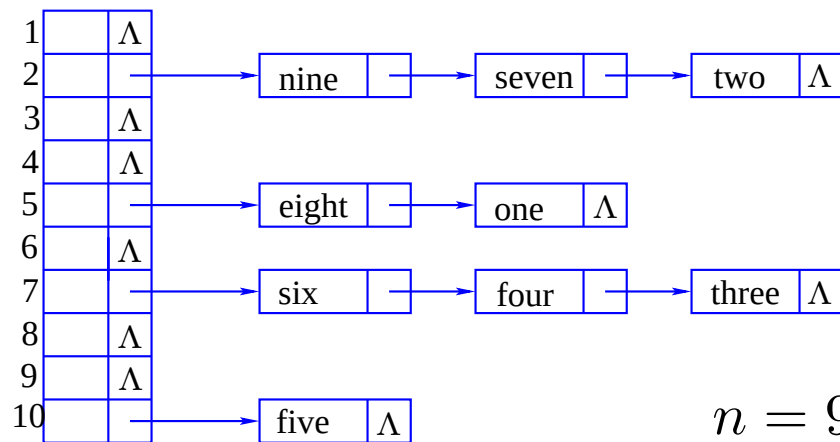
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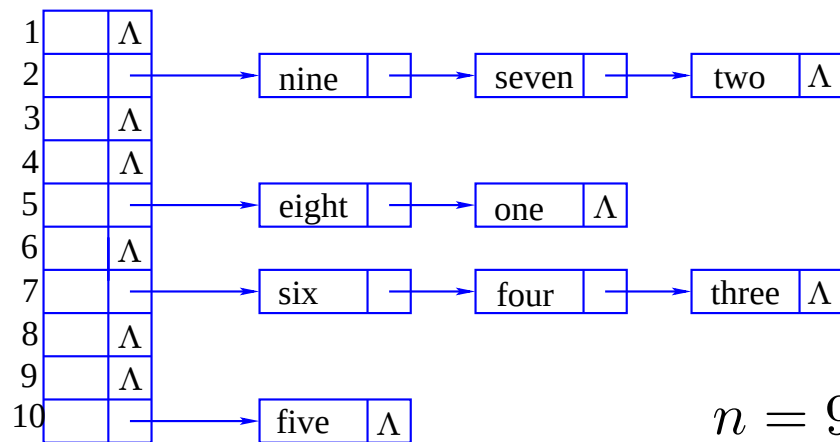
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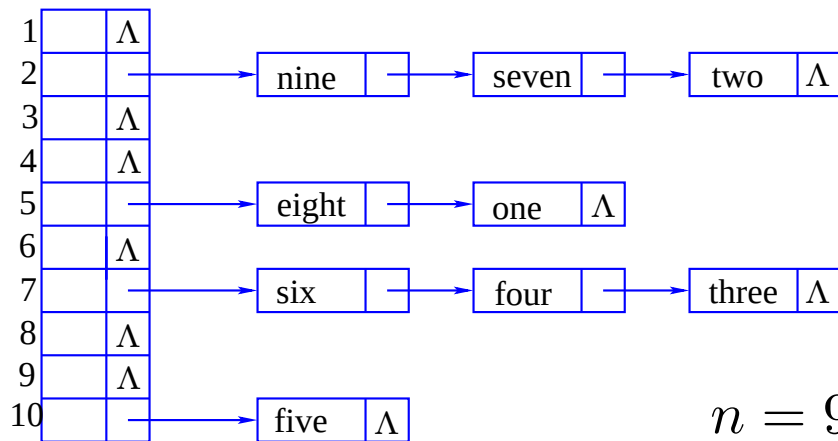
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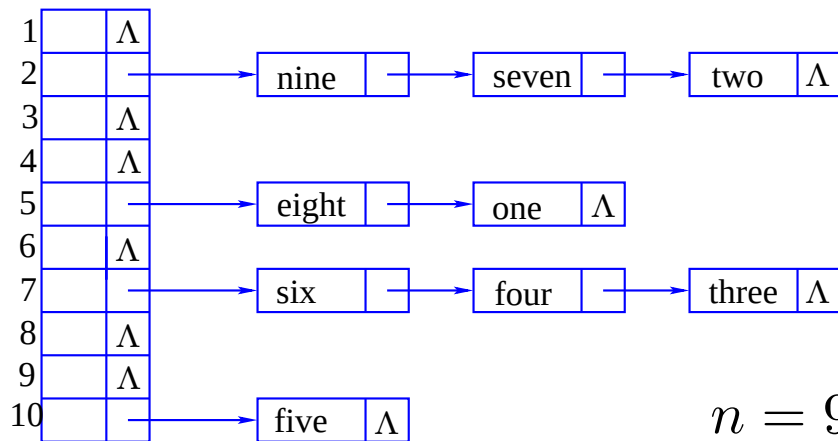
Chained hashing—example



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- Set up the hash table by entering keys: **two**, **five**, **three**, **four**, **six**, **one**, **seven**, **nine**, **eight** in this order.

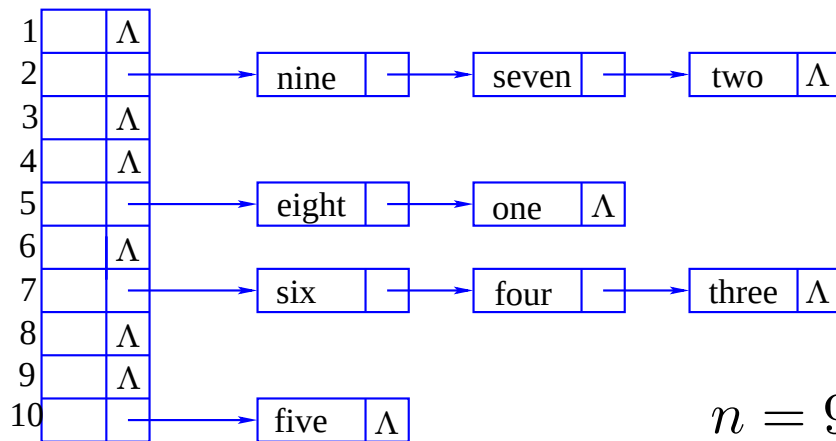
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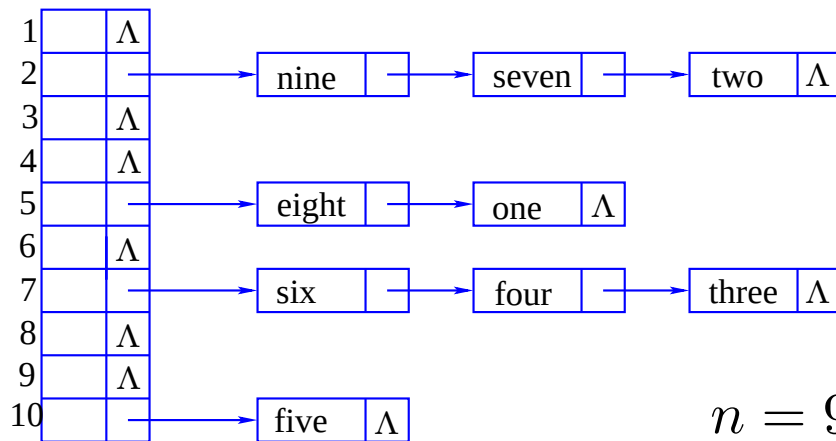
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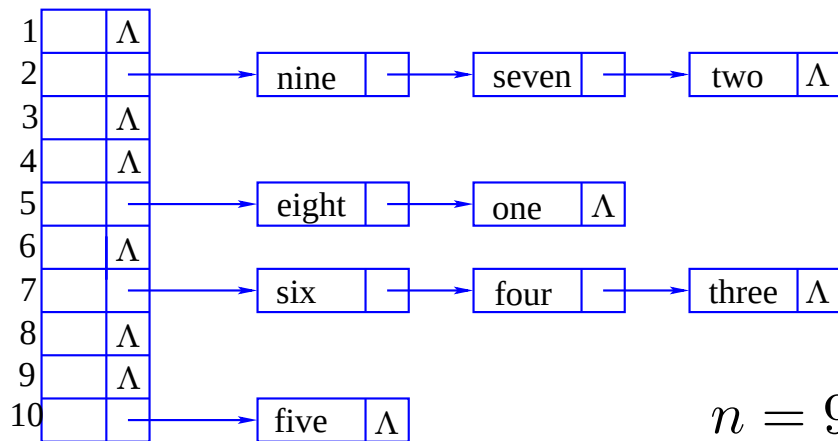
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- Initially there are $n = 0$ items in the store. Let us enter **two**, assume `id = "two"`, so, first hash it: `i = hash(id)` giving `i = 2`. Enter `id` at `++n`, in the store, by: `store[n] = <id, 0>`.

Chained hashing—example



- Set up the hash table by entering keys: **two**, **five**, **three**, **four**, **six**, **one**, **seven**, **nine**, **eight** in this order. Other orders will give logically equivalent tables, but their links will be different.
- Initially there are $n = 0$ items in the store. Let us enter **two**, assume `id = "two"`, so, first hash it: `i = hash(id)` giving `i = 2`. Enter `id` at `++n`, in the store, by: `store[n] = <id, 0>`. Check if the table entry **2** is open or filled by examining `d[i]`. If `d[i] != 0`, i.e. `d[i]` is already occupied, the link **0** in the node `<id, 0>` must be replaced by `d[i]` as in `<id, d[i]>`, and `d[i] == n`.

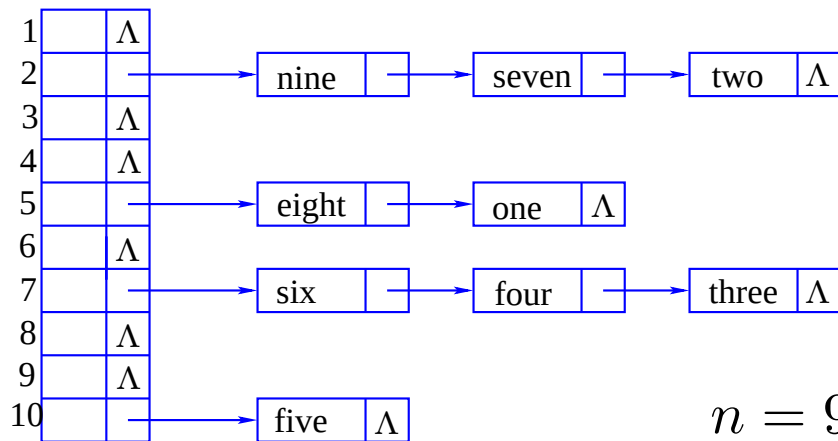
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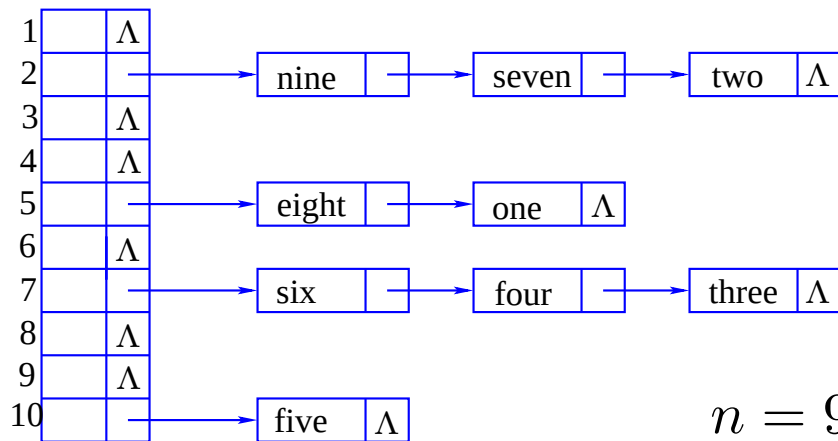


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- Now enter **five**.

Chained hashing—example



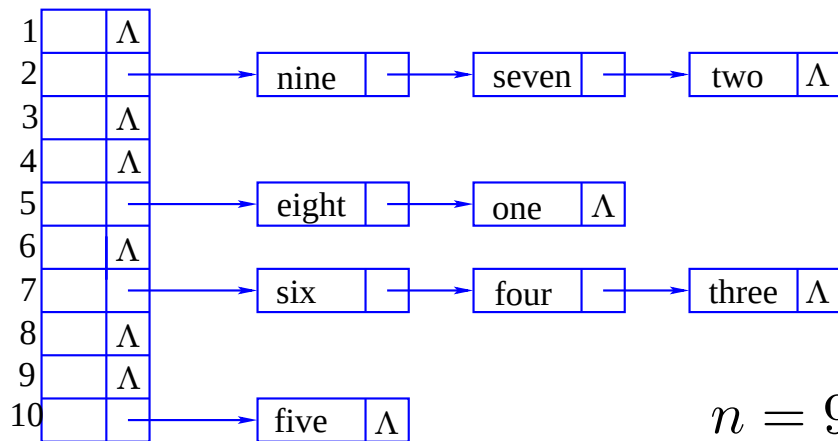
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Set up the hash table by entering keys: **two**, **five**, **three**, **four**, **six**, **one**, **seven**, **nine**, **eight** in this order.

- Now enter **five**.

There is now $n = 1$ item in the store. Entering **five**, assume $id = \text{"five"}$, so, first hash it: $i = \text{hash}(id)$ For this table $i = 10$. The data, id , may be placed at the next open place, $++n$, making $n = 2$ in the store, by: $\text{store}[n] = \langle id, 0 \rangle$ Check if the table entry i is open or filled by examining $d[i]$. Since $d[i] == 0$, i.e. $d[i]$ is *not* occupied, the link 0 in the node $\langle id, 0 \rangle$ is used as in $\langle id, 0 \rangle$, and $d[i] = n == 2$.

Chained hashing—example

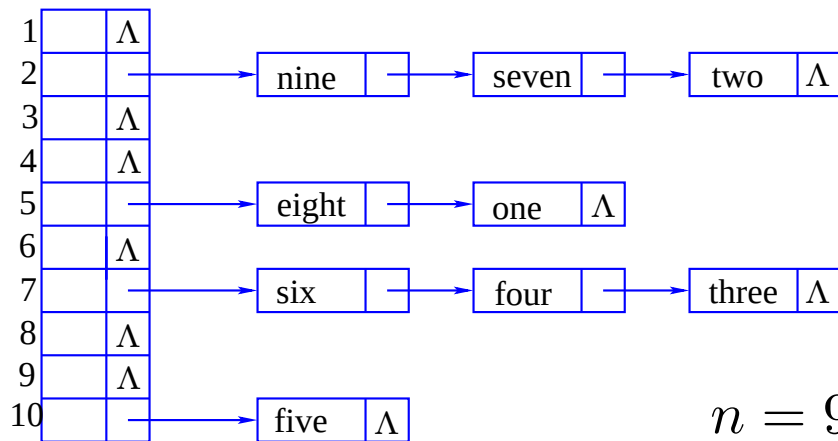


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- Next enter **three**.

Chained hashing—example

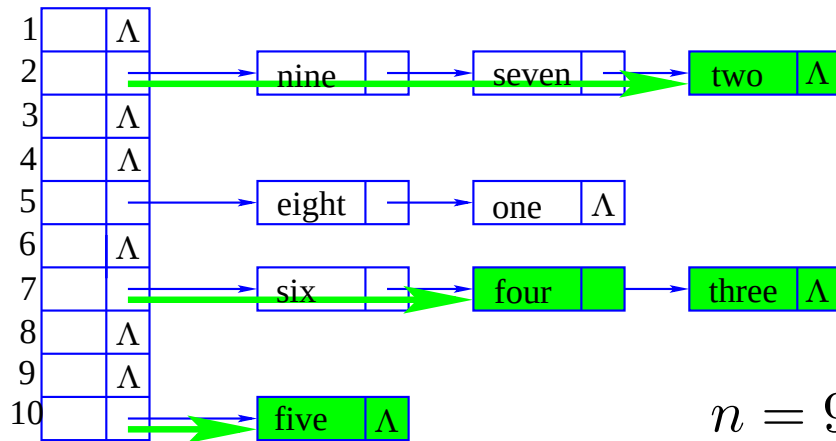


$n = 9$ and $m = 10$

Set up the hash table by entering keys: **two**, **five**, **three**, **four**, **six**, **one**, **seven**, **nine**, **eight** in this order.

- Next enter **three**.
- There are $n = 2$ items in the store. Entering **three**, put `id = "three"`, so, first hash it: `i = hash(id)`; For this table `i = 7`. The data, `id`, may be placed at the next open place, `++n`, making `n = 3` in the store, by: `store[n] = <id, 0>`; The table entry `d[i]` is still open, so the link `0` in the node `<id, 0>` is used and `d[i] = n == 3`;

Chained hashing—example: Enter **four**

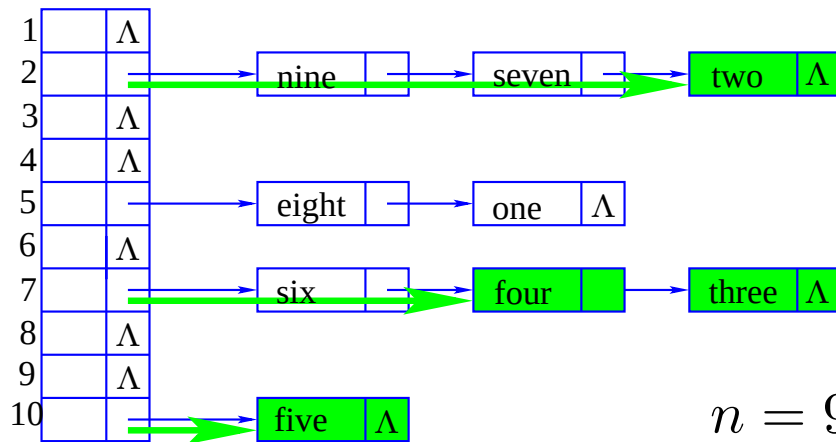


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Set up the hash table by entering keys: two, five, three, four, six, one, seven, nine, eight in this order.

- Now enter **four**.

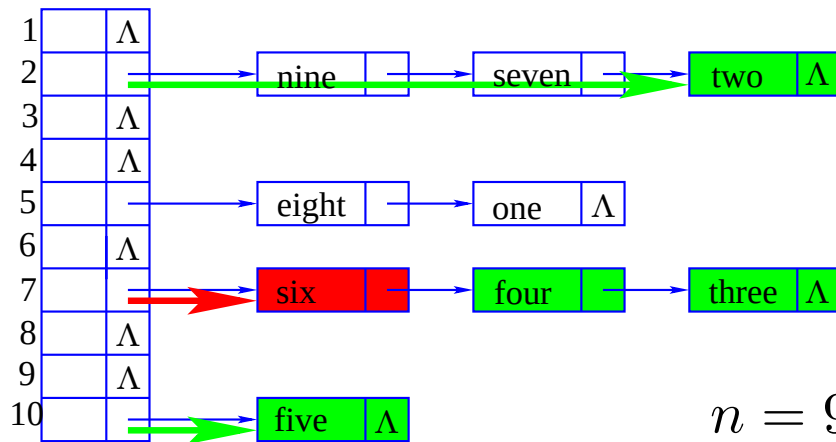
Chained hashing—example: Enter four



Set up the hash table by entering keys: two, five, three, four, six, one, seven, nine, eight in this order.

- Now enter **four**.
- There are now $n = 3$ item in the store. Entering **four**, assume $id = \text{"four"}$, so, first hash it: $i = \text{hash}(id)$; For this table $i = 7$. The data, id , may be placed at the next open place, $++n$, makes $n = 4$ in the store, by: $\text{store}[n] = \langle id, \emptyset \rangle$; Check if the table entry i is open or filled by examining $d[i]$. Since $d[i] \neq \emptyset$, i.e. $d[i]$ is *occupied*, the link $\emptyset$ in the node $\langle id, \emptyset \rangle$ must be replaced by $d[i]$ as in $\langle id, d[i] \rangle$; and $d[i] == n$;

Chained hashing—example: Enter **six**

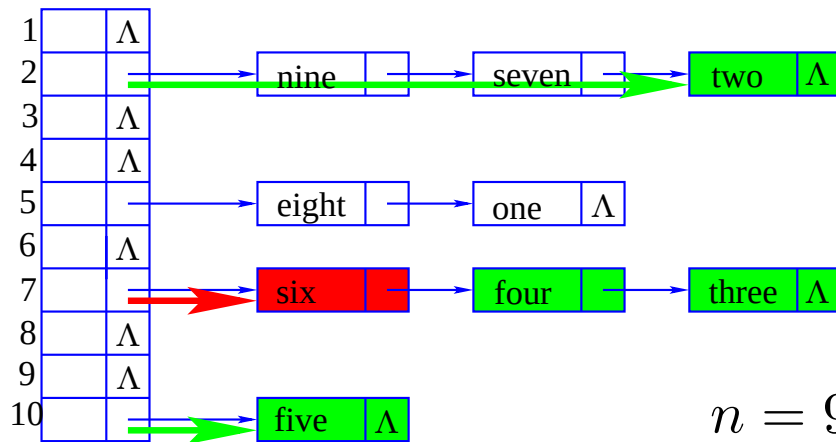


$$n = 9 \text{ and } m = 10$$

Set up the hash table by entering keys: two, five, three, four, six, one, seven, nine, eight in this order.

- Now enter **six**.

Chained hashing—example: Enter **six**



$n = 9$ and $m = 10$

Set up the hash table by entering keys: two, five, three, four, six, one, seven, nine, eight in this order.

- Now enter **six**.
- There are now $n = 4$ item in the store. Entering **six**, assume **id** = "six", so, first hash it: $i = \text{hash}(\text{id})$; For this table $i = 7$. The data, **id**, may be placed at the next open place, $++n$, makes $n = 5$ in the store, by: $\text{store}[n] = \langle \text{id}, \emptyset \rangle$; Check if the table entry i is open or filled by examining $d[i]$. Since $d[i] \neq \emptyset$, i.e. $d[i]$ is *occupied*, the link $\emptyset$ in the node $\langle \text{id}, \emptyset \rangle$ must be replaced by $d[i]$ as in $\langle \text{id}, d[i] \rangle$; and $d[i] == n$;

Chained hashing—hashInsert—array implementation

The hash table has m entries or slots that each contains the index of the key of the data that has been hashed to that slot.

Chained hashing—hashInsert—array implementation

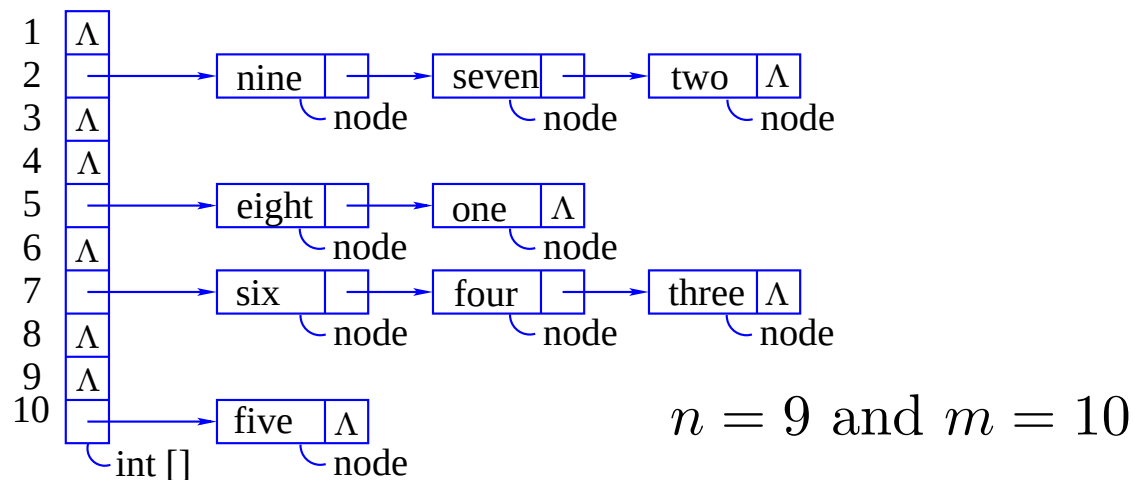
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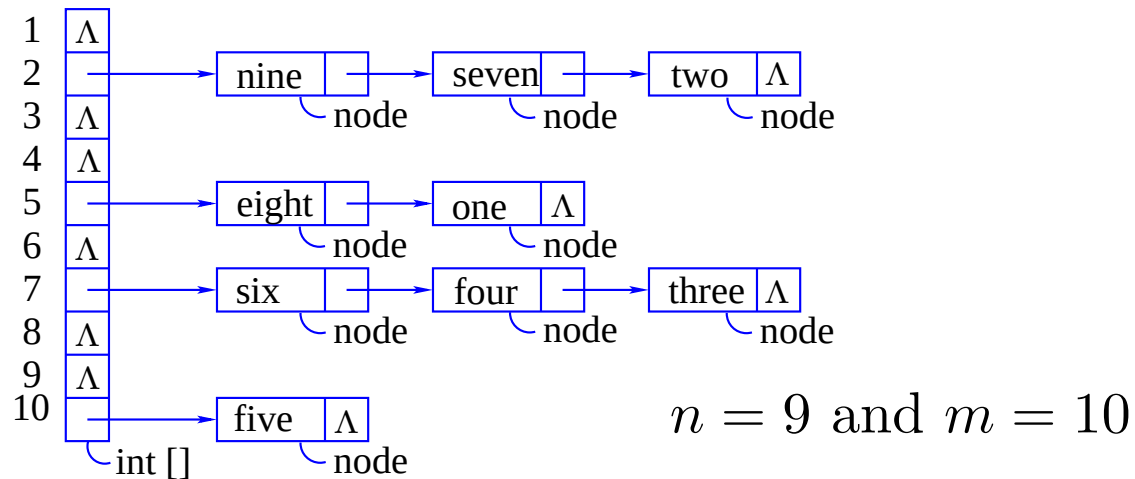
The data itself is stored in a storage pool of nodes.



Java-like pseudocode to create the table and the storage pool.

```
1 int d[] = new int [m+1];  
2 node store[] = new node [6*m +1];  
3 int n = 0;
```

Chained hashing—hashInsert—array implementation



```
1 int d[] = new int [m+1]; // Java-like pseudocode
2 node store[] = new node [6*m +1];
3
4 int n = 0;
5
6 void hashInsert(string id) {
7     ++n;
8     store[n].key = id;
9     i = hash(id);
10    store[n].link = d[i];
11    d[i] = n; }
```


Chained hashing—hashSearch

This method is called as in the example:

```
1 boolean found = hashSearch("ten");
```

The method follows

```
1 boolean hashSearch(string id) {  
2   int i = hash(id);  
3   i = d[i];  
4   while (i != 0 && store[i].key != id) {  
5     i = store[i].link; }  
6   return store[i].key == id; }
```

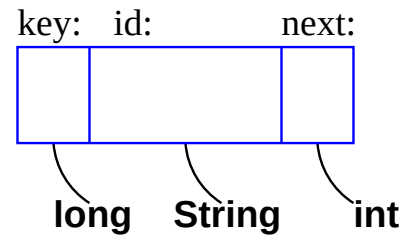
Chained hashing into an array of nodes—Java for the main class

The coding of some of the methods clarify some of our doubtful pseudo code constructs.

```
1 public class tryHash {
2 public static void main (String args[]) {
3 String data [] =
4 {"zero", "one", "two", "three", "four", "five",
5     "six", "seven", "eight", "nine", "ten",
6     "eleven", "twelve", "thirteen", "fourteen", "fifteen",
7     "sixteen", "seventeen"};
8 int k;
9 tableHash H;
10
11     H = new tableHash();
12
13     for (k = 0; k < data.length; k++)
14         H.hashInsert(data[k]);
15     H.display(); }}
```

Here we have declared a reference or variable `H` to a hash table, and then instantiated or created it using the constructor `tableHash()`. Next, we insert the data items one-by-one into the `H`, and finally display the table.

Chained hashing into an array of nodes—the **node** class in Java



```
1 public class node {
2
3     long key;
4     String id;
5     int next;
6
7     public node () {
8         key = 0L;
9         id = "";
10        next = 0; }
11
12    public node (long k, String s) {
13        key = k;
14        id = s;
15        next = 0; } }
```

Chained hashing into an array—the `tableHash()` constructor

```
1 public class tableHash {
2   int m = 10;
3   int d[] = new int [m+1];
4   node store[] = new node [6*m + 1];
5   int n = 0;
6
7   public tableHash () { // the table constructor
8
9     for (int i=0; i < m + 1; i++)
10       d[i] = 0;
11
12     for (int j = 0; j < 6*m +1; j++)
13       store[j] = new node(); }
```

First `m` is defined as 10, next space is reserved for the hash table of integers `d[]`. Space is made for `store[]`, and `n` is set to zero.

Chained hashing into an array—the `tableHash()` constructor

```
1 public class tableHash {
2   int m = 10;
3   int d[] = new int [m+1];
4   node store[] = new node [6*m + 1];
5   int n = 0;
6
7   public tableHash () { // the table constructor
8
9     for (int i=0; i < m + 1; i++)
10       d[i] = 0;
11
12     for (int j = 0; j < 6*m +1; j++)
13       store[j] = new node(); }
```

First `m` is defined as 10, next space is reserved for the hash table of integers `d[]`. Space is made for `store[]`, and `n` is set to zero.

In the `tableHash()` constructor the hash table is set to zeros and the pool of nodes, called `store[]`, is filled with instantiated nodes using the `node()` constructor.

A hashing function $h(id)$ in Java

```
1 int h (String id) {  
2   int hash = 0;  
3  
4   for (int ch = 0; ch < id.length(); ch++)  
5     hash = (hash + (int)id.charAt(ch)) % m;  
6  
7   return hash + 1; }
```

The function $h(id)$ simply sums the numerical or ASCII values of each character and gets $\text{mod } m$ at each step. The value of the running hash will lie in the range $[0..m-1]$. The addition of 1 puts the result back into the range $[1..m]$.

A method to display the entire hash table in Java

```
1
2 void display() {
3     int i, j;
4     System.out.printf("The_hash_table:\n");
5     for (i = 1; i <= m; i++) {
6         j = d[i];
7         System.out.printf("%d:_", j);
8         while (j > 0) {
9             System.out.printf("-->%s", store[j].id);
10            j = store[j].next; }
11            System.out.printf("-->%d\n", j); }
12    System.out.printf("-----\n"); }
```

The `display()` method runs through the “pointers” in `d[]` one-by-one and follows the “pointers” until the end of the chain is reached.

The `hashInsert(id)` and `hashSearch(id)` methods in Java

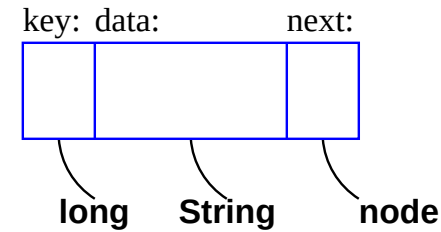
```
1 void hashInsert(String id) {
2   int i;
3   ++n;
4   store[n].key = (long)n;
5   i = h(id);
6   store[n].next = d[i];
7   store[n].id = id;
8   System.out.printf("insert_%s_at_%d\n" , id, n);
9   System.out.printf("%s\n" , store[n].id);
10  d[i] = n; }
11
12 boolean hashSearch(String id) {
13  int i = h(id);
14  i = d[i];
15  while (i != 0 && store[i].id != id) {
16    i = store[i].next; }
17  return store[i].id == id; } }
```


Chained hashing—hashInsert—list implementation

Now we alter the **node** so that it has a **next** pointer that is a **node** reference instead of an **integer**.

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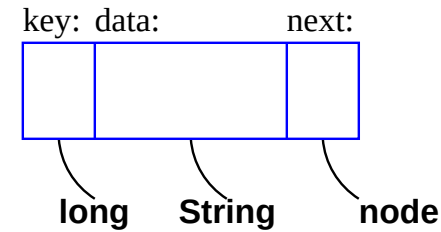


```
1 node d[] = new node [m+1];
2 int n = 0;
3
4 void hashInsert(long key, String id) {
5     node n = new node (key, id);
6     i = hash(id);
7     n.next = d[i];
8     d[i] = n; }
```

Note that the array `store[]` is no longer needed.

Chained hashing—hashInsert—list implementation

Now we alter the **node** so that it has a **next** pointer that is a **node** reference instead of an **integer**.



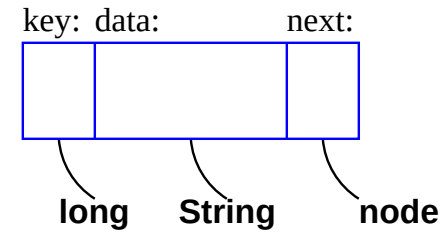
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3
4 void hashInsert(long key, String id) {
5     node n = new node (key, id);
6     i = hash(id);
7     n.next = d[i];
8     d[i] = n; }
```

Note that the array `store[]` is no longer needed.

The program creates new `nodes` on-the-fly as they are required.

Chained hashing—hashInsert—list implementation

Now we alter the **node** so that it has a **next** pointer that is a **node** reference instead of an **integer**.



```
1 node d[] = new node [m+1];
2 int n = 0;
3
4 void hashInsert(long key, String id) {
5     node n = new node (key, id);
6     i = hash(id);
7     n.next = d[i];
8     d[i] = n; }
```

Note that the array `store[]` is no longer needed.

The program creates new `nodes` on-the-fly as they are required.

The array `d[]` appears to be a list of nodes, but it is actually a list of references to nodes.

Chained hashing—hashSearch—using lists

This method is called as in the example:

```
1 boolean found = hashSearch(10L);
```

The method follows

```
1 boolean hashSearch(long key, String id) {  
2   int i = hash(key);  
3   node here = d[i];  
4   while (here != null && here.key != key)  
5       here = here.next;  
6   return here != null && here.key == key; }
```

A class for a node

Suppose we have a **node** class, the declaration:

```
1 node n;
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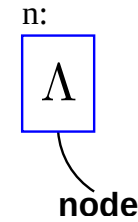
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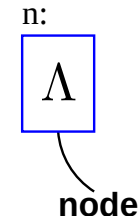


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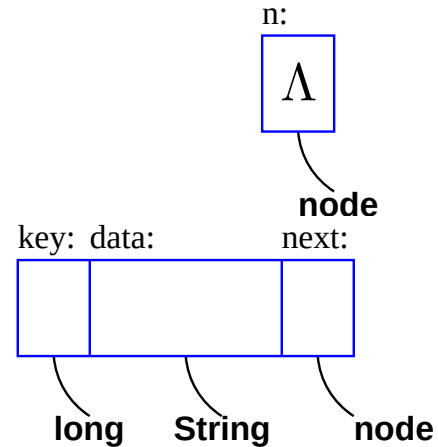
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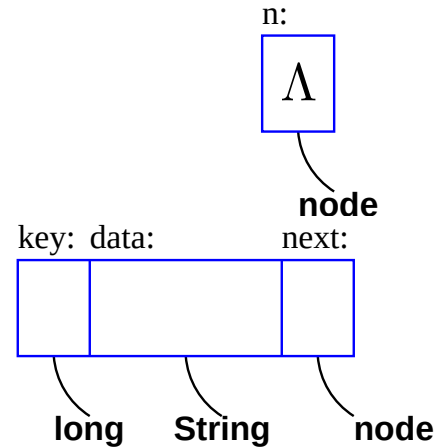
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When the method **n = new node();** is invoked an instance of the node is created and a pointer that points to that instance is placed into the variable **n**.

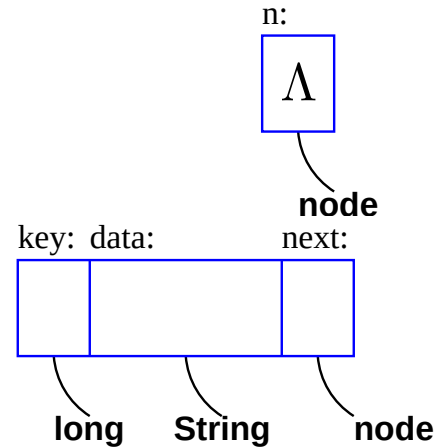
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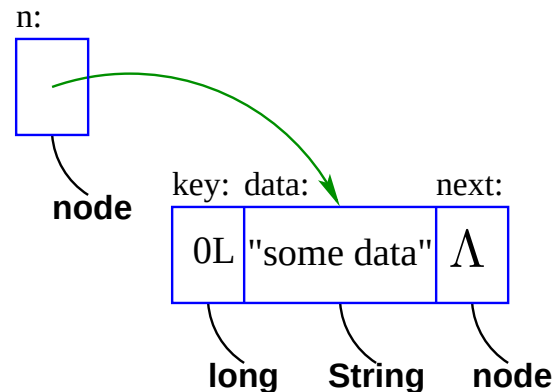
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1 node n;
```

It creates the variable **n** of type **node**

We depict the instance of a **node** graphically using



When the method **n = new node();** is invoked an instance of the node is created and a pointer that points to that instance is placed into the variable **n**.



class node

```
1 public class node{
2     long key;
3     String data;
4     node next;
5
6     public node(long k, String d){
7         key = k;
8         data = d;
9         next = null; }
10
11    public node(){
12        key = 0;
13        data = "";
14        next = null; }
15
16    public String toString() {
17        return "<" + key + ", " + data +
18                ", " + next + ">" } }
```

class linkedList

```
1 public class linkedList{
2
3     protected static int length;
4     protected node head;
5
6     public linkedList(){
7         head = new node (0, "head_node");
8         length=0; }
9
10    public void insert(node n){
11        n.next = head.next;
12        head.next = n;
13        length++; }
14
15    public int length(){
16        return length; }
```

class LinkedList continued

```
1 public boolean isEmpty(){
2     return length==0; }
3
4 public void clear(){
5     head.next = null;
6     length = 0; }
7
8 public void display(){
9     node here = head.next;
10    while (here != null) {
11        System.out.println(
12            "Key_=" + here.key +
13            "_Data_=" + here.data);
14        here = here.next; }
15    System.out.println("length=" + length); } }
```

class tryList: Get going

```
1 import java.lang.Math.*;
2 import java.io.*;
3
4 public class tryList {
5     public int n = 2;
6     public static void main(String args[]) {
7         linkedList L = new linkedList();
8         System.out.println(
9             "L.Head.Key=_ " + L.head.key +
10            "_L.Head.Data=_ " + L.head.data +
11            "_L.Length=_ " + L.length +
12            "_L.Head.Next=_ " + L.head.next);
13         node item;
14         long id;
15         String studentName;
16         L.display();
17         L.head.next = null; // L.clear(); is better
```


class tryList Check the arguments

```
1 String theFileName = args[0];
2 if (args.length == 1) {
3     try {
4         FileInputStream inFileName
5             = new FileInputStream (theFileName);
6         DataInputStream inFile
7             = new DataInputStream (inFileName);
8         System.out.println(
9             "File:_" + theFileName
10             + "_successfully_opened");
11         // proceed with processing
```

class tryList Do the processing

```
1  int lineNumber = 0;
2  // Process first line
3  //---the header---differently
4  System.out.println(inFile.readLine());
5  // // readLine() deprecated
6  // Process rest of file
7  while (inFile.available() != 0){
8      String inString = inFile.readLine();
9      String outString; // readLine() deprecated
10     int commaPos = inString.indexOf(",");
11     // One space: handle Da Costa, Van Wyk, Le Roux, ...
12     // Not treated: van der Merwe, de la Querra, etc.
13     // Before processing
14     //2410832 VAN ZITTERS, ST ... 000000
15     // Final output for this record:
16     // Key = 2410832 Data = Van Zitters, ST
17     int spacePos = inString.substring
18         (10, commaPos).indexOf("_") + 10;
19     int minusPos = inString.substring
20         (10, commaPos).indexOf("-") + 10; }
```

class tryList Continue processing

```
1 if (spacePos > 10)
2   outString = inString.substring(0, 10)
3   + inString.substring(10, spacePos).toLowerCase()
4   + inString.substring(spacePos, spacePos+2)
5   + inString.substring(spacePos+2, commaPos).toLowerCase()
6   + inString.substring(commaPos, inString.length());
7 else if (minusPos > 10) {
8   System.out.println(minusPos);
9   outString = inString.substring(0, 10)
10  + inString.substring(10, minusPos+1).toLowerCase()
11  + inString.substring(minusPos+1, minusPos+2)
12  + inString.substring(minusPos+2, commaPos).toLowerCase()
13  + inString.substring(commaPos, inString.length()); }
14 else outString = inString.substring(0, 10)
15   + inString.substring(10, commaPos).toLowerCase()
16   + inString.substring(commaPos, inString.length());
17 studentName = outString.substring(9, 71).trim();
18 System.out.println(outString + "_" + outString.length());
```

class tryList Build linked list

```
1  // Build the linked list:
2  id = (long)Integer.parseInt(
3      outString.substring(1,8));
4  // first 7 characters: the student number
5  item = new node (id, studentName);
6  L.insert (item);
7  }
8  inFile.close();
9  L.display(); }
10 catch (Exception e) {
11     System.err.println("File:_" + theFileName
12         + "_error_occurred"); } }
13 else {
14     System.err.println("There_are_" + args.length
15         + "parameters,_there_should_be_1"); } } }
```

The **lookup** method

```
1 public node lookup (long key) { //tiny
2   node here = this.head.next,
3     prev = here;
4   while (here != null && here.key != key) {
5     prev = here;
6     here = here.next; }
7   return prev; }
```

The Makefile

```
1 % make $ mean \$ in lstlisting
2 OBJ = tryList
3 LIST = linkedList
4 DEP = node
5 DATA = cos224.list
6
7 all:$(DEP).class, $(LIST).class
8     -rm errors
9     javac -deprecation $(OBJ).java 2> errors
10    -rm output
11    java $(OBJ) $(DATA) > output
12
13 $(DEP).class, $(LIST).class:
14     javac -deprecation $(LIST).java
15     javac -deprecation $(DEP).java
16
17 clean:
18     -rm *~ *.class errors output
```

Improved `class linkedList`

```
1 public class linkedList{
2
3     protected static int length;
4     protected node head;
5
6     public linkedList(){
7         head = new node (0, "head_node");
8         length=0; }
9
10    public void insert(node n){
11        n.next = head.next;
12        head.next = n;
13        length++; }
14
15    public int length(){
16        return length; }
```

Improved class `LinkedList` continued

```
1 public boolean isEmpty(){
2     return length==0; }
3
4 public void clear(){
5     head.next = null;
6     this.length = 0; }
7
8 public void display(){
9     node here = this.head.next;
10    while (here != null) {
11        System.out.println(
12            "Key_=_ " + here.key +
13            "_Data_=_ " + here.data);
14        here = here.next; }
15    System.out.println(
16        "length=" + this.length); } }
```


The `class node` revised

This version of `class node` is not properly encapsulated—one can access parts of the class that should be invisible outside of the `node` class.

```
1 public class node{
2     long key;
3     String data;
4     node next;
5
6     public node(long k, String d){
7         key = k;
8         data = d;
9         next = null; }
10
11    public node(){
12        key = 0;
13        data = "";
14        next = null; }
15
16    public String toString() {
17        return "<" + key + ",_" + data + ",_" + next + ">" } }
```

For example the `insert` method in the `linkedList` class can access the next `next` field directly.

```
1 public void insert(node n){
2     n.next = head.next;
3     head.next = n;
4     length++;
5 }
```

The class `node` revised

The fields inside `node` should be made `private` or `protected` and methods to `get` and `set` them should be supplied.

```
1 public class node{
2     private long key;
3     private String data;
4     private node next;
5
6     public node(long k, String d){
7         key = k;
8         data = d;
9         next = null; }
10
11     public node(){
12         key = 0;
13         data = "";
14         next = null; }
15
16     public void setKey(long k) {
17         key = k; }
18
19     public long getKey() {
20         return key; } . . . }
```

```
1     public long getKey() {
2         return key; }
3
4     public String getData() {
5         return data; }
6
7     public node getNext() {
8         return next; }
9
10    public void setKey(long k) {
11        key = k; }
12
13    public void setData(String d) {
14        data = d; }
15
16    public void setNext(node n) {
17        next = n; }
```

The class node revised

Now the `LinkedList` method `insert` becomes

```
1 public void insert(node n){  
2     \ n.next = head.next;  
3     n.setNext(head.getNext());  
4  
5     \ head.next = n;  
6     head.setNext(n);  
7  
8     length++; }
```

Linked list—insert (node n)

Start with an instantiated list

```
1 linkedList L;  
2 L = new linkedList();
```

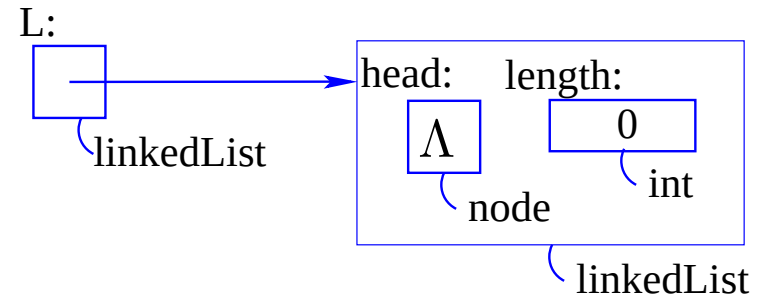
Insert a "first" node

```
1 node n = new node(1L, "first")  
2 L.insert(n);
```

Insert a "second" node

```
1 node n = new node(2L, "second")  
2 L.insert(n);
```

```
1 public void insert(node n){  
2   if (head == null)  
3     head = n;  
4   else {  
5     n.next = head;  
6     head = n;  
7     length++; }  
}
```



Linked list—insert (node n)

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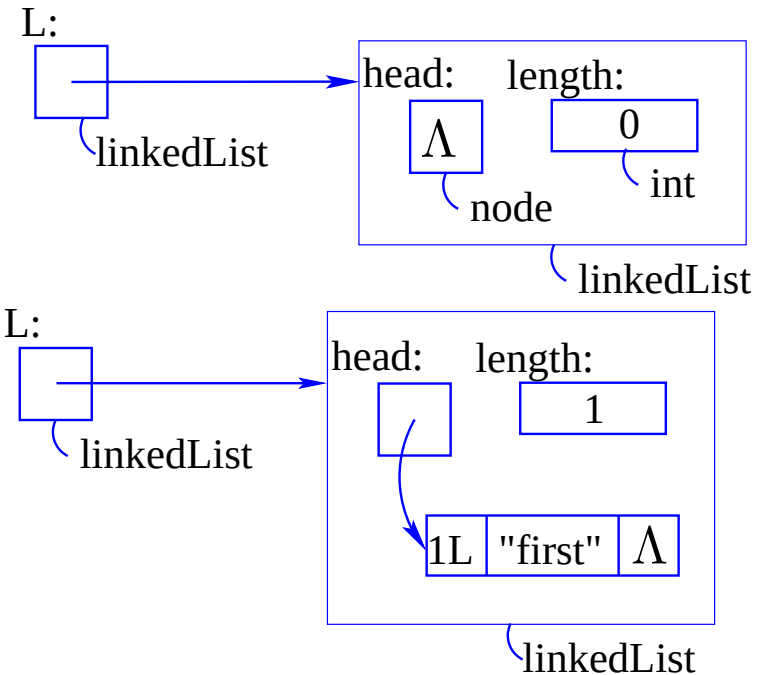
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1 public void insert(node n){  
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}
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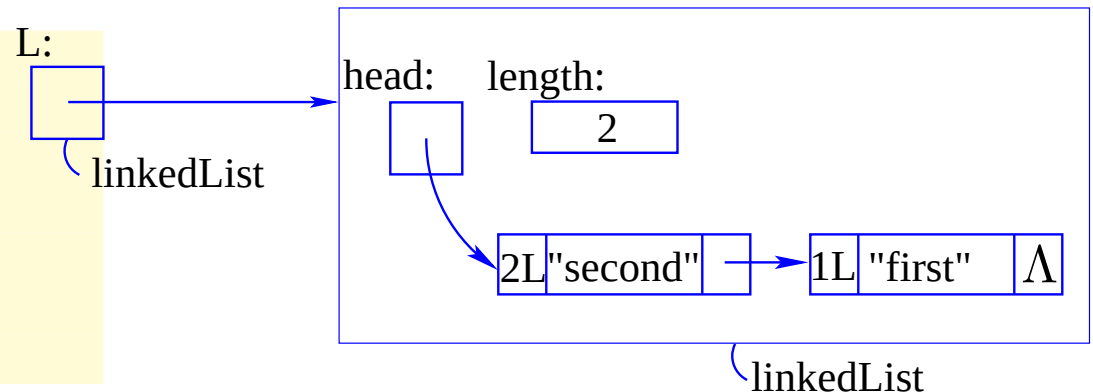
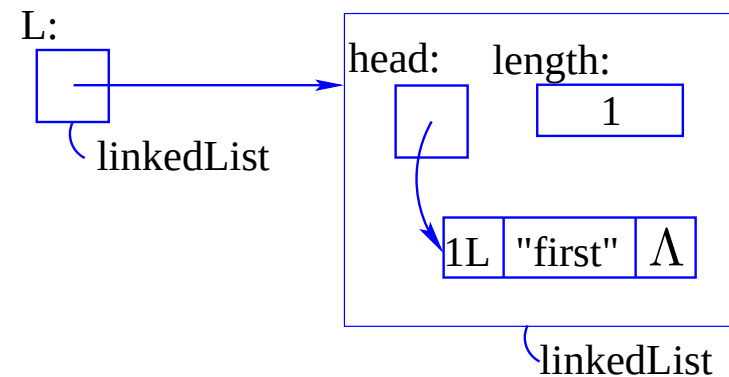
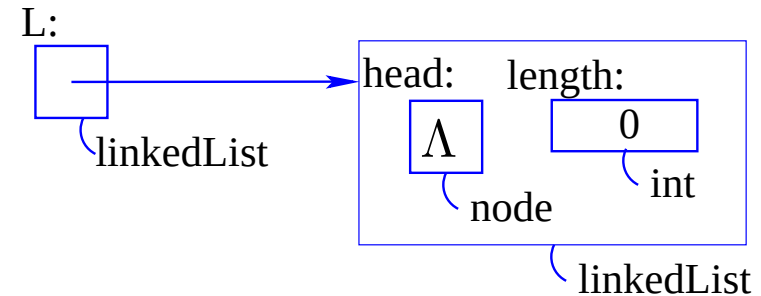
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}
```



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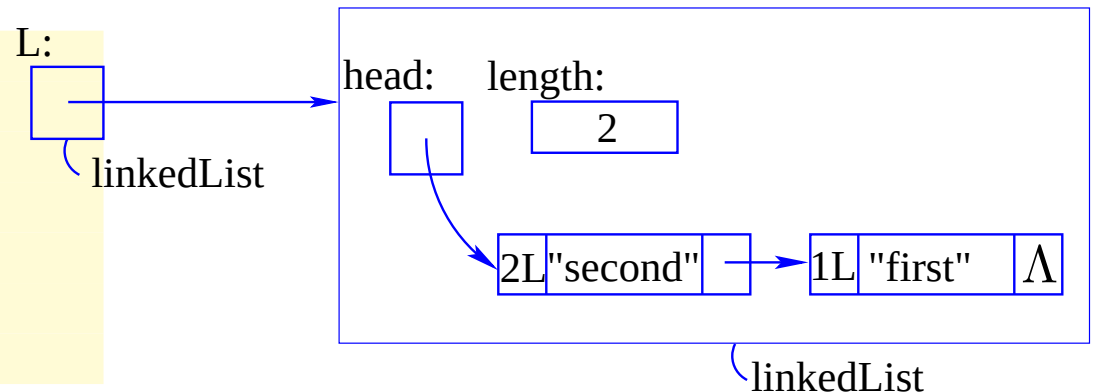
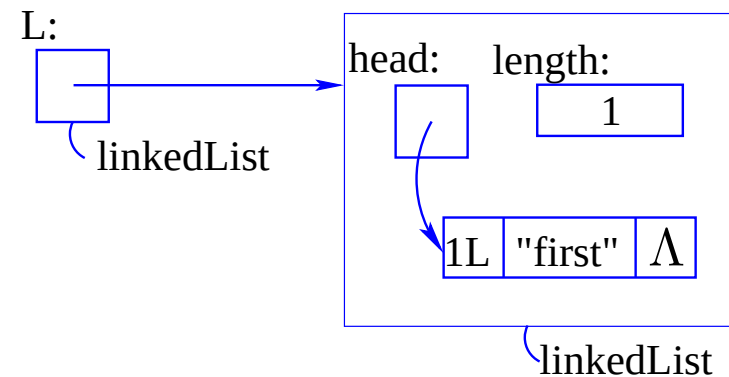
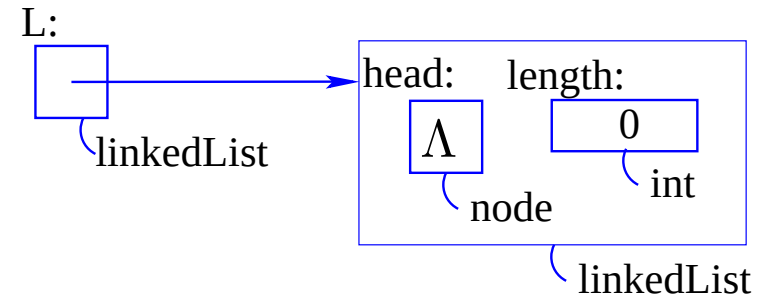
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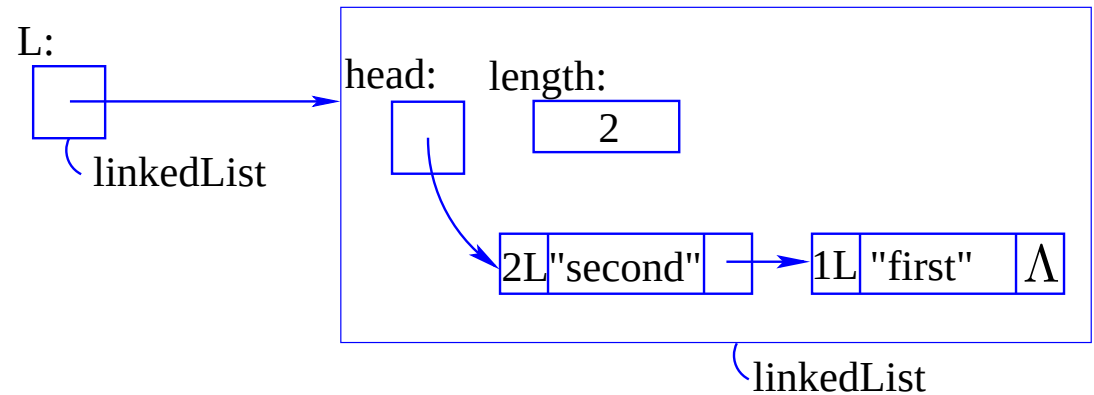
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1 node n = new node(2L, "second")  
2 L.insert(n);
```

```
1 public void insert(node n){  
2   if (head == null)  
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}
```



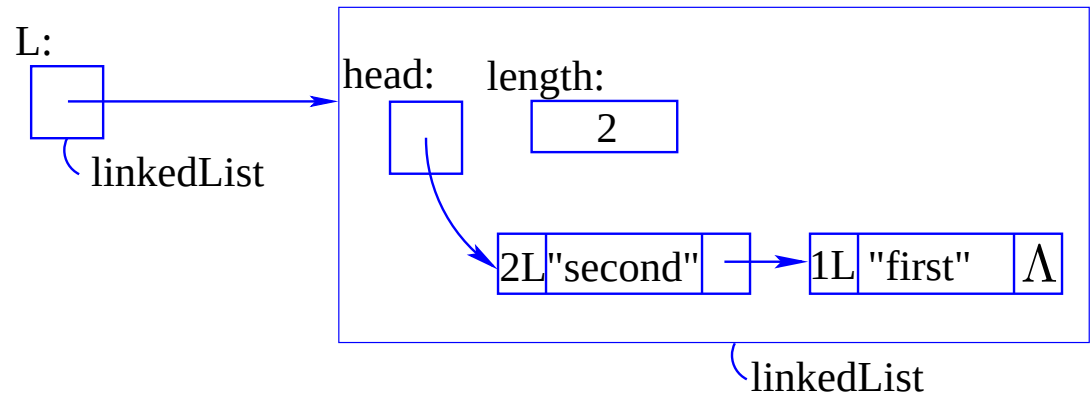
Linked list—no dummy head: **remove (long Key)**

Start with a list of two elements. Consider removing either of the nodes.



Linked list—no dummy head: remove (long Key)

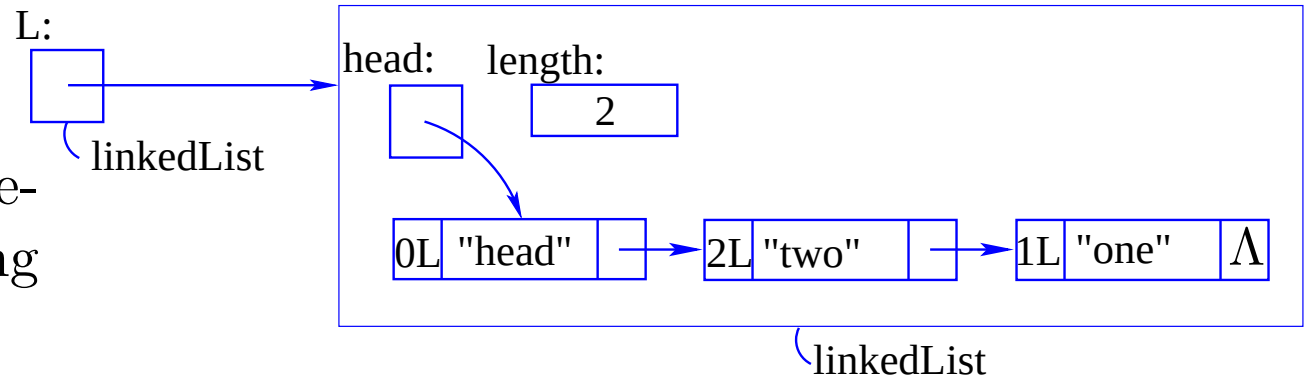
Start with a list of two elements. Consider removing either of the nodes.



```
1 public node remove(long Key){
2   node here = head, previous = here;
3   while (here != null && here.key != Key) {
4     previous = here;
5     here = here.next;}
6   if (here != null) {
7     if (here == head)
8       head = here.next;
9     else
10      previous.next = here.next;
11    length--; }
12  return here;}
```

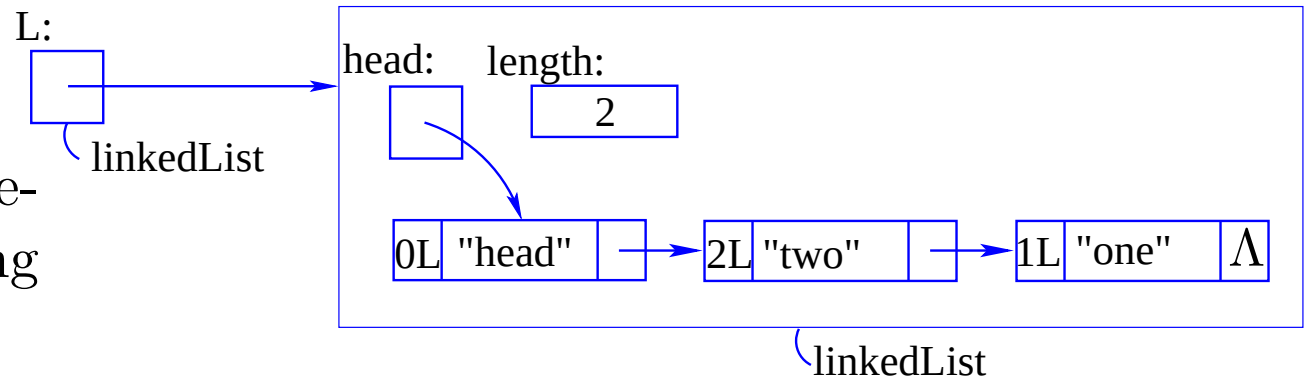
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Linked list with dummy head—remove (long Key)

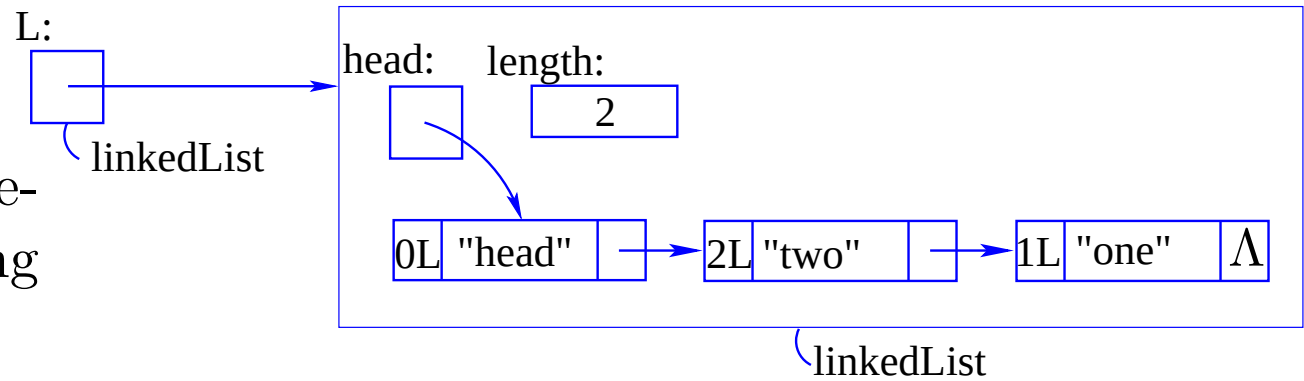
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4     previous = here;
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Linked list with dummy head—remove (long Key)

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```

The code is much simpler.

Stacks

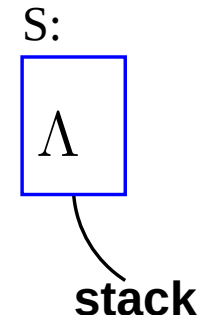
A stack behaves like a linked list where operations are allowed only at the head of the list.

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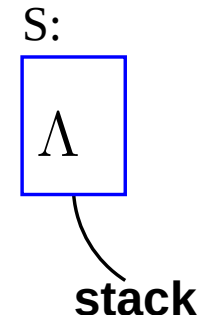


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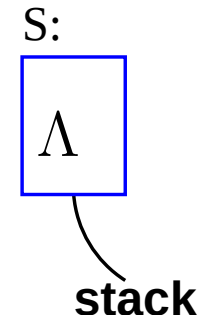
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S = new stack(n);



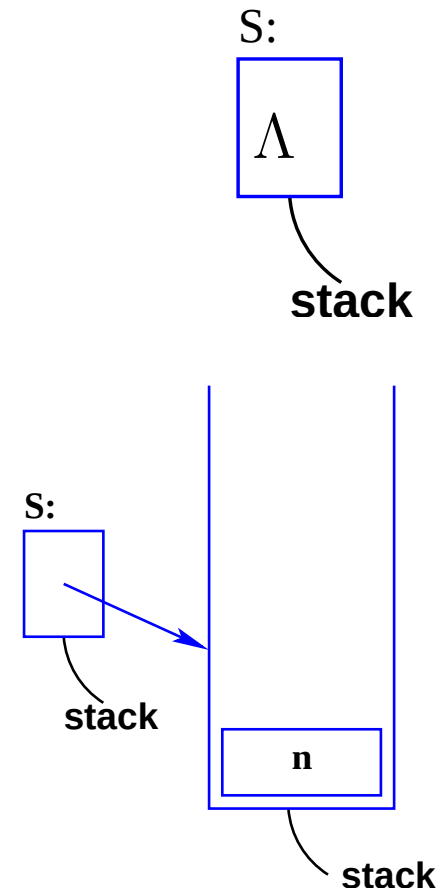
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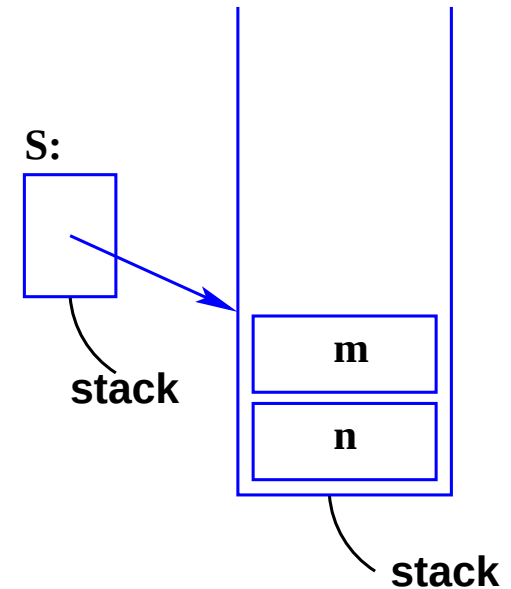


Stacks—a similar stack created differently

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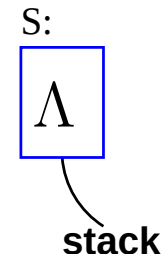
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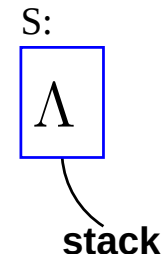
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Suppose that a **stack** variable called **S** is created by

```
public static stack S;
```

If **n** is of type **node**, then a new instance of a stack with *nothing* placed on the stack is created by

```
S = new stack();
```



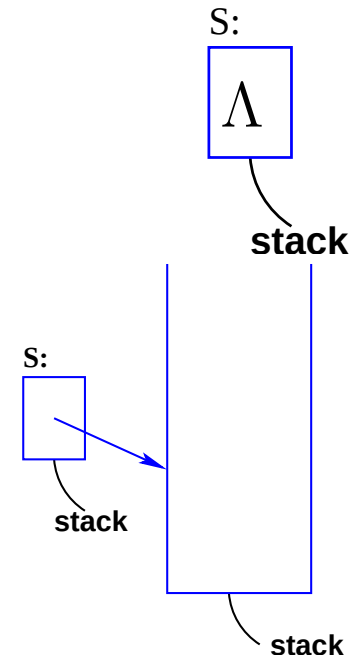
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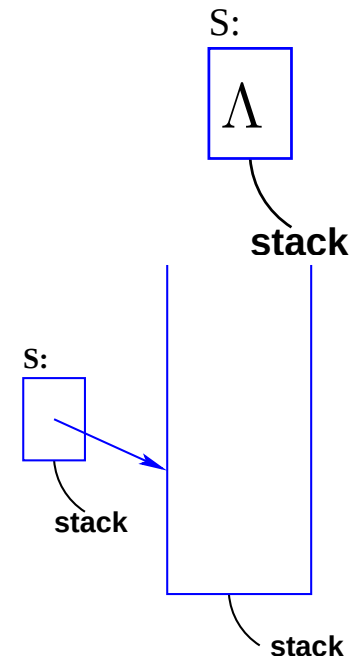
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If **n** is a **node**, then it may be pushed onto the stack with **S.push(n)**;



Stacks—a similar stack created differently

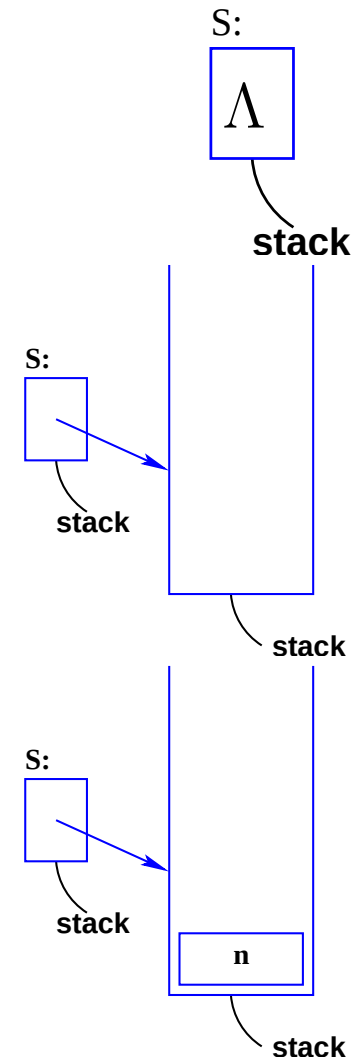
Suppose that a **stack** variable called **S** is created by

```
public static stack S;
```

If **n** is of type **node**, then a new instance of a stack with *nothing* placed on the stack is created by

```
S = new stack();
```

If **n** is a **node**, then it may be pushed onto the stack with **S.push(n)**;

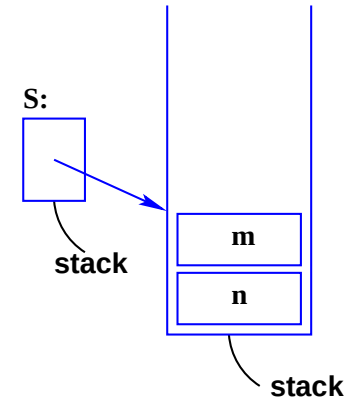


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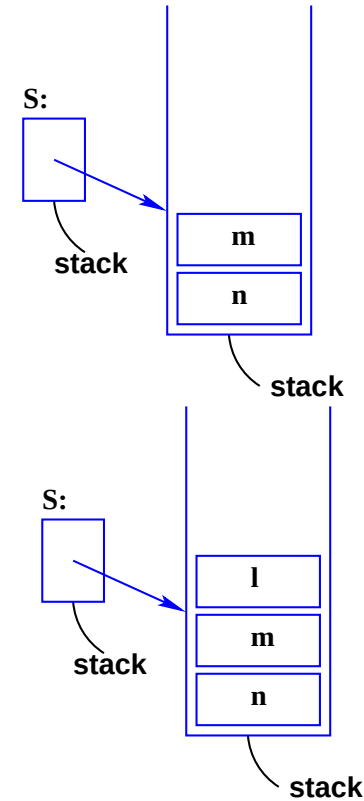
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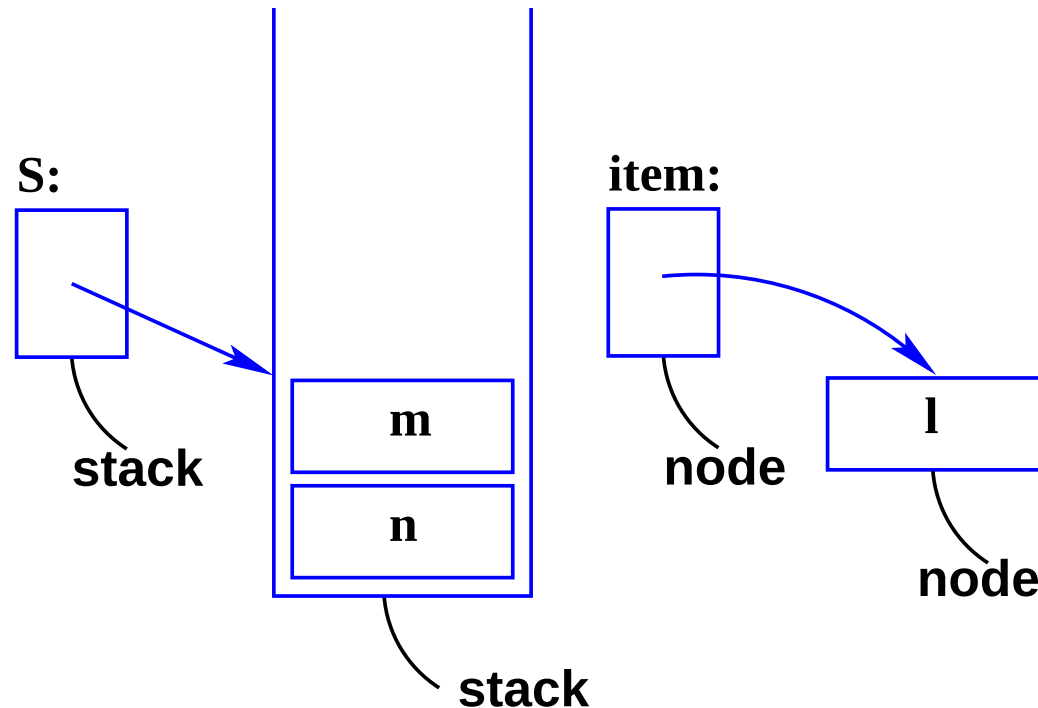
Stacks—`S.push(l)` and `item = S.pop()`;

If the stack is not empty, the function method

`node pop()`

returns the value of uppermost item on the stack and also deletes it off the stack. Now create a variable called `item` of type `node` that points to the node that was removed from the top of the stack.

`node item = S.pop();`



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`S = new stack();`
creates an instance of an empty stack to which **S** points.

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`item = S.pop();`
which deletes it from the top of the stack.

Stacks implemented with a fixed array

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- The `boolean isFull()` method is questionable because when a push is being done the node being stacked has already been allocated memory.

A stack implemented with a fixed array

Build a stack of nodes.

```
1 public class stack {
2   public static node [] d = new node [1<<20+1];
3   public static int top;
4
5   public stack() {
6     top = 0; }
7
8   public void push (node n) {
9     d[++top] = n; }
10
11  public node pop() {
12    if (top > 0)
13      return d[top--];
14    else
15      return null; } }
```

A stack implemented as linked nodes

Use the deprecated non Object Oriented **node** class.

```
1 public class stack {  
2  
3     public static node top;  
4  
5     public stack() {  
6         top = null; }  
7  
8     public void push (node n) {  
9         n.next = top;  
10        top = n; }  
11  
12    public node pop() {  
13        node value = top;  
14        top = top.next;  
15        return value; } }
```

A stack implemented as linked nodes

Use the Object Oriented **node** class.

```
1 public class stack {  
2  
3     public static node top;  
4  
5     public stack() {  
6         top = null; }  
7  
8     public void push (node n) {  
9         n.setNext(top); // n.next = top;  
10        top = n; }  
11  
12    public node pop() {  
13        node value = top;  
14        top = top.getNext(); // top = top.next;  
15        return value; } }
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Queues implemented with a fixed array

- A queue is a first-in-first-out (FIFO) data structure whose elements are served when they reach the front but they enter the queue at the rear.

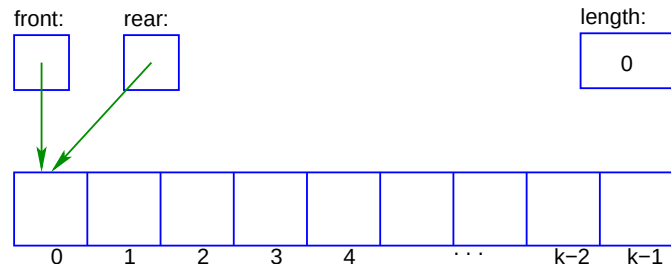
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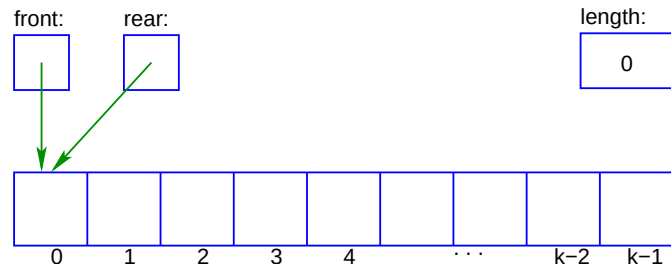
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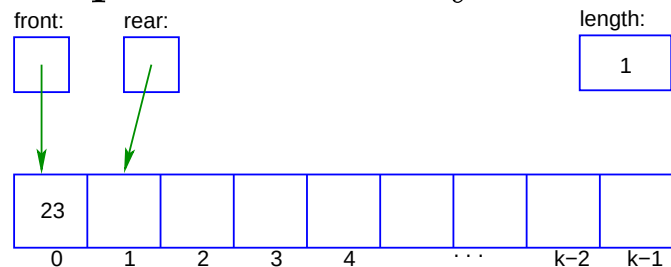


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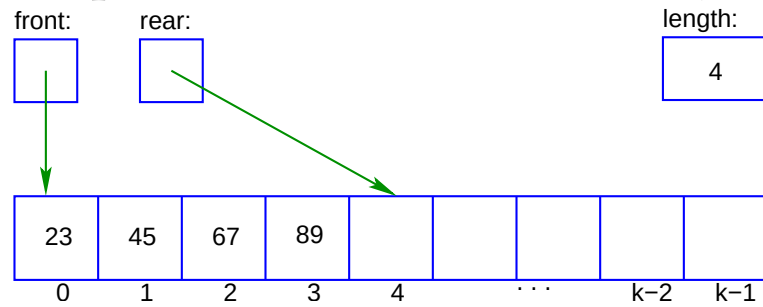


- A queue with only the node 23 entered.



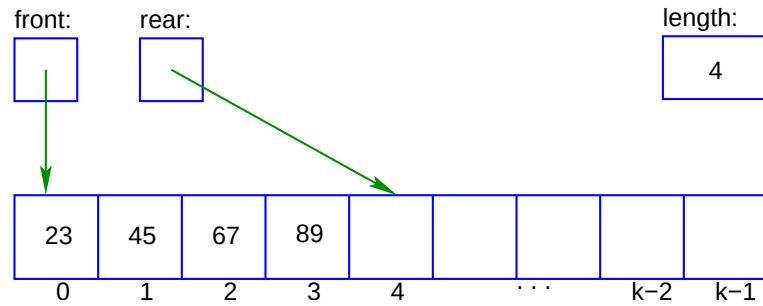
Queues implemented with fixed arrays

- A queue with the nodes 23, 45, 67 and 89 entered in this order.

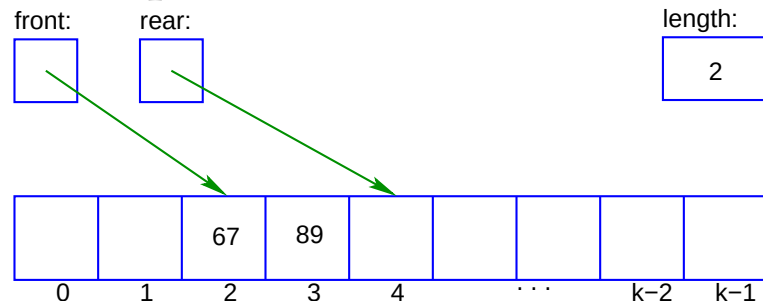


Queues implemented with fixed arrays

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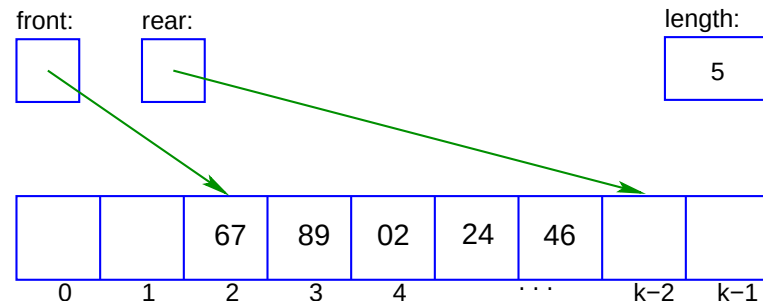


- The queue where the nodes 23, 45 have been dequeued or “served”.



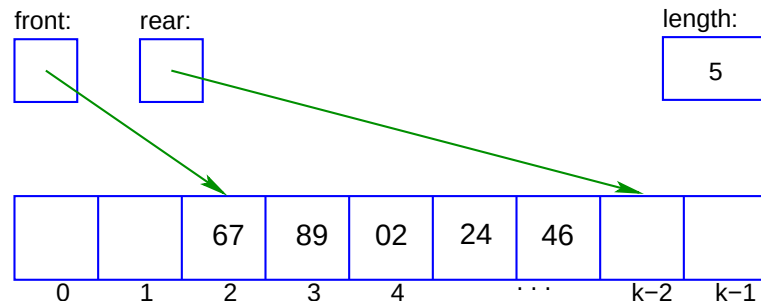
Queues implemented with fixed arrays

- The previous queue where the nodes 02, 24 and 46 have also been entered.



Queues implemented with fixed arrays

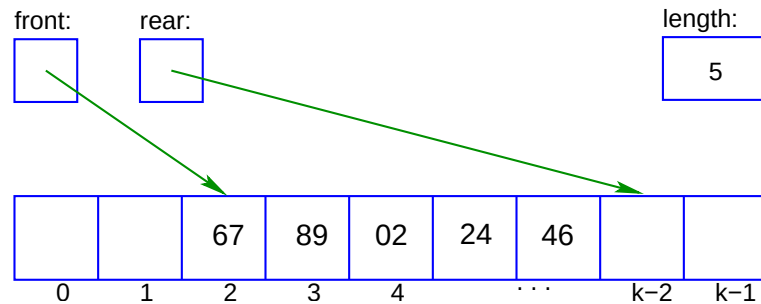
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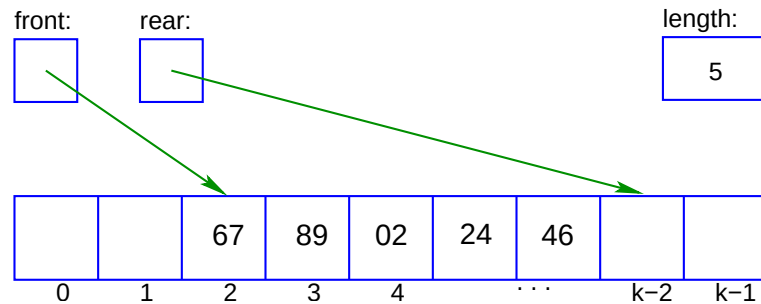
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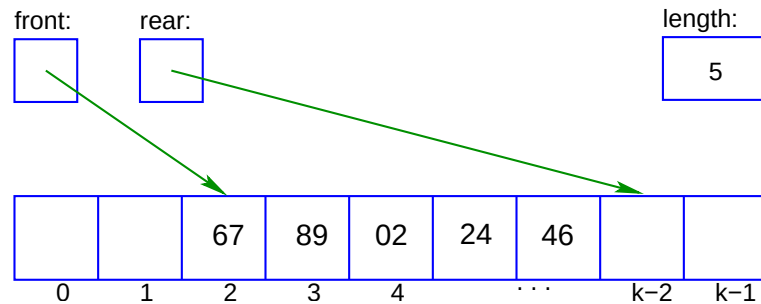
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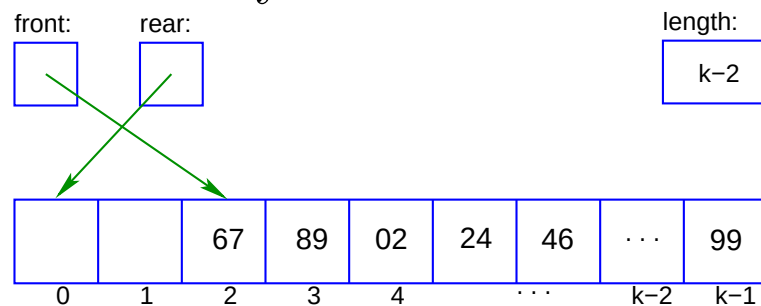
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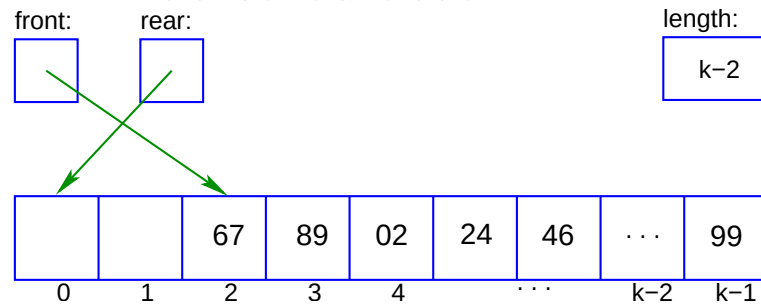
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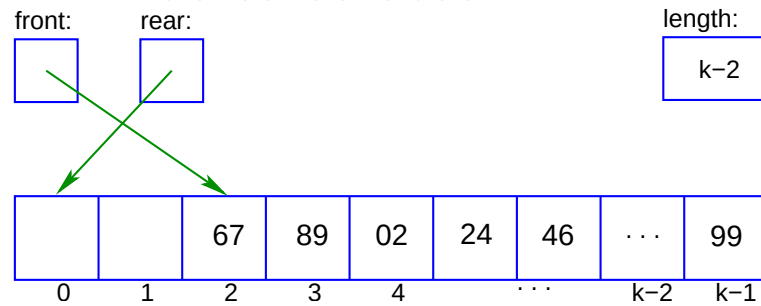
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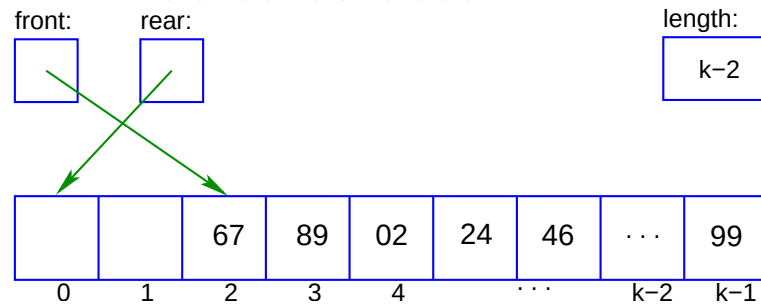


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rear = (rear + 1) % k; and also **length++;**

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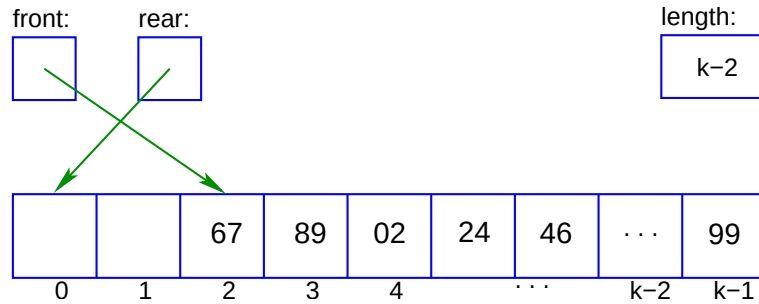


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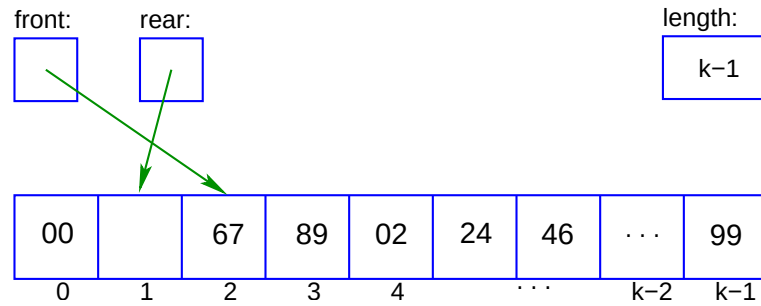


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- Note that a 1 is *added* in both cases.

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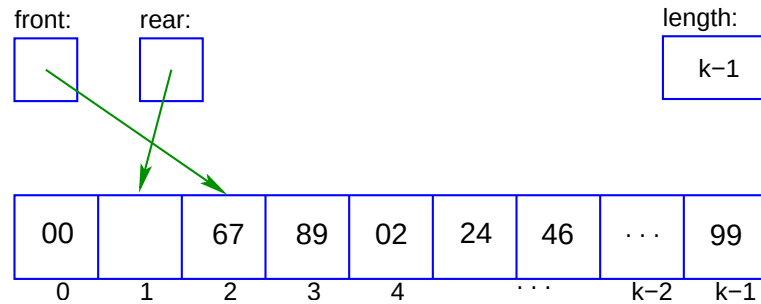
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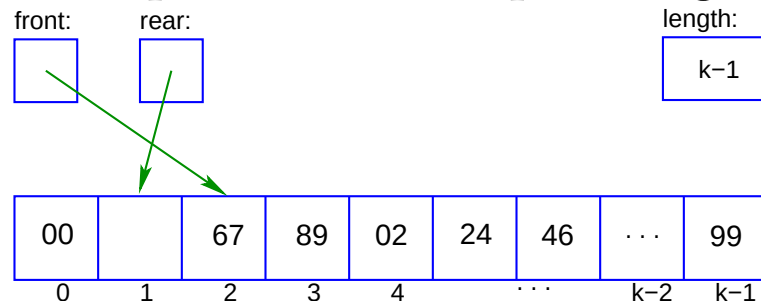


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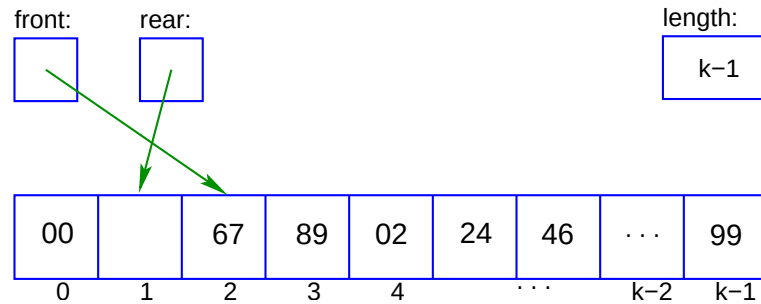


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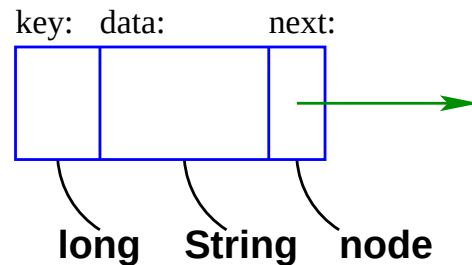
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- Here **length** $= (\text{rear} - \text{front} + k) \% k$; calculates length when there is no counter.

A queue implemented with linked nodes

- Queues may be implemented with a linked list of nodes with a pointer to the serving or leaving end called **front**, and with another pointer to the joining end named **rear**.

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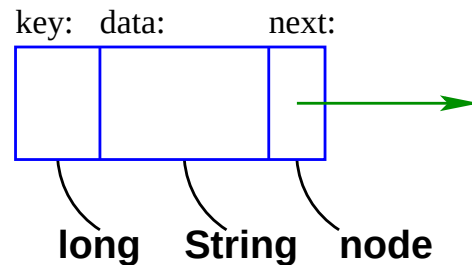
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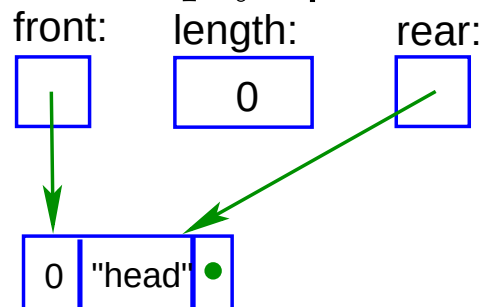
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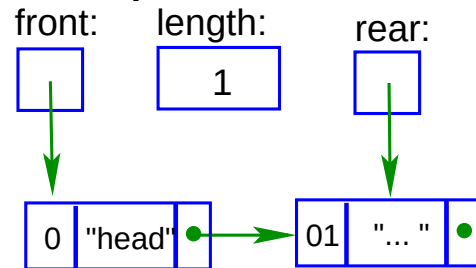


- An empty **queue**

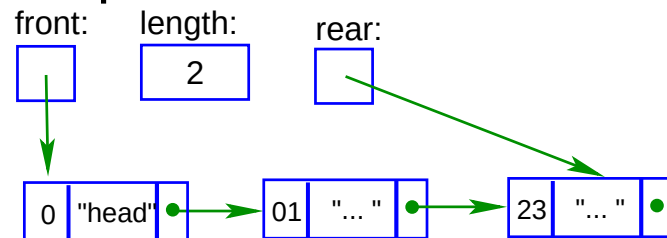


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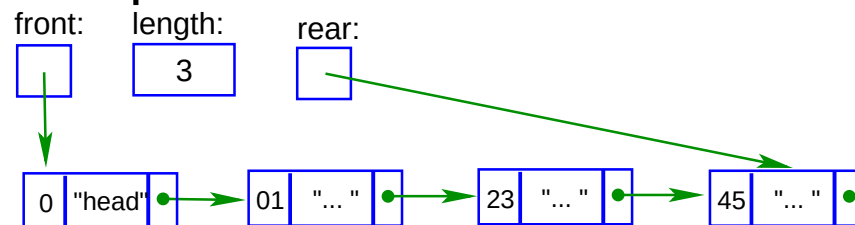
- An queue with one node:



- A queue with two linked nodes.

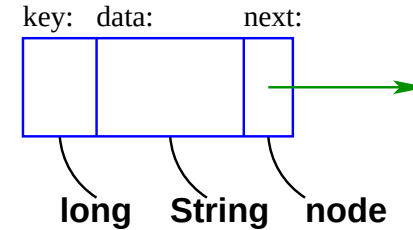


- An queue with three linked nodes.



Queue's node class—elements private

A queue's node



```
1 public class node{
2
3 private long key;
4 private String data;
5 private node next;
6
7 public node(){
8     key = 0;
9     data = "";
10    next = null; }
11
12 public node(long k, String d){
13     key = k;
14     data = d;
15     next = null; }
16
17 public long getKey() {
18     return key }
19     ...
```

Queue's **node**—methods **public**

```
1 public void setKey(long k) {  
2     key = k; }  
3  
4 public String getData() {  
5     return data; }  
6  
7 public void setData(String d) {  
8     data = d; }  
9  
10 public node getNext() {  
11     return next; }  
12  
13 public void setNext(node n) {  
14     next = n; }  
15  
16 public String toString() {  
17     return "<" + key + ",_" + data + ",_" + next + ">" } }
```

The queue itself

```
1 public class queue{
2
3 private int length;
4 private node front;
5 private node rear;
6
7 public queue(){
8     rear = new node (0, "head_node");
9     front = rear;
10    length=0; }
11
12 public queue(long k, String d){
13     rear = new node (k, d);
14     front = rear;
15     length=0; }
16
17 public int size(){
18     return length; }
19
20 public boolean isEmpty(){
21     return length==0; }
```

The basic manipulations of a queue

```
1 public boolean isFull(){
2     return length==N; }
3
4 public void enqueue(node n){
5     rear.setNext(n);
6     rear = n;
7     length++; }
8
9 public node dequeue(){
10    node n;
11    if (isEmpty())
12        return null;
13    else {
14        n = front;
15        front = front.getNext();
16        length--;
17        return n; } }
18
19 public node topQueue(){
20     if (isEmpty())
21         return null;
22     else {
23         return front; } }
```

void display() is useful for debugging

```
1 public void display(){
2     node here = front;
3     int listed = 0;
4     while (here != rear && listed++ < 6) {
5         System.out.println(here);
6         here = here.getNext(); }
7     if (length > 0)
8         System.out.println("Queue's_length=_ " + length);
9     else
10        System.out.println("Queue_is_Empty"); } }
```

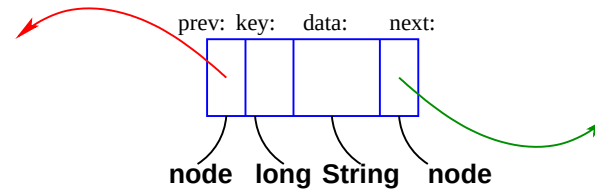

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```

The necessity of a **void display()** method is questionable, but we find it useful for debugging our **queues**.

The node of a deque

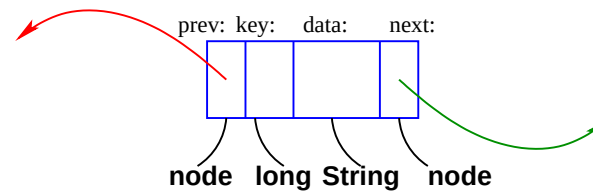
A node for a deque



The node points backwards with its **prev** pointer and forwards with its **next** pointer.

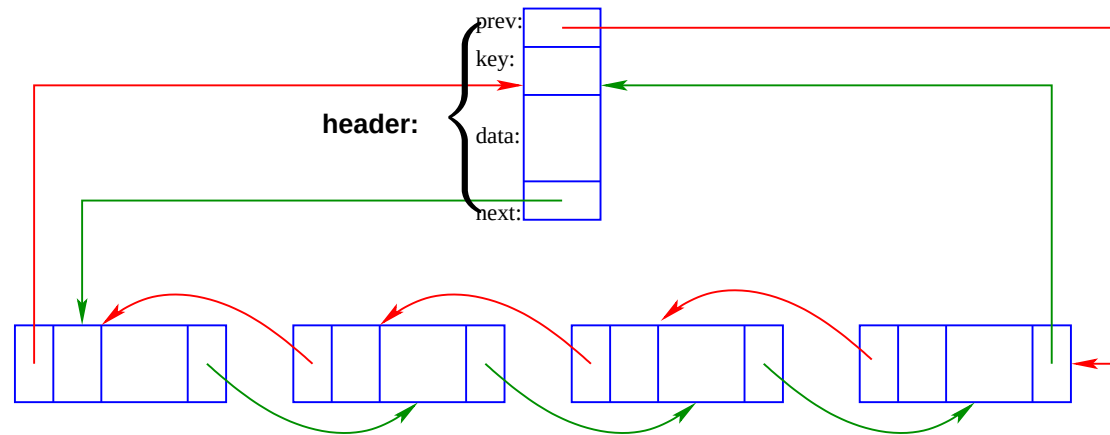
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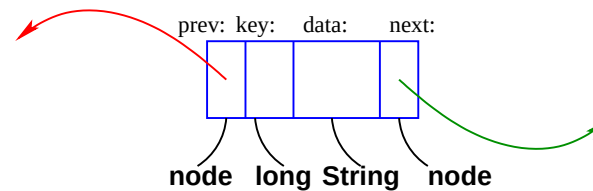
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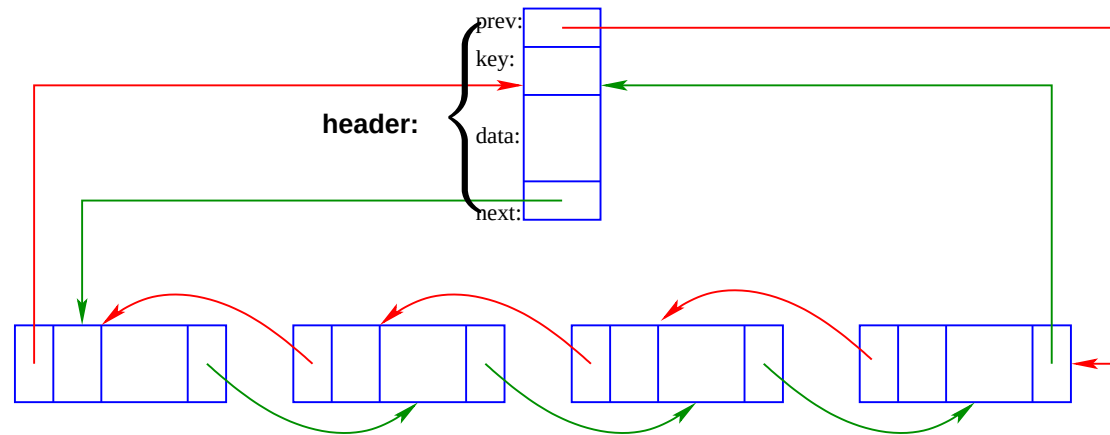
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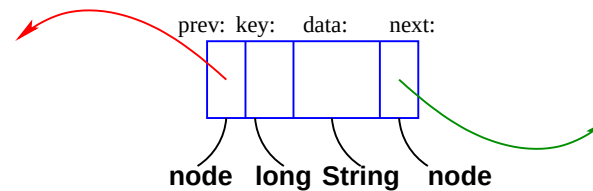
A deque



The “sentinel” node’s **next** pointer points to the front of the deque and the sentinel’s **prev** pointer points to the the rear of the deque.

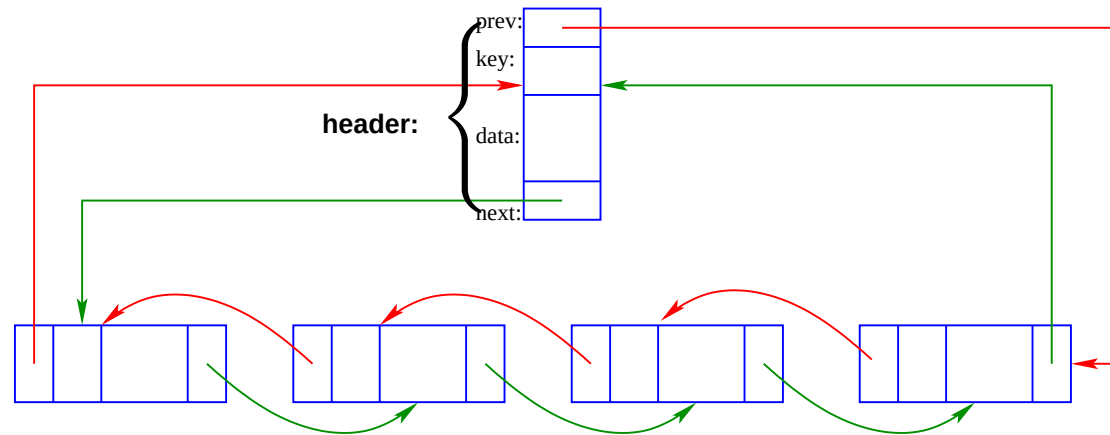
The node of a deque

A node for a deque



The node points backwards with its **prev** pointer and forwards with its **next** pointer.

A deque

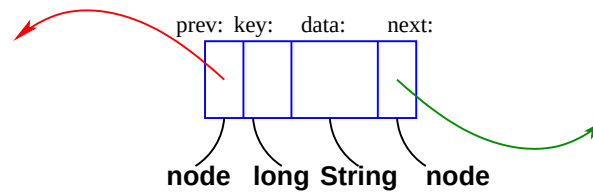


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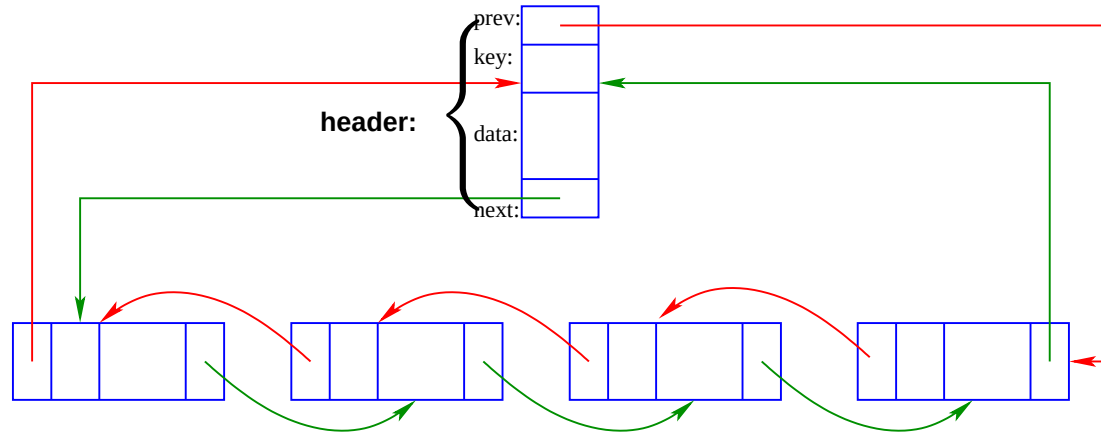
The deque has methods for adding and removing elements at both ends.

Insert a **node** to the front of a **deque**

A deque node



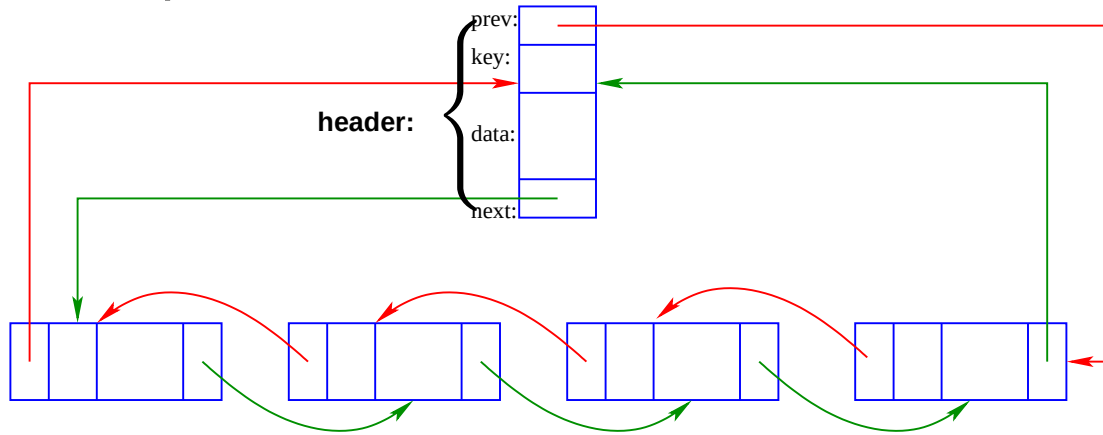
A deque



```
1 public void insertFirst(node n){  
2     n.setNext(header.getNext());  
3     n.setPrev(header);  
4     (header.getNext()).setPrev(n);  
5     header.setNext(n);  
6     length++; }
```

Insert a **node** to the rear of a **deque**

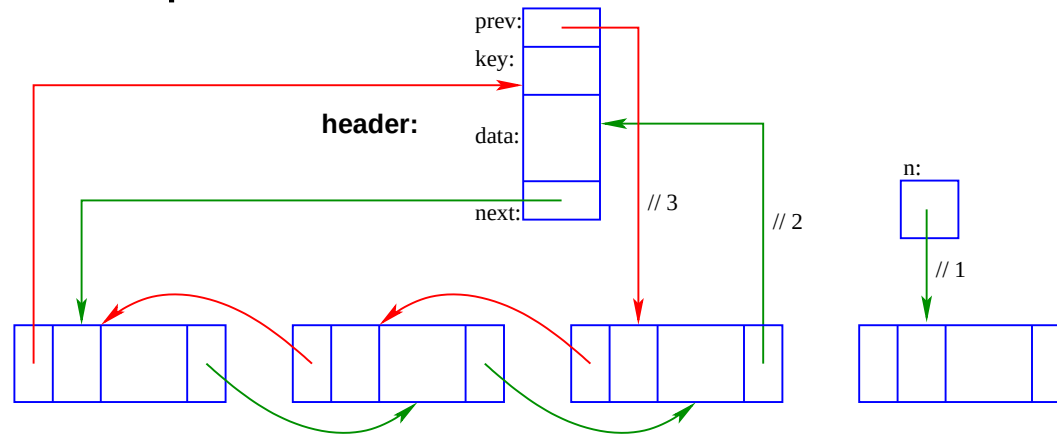
A deque



```
1 public void insertLast(node n){  
2     n.setPrev(header.getPrev());  
3     n.setNext(header);  
4     (header.getPrev()).setNext(n);  
5     header.setPrev(n);  
6     length++; }
```

Remove a node from the rear of a deque

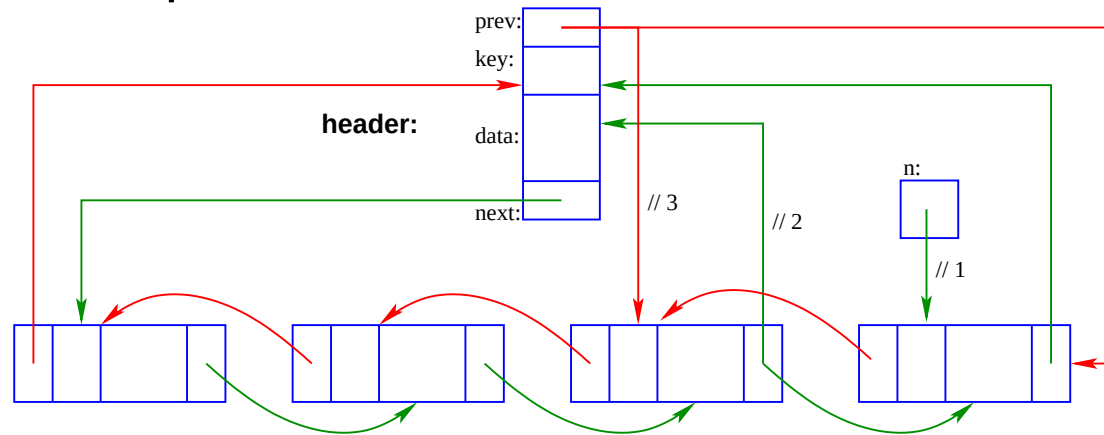
A deque



```
1 public node removeLast(){
2   node n = null;
3   if (!isEmpty()) {
4     n = header.getPrev(); // 1
5     header.setPrev(n.getPrev()); // 2
6     (n.getPrev()).setNext(header); // 3
7     n.setPrev(null);
8     n.setNext(null);
9     length--; }
10  return n; }
```


Remove a node from the rear of a deque

- A deque



-

```
1 public node removeLast(){
2   node n = null;
3   if (!isEmpty()) {
4     n = header.getPrev(); // 1
5     header.getPrev(n.getPrev(n)); // 2
6     (n.getPrev()).setNext(header); // 3
7     n.setPrev(null);
8     n.setNext(null);
9     length--; }
10  return n; }
```

Binary search trees (BST)

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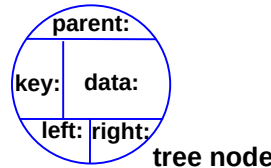
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Binary search trees (BST)

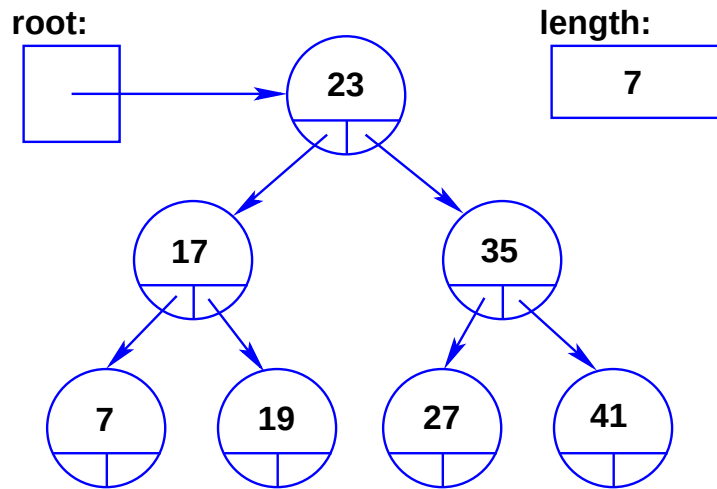
- A binary search tree (BST) is an efficient data structure for storing and retrieving data.
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- The number of nodes in a complete binary tree up to but excluding *level-k* is $n = 2^{k+1} - 1 - 2^k = 2^k - 1$.
- Since the height of a proper binary tree is thus k , it follows that the maximum search length is k , but $k = \log n + 1$, i.e. $O(\log n)$.

The nodes of a tree



- A tree node
- The nodes in a binary **tree** have a **left** and a **right** forward pointer, a back link to the **parent** is useful. The element also should contain a **key** and **data** field.
- In a *binary search tree* each node's **left** pointer points to a node such that **left.key** < **key** when **left** $\neq$ **null** and **right** and a pointer that points to a node such that **right.key** $\geq$ **key** when **right** $\neq$ **null**.

A proper binary search tree



```
1 public class tryTree{
2 public static void main(String args[]){
3 tNode tItem;
4 tree T = new tree ();
5
6 T.display ();
7 T.insert (new tNode (23, "twenty_three"));
8 T.insert (new tNode (17, "seventeen"));
9 T.insert (new tNode (7, "seven"));
10 T.insert (new tNode (35, "thirty_five"));
11 tItem = new tNode (41, "fourty_one");
12 T.insert (new tNode (27, "twenty_seven"));
13 T.insert (tItem);
14 T.insert (19, "nineteen");
15 T.display (); } }
```

A class for a tNode

```
1 public class tNode{
2
3 private long key; private String data;
4 private tNode left; private tNode parent;
5 private tNode right;
6
7 public tNode() {
8     key = 0;
9     data = "";
10    parent = null;
11    left = null;
12    right = null; }
13
14 public tNode(long k, String d) {
15     key = k;
16     data = d;
17     parent = null;
18     left = null;
19     right = null; }
20
21 public long getKey() {
22     return key; }
23
24 public void setKey(long k) {
25     key = k; }
26 // continued on next slide
```

The tNode class (continued)

```
1 public String getData() {  
2     return data; }  
3 public void setData(String d) {  
4     data = d; }  
5  
6 public tNode getLeft() {  
7     return left; }  
8 public void setLeft(tNode n) {  
9     left = n; }  
10  
11 public tNode getRight() {  
12     return right; }  
13 public void setRight(tNode n) {  
14     right = n; }  
15  
16 public tNode getParent() {  
17     return parent; }  
18 public void setParent(tNode n) {  
19     parent = n; } }
```

Defining a tree

- An empty **tree** has a **root** set to **null** and a counter **length** set to 0 by the constructor.

root:

null

length:

0

```
1 public class tree {  
2   private int length;  
3   private tNode root;  
4  
5   public tree() {  
6     root = null;  
7     length=0; } }
```

- An instance of the tree is defined in the user program by:

```
1 tree T = new tree();
```

- The first **node** in a tree of type **tNode** is pointed to by **root**.
- A subsequent node **n** is added to the left subtree if **n.key < root.key** or otherwise to its right.

Inserting a node into an empty tree

- An empty **tree** has a **root** set to **null** and a counter **length** set to 0 by the constructor.

```
1 tree T = new tree();
```

root:

null

length:

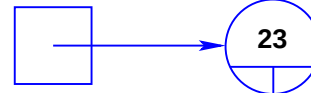
0

- The tree after inserting a node of type **tNode** with **key = 23** using:

```
1 T.insert(new tNode(23, "twenty_three"));
```

```
1 public class tree {  
2   ...  
3   public void insert(tNode n) {  
4     if (root == null) {  
5       root = n;  
6       root.setParent(null); }  
7   else  
8     root.insert(n); }  
}
```

root:



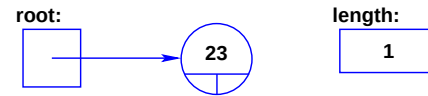
length:

1

- **root.insert(tNode)** in the **tNode** class inserts the **tNode** **n** into the “tree” starting at **root**.

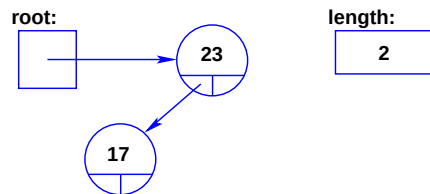
Inserting a second node into the tree

- Starting with a tree with a single node at the root



- and after inserting a node with **key = 17** using:

```
1 T.insert(new tNode(17, "seventeen"));
```



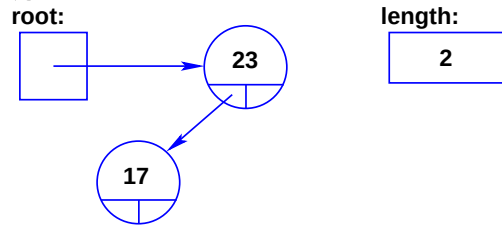
- The code that does this lies in the **tNode** class.

```
1 public void insert(tNode n) {  
2     if (n.key < key) // n into left subtree  
3         if (left == null) {  
4             n.parent = this;  
5             left = n; }  
6     else  
7         left.insert(n); ... }
```

- If **left != null**, then **left.insert(n)** is called, as happens when a node with **key = 7** is entered.

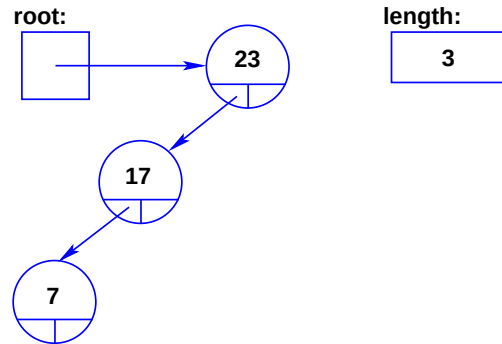
Adding a node with key = 7

- Start with the two node tree:



- Insert a node with key = 7 using:

```
1 T.insert(new tNode(7, "seven"));
```



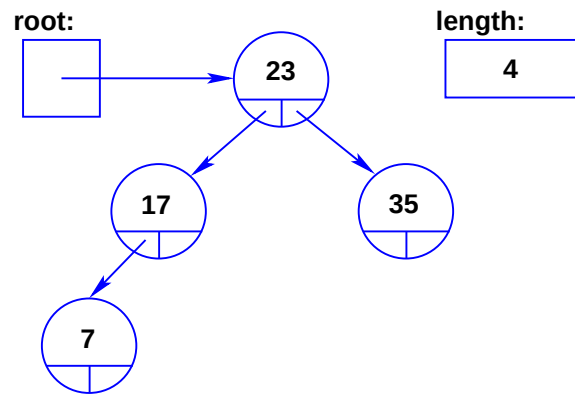
The pointers down the left are followed until a **null** is found.

```
1 public void insert(tNode n){
2     if (n.key < key) // n into left subtree
3         if (left == null){
4             n.parent = this;
5             left = n;
6         }
7     else
8         left.insert(n);
```


Adding the node with key = 35

- Insert a node with key = 35 using:

```
1 T.insert(new tNode(35, "thirty_five"));
```



The **right** pointer of the root node is **null** and **n** is inserted there.

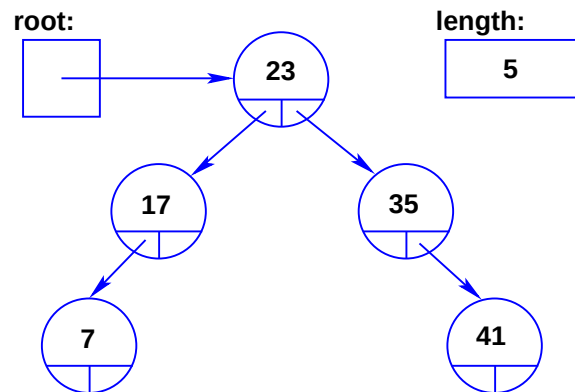
•

```
1 public void insert(tNode n) {
2     if (n.key < key) // n into left subtree
3         if (left == null) {
4             n.parent = this;
5             left = n; }
6     else
7         left.insert(n);
8     else // n into right subtree
9         if (right == null) {
10            n.parent = this;
11            right = n; }
12    else
13        right.insert(n); }
```

Adding the node with key = 41

- Insert a node with key = 41 using:

```
1 T.insert(new tNode(41, "fourty_one"));
```



The **right** pointer of the root node is not **null** and the pointer is followed to **35** whose right pointer is **null**. where **n** is inserted.

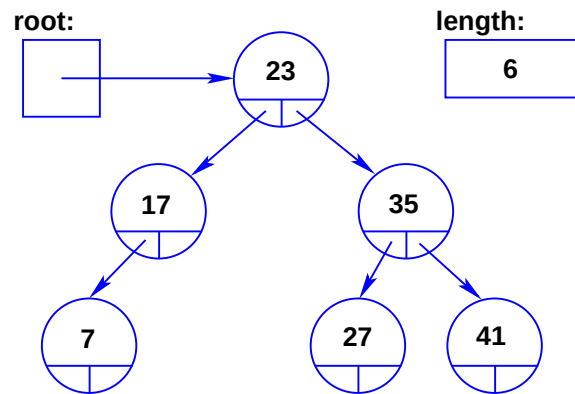
•

```
1 public void insert(tNode n) {
2     if (n.key < key) // n into left subtree
3         if (left == null) {
4             n.parent = this;
5             left = n; }
6     else
7         left.insert(n);
8     else // n into right subtree
9         if (right == null) {
10            n.parent = this;
11            right = n; }
12    else
13        right.insert(n); }
```

Adding the node with key = 27

- Insert a node with key = 27 using:

```
1 T.insert(new tNode(27, "twenty_seven"));
```



The **right** pointer of the root node is not **null** and the pointer is followed to **35** whose left pointer is **null**. where **n** is inserted.

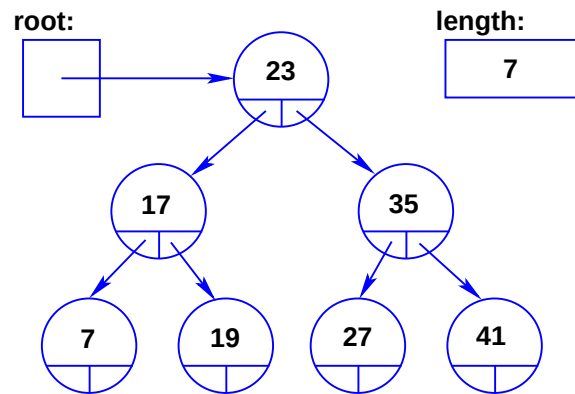
•

```
1 public void insert(tNode n) {
2     if (n.key < key) // n into left subtree
3         if (left == null) {
4             n.parent = this;
5             left = n; }
6     else
7         left.insert(n);
8     else // n into right subtree
9         if (right == null) {
10            n.parent = this;
11            right = n; }
12    else
13        right.insert(n); }
```

Adding the node with key = 19

- Insert a node with key = 19 using:

```
1 T.insert(new tNode(19, "nineteen"));
```



The **left** pointer of the root node is not **null** and is followed to **17** whose right pointer is **null** where **n** is inserted.

•

```
1 public void insert(tNode n) {
2     if (n.key < key) // n into left subtree
3         if (left == null) {
4             n.parent = this;
5             left = n; }
6     else
7         left.insert(n);
8     else // n into rightsubtree
9         if (right == null) {
10            n.parent = this;
11            right = n; }
12    else
13        right.insert(n); }
```

The tree class

```
1 public class tree{
2 private static final int N = 1000000;
3 private static int listed = 0;
4
5 private static int length;
6 private tNode root;
7
8 public tree() {
9     root = null;
10    length=0; }
11
12 public void setRoot(tNode n) {
13     root = n; }
14 public tNode getRoot() {
15     return root; }
16
17 public void insert(tNode n) {
18     if (root == null) {
19         root = n;
20         trace(n, "entered_at_root"); }
21     else
22         insert(root, n); }
23 // two more insert methods on next slide
```

insert methods entirely in tree class

```
1 public void insert(long k, String d){
2   tNode n = new tNode(k, d);
3   if (root == null){
4     root = n;
5     trace(n, "entered_at_root"); }
6   else
7     insert(root, n);
8   length++; // Why not at the start? }
9
10 public void insert(tNode here, tNode n) {
11   if (n.getKey() < here.getKey()) // n goes left
12     // n must go into left subtree
13     if (here.getLeft() == null) {
14       n.setParent(here);
15       here.setLeft(n); }
16   else insert(here.getLeft(), n);
17   else if (here.getRight() == null) { // n goes right
18     n.setParent(here);
19     here.setRight(n); }
20   else insert(here.getRight(), n); }
```

Methods in class tree

```
1 \begin{verbatim}
2 public void tDelete(long key){
3     . . . follows later
4     length--; }
5
6 public int size(){
7     return length; }
8
9 public boolean isEmpty(){
10    return length==0; }
11
12 public boolean isFull(){
13    return length==N; }
```

The $\ell N r$ display method for class tree

```
1 public void display() {
2     if (this == null)
3         System.out.println("Tree_is_empty.");
4     else
5         display(root); }
6
7 public void display(tNode n) { // left Node right
8                                 // i.e. lNr
9     if (n != null) {
10        display(n.getLeft()); // left
11        System.out.print( // Node
12            "In_Tree: _Node_number_" + listed++ +
13            "_Key_=" + n.getKey() +
14            "_Data_=" + n.getData());
15        if (n.getParent() == null)
16            System.out.println("_Parent.key=_null");
17        else
18            System.out.println("_Parent.key_="
19                                + (n.getParent()).getKey());
20        display(n.getRight()); } } }
```


A tree class where nodes are trees

```
1 public class tree {
2     private static final int N = 1000000;
3     private static int length = 0;
4     private static tree root = null;
5     // tree nodes
6     private long key; private String data;
7     private tree parent;
8     private tree left; private tree right;
9
10    public tree() {
11        key = 0;
12        data = "";
13        parent = null;
14        left = null;
15        right = null; }
16
17    public tree(long k, String d) {
18        key = k;
19        data = d;
20        parent = null;
21        left = null;
22        right = null; } }
```

A tree class where nodes are trees

```
1 public void insertNode(tree n) {
2     if (root == null){
3         root = n;
4         trace(n, "entered_at_root"); }
5     else
6         root.insert(n);
7     length++; }
8
9 public void insertNode(long key, String data) {
10    tree n = new tree(key, data);
11    if (root == null) {
12        root = n;
13        trace(n, "entered_at_root"); }
14    else
15        root.insert(n);
16    length++; }
```

A tree class where nodes are trees

```
1 public void setRoot(tree n) {  
2     root = n; }  
3 public tree getRoot() {  
4     return root; }  
5  
6 public void tDelete(long key) {  
7     length--; }  
8  
9 public int size() {  
10    return length; }  
11  
12 public boolean isEmpty() {  
13    return length==0; }  
14 public boolean isFull() {  
15    return length==N; }
```

A tree class where nodes are trees

```
1 public long getKey() {  
2     return key; }  
3 public void setKey(long k) {  
4     key = k; }  
5  
6 public String getData() {  
7     return data; }  
8 public void setData(String d) {  
9     data = d; }  
10  
11 public tree getLeft() {  
12     return left; }  
13 public void setLeft(tree n) {  
14     left = n; }
```

A tree class where nodes are trees

```
1 public tree getRight() {  
2     return right; }  
3 public void setRight(tree n) {  
4     right = n; }  
5  
6 public tree getParent() {  
7     return parent; }  
8 public void setParent(tree n) {  
9     parent = n; }
```

A tree class where nodes are trees

```
1 public void insert(tree n) {
2     if (n.key < key)
3         // n must go into left subtree
4         if (left == null){
5             n.parent = this;
6             trace(n, "entered_at_here.L");
7             left = n; }
8     else {
9         trace(n, "insert(here.L,_n)");
10        left.insert(n); }
11 else
12     // n must go into rightsubtree
13     if (right == null){
14         n.parent = this;
15         trace(n, "entered_at_here.R");
16         right = n; }
17     else {
18         trace(n, "insert(here.R,_n)");
19         right.insert(n); } }
```

A tree class where nodes are trees

```
1 public void displayTree() {
2     if (root == null)
3         System.out.println("Tree_is_empty.");
4     else
5         root.display();
6     System.out.println(listed + "_nodes_in_tree."); }
7
8 public void display(){// Node left right
9     // i.e. Nlr
10    if (left != null) left.display(); // left
11    System.out.print("// N
12        "In_Tree:_Node_number_" + listed++ +
13        "_Key=_ " + key +
14        "_Data=_ " + data);
15    if (parent == null)
16        System.out.println("_Parent.key=_null");
17    else
18        System.out.println("_Parent.key=_ "
19            + parent.key);
20    if (right != null) right.display(); } // right
21
22 public void trace(tree n, String message) {
23     System.out.println("tree:_ "
24         + n.key
25         + "_ " + message); } }
```

A method for removing nodes from a tree

```
1 public void Remove(long k) {
2     tree n = find(k);
3     tree here;
4     if (n != null)
5         if (n.left == null) // one right child
6             if (n.parent == null){
7                 root = n.right;
8                 if (root != null)
9                     root.parent = null;
10            }
11            else if (n.key < (n.parent).key)
12                (n.parent).left = n.right;
13            else
14                (n.parent).right = n.right;
15        else if (n.right == null) // one left child
16            if (n.parent == null){
17                root = n.left;
18                if (root != null)
19                    root.parent = null;
20            }
21            else if (n.key < (n.parent).key)
22                (n.parent).left = n.left;
23            else
24                (n.parent).right = n.left;
```


A method for removing nodes from a tree

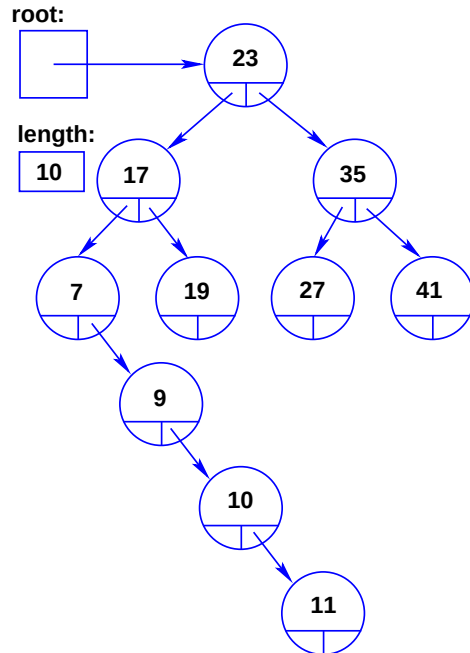
```
1  else { // both children not null
2      here = n.left;
3      while (here.right != null)
4          here = here.right;
5      if (here.parent == null){
6          root = here.left;
7          if (root != null)
8              root.parent = null;
9      }
10     else if ( here.key < (here.parent).key)
11         (here.parent).left = here.left;
12     else
13         (here.parent).right = here.left;
14     n.key = here.key;
15     n.data = here.data; }
16 else
17     System.out.println("Node_" + k + "_not_in_tree"); }
```

Remove node from a tree—more readable

```
1 public void remove(long k) {
2     tree n = find(k);
3     tree here;
4     if (n != null)
5         if (n.left == null) // one right child
6             updateTree(n, n.right);
7         else if (n.right == null) // one left child
8             updateTree(n, n.left);
9         else { // both children not null
10            here = n.left;
11            while (here.right != null)
12                here = here.right;
13            updateTree(here, here.left);
14            n.key = here.key;
15            n.data = here.data; }
16     else
17         System.out.println("Node_" + k + "_not_in_tree"); }
18
19 public void updateTree (tree n, tree p){
20     if (n.parent == null){
21         root = p;
22         if (root != null)
23             root.parent = null; }
24     else if (n.key < (n.parent).key)
25         (n.parent).left = p;
26     else
27         (n.parent).right = p; }
```

Explanation of removal of nodes from a tree

Consider the tree

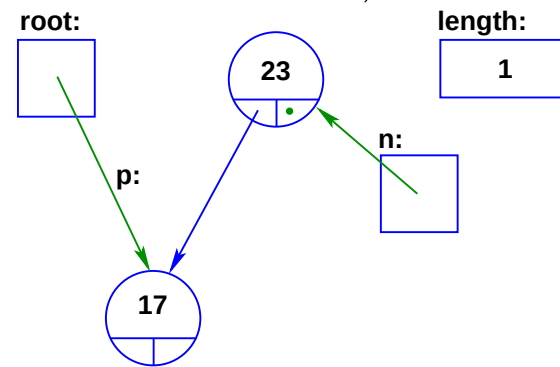
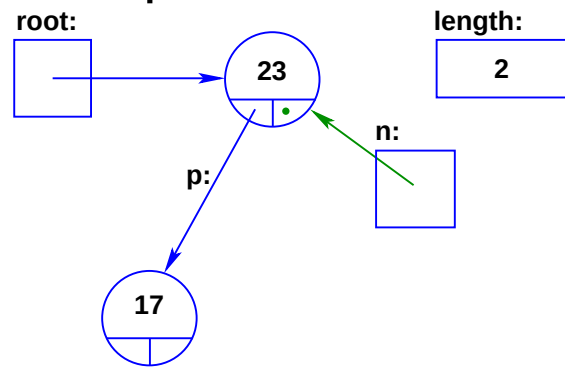


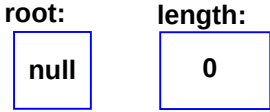
1. Removing elements with zero children is the easiest: Simply replace the pointer to the leaf with a null. We will see later that *this case can be ignored* because it is subsumed by the case where any one of the pointers is a null. Initially, the nodes with keys 11, 19, 27 and 41 are such nodes.

2. To remove a node with only one child: The non-null pointer emanating from the node to be deleted replaces the pointer pointing to the node that must be removed. Initially, the nodes with keys 7, 9, and 10 are such nodes.
3. To remove a node with two children: Find its biggest (smallest) child in its left (right) subtree, copy that biggest (smallest) child over the node to be deleted and then remove the node that was copied as usual. The copied node cannot have two children.

Remove using `update(n, p)`

- The method `update(n, p)` is given a non-null node `n` to be removed and a pointer `p` to the node to replace it.
- If `n.parent == null` then `n` is the root node, i.e. it has no parent:



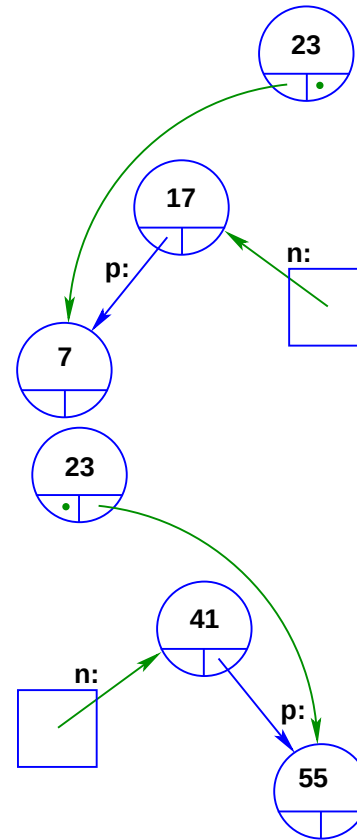
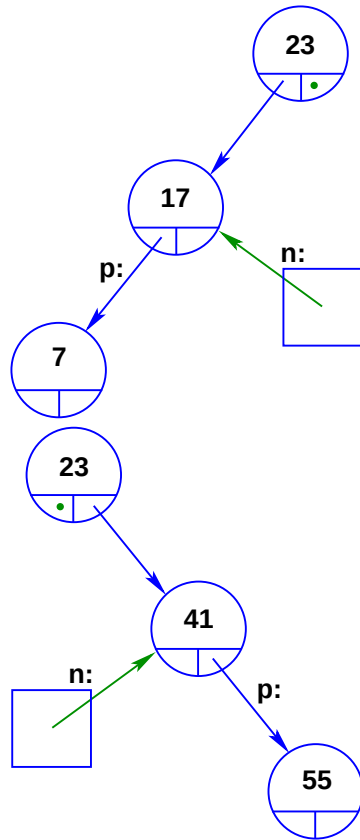
- Then `root = p;` suffices, but `p` could become `null`. This would make the `root == null`,
 otherwise the new node, now pointed to by `root`, must have its `parent` set to `null`, i.e.

```
1 if (root != null)
2   root.parent = null;
```

Remove using `update(n, p)`

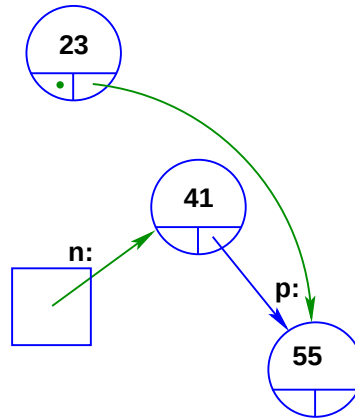
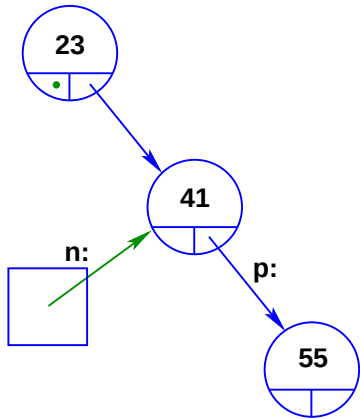
- The parent may have its `left` pointing to `n`. Test this using:

```
1  else if (n.key < (n.parent).key)
2    (n.parent).left = p;
3  else
4    (n.parent).right = p;
```



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Traversing trees—depth-first

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1 public void NlrTraverse (tree t) {  
2     visit(t);  
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- A **visit** to a node could entail printing its value or processing it somehow.

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```
1 queue Q = new queue();
2 public void BFS(tree t) {
3     Q.enqueue(t);
4     while (!Q.isEmpty()) {
5         n = Q.dequeue();
6         Q.enqueue(n.left);
7         Q.enqueue(n.right);
8         visit(n); } }
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For every node its key is greater or equal to the key of any of its descendants.

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- *buildUp(node X[], int sizeX)*;
- *heapify(int i)*;
- *sort(node X[], int sizeX)*;

Heaps—Constructors, gets and sets

```
1 public class Heap() {  
2     private int final maxSize = 1 << 20;  
3     private int n = 0; // n is the size of the heap  
4     private node[] A = new node[maxSize+1];  
5  
6     public Heap(){ }  
7  
8     int getSize(){  
9         return n; }  
10    void setSize(int i){  
11        n = i; }  
12  
13    ... }
```

Heaps—*heapify(int i)*

```
1 public node heapify(int i) {  
2  
3      $\ell$  = left(i); r = right(i);  
4     if ( $\ell$  <= n && A[ $\ell$ ].key > A[i].key)  
5         top =  $\ell$ ;  
6     else  
7         top = i;  
8  
9     if (r <= n && A[r].key > A[top].key)  
10         top = r;  
11  
12     if (top != i) {  
13         swap(A[i], A[top]); // this is not Java  
14         heapify(top); } }
```

Heaps—*heapify(int i)*—a bit faster

```
1 public node heapify(int i) {  
2  
3      $\ell = 2 \times i$  ;  $r = \ell + 1$ ;  
4     if ( $\ell \leq n$  &&  $A[\ell].key > A[i].key$ )  
5         top =  $\ell$ ;  
6     else  
7         top = i;  
8  
9     if ( $r \leq n$   
10        &&  $A[r].key > A[top].key$ )  
11        top = r;  
12  
13     if (top != i) {  
14         swap( $A[i]$ ,  $A[top]$ );  
15         if (left(top) <= n)  
16             heapify(top); } }
```

Heaps—*heapify(int i)*—iterative

First, we see if `child = left(i)` is within the heap. Next, make `child += 1` if the right child is bigger than the left child, then `break` if `A[i] >= A[child]` or carry on by swapping `A[i]` and `A[child]` and repeat another level.

```
1 public node heapify(int i) {
2   int child = left(i);
3   while (child <= n)
4     if (child < n
5         && A[child] < A[child+1])
6       child++;
7     if (A[child] < A[i]){
8       swap(A[i], A[child]); // not Java
9       i = child;
10      child = left(i); }
11   else
12     break } }
```


Heaps—*insert(node N)*

```
1 public node insert(node N) {  
2  
3     n++;  
4     i = n;  
5     while (i > 1 && A[parent(i)].key < N.key) {  
6         A[i] = A[parent(i)];  
7         i = parent(i); }  
8     A[i] = N; }
```

Heaps—*node removeMax()*

```
1 public node remove() {  
2     if (n < 1)  
3         return null;  
4     else {  
5         top = A[1];  
6         A[1] = A[n];  
7         n--;  
8         heapify(1);  
9         return top; } }
```

Heaps—*node buildUp(node X[], int sizeX)*

Pseudocode:

```
1 // pre: given an array X of sizeX nodes
2 // post: node X[] is copied into the
3 // heap array, node A[] which is
4 // turned into a heap of size n.
5 public void buildUp(node X[], int sizeX) {
6
7     for (i = 1; i <= sizeX; i++)
8         A[i] = X[i];
9
10    n = sizeX;
11    for (i = n/2; i > 0; i--)
12        heapify(i); }
```

Heaps—*node [] sort(node X[], int sizeX)*

```
1 // pre: Given array X[] of sizeX elements
2 // n becomes the size of the heap.
3 // post: The array node A[] is sorted.
4
5 public node [] sort(node X[], int sizeX) {
6     buildUp(X, sizeX);
7     for (i = n; i>1; i--){
8         swap(A[1],A[i]);
9         n--;
10        heapify(1); }
11    n = sizeX;
12    return A;}
```

Note that the heap is both created and “destroyed” in the process of sorting using **buildUp** and **sort**.

The procedure invocation **swap(A[1], A[i])** needs to be adapted to your programming language. Try **swap(A, 1, i);** in Java.

Heaps—*node sort(node X[], int sizeX)*

```
1 // pre: Given array X[] of sizeX elements
2 // n becomes the size of the heap.
3 // post: The array node A[] is sorted.
4
5 public void sort(node X[], int sizeX) {
6     buildUp(X, sizeX);
7     for (i = n; i>1; i--){
8         swap(A[1],A[i]);
9         n--;
10        heapify(1); }
11    copy A back to X;}
```

Note that the heap is both created and destroyed in the process of sorting using **buildUp** and **sort**.

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- In a heap of n nodes call the total time to run `buildUp`, T .

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- and it follows that $T = O(n)$.

$$S_H = \sum_{h=1}^H h2^{-h-1} < 1$$

$$\begin{aligned} S_H &= \sum_{h=1}^H h2^{-h-1} \\ &= 1.2^{-2} + 2.2^{-3} + \dots + (H-1).2^{-H} + H.2^{-H-1} \\ 2S_H &= 1.2^{-1} + 2.2^{-2} + 3.2^{-3} + \dots + H.2^{-H} \end{aligned}$$

Subtracting S_H from $2S_H$ gives

$$\begin{aligned} S_H &= \underbrace{2^{-1} + 2^{-2} + 2^{-3} + \dots + 2^{-H}}_{0.111\dots 11_2} - H.2^{-H-1} \\ &= 0.111\dots 11_2 - H.2^{-H-1} \\ &= 1 - 2^{-H} - H.2^{-H-1} \\ &< 1 \end{aligned}$$

An n -node heap has height $H = \lfloor \log_2 n \rfloor$

Suppose that the root of a heap has height 0.

n	height	$\lfloor \log_2 n \rfloor$
1	0	0
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3	1	1
4	2	2
5	2	2
6	2	2
7	2	2
8	3	3
9	3	3
10	3	3
11	3	3
12	3	3
13	3	3
14	3	3
15	3	3
16	4	4
31	4	4
32	5	5
64	6	6
127	6	6
255	7	7
256	8	8
511	8	8
1023	9	9
1024	10	10
2048	11	11

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- The height of the i^{th} node in a heap is $\lfloor \log_2 i \rfloor$.

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Heaps—applications

Heaps have many applications in other algorithms.²

Sorting	Heap sort is optimal $\sim 2N \log N$
Data compression	Combine smallest two elements in Huffman coding
Event-driven simulation	Put most important request first
Physics simulation	Next particle to collide
Dijkstra's shortest path	Finds closest node
Prim's MST algorithm	Choosing edge with minimal weight
Number theory	Sum of powers
Artificial intelligence	A^* search
Statistics	Maintain largest K values in a sequence
Finding the median	Compare using quicksort for median
Operating systems	Prioritize interrupts, load balancing
Discrete optimization	Bin packing, scheduling
Bayesian spam filtering	Filter messages with highest spam weight
Reducing roundoff error	How is heap used?

²The list was inspired by [Sedgewick and Wayne, 2011]

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Which is clearly *less* than 4096.

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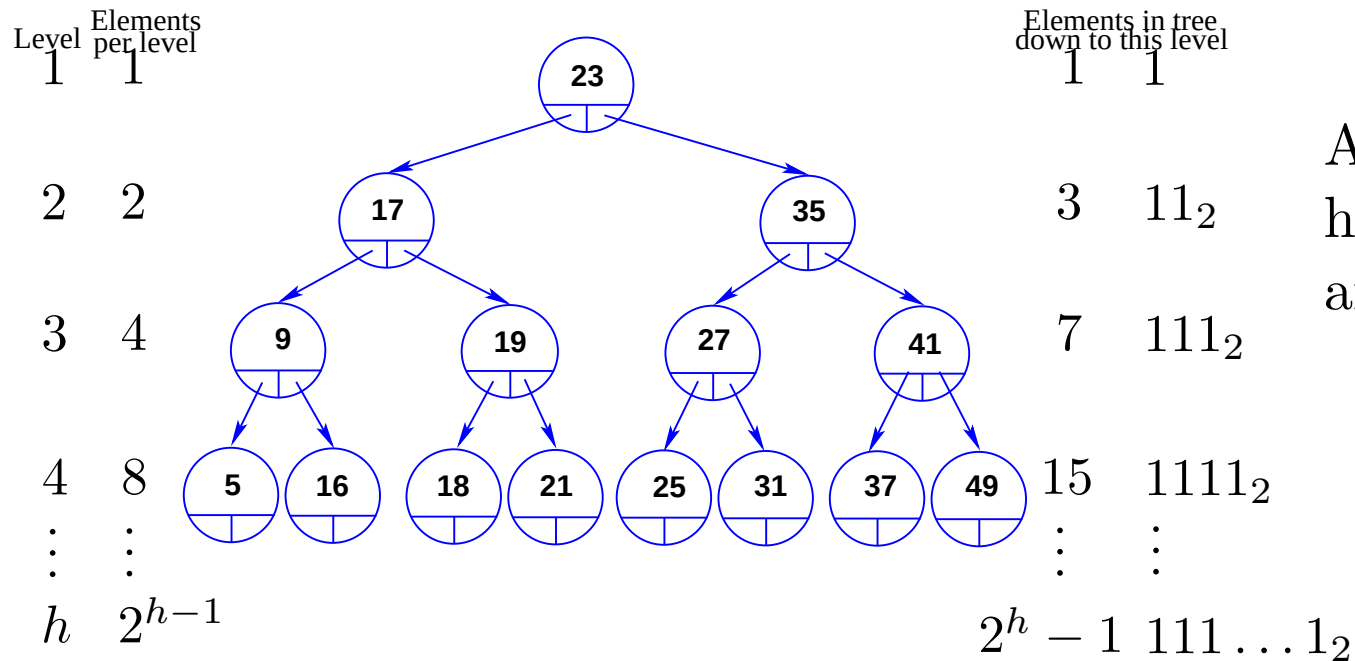
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A complete binary search tree

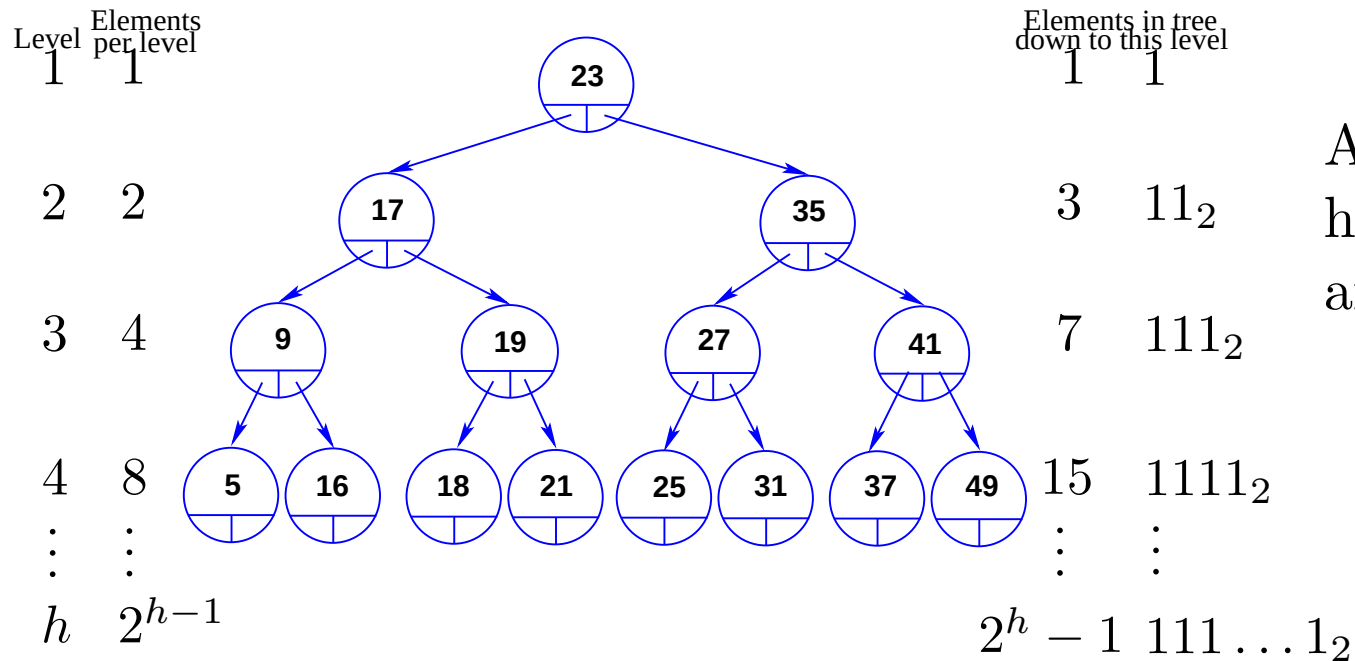


A tree with h levels, has $N = 2^h - 1$ nodes, and $h = \log_2(N + 1)$.

The number of probes to visit each node once in a complete BST is

$$T = \sum_{r=1}^h r 2^{r-1}, \text{ and } 2T = \sum_{r=1}^h r 2^r$$

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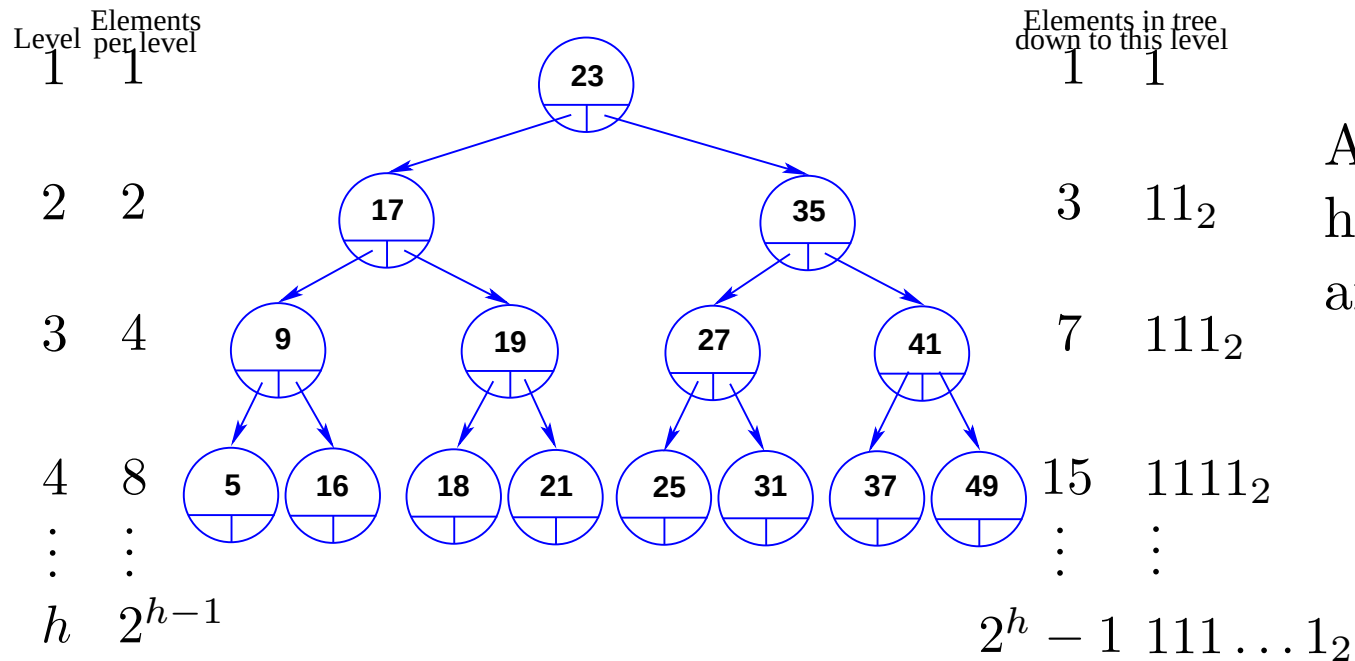
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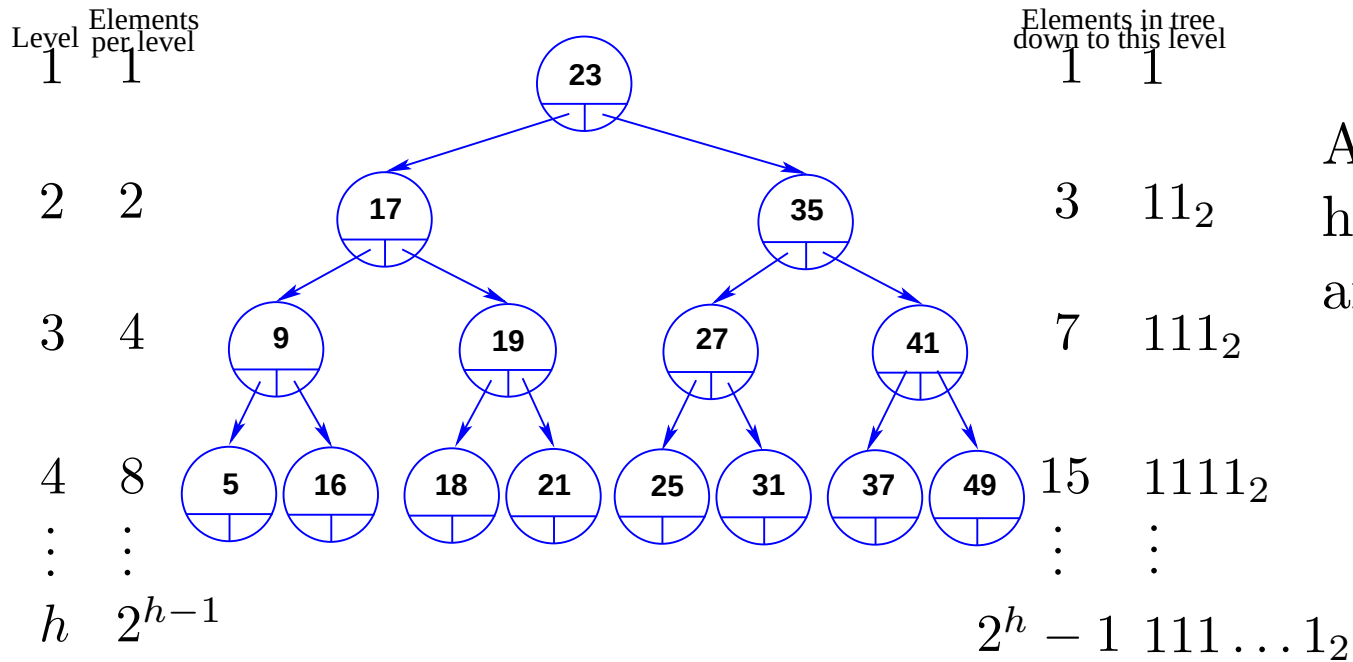
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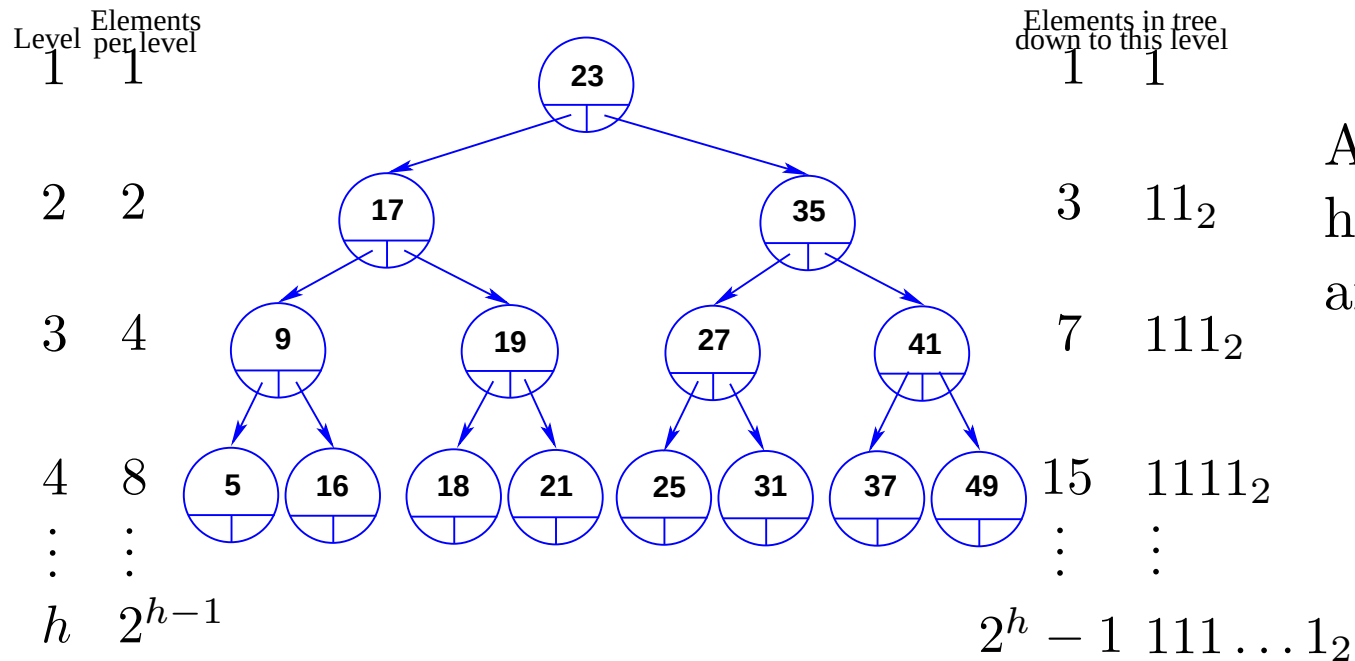


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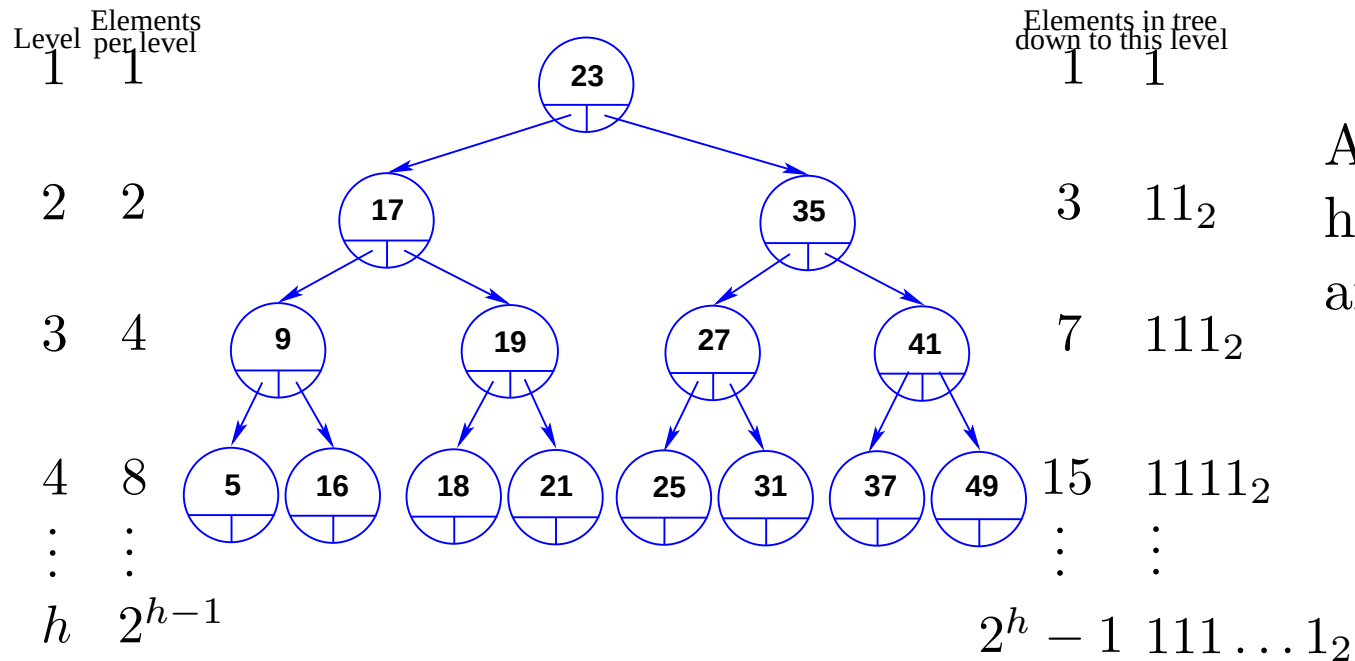


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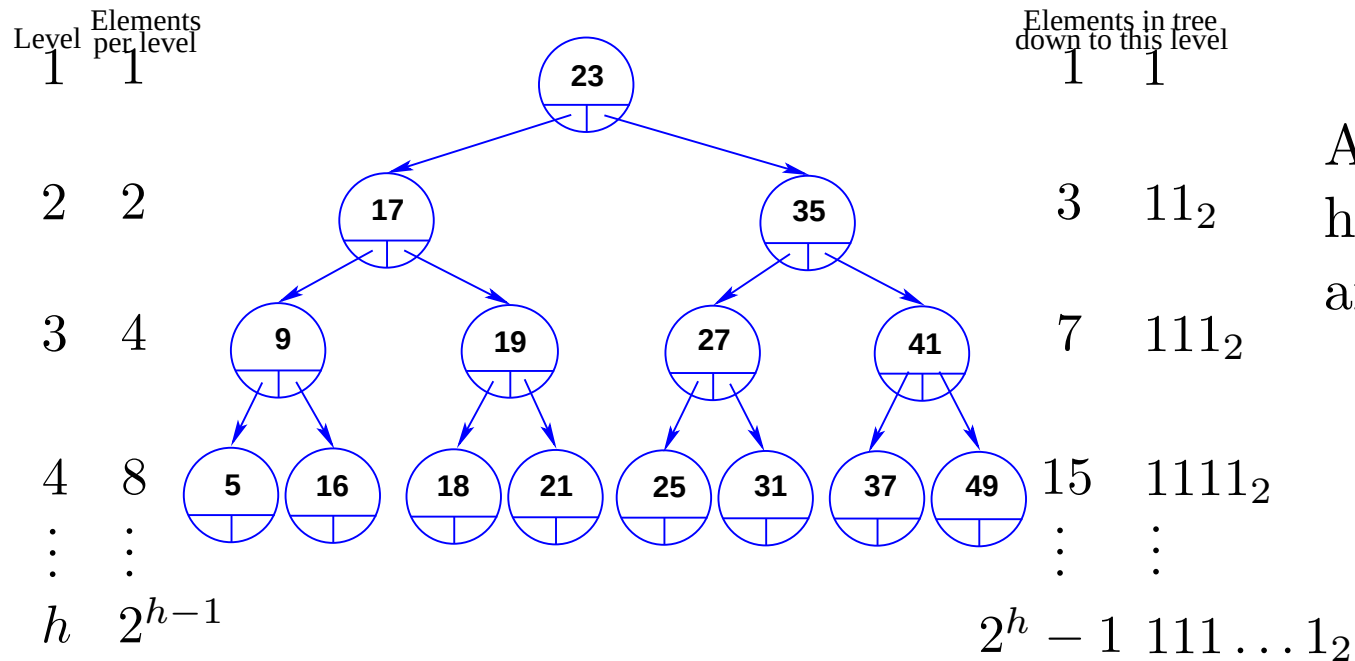


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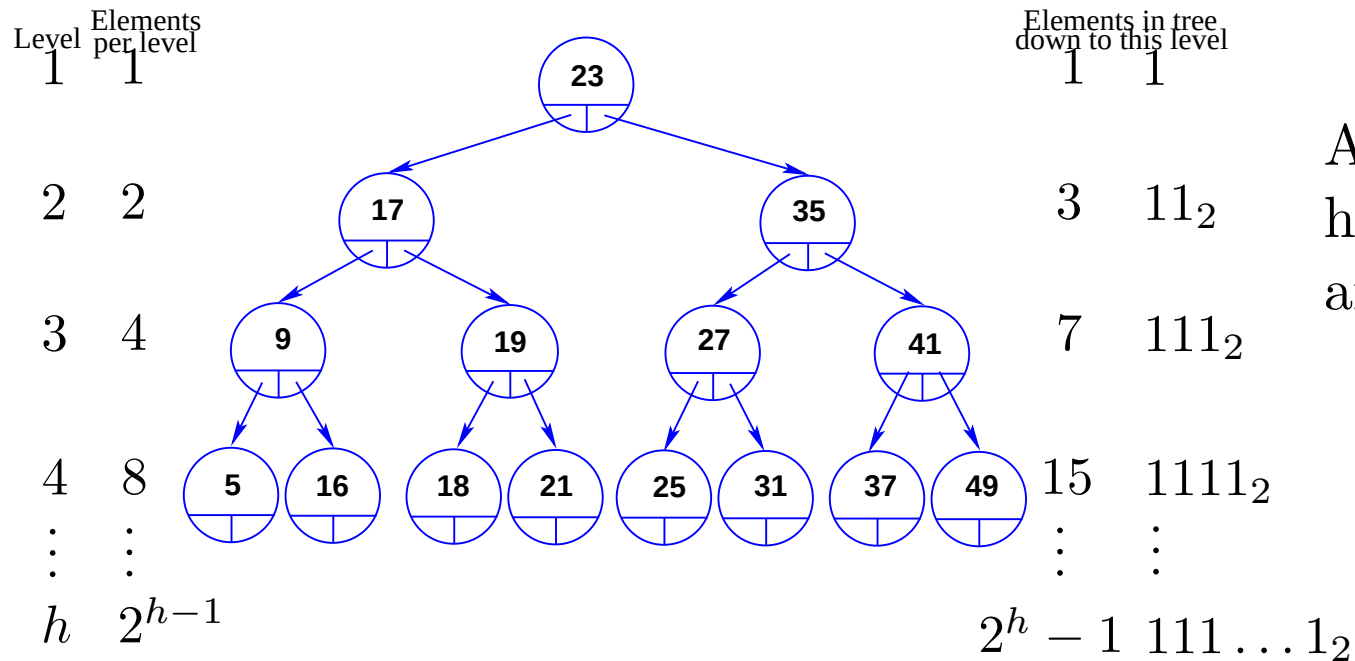


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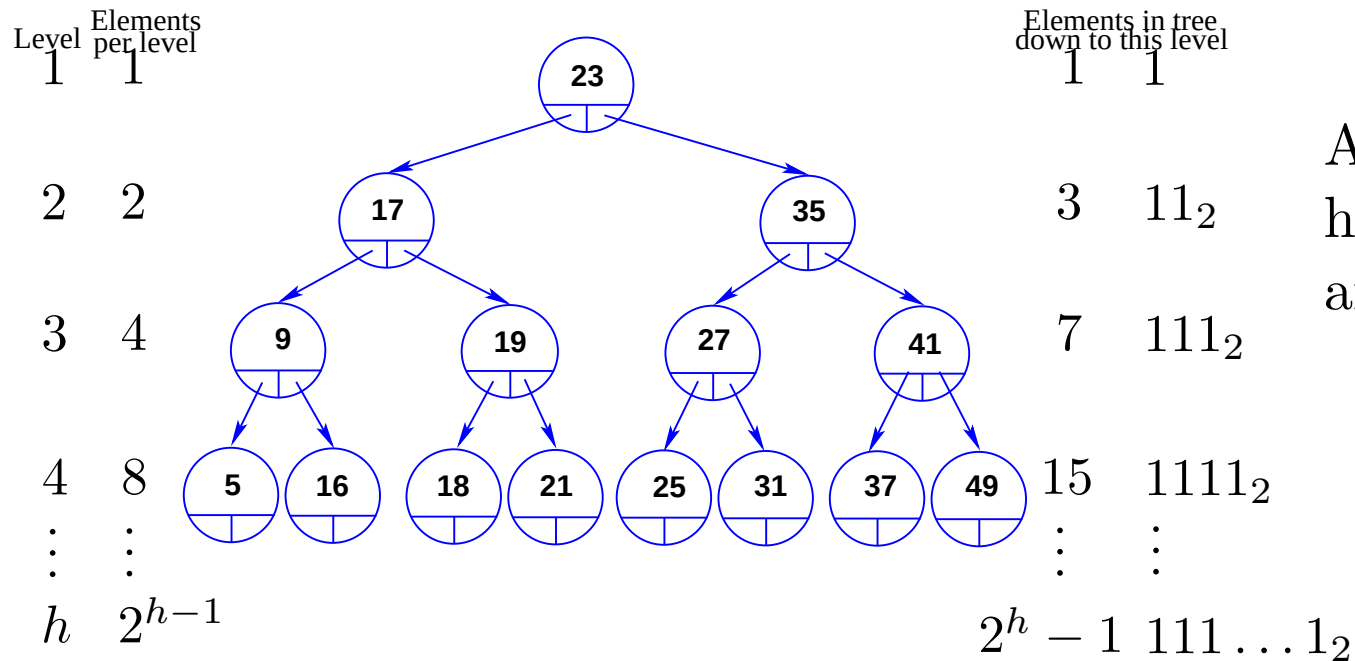


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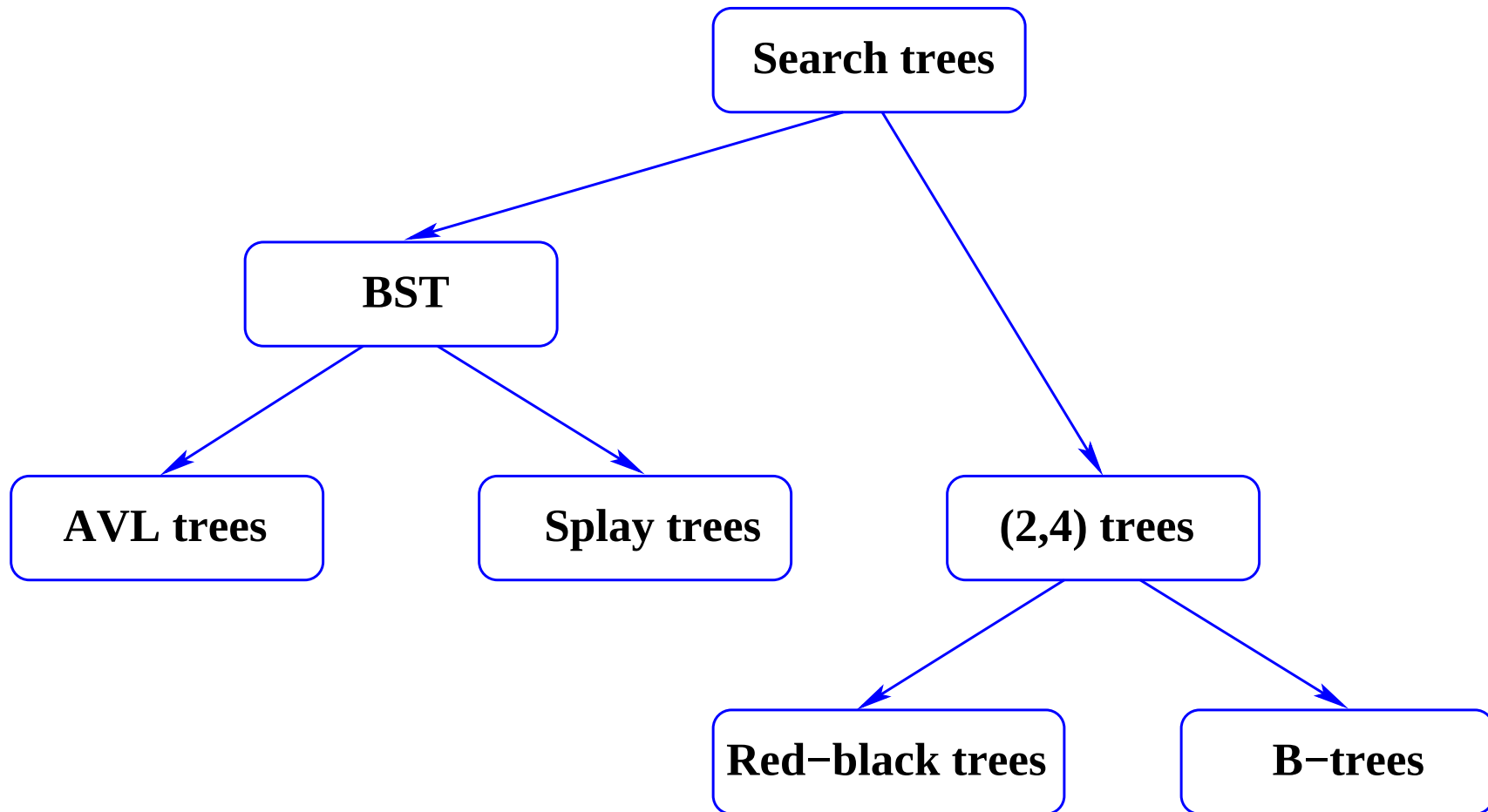


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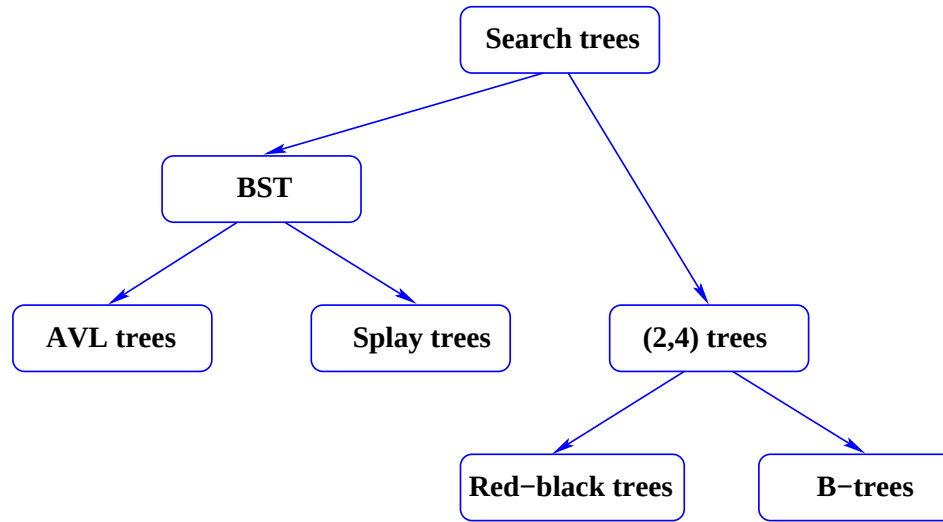
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Search trees—a hierarchy



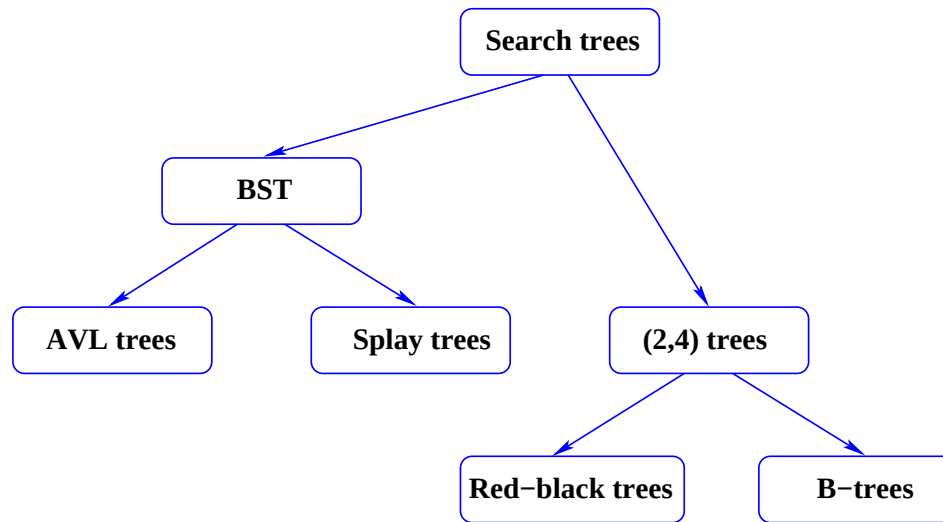
Search trees



Search trees implement the dictionary abstract data structure (ADT) and have the methods

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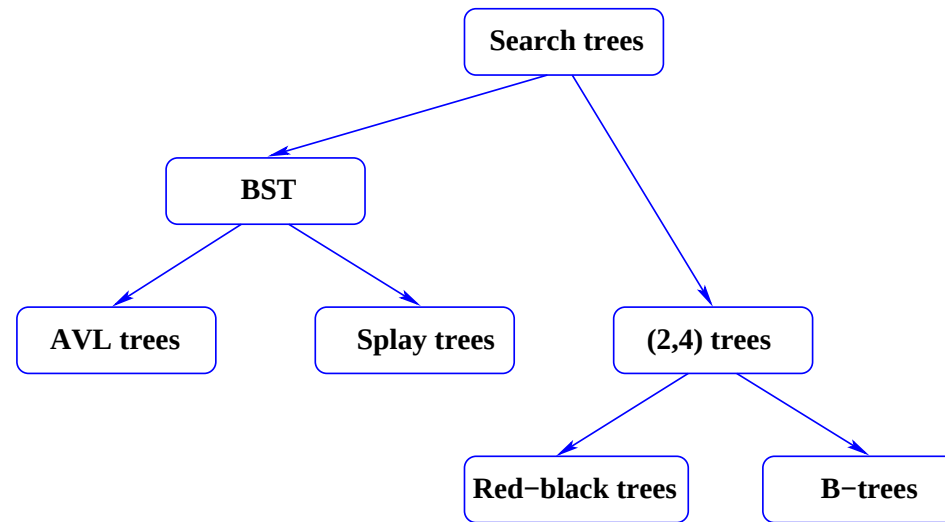
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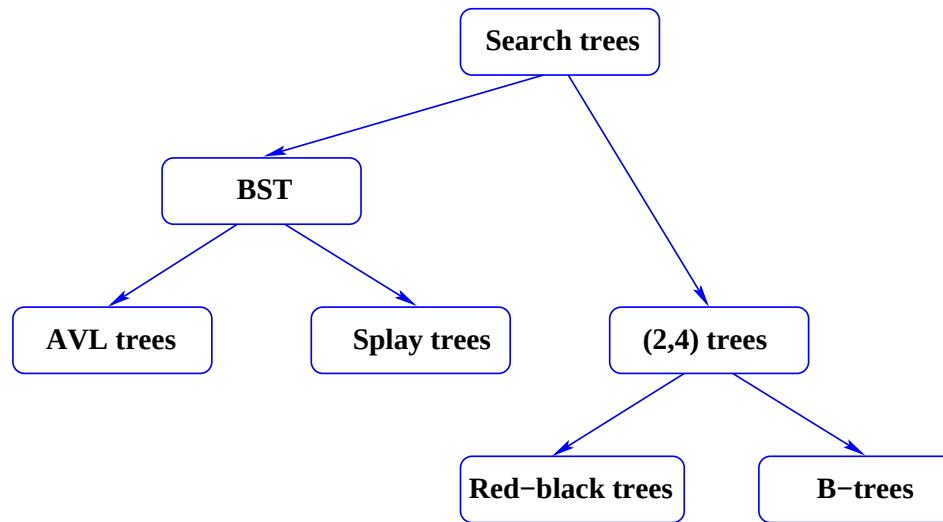
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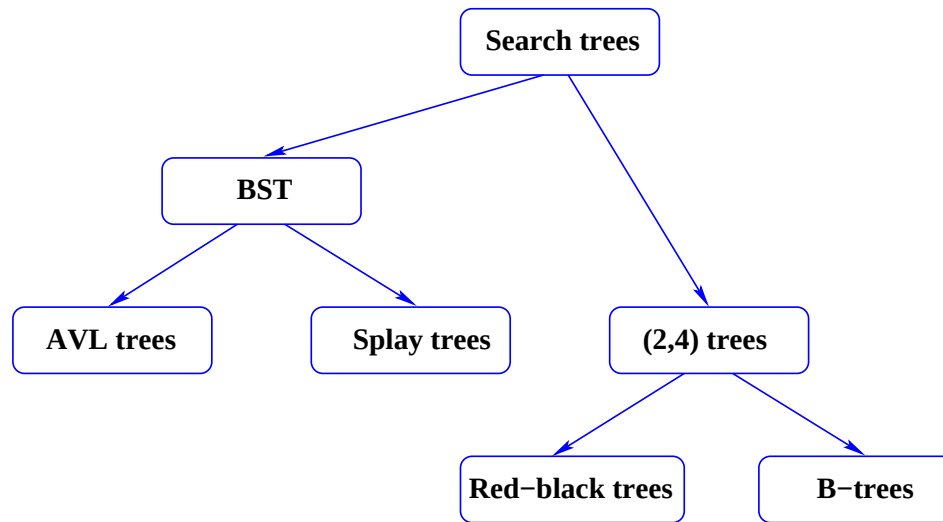
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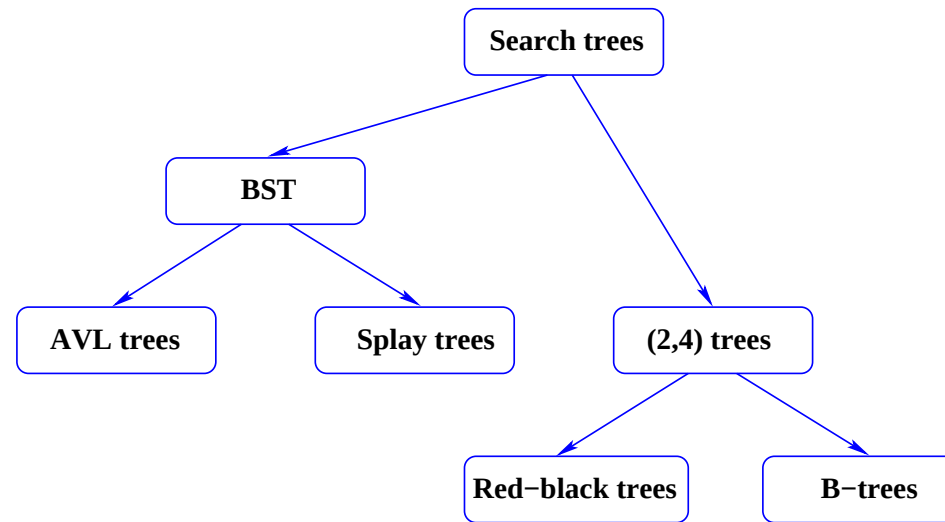
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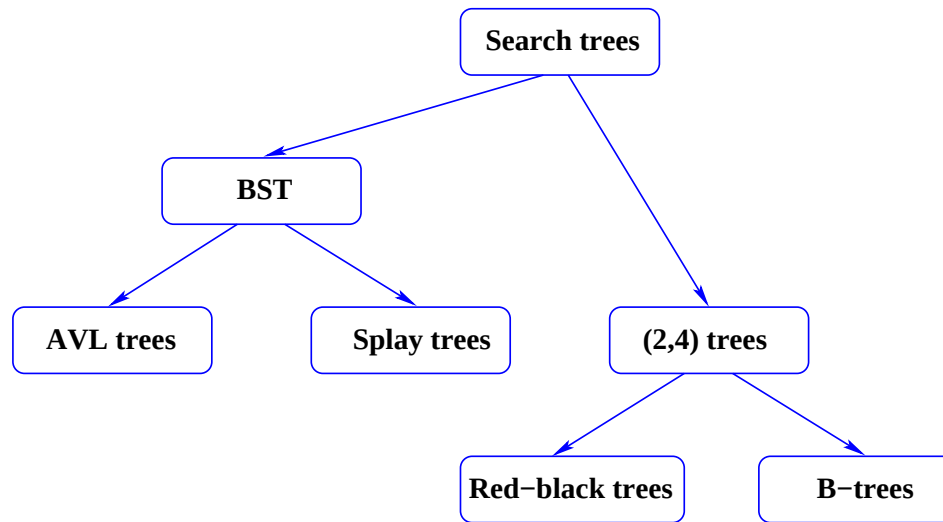
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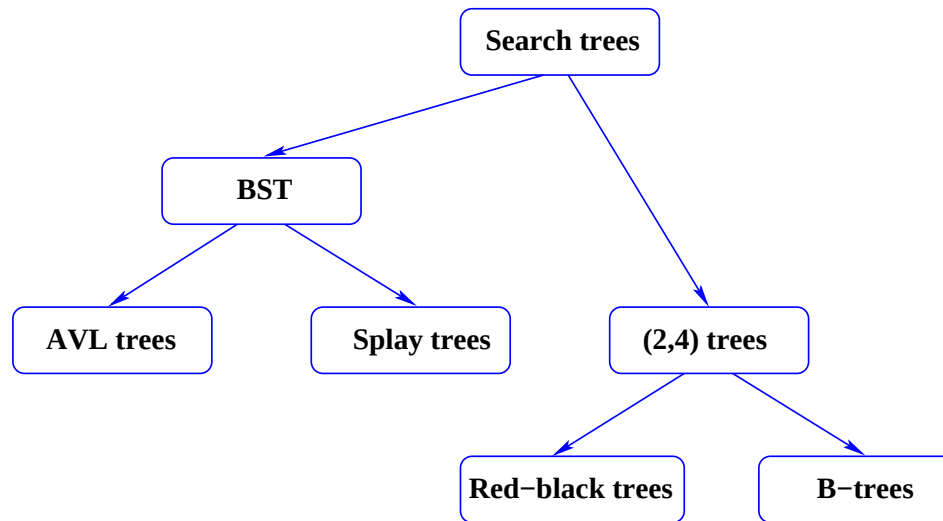
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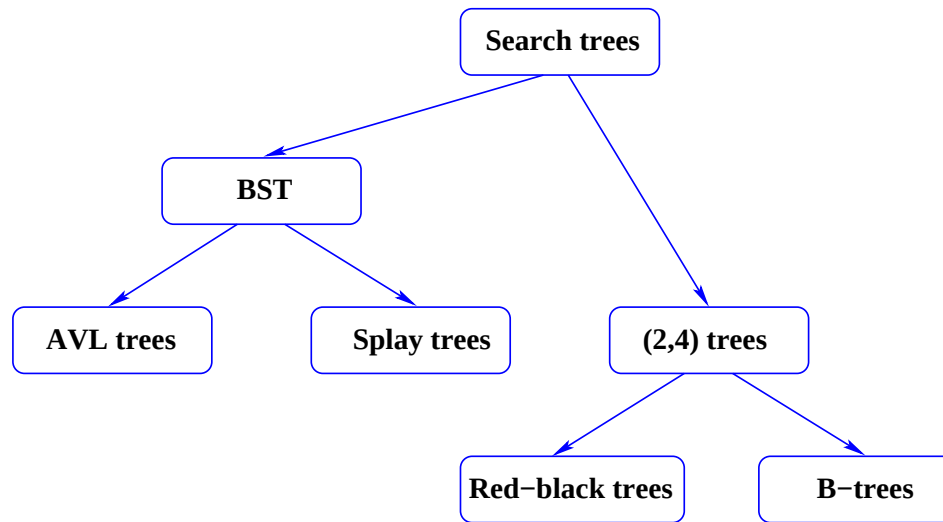
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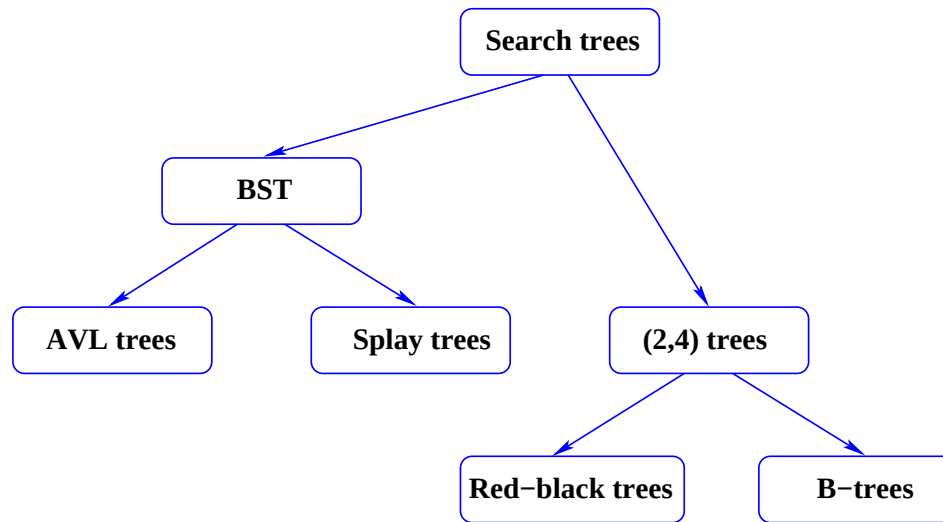
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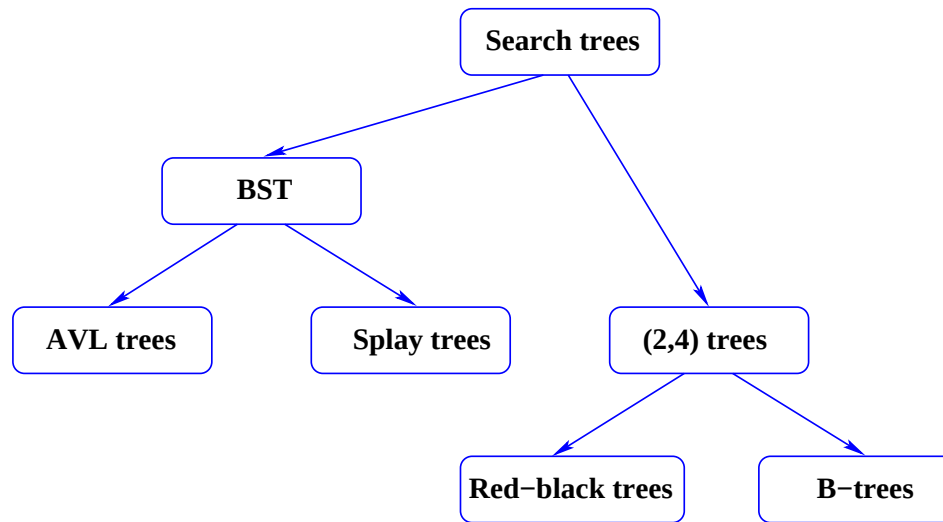
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- Retrieval in a balanced BST takes $O(\log n)$ time and in an unbalanced BST retrieval can be as bad as $O(n)$.
- So, much effort is made to keep BSTs balanced.

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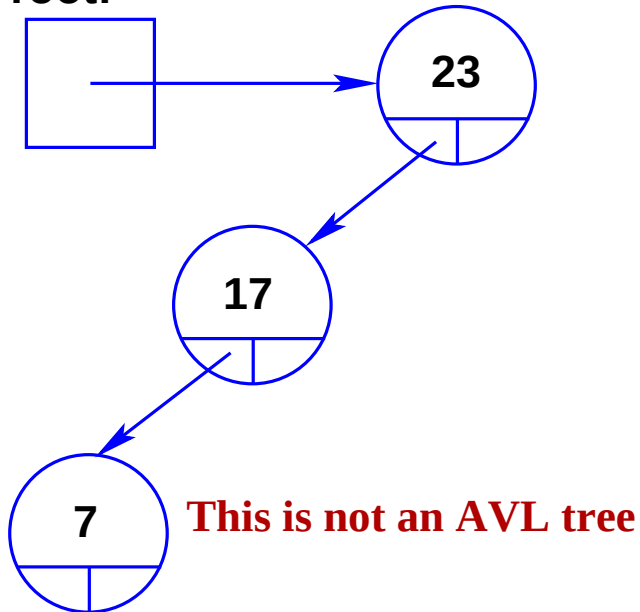
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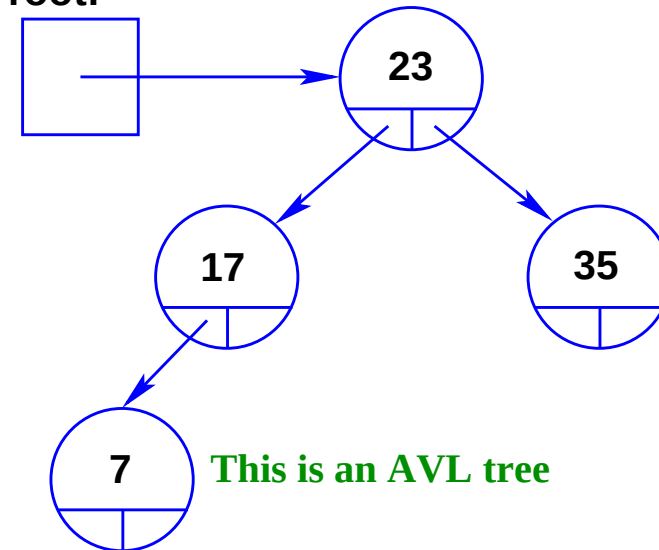
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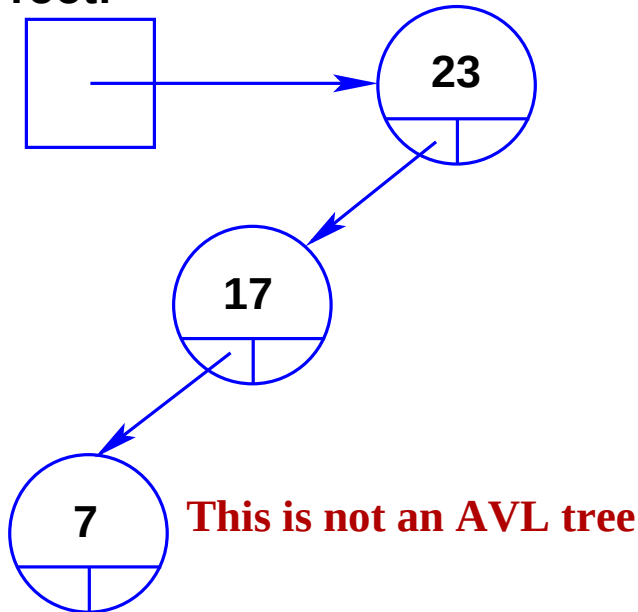
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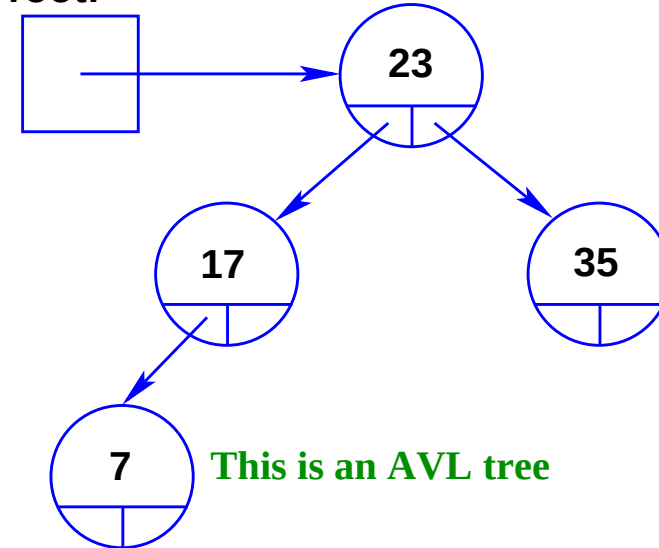
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- Since AVL trees are balanced the retrieval of nodes is at worst $\log_2 n$

AVL trees—height of nodes

- A complete height- h BST has $n = 2^h - 1$ nodes

Consider a complete BST.

Level	Nodes on level	Total
1	1	1
2	2	3
3	4	7
4	8	15
5	16	31
6	32	63
...	...	...
$h - 1$	2^{h-2}	$2^{h-1} - 1$
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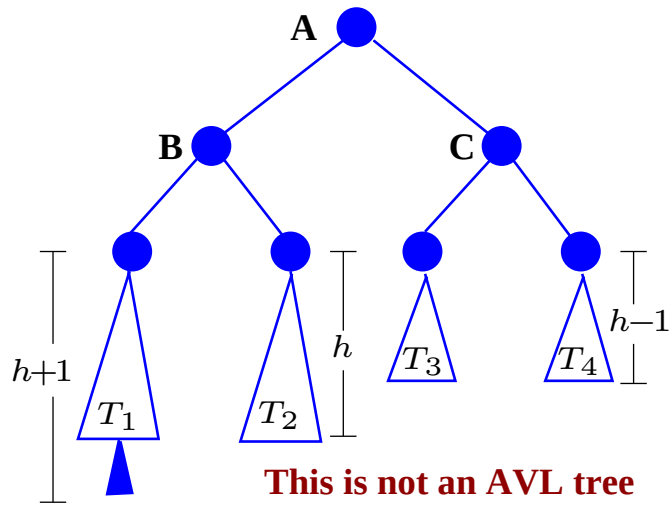
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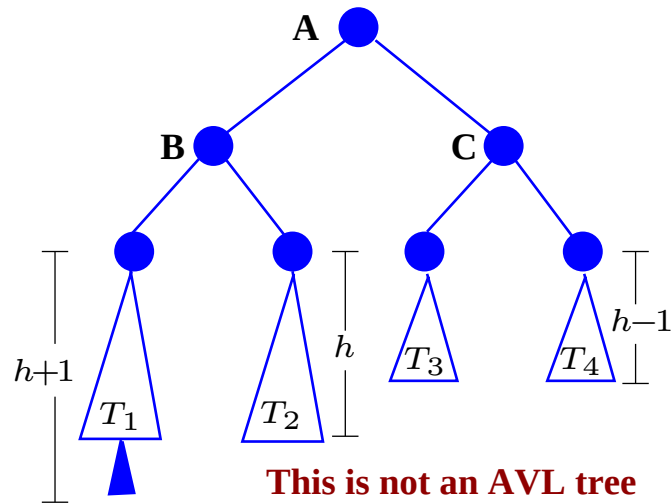
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- An AVL tree is “almost” complete so its maximum height is $O(\log n)$.

Turning a non-AVL tree into an AVL tree

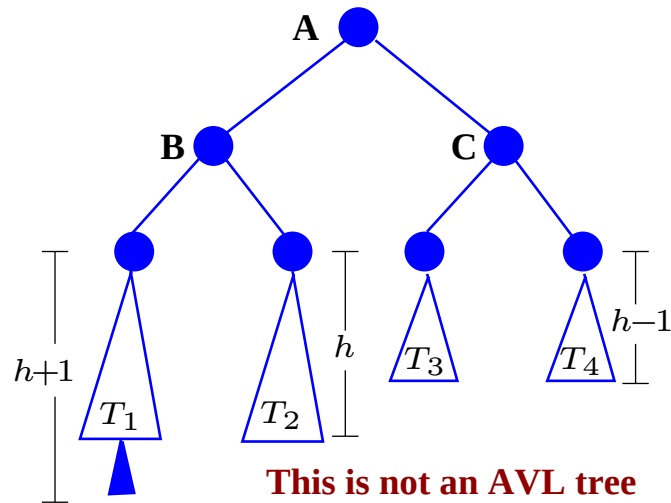


Turning a non-AVL tree into an AVL tree



Node B does not violate the AVL property, and neither does node C , but node A has a child B with height $h + 2$ and a child C with height h . The difference in the heights is 2 which violates the AVL property.

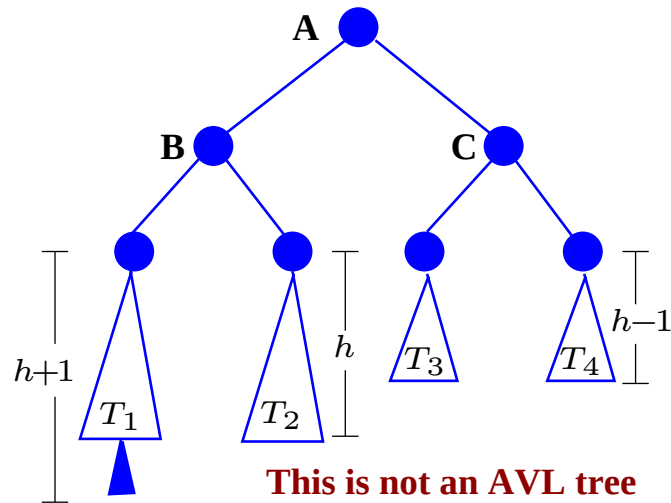
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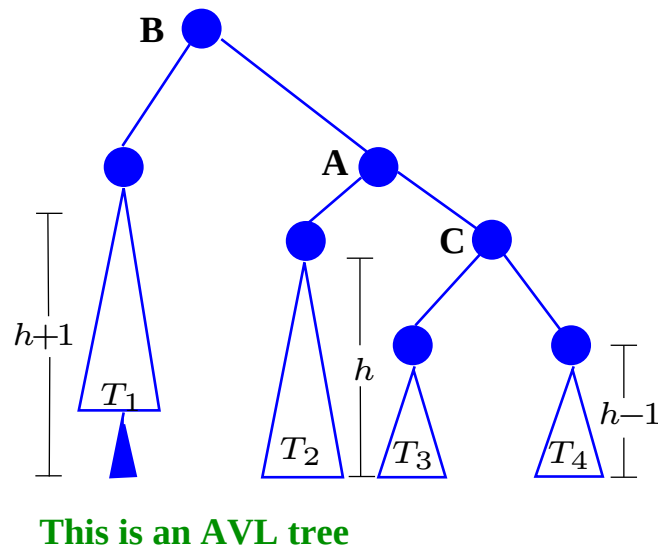
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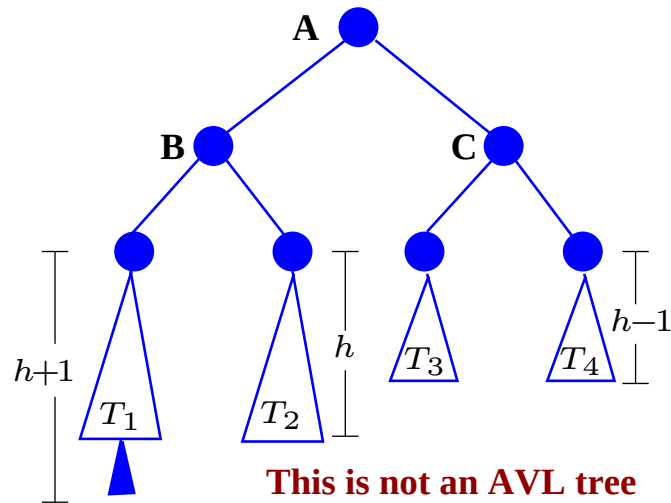


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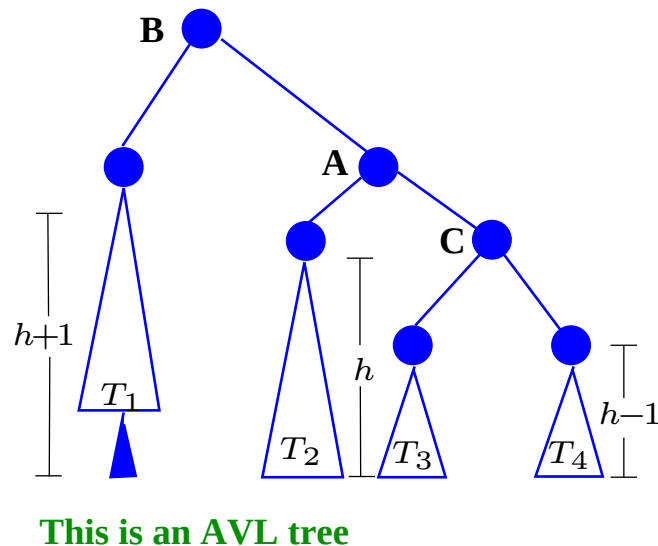


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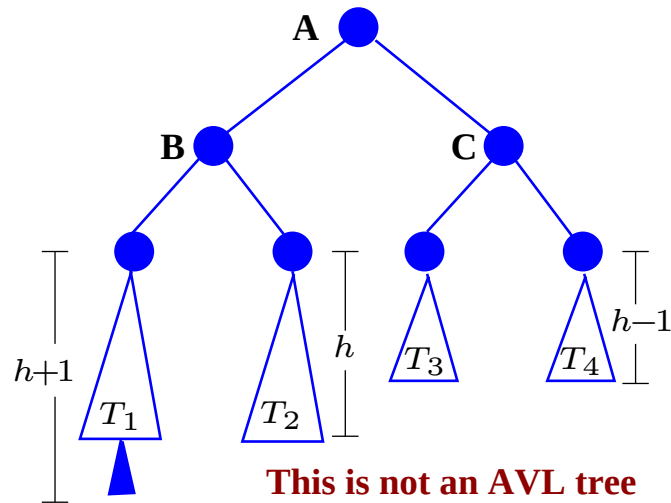
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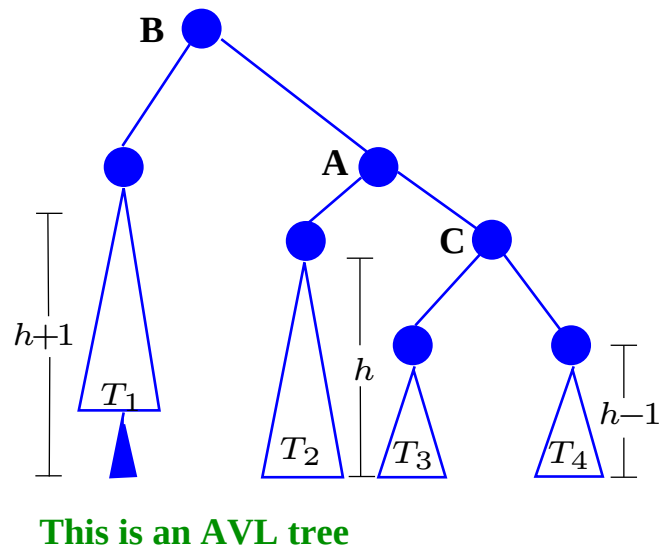
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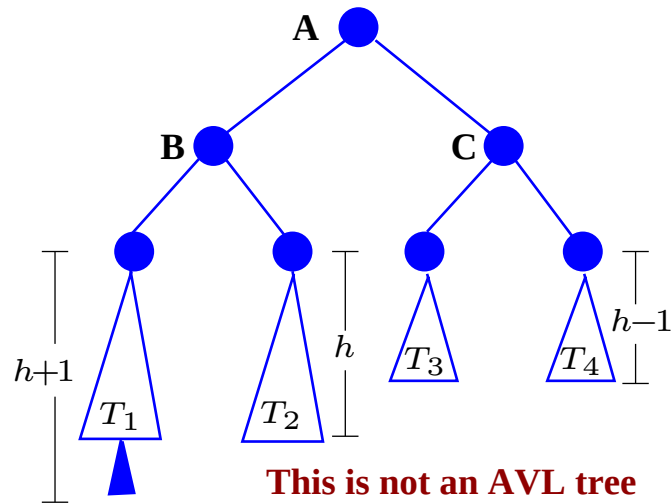
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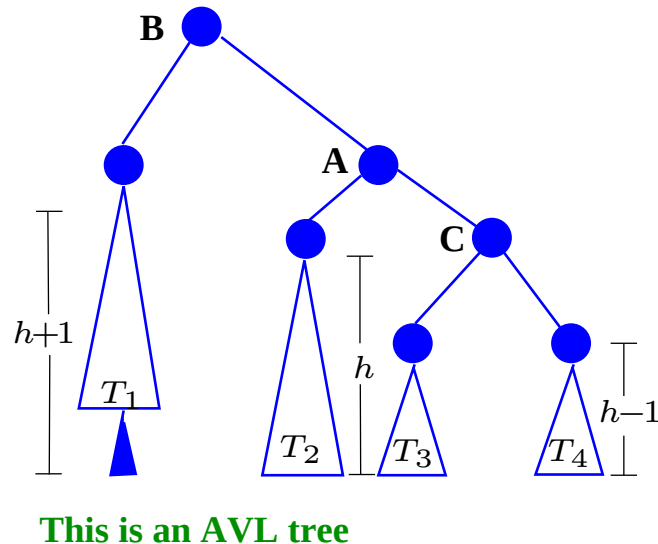
```
B = A.left;
```

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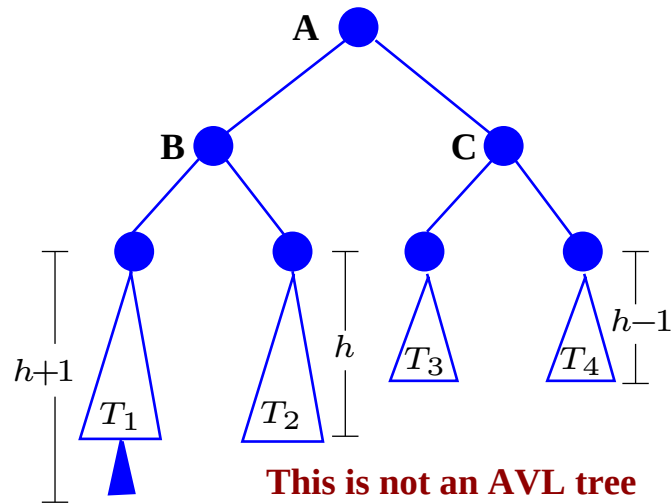
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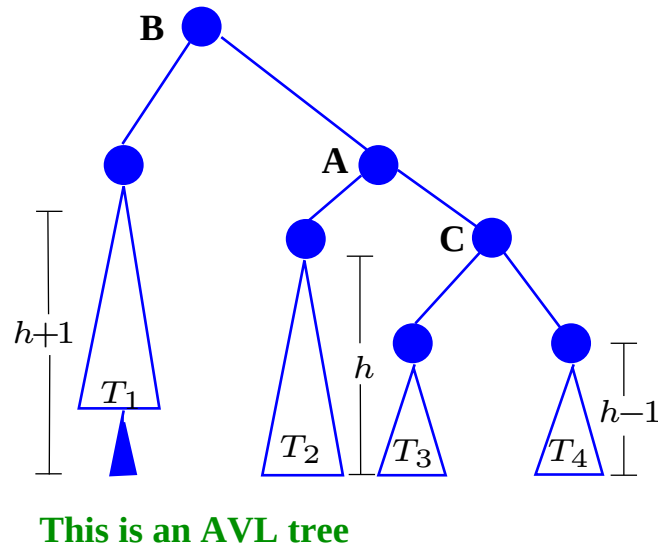
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B = A.left;  
A.left = B.right;
```

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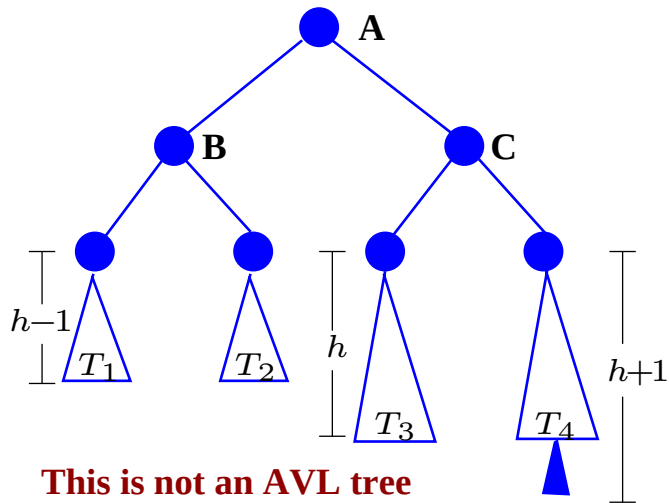


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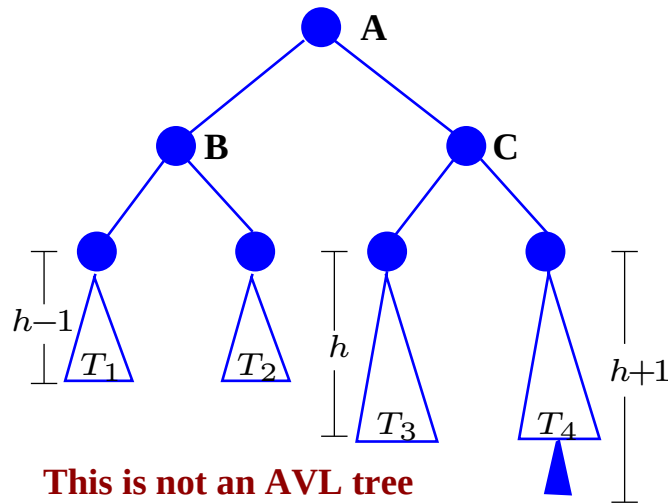
```
B = A.left;  
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```

None of the nodes now violates the AVL property.

Turning a non-AVL tree into an AVL tree—mirror

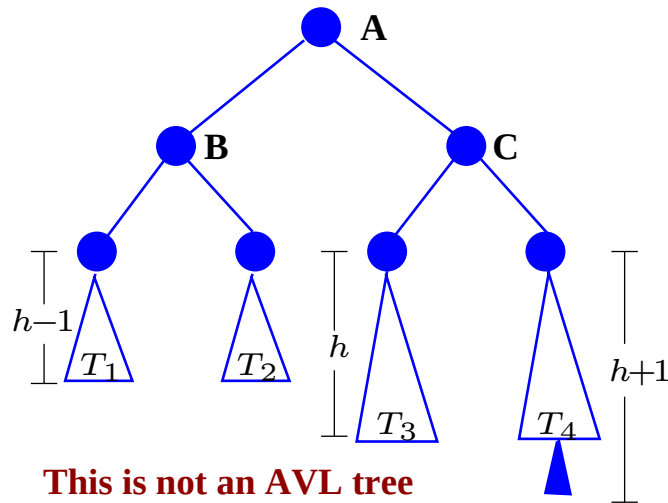


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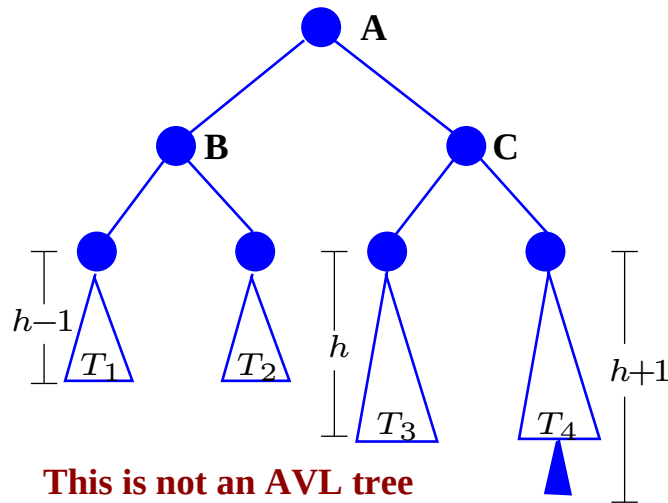
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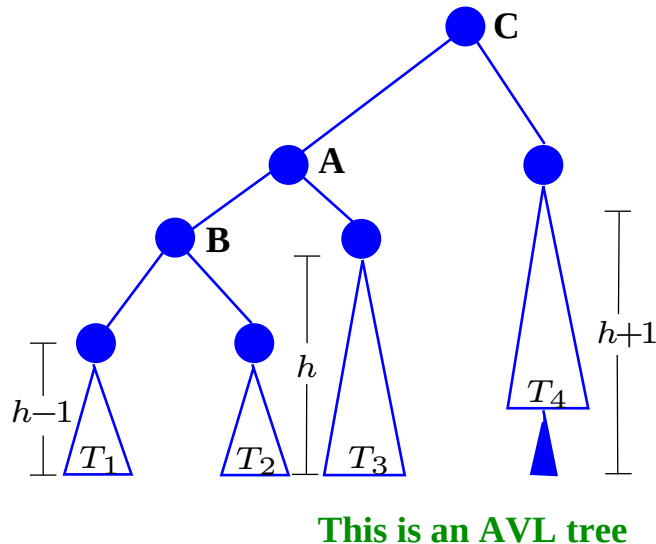
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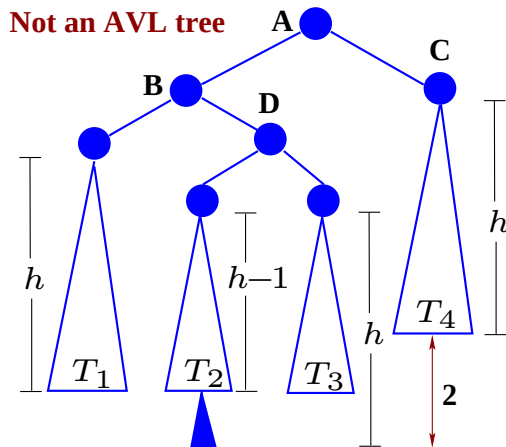


Given only A , the code is:

```
1  C = A.right;  
2  A.right = C.left;  
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```

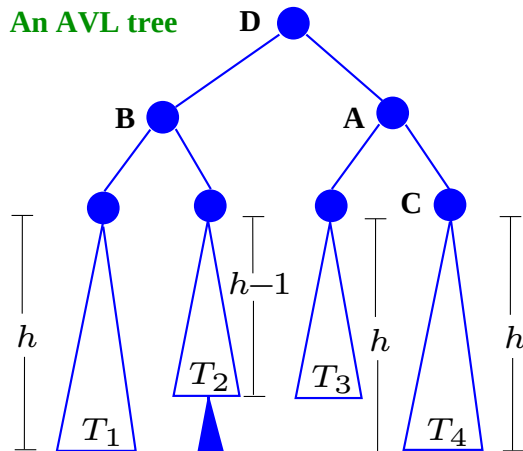
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AVL trees—double rotations



Node A violates the AVL property. The other internal nodes do not. A double rotation is required to turn this into an AVL tree.

After a double rotation, in which D becomes the root, the tree stays a BST and becomes an AVL tree:

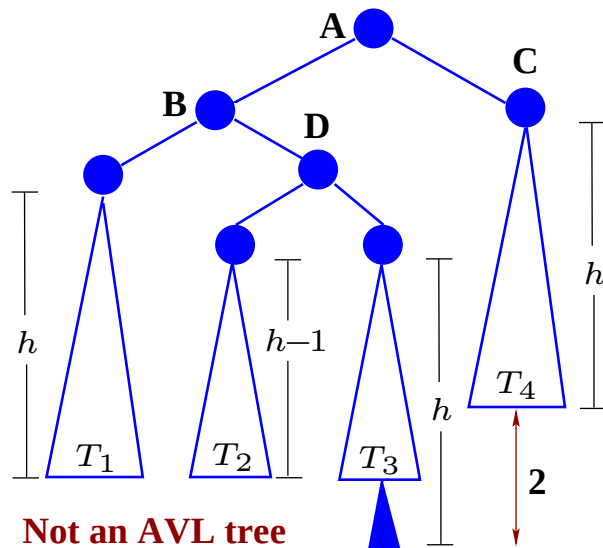


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```
1  B = A.left;  
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4  D.left = B;  
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6  D.right = A;
```

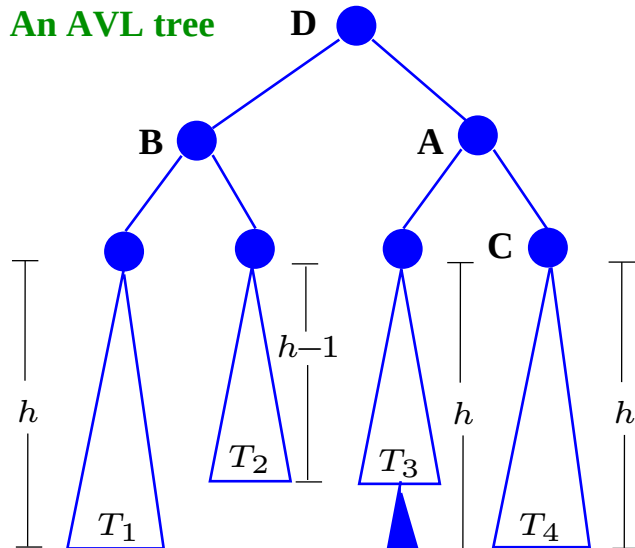
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AVL trees—double rotations: same code



Insertion at T_3 instead of at T_2 results in: Node A violates the AVL property. The other internal nodes do not. A double rotation is required to turn this into an AVL tree.

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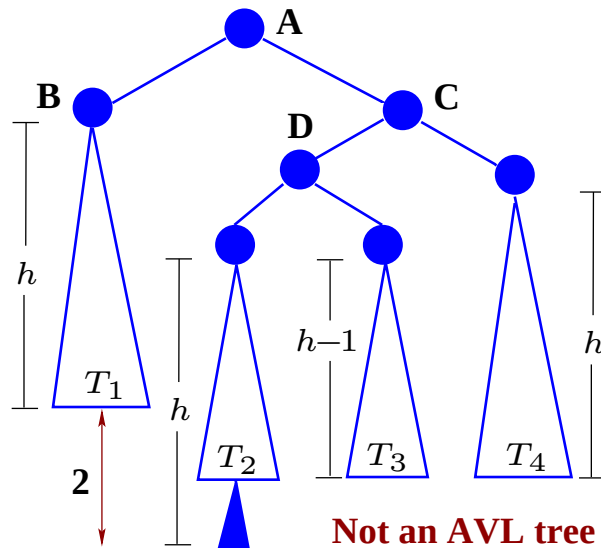


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```
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2  D = B.right;  
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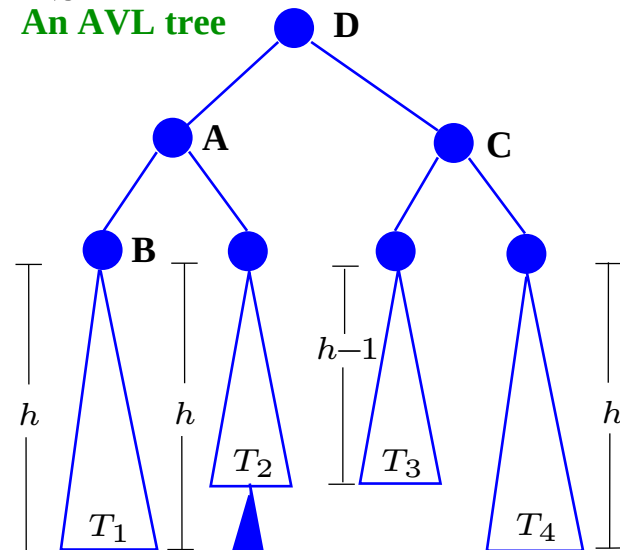
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AVL trees—double rotations: mirror



Insertion at T_2 now results in: Node A violating the AVL property. The other internal nodes do not. A double rotation is required to turn this into an AVL tree.

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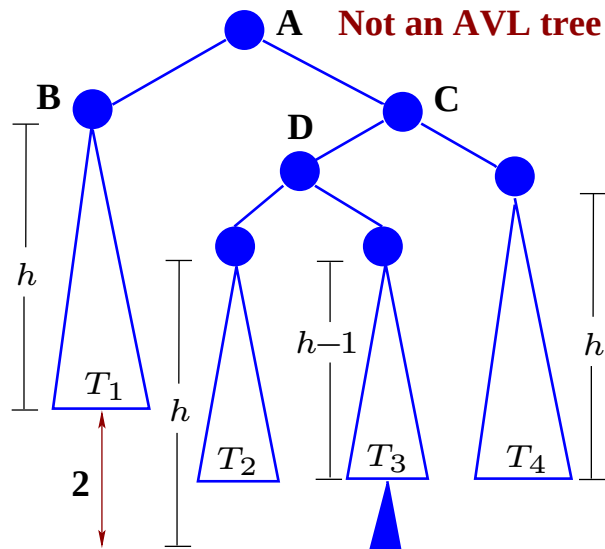


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```
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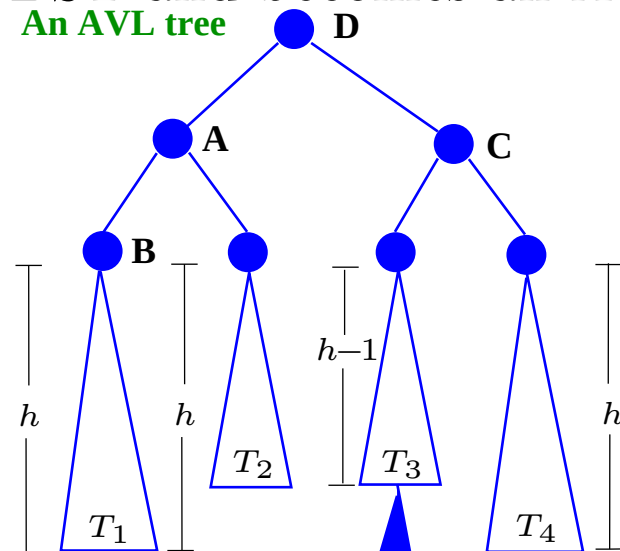
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An insertion at T_3 now results in node **A** violating the AVL property. The other internal nodes do not. A double rotation is required to turn this into an AVL tree.

After a double rotation, in which **D** becomes the root, the tree stays a BST and becomes an AVL tree:



Given only **A**, the code is:

```
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2  D = C.left;  
3  A.right = D.left;  
4  D.left = A;  
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- From this position work bottom-up to the critical node readjusting balance factors and perform a rotation if necessary.

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- Experiments have shown that AVL trees yield an average search time of $\log_2 n + 0.25$ comparisons.
- A disadvantage is the extra space, and bookkeeping required for handling the balance factors.

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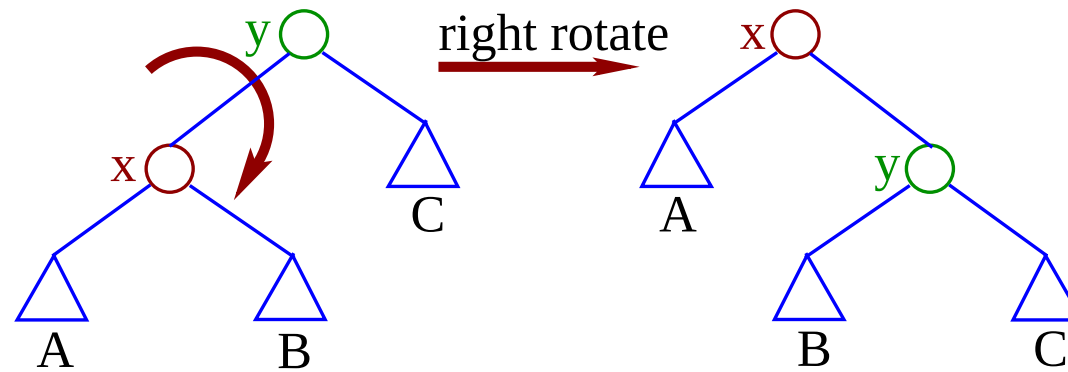
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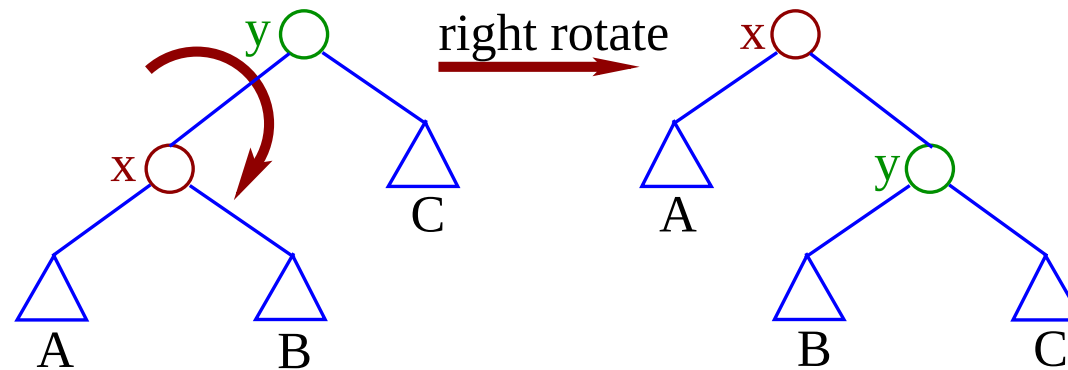
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Splay tree rotations—right rotation



Rotation of the edge joining node x and node y .
Each triangle represents a subtree.

Splay tree rotations—right rotation

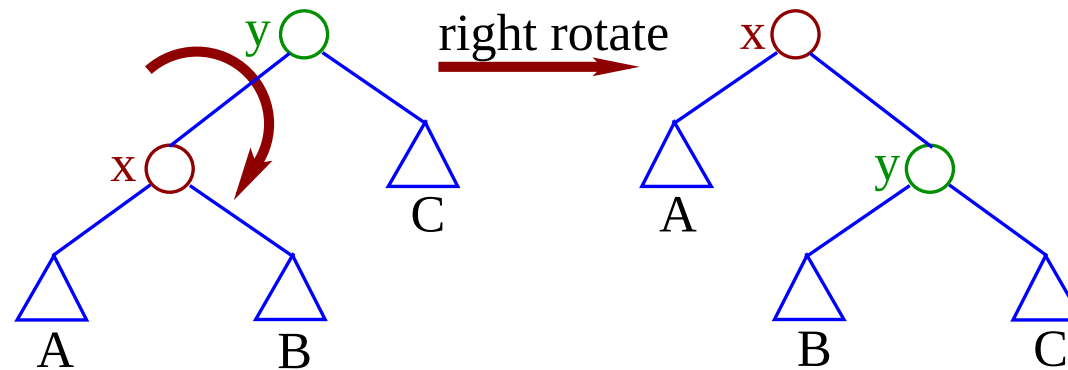


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Note that the subtrees are in the same $\ell N r$ order before and after a rotation. Rotation does not affect the ordering of the nodes in the subtrees, A , B , and C .

Splay tree rotations—right rotation



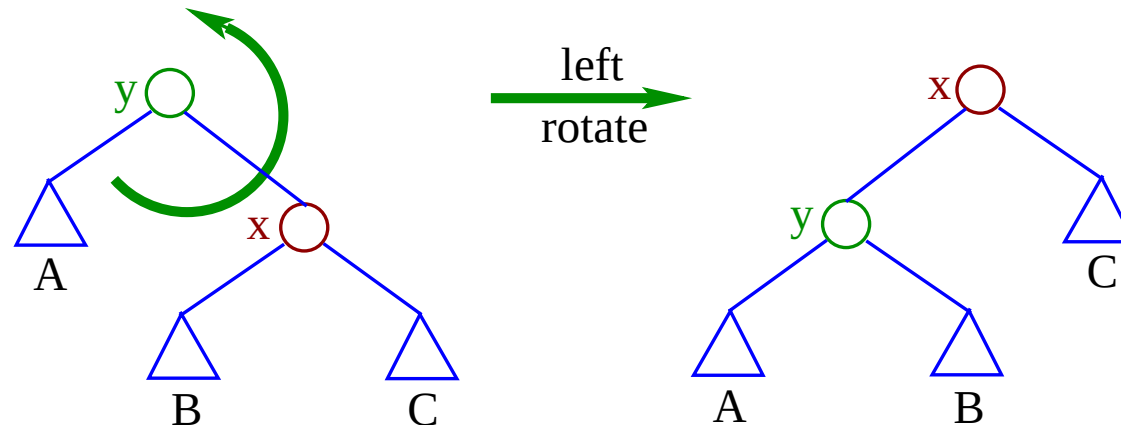
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Note that the subtrees are in the same $\ell N r$ order before and after a rotation. Rotation does not affect the ordering of the nodes in the subtrees, A , B , and C . Assuming the figure on the *left* and pointers to nodes x and y , the code to execute a *right* rotation is:

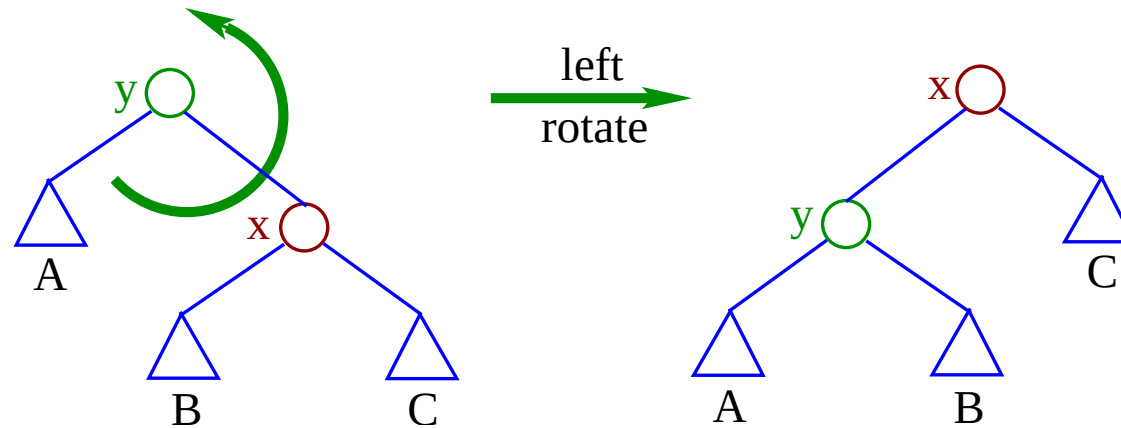
```
1 subtree B = x.right;  
2   x.right = y;  
3   y.left = B;  
4   y.parent.(left or right) = x;  
5   x.parent = y.parent;  
6   y.parent = x;
```

Splay tree rotations—left rotation



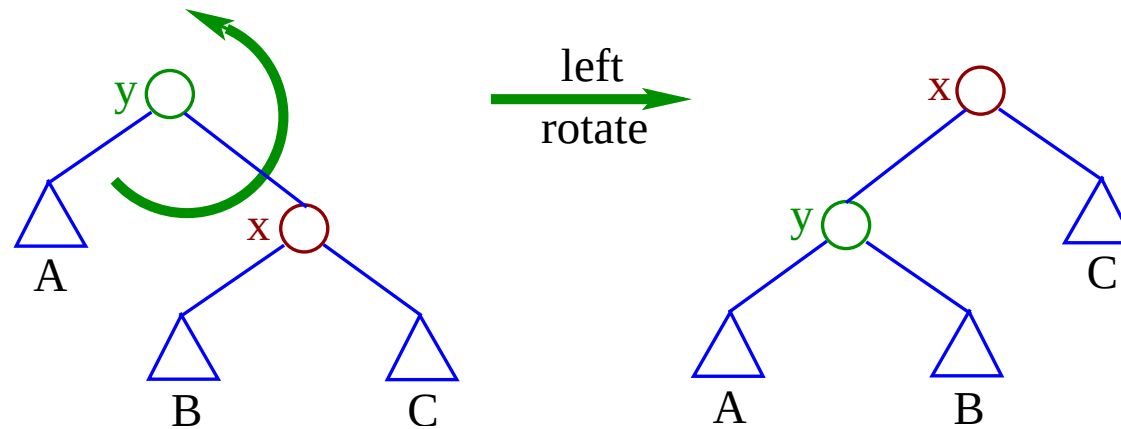
Rotation of the edge joining node x and node y .

Splay tree rotations—left rotation



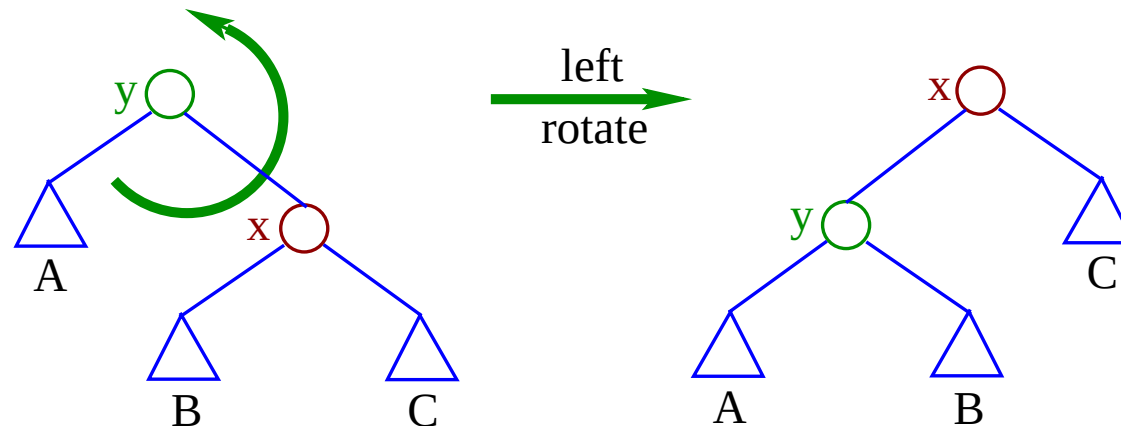
Rotation of the edge joining node x and node y . Triangles represent subtrees.

Splay tree rotations—left rotation



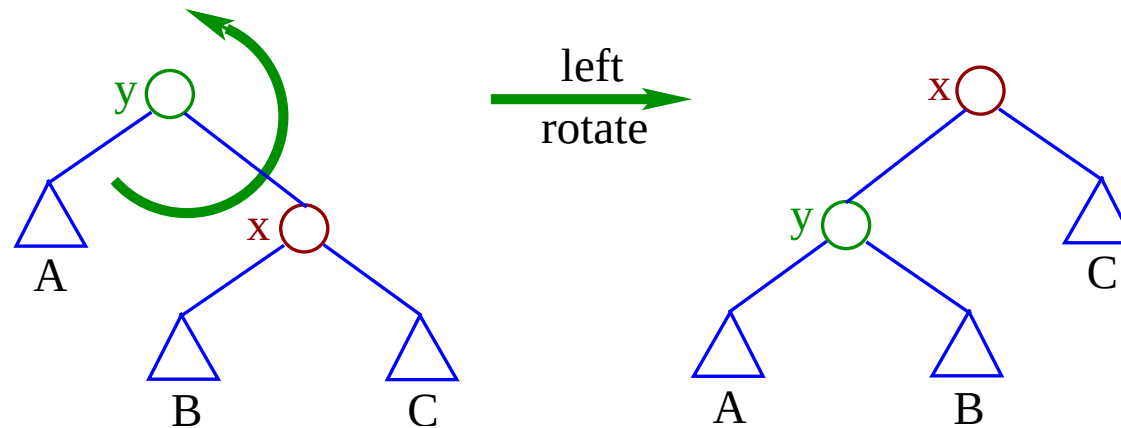
Rotation of the edge joining node x and node y . Triangles represent subtrees. Subtrees are in the same $\ell N r$ order before and after a rotation. Rotation preserves the ordering of the nodes in the subtrees, A , B , and C .

Splay tree rotations—left rotation



Rotation of the edge joining node x and node y . Triangles represent subtrees. Subtrees are in the same $\ell N r$ order before and after a rotation. Rotation preserves the ordering of the nodes in the subtrees, A , B , and C . Assuming the figure on the *left* and pointers to nodes x and y , the code to execute a *right* rotation is:

Splay tree rotations—left rotation



Rotation of the edge joining node x and node y . Triangles represent subtrees. Subtrees are in the same $\ell N r$ order before and after a rotation. Rotation preserves the ordering of the nodes in the subtrees, A , B , and C . Assuming the figure on the *left* and pointers to nodes x and y , the code to execute a *right* rotation is:

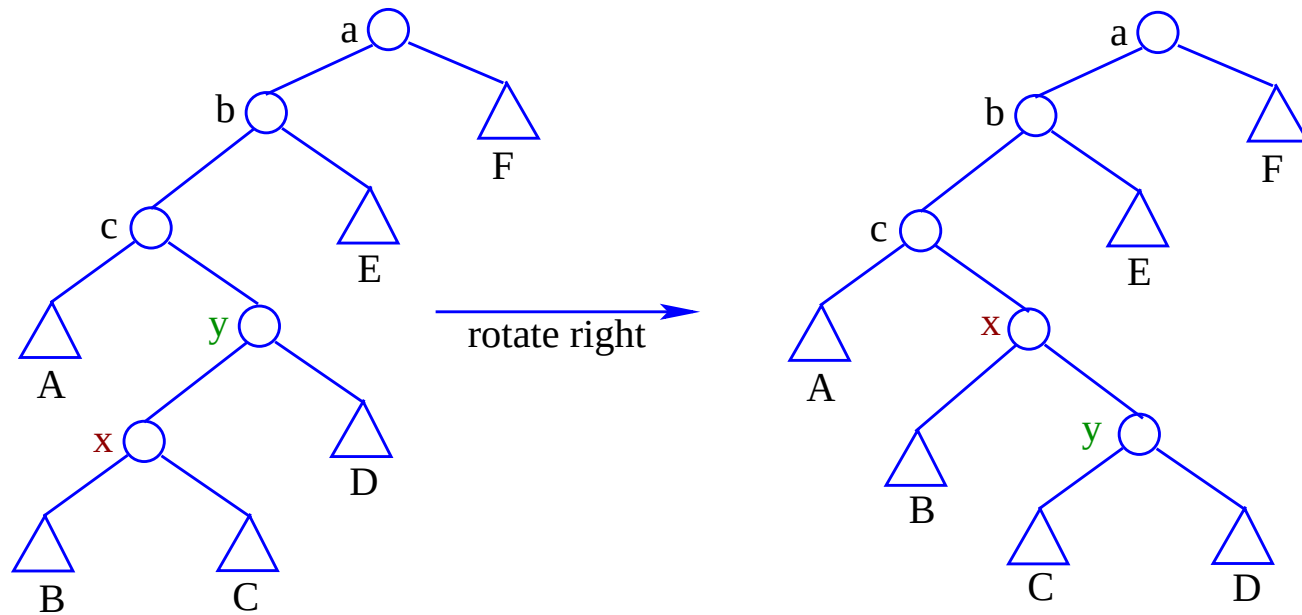
```
1 subtree B = x.left;  
2 x.left = y;  
3 y.right = B;  
4 y.parent.(left or right) = x;  
5 x.parent = y.parent;  
6 y.parent = x;
```


Splay tree rotations

- A single rotation around $x - y$ embedded inside the tree.

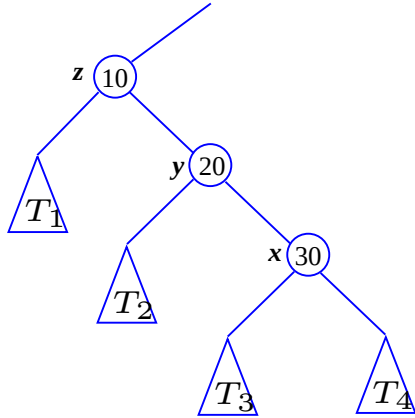
Splay tree rotations

- A single rotation around $x - y$ embedded inside the tree.

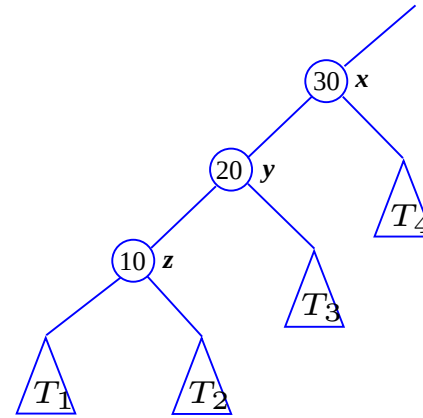


Splay trees

Before the *zig* operation:



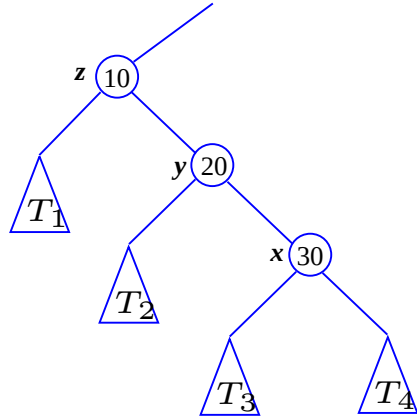
After the *zig* operation:



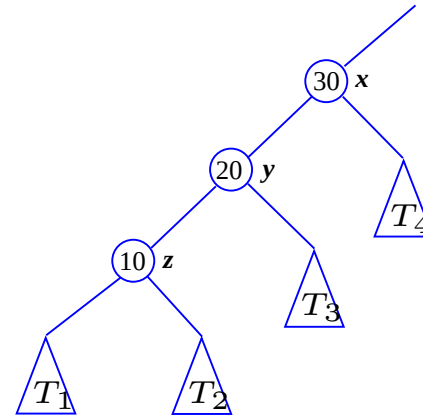
- Splay trees restructure the tree so that each time a node x is accessed it is “*moved to the root*” by doing rotations bottom-up along the path of access by one three splaying steps until x reaches the root.
 1. *zig*, which reduces x ’s height by one.

Splay trees

Before the *zig* operation:



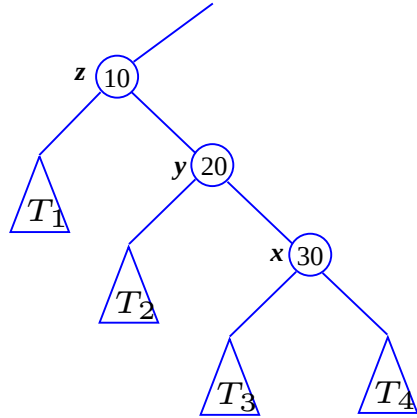
After the *zig* operation:



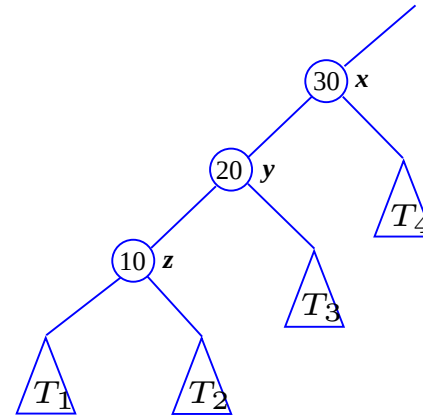
- Splay trees restructure the tree so that each time a node x is accessed it is “*moved to the root*” by doing rotations bottom-up along the path of access by one three splaying steps until x reaches the root.
 1. *zig*, which reduces x ’s height by one.
 2. *zig-zig*, and

Splay trees

Before the *zig* operation:



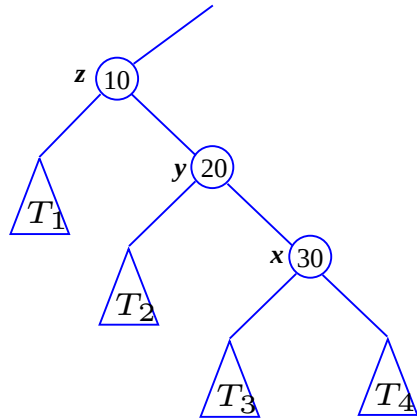
After the *zig* operation:



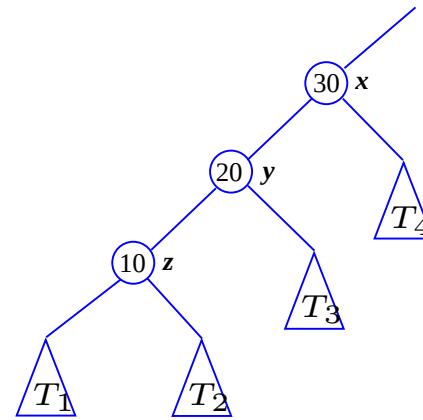
- Splay trees restructure the tree so that each time a node x is accessed it is “*moved to the root*” by doing rotations bottom-up along the path of access by one three splaying steps until x reaches the root.
 1. *zig*, which reduces x ’s height by one.
 2. *zig-zig*, and
 3. *zig-zag* and their mirrored symmetries.

Splay trees

Before the *zig* operation:



After the *zig* operation:



- Splay trees restructure the tree so that each time a node x is accessed it is “*moved to the root*” by doing rotations bottom-up along the path of access by one three splaying steps until x reaches the root.
 1. *zig*, which reduces x ’s height by one.
 2. *zig-zig*, and
 3. *zig-zag* and their mirrored symmetries.
- *Zig-zig* and *zig-zag* reduce x ’s height by two, i.e. each splay x moves 1 or 2 closer to the root.

The Splaying steps *zig*, *zig-zig* and *zig-zag*

- To move x to the root:

The Splaying steps *zig*, *zig-zig* and *zig-zag*

- To move x to the root:

Case 1: *zig*: If $y = \text{parent}(x)$ is the root, rotate along the edge $x - y$.

The Splaying steps *zig*, *zig-zig* and *zig-zag*

- To move x to the root:

Case 1: *zig*: If $y = \text{parent}(x)$ is the root, rotate along the edge $x - y$.

Case 2: *zig-zig*: If $y = \text{parent}(x)$ is not the root and x and y are *both* left children, or *both* right children: let $z = \text{parent}(y)$, first rotate the edge $y - z$ and then rotate the edge $x - y$.

The Splaying steps *zig*, *zig-zig* and *zig-zag*

- To move x to the root:

Case 1: *zig*: If $y = \text{parent}(x)$ is the root, rotate along the edge $x - y$.

Case 2: *zig-zig*: If $y = \text{parent}(x)$ is not the root and x and y are *both* left children, or *both* right children: let $z = \text{parent}(y)$, first rotate the edge $y - z$ and then rotate the edge $x - y$.

Case 3: *zig-zag*: If $y = \text{parent}(x)$ is not the root and x is a left child and y a right child, or vice versa, letting $z = \text{parent}(y)$, first rotate the edge $x - y$ and afterwards rotate $x - z$.

The Splaying steps *zig*, *zig-zig* and *zig-zag*

- To move x to the root:

Case 1: *zig*: If $y = \text{parent}(x)$ is the root, rotate along the edge $x - y$.

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- Splaying a node x at depth d takes $\Theta(d)$ time which is proportional to the time that it takes to access x .

The Splaying steps *zig*, *zig-zig* and *zig-zag*

- To move x to the root:

Case 1: *zig*: If $y = \text{parent}(x)$ is the root, rotate along the edge $x - y$.

Case 2: *zig-zig*: If $y = \text{parent}(x)$ is not the root and x and y are *both* left children, or *both* right children: let $z = \text{parent}(y)$, first rotate the edge $y - z$ and then rotate the edge $x - y$.

Case 3: *zig-zag*: If $y = \text{parent}(x)$ is not the root and x is a left child and y a right child, or vice versa, letting $z = \text{parent}(y)$, first rotate the edge $x - y$ and afterwards rotate $x - z$.

- Splaying a node x at depth d takes $\Theta(d)$ time which is proportional to the time that it takes to access x .
- Note that each splay step preserves the inorder properties of the tree.

Splay trees—a left-left *zig-zig* step-by-step

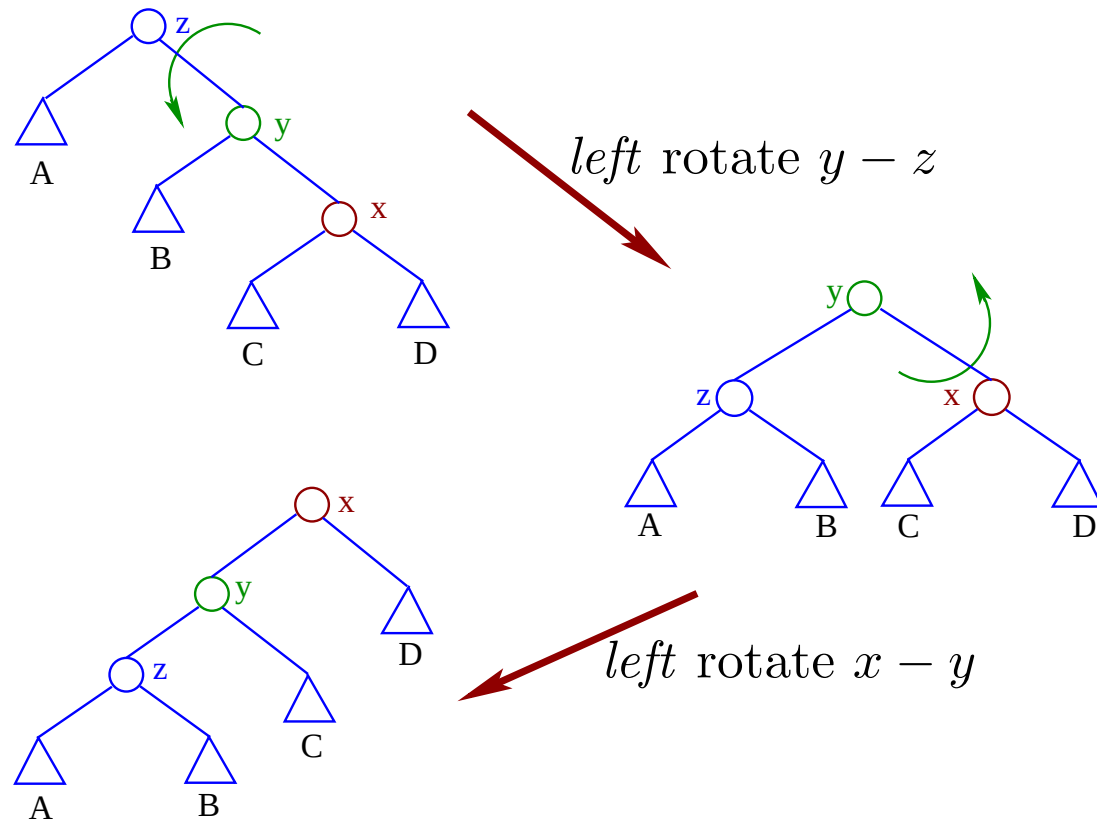
A *zig-zig* is done using two rotations:

Splay trees—a left-left *zig-zig* step-by-step

A *zig-zig* is done using two rotations: If $y = \text{parent}(x)$ is not the root and x and y are *both* left³ children: let $z = \text{parent}(y)$, first left rotate the edge $y - z$ and then *left* rotate the edge $x - y$.

Splay trees—a left-left *zig-zig* step-by-step

A *zig-zig* is done using two rotations: If $y = \text{parent}(x)$ is not the root and x and y are *both* left³ children: let $z = \text{parent}(y)$, first left rotate the edge $y - z$ and then *left* rotate the edge $x - y$.



Splay trees—a right-right *zig-zig* in 2 steps

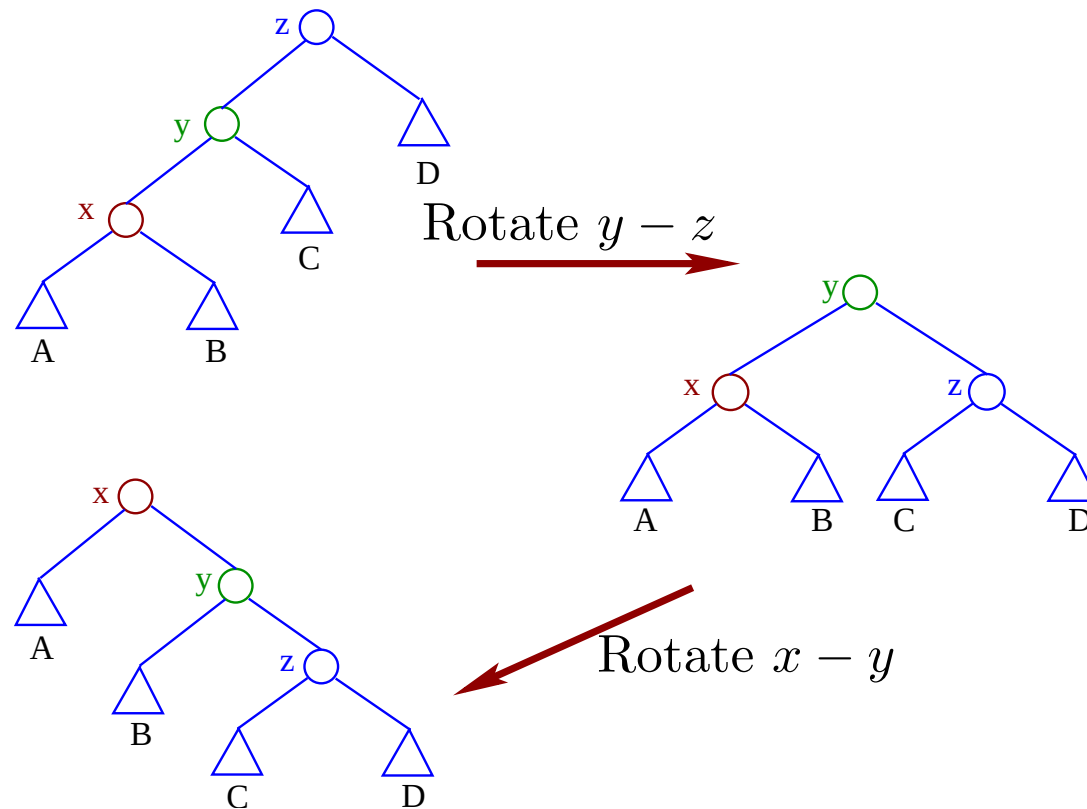
A *zig-zig* is done using two rotations:

Splay trees—a right-right *zig-zig* in 2 steps

A *zig-zig* is done using two rotations: If $y = \text{parent}(x)$ is not the root and x and y are *both* right⁴ children: let $z = \text{parent}(y)$, first rotate the edge $y - z$ and then rotate the edge $x - y$.

Splay trees—a right-right *zig-zig* in 2 steps

A *zig-zig* is done using two rotations: If $y = \text{parent}(x)$ is not the root and x and y are *both* right⁴ children: let $z = \text{parent}(y)$, first rotate the edge $y - z$ and then rotate the edge $x - y$.



⁴Or *both* children are left. See the previous slide.

Splay trees—a left-right *zig-zag* in 2 steps

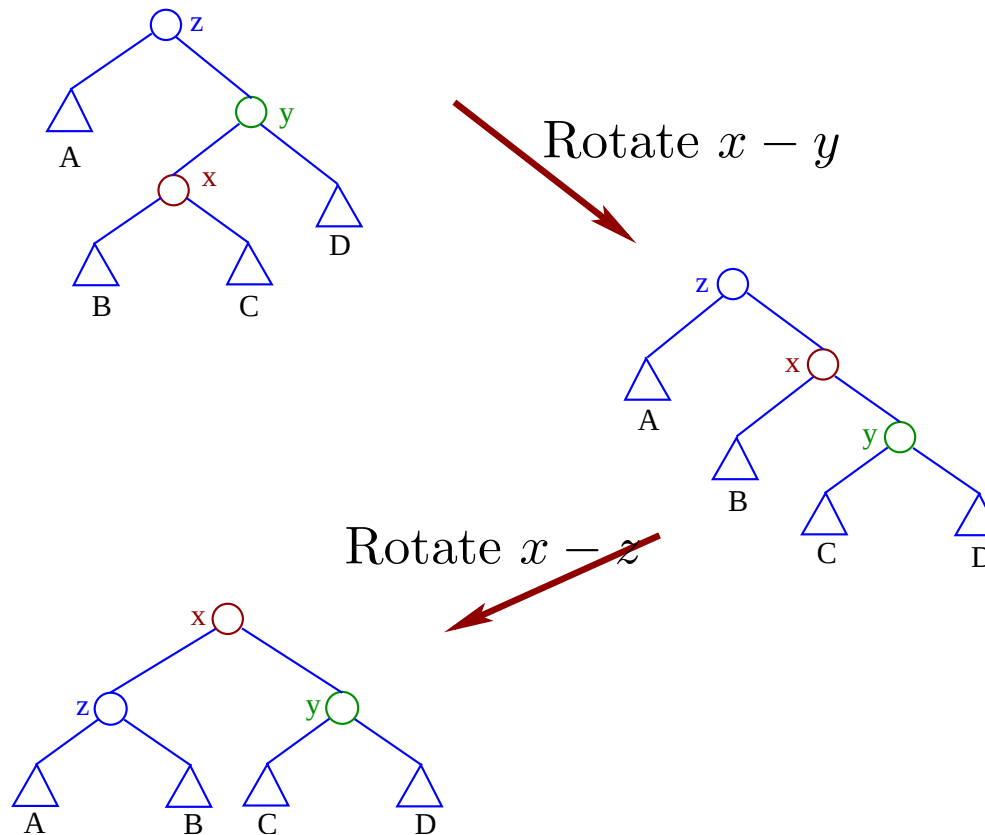
A *zig-zag* employs two rotations:

Splay trees—a left-right *zig-zag* in 2 steps

A *zig-zag* employs two rotations: If $y = \text{parent}(x)$ is not the root and x is a left child and y a right child⁵ letting $z = \text{parent}(y)$, first rotate the edge $x - y$ and afterwards rotate $x - z$.

Splay trees—a left-right *zig-zag* in 2 steps

A *zig-zag* employs two rotations: If $y = \text{parent}(x)$ is not the root and x is a left child and y a right child⁵ letting $z = \text{parent}(y)$, first rotate the edge $x - y$ and afterwards rotate $x - z$.



⁵Or vice versa, see next slide.

Splay trees—a right-left *zig-zag* in 2 steps

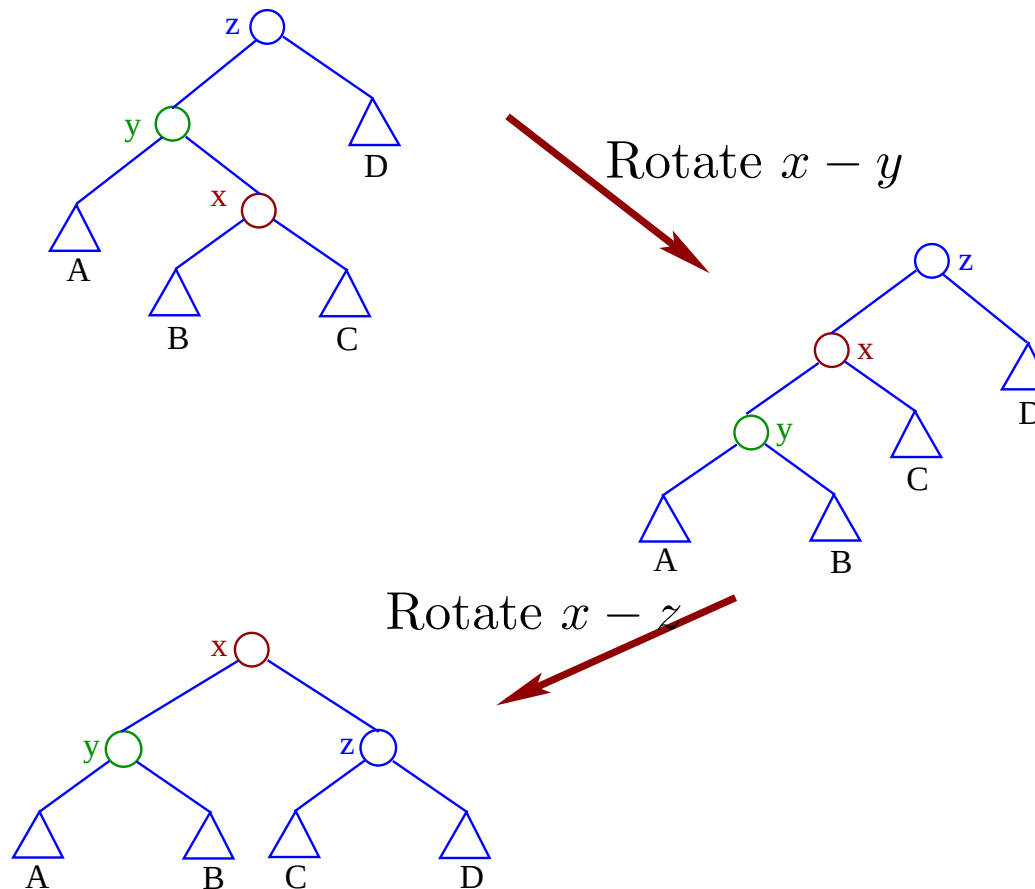
A *zig-zag* is done in two rotations:

Splay trees—a right-left *zig-zag* in 2 steps

A *zig-zag* is done in two rotations: If $y = \text{parent}(x)$ is not the root and x is a right child and y a left child⁶ letting $z = \text{parent}(y)$, first rotate the edge $x - y$ and afterwards rotate $x - z$.

Splay trees—a right-left *zig-zag* in 2 steps

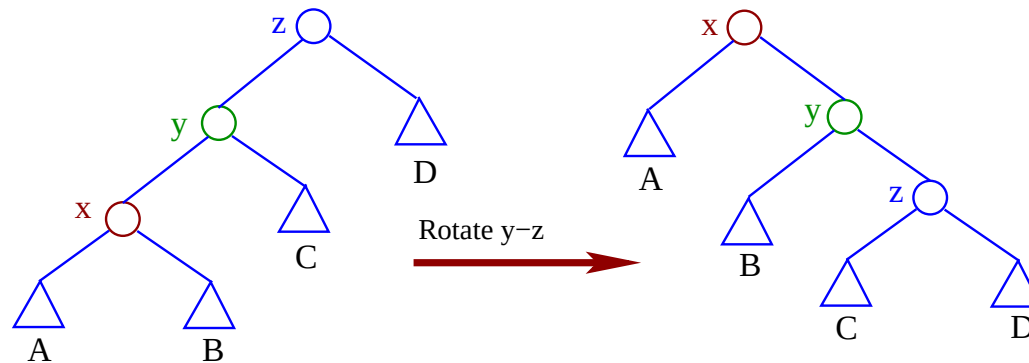
A *zig-zag* is done in two rotations: If $y = \text{parent}(x)$ is not the root and x is a right child and y a left child⁶ letting $z = \text{parent}(y)$, first rotate the edge $x - y$ and afterwards rotate $x - z$.



⁶Or vice versa, see previous slide.

Splay trees—a left-left *zig-zig* step-by-step

A direct *zig-zig* is done using two rotations: If $y = \text{parent}(x)$ is not the root and x and y are *both* left⁷ children: let $z = \text{parent}(y)$, first rotate the edge $y - z$ around y and then rotate the edge $x - y$ around x .



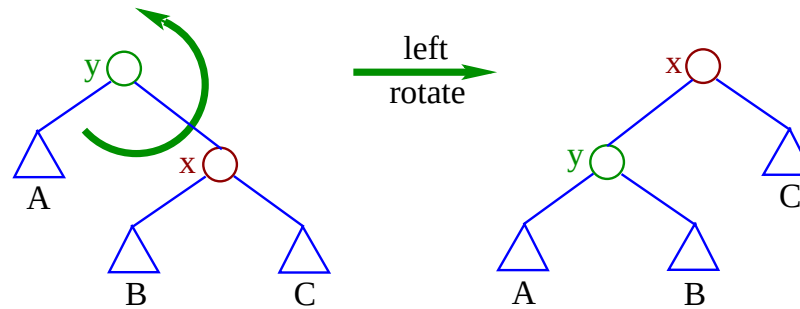
Assume that we know x , y and z . Why can we assume this? The code to do a zig-zig:

```
1  rightRotate(y, z);  
2  rightRotate(x, y);
```

⁷Or *both* children are right. See the next slide.

Splay trees—code for left rotation

Consider the figure

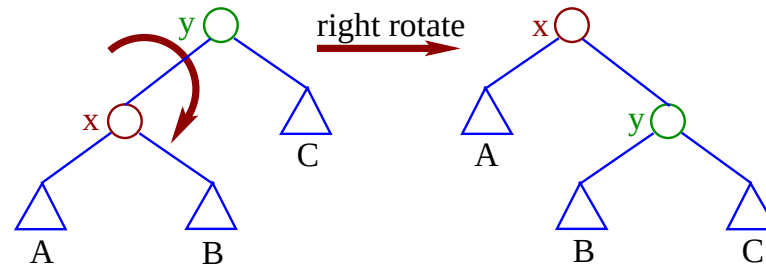


Assume that we know x and y . The code for a left rotation:

```
1 rotateLeft(tree x, tree y) {  
2   // pre: y == x.parent != null && x != null  
3   // && x.key >= y.key  
4   tree C = x.left;  
5   x.left = y;  
6   y.right = C;  
7   x.parent = y.parent;  
8   y.parent = x;  
9   if (x.parent != null)  
10      if ((x.parent).key > x.key)  
11         (x.parent).left = x;  
12      else  
13         (x.parent).right = x; }
```

Splay trees—code for right rotation

Consider the figure



Assume that we know x and y . The code for a right rotation:

```
1 rotateRight(tree x, tree y) {
2   // pre: x != null
3   // && y == x.parent != null
4   // && x.key < y.key
5   tree B = x.right
6   x.right = y;
7   y.left = B;
8   x.parent = y.parent;
9   y.parent = x;
10  if ((x.parent).key > x.key)
11    (x.parent).left = x;
12  else
13    (x.parent).right = x; }
```

Splay trees—code for automatic rotation

Do a *left* rotation when $x.\text{key} \geq (x.\text{parent}).\text{key}$, i.e. y is a *right* child—otherwise do a *right* rotation.

Left rotation:

```
1 rotateLeft(tree x, tree y) {
2 //pre: && x != null
3 // && y == x.parent != null
4 // && x.key >= y.key
5 tree C = x.left;
6   x.left = y;
7   y.right = C;
8   x.parent = y.parent;
9   y.parent = x;
10  if ((x.parent).key > x.key)
11    (x.parent).left = x;
12  else (x.parent).right = x; }
```

Right rotation:

```
1 rotateRight(tree x, tree y) {
2 // pre: x != null
3 // && y == x.parent != null
4 // && x.key < y.key
5 tree B = x.right
6   x.right = y;
7   y.left = B;
8   x.parent = y.parent;
9   y.parent = x;
10  if ((x.parent).key > x.key)
11    (x.parent).left = x;
12  else (x.parent).right = x; }
```

Only one argument is needed: the pivot of the rotation, x .

Splay trees—code for automatic rotation

Do a *left* rotation when $x.key \geq (x.parent).key$, i.e. y is a *right* child—otherwise do a *right* rotation.

Left rotation:

```
1 rotateLeft(tree x, tree y) {  
2 //pre: && x != null  
3 // && y == x.parent != null  
4 // && x.key >= y.key  
5 tree C = x.left;  
6   x.left = y;  
7   y.right = C;  
8   x.parent = y.parent;  
9   y.parent = x;  
10  if ((x.parent).key > x.key)  
11    (x.parent).left = x;  
12  else (x.parent).right = x; }
```

Right rotation:

```
1 rotateRight(tree x, tree y) {  
2 // pre: x != null  
3 // && y == x.parent != null  
4 // && x.key < y.key  
5 tree B = x.right  
6   x.right = y;  
7   y.left = B;  
8   x.parent = y.parent;  
9   y.parent = x;  
10  if ((x.parent).key > x.key)  
11    (x.parent).left = x;  
12  else (x.parent).right = x; }
```

Only one argument is needed: the pivot of the rotation, x .

$y = x.parent$; // assuming that $x \neq \text{null}$

Splay trees—code for automatic rotation

Do a *left* rotation when $x.\text{key} \geq (x.\text{parent}).\text{key}$, i.e. y is a *right* child—otherwise do a *right* rotation.

Left rotation:

```
1 rotateLeft(tree x, tree y) {
2 //pre: && x != null
3 // && y == x.parent != null
4 // && x.key >= y.key
5 tree C = x.left;
6   x.left = y;
7   y.right = C;
8   x.parent = y.parent;
9   y.parent = x;
10  if ((x.parent).key > x.key)
11    (x.parent).left = x;
12  else (x.parent).right = x; }
```

Right rotation:

```
1 rotateRight(tree x, tree y) {
2 // pre: x != null
3 // && y == x.parent != null
4 // && x.key < y.key
5 tree B = x.right
6   x.right = y;
7   y.left = B;
8   x.parent = y.parent;
9   y.parent = x;
10  if ((x.parent).key > x.key)
11    (x.parent).left = x;
12  else (x.parent).right = x; }
```

Only one argument is needed: the pivot of the rotation, x .

$y = x.\text{parent};$ // assuming that $x \neq \text{null}$

The temporary nodes **B** or **C** are not needed.

Splay trees—code for automatic rotation

Do a *left* rotation when $x.key \geq (x.parent).key$, i.e. y is a *right* child—otherwise do a *right* rotation.

Left rotation:

```
1 rotateLeft(tree x, tree y) {  
2 //pre: && x != null  
3 // && y == x.parent != null  
4 // && x.key >= y.key  
5 tree C = x.left;  
6   x.left = y;  
7   y.right = C;  
8   x.parent = y.parent;  
9   y.parent = x;  
10  if ((x.parent).key > x.key)  
11    (x.parent).left = x;  
12  else (x.parent).right = x; }
```

Right rotation:

```
1 rotateRight(tree x, tree y) {  
2 // pre: x != null  
3 // && y == x.parent != null  
4 // && x.key < y.key  
5 tree B = x.right  
6   x.right = y;  
7   y.left = B;  
8   x.parent = y.parent;  
9   y.parent = x;  
10  if ((x.parent).key > x.key)  
11    (x.parent).left = x;  
12  else (x.parent).right = x; }
```

Only one argument is needed: the pivot of the rotation, x .

$y = x.parent$; // assuming that $x \neq \text{null}$

The temporary nodes **B** or **C** are not needed. Rotation is determined by $\text{arity}(x)$, i.e. the *leftness* or *rightness* of x , given by $x.key < y.key$.

zig, zig-zig and zig-zag with **rotate(x)**

- Splaying a node x is a sequence of rotations that moves it to the root.

zig, *zig-zig* and *zig-zag* with **rotate(x)**

- Splaying a node x is a sequence of rotations that moves it to the root.
- Move x to the root by repeating:

Case 1: *zig*: If $\text{parent}(x)$ is the root, **rotate(x)**.

zig, *zig-zig* and *zig-zag* with **rotate(x)**

- Splaying a node x is a sequence of rotations that moves it to the root.
- Move x to the root by repeating:

Case 1: *zig*: If $\text{parent}(x)$ is the root, **rotate(x)**.

Case 2: *zig-zig*: If $\text{parent}(x)$ is not the root and x and $\text{parent}(x)$ are both *left* children, or both *right* children: first the **rotate(parent(x))** and then **rotate(x)**.

zig, *zig-zig* and *zig-zag* with **rotate(x)**

- Splaying a node x is a sequence of rotations that moves it to the root.
- Move x to the root by repeating:

Case 1: *zig*: If $\text{parent}(x)$ is the root, **rotate(x)**.

Case 2: *zig-zig*: If $\text{parent}(x)$ is not the root and x and $\text{parent}(x)$ are both *left* children, or both *right* children: first the **rotate(parent(x))** and then **rotate(x)**.

Case 3: *zig-zag*: If $\text{parent}(x)$ is not the root and x is a left child and $\text{parent}(x)$ a right child, or vice versa, first **rotate(x)** and afterwards **rotate(x)**.

zig, zig-zig and zig-zag with rotate(x)

- Splaying a node x is a sequence of rotations that moves it to the root.
- Move x to the root by repeating:

Case 1: *zig*: If $\text{parent}(x)$ is the root, $\text{rotate}(x)$.

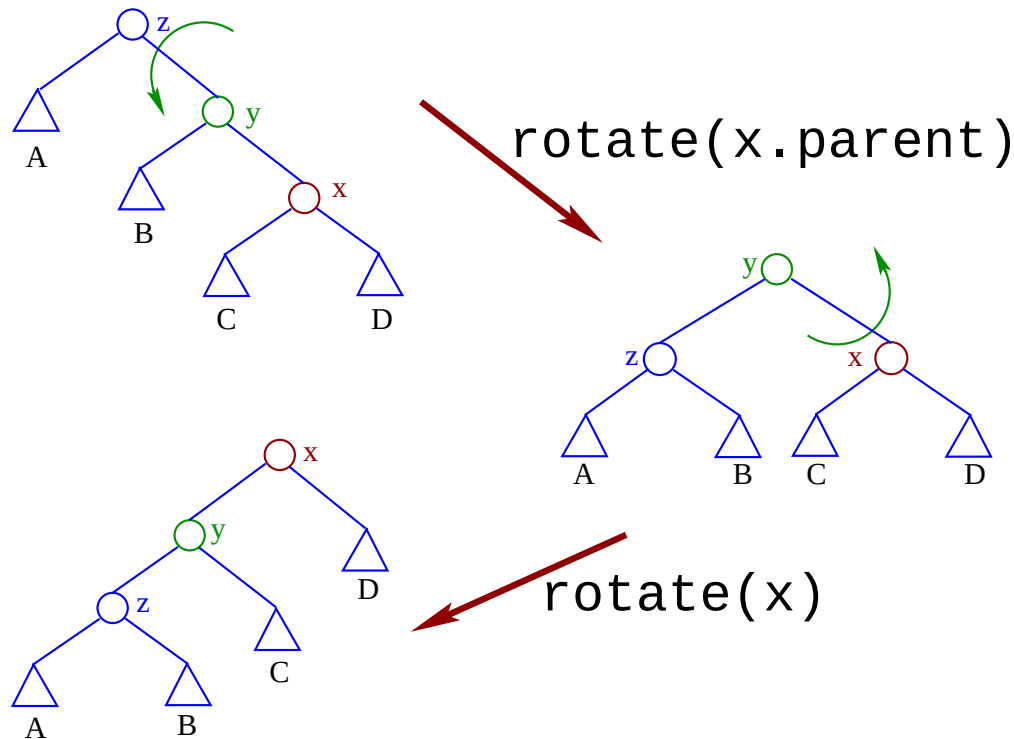
Case 2: *zig-zig*: If $\text{parent}(x)$ is not the root and x and $\text{parent}(x)$ are both *left* children, or both *right* children: first the $\text{rotate}(\text{parent}(x))$ and then $\text{rotate}(x)$.

Case 3: *zig-zag*: If $\text{parent}(x)$ is not the root and x is a left child and $\text{parent}(x)$ a right child, or vice versa, first $\text{rotate}(x)$ and afterwards $\text{rotate}(x)$.

- Splaying a node x at depth d takes $\Theta(d)$ time which is proportional to the time that it takes to access x .
- Note that each splay step preserves the inorder properties of the tree.

Splay trees—a left-left *zig-zig* step-by-step

A *zig-zig* is done using two rotations: If $\text{parent}(x)$ is not the root and x and $\text{parent}(x)$ are both *right*⁸ children: first $\text{rotate}(x.\text{parent})$ and then $\text{rotate}(x)$.

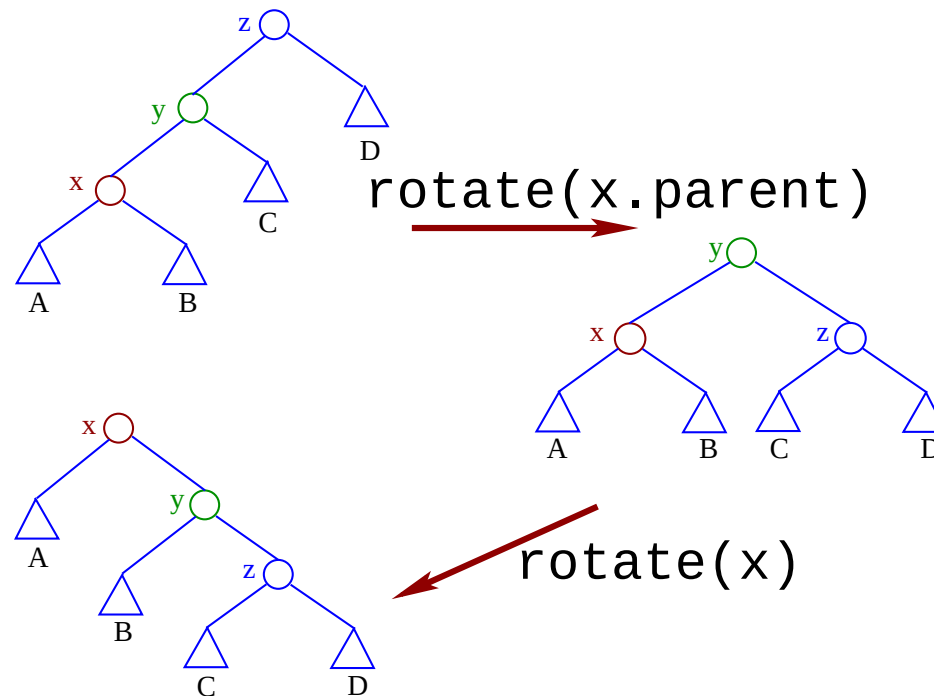


It is called a “left-left” *zig-zig* because both rotations are to the left, i.e. counter-clockwise or anti-clockwise.

⁸Or both are *left* children. See the next slide.

Splay trees—a right-right *zig-zig* in 2 steps

A *zig-zig* is done using two *clockwise* rotations: If $\text{parent}(x)$ is not the root and x and $\text{parent}(x)$ are both *left*⁹ children: first $\text{rotate}(y)$ and then $\text{rotate}(x)$ again.

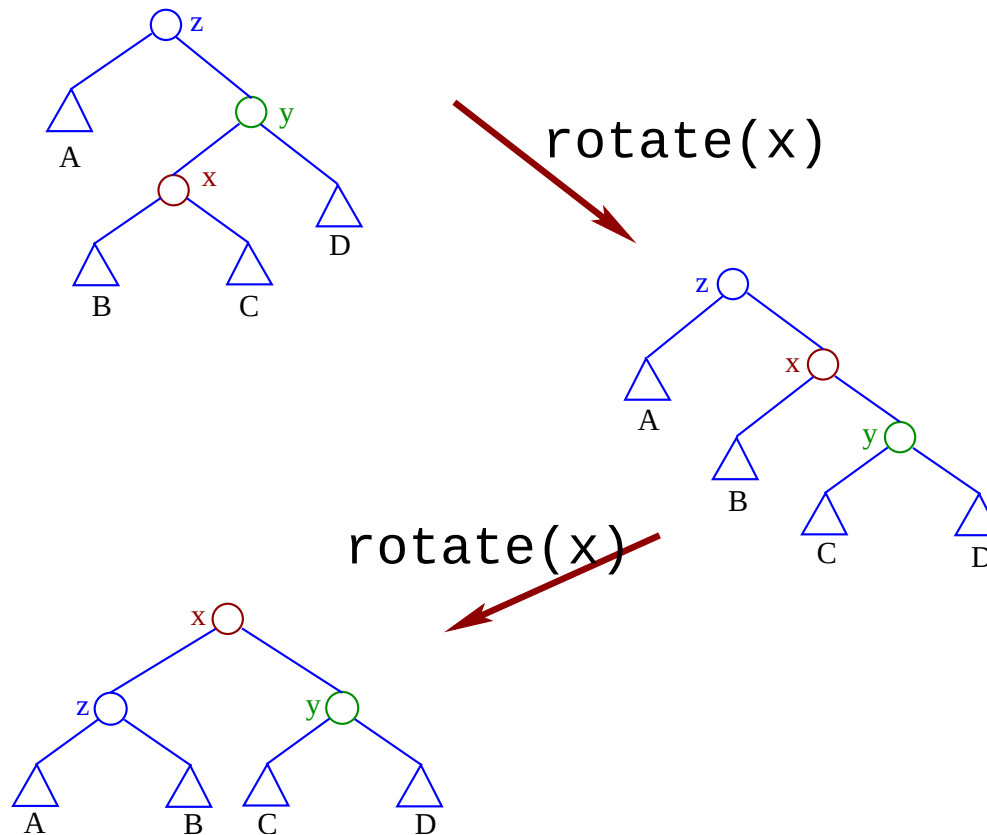


Notice that the code for a “right-right” *zig-zig* is the same as that for a “left-left” one. $\text{rotate}(x)$ decides the direction.

⁹Or both are *right* children. See the previous slide.

Splay trees—a left-right *zig-zag* in 2 steps

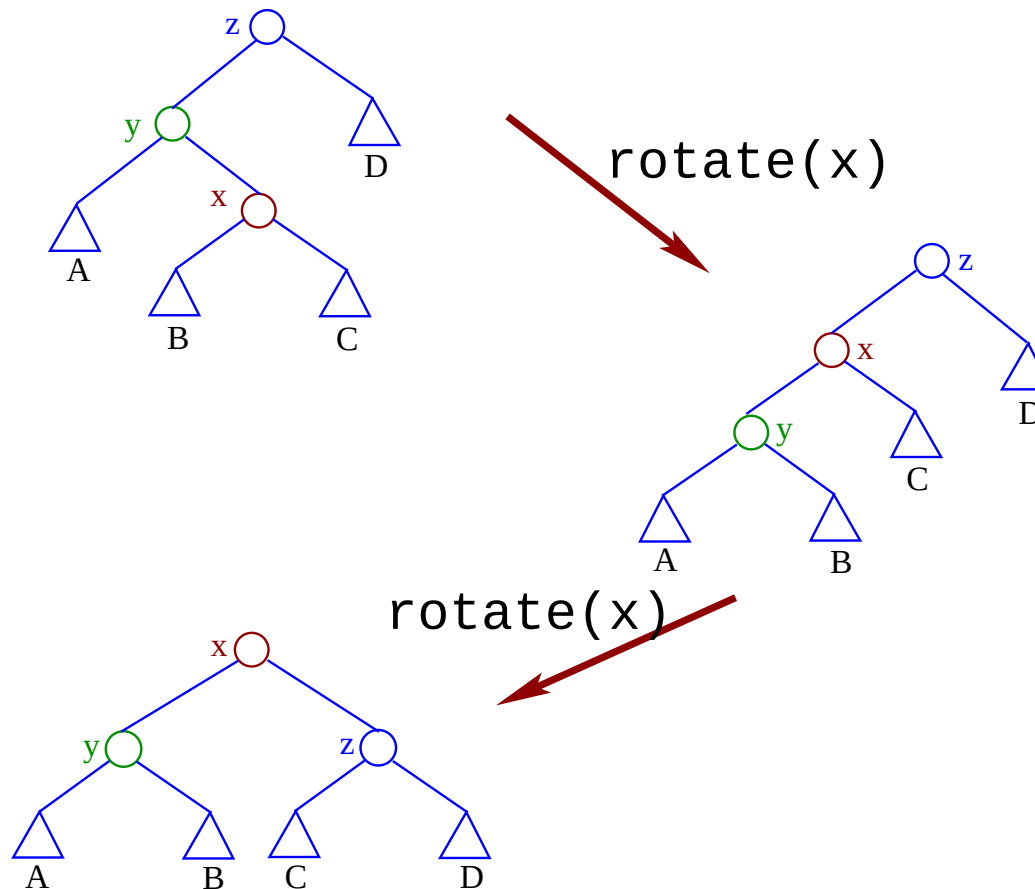
A *zig-zag* employs two rotations: If $\text{parent}(x)$ is not the root and x is a *left* child and $\text{parent}(x)$ a *right* child¹⁰: first $\text{rotate}(x)$ and then $\text{rotate}(x)$ again.



¹⁰Or vice versa, see next slide.

Splay trees—a right-left *zig-zag* in 2 steps

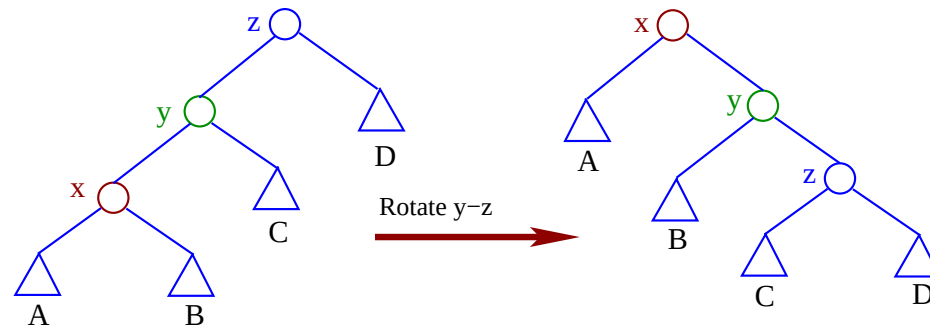
A *zig-zag* is done in two rotations: If $\text{parent}(x)$ is not the root and x is a *right* child and y a *left* child¹¹ first $\text{rotate}(x)$ and afterwards $\text{rotate}(x)$ again.



¹¹Or vice versa, see previous slide.

Splay trees—a left-left *zig-zig* step-by-step

A direct *zig-zig* is done using two rotations: If $y = \text{parent}(x)$ is not the root and x and y are *both* left¹² children: let $z = \text{parent}(y)$, first rotate the edge $y - z$ and then rotate the edge $x - y$.



Assume that we know x , y and z . Why can we assume this? The code to do a zig-zig:

```
1 leftRotate(y, z);  
2 leftRotate(x, y);
```

This becomes

```
1 rotate(x.parent);  
2 rotate(x);
```

¹²Or *both* children are right. See the next slide.

Splay trees—rotation around the pivot

```
1 rotate(tree x) {
2 // pre: x != null
3 tree y = x.parent;
4 if (y != null) // x is not the root
5     if (x.key < y.key) { // right rotation
6         y.left = x.right;
7         x.right = y;
8     }
9     else { // x is right child:
10        left rotation
11        y.right = x.left;
12        x.left = y;
13    }
14    if (y.parent != null)
15        if (y.key < (y.parent).key)
16            (y.parent).left = x;
17        else
18            (y.parent).right = x;
19    x.parent = y.parent;
20    y.parent = x; }
```

- In our pseudo-code we have used the expression `x.parent` instead of `getParent(x)` to clarify our coding.
- Next we tackle the code for `splay(x)` given only the tree node `x`.

Splay trees—`splay(x)`—1st version

```
1 void splay(tree x) {
2     while (x.parent != null) { // x is root
3         if ((x.parent).parent == null)
4             rotate(x); // x.parent is root: zig
5         // x.parent has a non-root parent
6         else if (x.key < (x.parent).key) // x is left child
7             if ((x.parent).key < ((x.parent).parent).key)
8                 rotate(x.parent); // left-left: zig-zig
9             else
10                rotate(x); // left-right: zig-zag
11        // x is a right child
12        else if ((x.parent).key < ((x.parent).parent).key)
13            rotate(x); // right-left: zig-zag
14        else { // right-right: zig-zig
15            rotate(x.parent); }
16        rotate(x); }
```

Take care this has some errors. Can you spot them?

Splay trees—`splay(x)`—2nd version

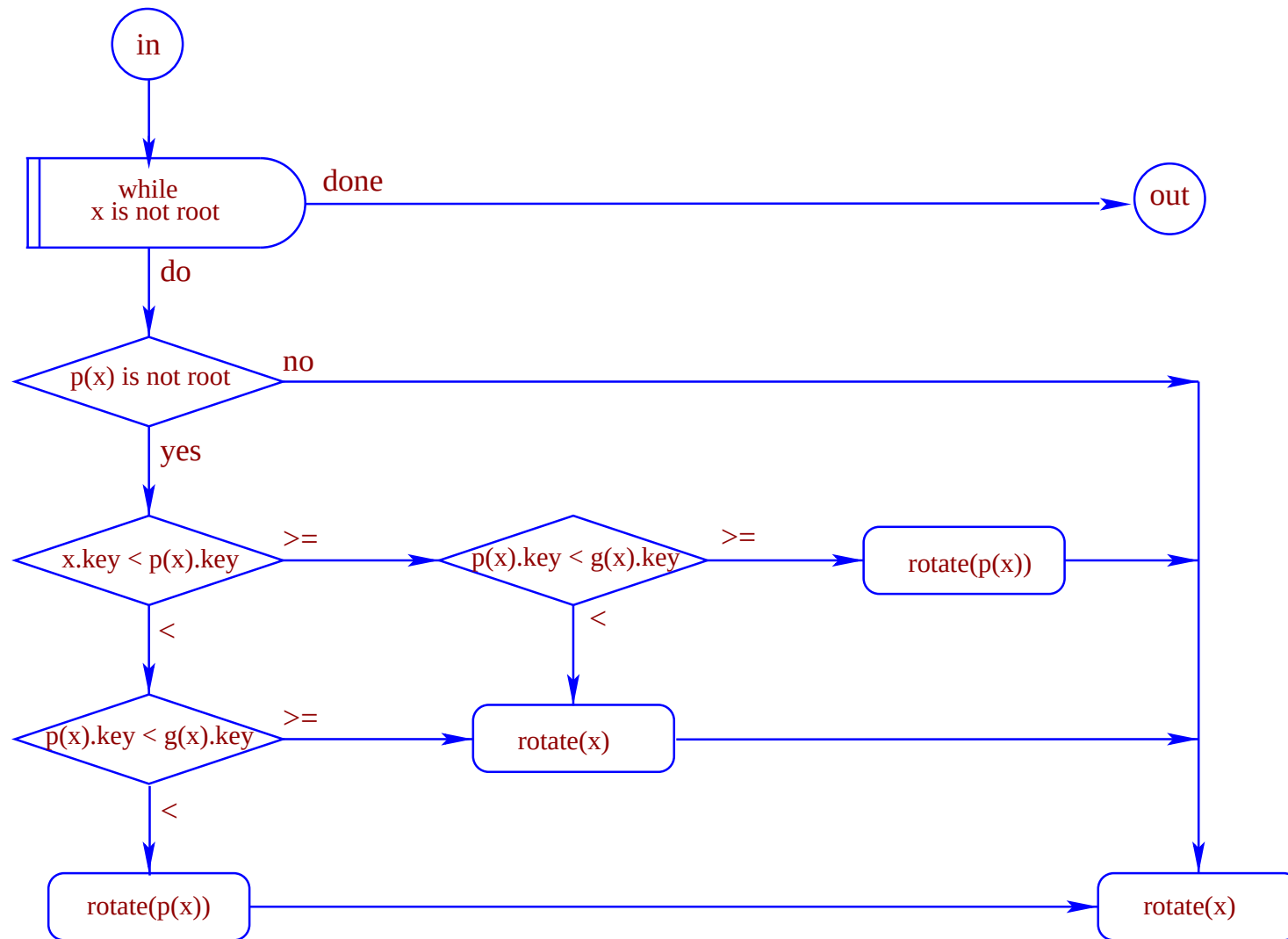
```
1 void splay(tree x) {
2     // repeat while x is not the root node
3     while (x.parent != null) { // y -- x
4         if ((x.parent).parent != null) { // z -- y -- x
5             // x.parent has a non-root parent
6             if (x.key < (x.parent).key)
7                 // x is a left child
8                 if ((x.parent).key < ((x.parent).parent).key)
9                     // x.parent is a left child: left-left zig-zig
10                    rotate(x.parent);
11                else // left-right: zig-zag
12                    rotate(x);
13            else // x is a right child
14                if ((x.parent).key < ((x.parent).parent).key)
15                    // x.parent is a left child: right-left zig-zig
16                    rotate(x.parent);
17                else // right-right: zig-gig
18                    rotate(x);
19            rotate(x); } } }
```

Still error ridden. Can you spot the errors?

Splay trees—**splay(x)**—tidier

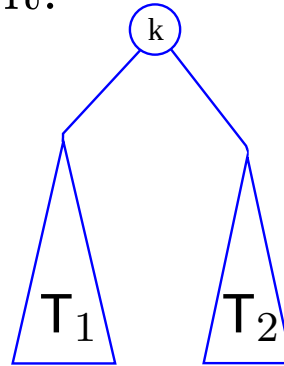
```
1 void splay(tree x) {
2     // repeat while x is not the root node
3     while (x.parent != null) { // y -- x
4         if ((x.parent).parent != null) // z -- y -- x
5             if (x.key < (x.parent).key)
6                 if ((x.parent).key < ((x.parent).parent).key)
7                     rotate(x.parent);
8                 else // left-right: zig-zag
9                     rotate(x);
10            else // right
11                if ((x.parent).key < ((x.parent).parent).key)
12                    rotate(x.parent);
13                else // right-left: zig-gig
14                    rotate(x);
15            rotate(x); } }
```

Splay trees—**splay(x)**—flow chart



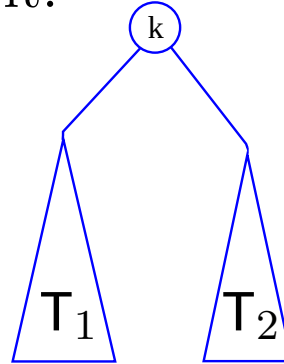
delete(k) from a *splay tree*

- First `find(k)` and splay it.



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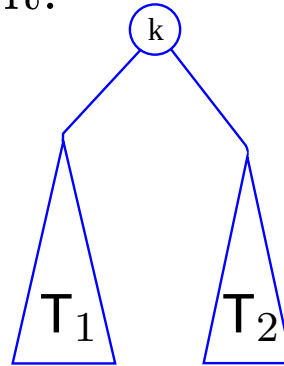
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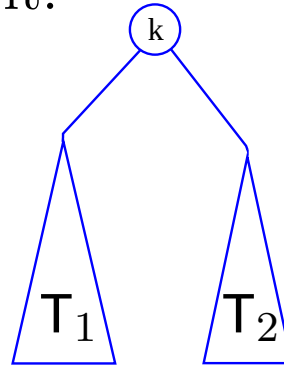
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- Note that all the keys in T_1 are less than k and all the keys in T_2 are greater than k .

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- Note that all the keys in T_1 are less than k and all the keys in T_2 are greater than k .
- Notice that the *greatest* key in T_1 is a *right* child but has no *right* child: so deleting or removing the *greatest* key from T_1 is simple: move its *left* key to its parent's *right* key.

- **Theorem:** Any sequence of m operations on an initially empty splay tree that never contains more than n nodes takes $O(m \log n)$

Splay trees—complexity

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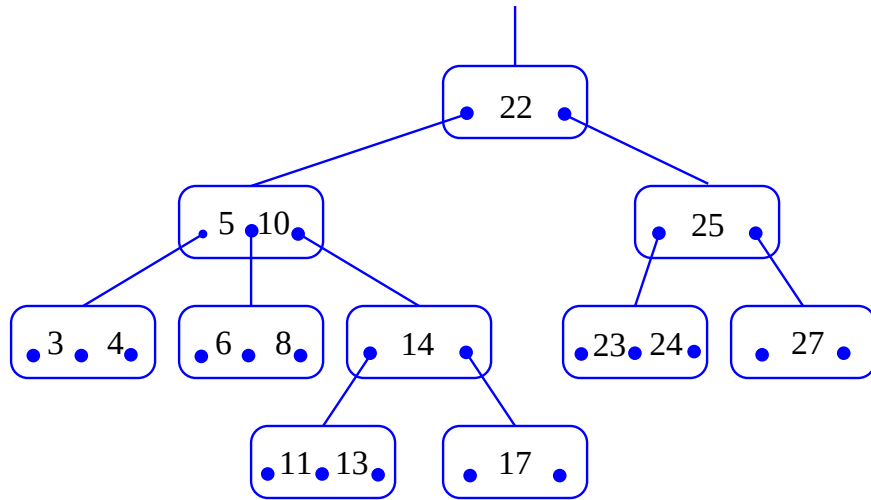
- See Goodrich and Tamassia for more slides in the SunLab at `../.. /notes/ds/SplayTrees.pdf` when you are in your home directory.

More up-to-date versions are on the web site that Goodrich and Tamassia maintain for the book.

These notes are superficial. Consult the book, pages 440–550 (4th edition).

Multiway trees

Before discussing $(2 - 4)$ trees we first consider multiway trees. The figure below is a multiway tree.



An n entry multiway tree has $n + 1$ leaves—external nodes. It is proved by induction.

$P[1]$: A multiway tree with one node has 2 leaves.

$P[k] \rightarrow P[k + 1]$: Assume $P[k]$, i.e. T_k is a multiway tree with k entries and has $k + 1$ leaves. Inserting a node into T_k at any leaf will replace one leaf and add two, i.e. the new tree T_{k+1} will have $k + 2$ leaves.

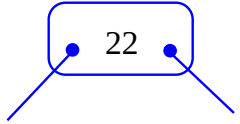
So, $\forall n P[n]$.

Red-black trees

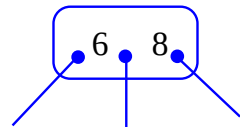
- A $(2 - 4)$ tree has an update—search, insert and remove—time of $O(\log n)$.
- A *red-black tree* has an update time of $O(\log n)$, i.e. it needs a constant number of changes to do an update.
- A *red-black tree* is a BST with nodes coloured *red* or *black* using the following rules:
 1. The root is black.
 2. Every external node is black.
 3. The children of a red node are black.
 4. External nodes have the same *black depth*. i.e. the number of black ancestors minus one.
- The following slides shows how to convert nodes from a $2 - 4$ tree into a *red-black* tree

Converting a 2 – 4 tree into a red-black tree

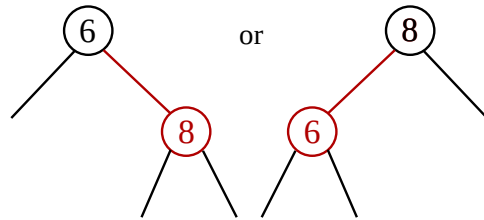
The figures below illustrate the conversion of nodes from a 2 – 4 tree into those for a red-black tree.



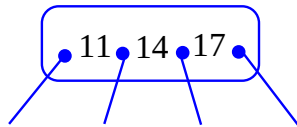
$\Rightarrow$



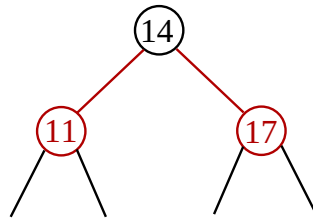
$\Rightarrow$



or

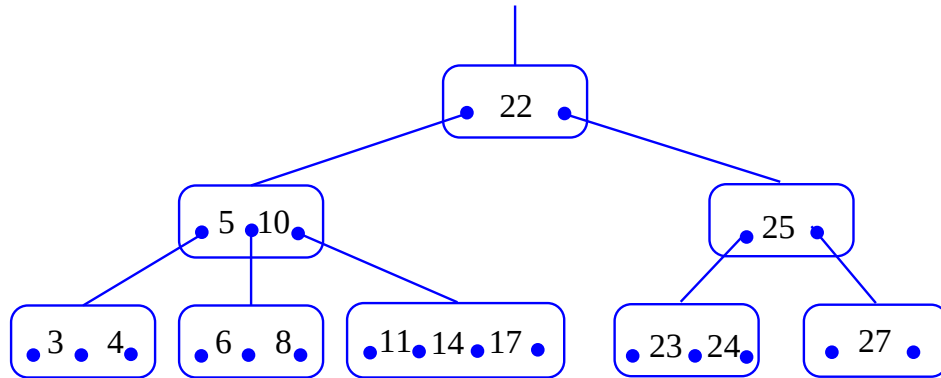


$\Rightarrow$

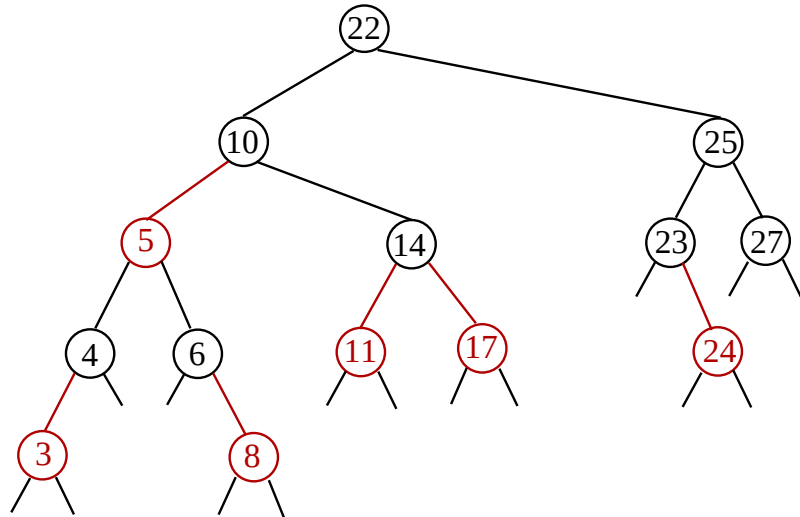


Converting a 2 – 4 tree into a red-black tree

We convert the following into a *red-black* tree:

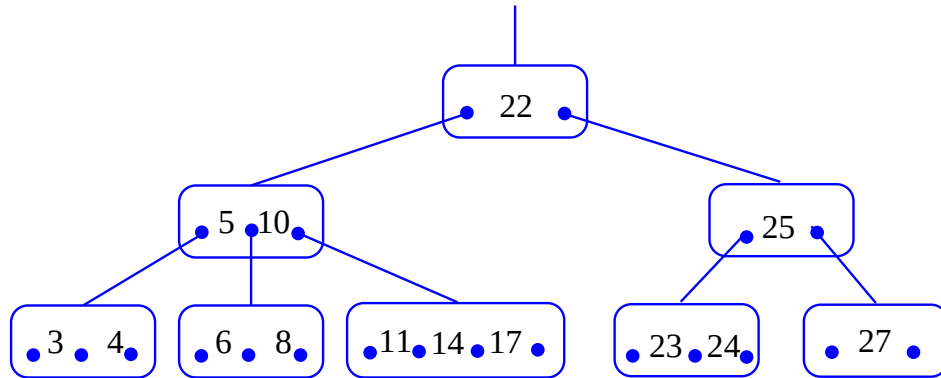


The *red-black* tree follows.

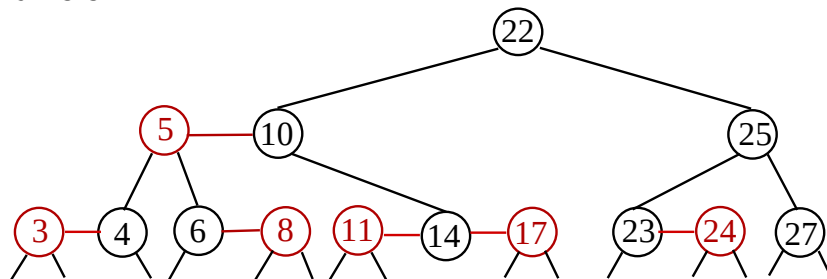


Converting a 2 – 4 tree into a *vertical-horizontal-red-black* tree

We convert the following into a *red-black* tree:

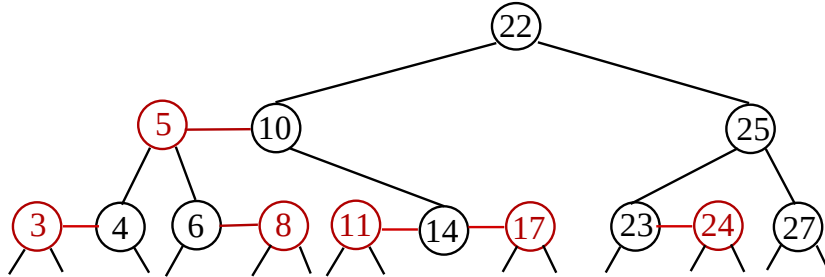


The vertical-horizontal *red-black* tree looks more like the original (2 – 4) tree.

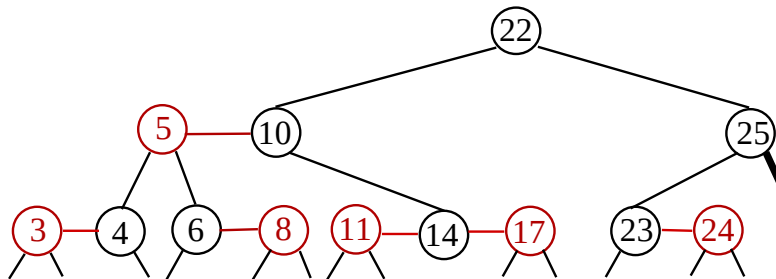


Deletion from a *red-black* tree

Consider removing 27 from the *red-black* tree below.



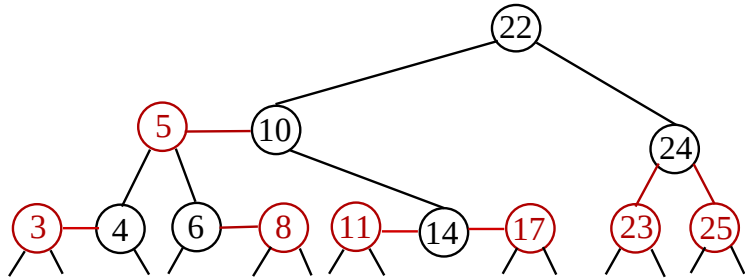
The resultant tree below has lost its black-depth property. Since 25's right child—a *double black*—now has a black depth of 2, it is no longer a *red-black* tree.



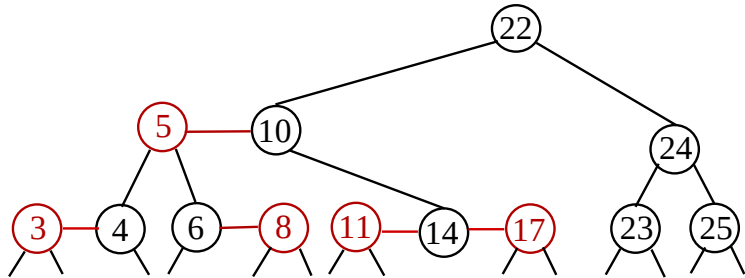
It is easy to repair the *double-black* error by restructuring 23 – 25 – 24 as three separate black 2-nodes with 24 on top and the black-depth property is restored.

Repairing the *double black* at 25

It is easy to repair the *double-black* error by restructuring $23 - 25 - 24$ first into the 4-node $23 - 24 - 25$

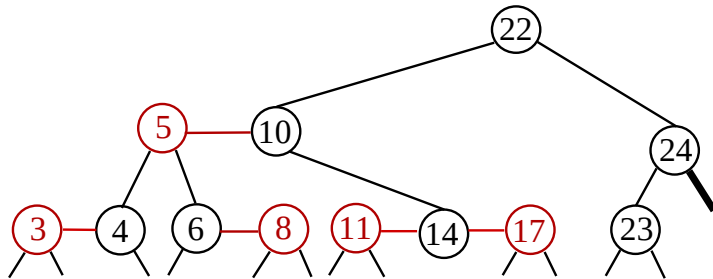


The leaves of 23 and 24 now have the incorrect *black depth* of 2. Subsequently, this 4-node $23 - 24 - 25$ is recoloured into three separate black 2-nodes with 24 on top. All the leaves now have a *black depth* of 3.

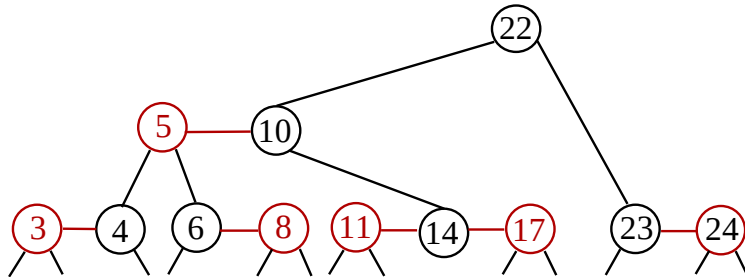


Non-trivial deletions from this *red-black* tree

Deletion of either 23 or 25 from this *red-black* tree cause a double black to hang from node 24. Let us remove 23. The double black hangs from node 24.

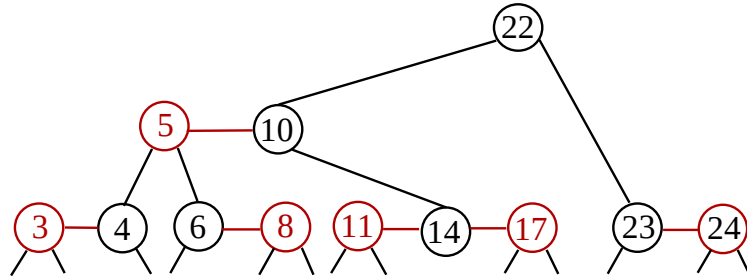


The red-black tree is repaired fixed by first amalgamating 24 and 24 into a 3-node and then the next step becomes obvious.

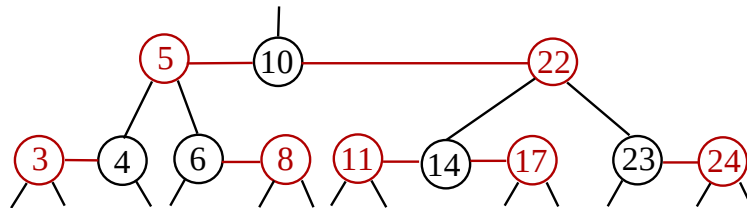


Easy deletions from this *red-black* tree

Amalgamating 24 and 24 into a 3-node makes the next step seem obvious.



Move the root 22 down to form the 4-node 5 – 10 – 22.



Since they are all red, removing 3, 8, 11, 17 or 24, has no effect on the depths of the leaves of the tree.

- A tree where all the nodes have *at least* a children and *at most* b children is a B-tree when
 $2 \leq a \leq (b + 1)/2$.
All the leaves have the same depth.

(a, b) Trees

- A tree where all the nodes have *at least* a children and *at most* b children is a B-tree when
 $2 \leq a \leq (b + 1)/2$.
All the leaves have the same depth.
- The root may have 0 children when the tree is empty.
- Otherwise the root must have at least 2 children.
- If there are less than a entries, they all lie in the root node, such that the root may have less than a children.

The height of an (a, b) tree T with n entries is $\Omega(\log_b n)$ and $O(\log_a n)$

- The height of an (a, b) tree is bounded from above by $O(\log_a n)$ and from below by $\Omega(\log_b n)$.
- The number of external nodes n' of T is at least $2a^{h-1}$: If the root has only one entry and the height, $h = 1$, then there are 2 leaves. If $h = 2$ and the root has at least a children and each child on the second level has a single entry, then there are at least $2a$ children on the second level. The third level will have at least $2a^2$ leaves.
- The number of external nodes is *at most* b^h where h is the height of the tree.
- An n -entry mutiway tree has $n + 1$ leaves, so

$$\begin{aligned} 2a^{h-1} &\leq n + 1 \leq b^h \\ \log_2(2a^{h-1}) &\leq \log_2(n + 1) \leq \log_2(b^h) \\ 1 + (h - 1) \log_2 a &\leq \log_2(n + 1) \leq h \log_2 b \end{aligned}$$

Upper and lower bounds for the height of an (a, b) tree

Since

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and

$$\begin{aligned} h &\geq \frac{\log_2(n + 1)}{\log_2 b} \\ &= \log_b(n + 1) \\ h &\text{ is } \Omega(\log_b n) \end{aligned}$$

Searching in an (a, b) tree

- Each node of an (a, b) tree has at least $a - 1$ entries and at most $b - 1$ entries.

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- Searching for an entry in an (a, b) tree T will take at worst $O(f(b) \log_a n)$ time.
- Since b is constant, a is also, and the search time is $O(\log n)$, which is independent of the specific search method.

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- d is chosen such that $d - 1$ keys and d references can be stored in a single block of size B .
- So each time a node is accessed a single disk transfer is done.
- The most transfers that ever need to be done is $O(\log_a n) = O(\log_{\lceil d/2 \rceil} n)$ and it will occupy $O(n/B)$ blocks.

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- During **insertion** *overflow* might occur.

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- During **deletion** *underflow* could occur.

Overflow and underflow in a *B*-tree

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Sorting—an $O(n^3)$ method

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- Suppose that the first $i - 1$ elements are in order, i.e. $A[1] \leq A[2] \leq A[3] \leq \dots \leq A[i - 1]$ insert the i -th one, $A[i]$ in its correct position relative to the elements we have already considered. Carry on until $i = n$.

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- Create now and find $j \in [1..i - 1]$ for which $now \leq A[j]$.

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1      now = A[i];  
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3          if (now <= A[j])  
4              insert(A, i, j); }
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- The **void** procedure `insert(A, i, j)` inserts $A[i]$ into the array $A[1..i]$ at j , by moving each element down one place and inserting now at $A[j]$:

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1 void insert (int A[], int i, int j) {  
2     int now = A[i], k;  
3     for (k = i; k >= j; k--)  
4         A[k] = A[k-1];  
5     A[j] = now; }
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- There are worse ways to sort. Can you think of sorting method that has exponential time complexity?
- We will next show another bad algorithm that takes $O(n^2)$ time.

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- Suppose that the first $i - 1$ elements are in their *correct* positions, i.e. $A[1] \leq A[2] \leq A[3] \leq \dots \leq A[i - 1]$. Find the i -th element in the rest of the array so that, $A[i]$ will be in its final position when it is placed. Carry on until $i = n$.

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- This is easy to do: Simply find the smallest element in the list $A[i..n]$ and when it is found swap it with whatever is at $A[i]$.

Sorting—an $O(n^2)$ method

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- After the swap $A[1..i]$ is completely sorted.
- Note this is again a disguised proof by mathematical induction that this sorting method is actually correct.

Sorting in $O(n^2)$ —the code

- The $i \in [1..n)$ in the outer loop points to the final position of the smallest element. It sets **smallest** to the $A[i]$, and preserves the index. The j -loop then iterates through all of the elements in $A[i+1..n]$ to get the smallest element.

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1  for (i = 1; i <= n-1; i++) {  
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4      for (j = i+1; j <= n; j++) {  
5          if (smallest > A[j]){  
6              smallest = A[j];  
7              indexSmallest = j;} }  
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- This time there are two nested loops of similar complexity to those of the previous sorting method. All the other lines are $O(1)$, so the algorithm has a complexity of $O(n^2)$.
- Although it improves the previous algorithm by an ‘order’ it is not good enough to earn our respect.

Merge sort—an $O(n \log n)$ algorithm

- Given a list of $to - from + 1$ objects:
 $A[from], A[from + 1], A[from + 2], \dots, A[to]$ that can be totally ordered, arrange them such that
 $A[from] \leq A[from + 1] \leq \dots \leq A[to]$.

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- Solving this, we will find that $T(n) = O(n \log n)$.

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$$T(n) = 2^iT(1) + icn, \quad \text{when } n = 2^i,$$

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$$\begin{aligned}T(n) &= 2^iT(1) + icn, && \text{when } n = 2^i, \text{ so } i = \log_2 n. \\&= icn, && \text{since } T(1) = 0.\end{aligned}$$

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When $2^i = n$, i.e. $i = \log_2 n$

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When $2^i = n$, i.e. $i = \log_2 n$, then $T(n) = icn$ and $T(n) = cn \log_2 n$,
i.e. $T(n)$ is $O(n \log n)$.

mergesort—the $O(n \log n)$ algorithm

- Given a list $A[from..to]$ Divide it into two almost equal parts at the midpoint p : $A[from..p]$ and $A[p + 1..to]$ and sort each part separately.

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```
1 public void mergeSort(int A[], int from, int to) {  
2   int p;  
3   if (from + 1 >= to) {  
4     if (A[from] <= A[to])  
5       return;  
6     swap(A, from, to); }  
7   else {  
8     p = (from+to)/2;  
9     mergeSort (A, from, p);  
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- We will now discuss the merging process.

The merging process—start merging

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1 public void merge(int A[], int from, int p, int to) {  
2   int merged[] = new int [to - from + 1];  
3   int i = from, j = p+1, k = 0;  
4   while (i <= p && j <= to)  
5     if (A[i] < A[j])  
6       merged[k++] = A[i++];  
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- At some stage either $i > p$, or $j > to$.
- We will first discuss the case where $i > p$, then all of $A[from..p]$ and only some of $A[p + 1..to]$ has been copied into $merged[0..k]$.

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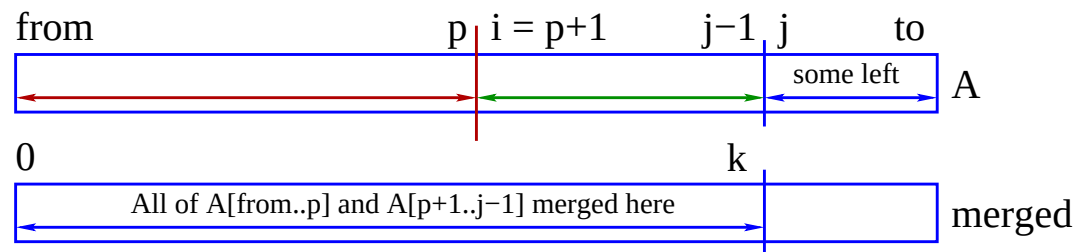
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- We will first discuss the case where $i > p$, then all of $A[from..p]$ and only some of $A[p + 1..to]$ has been copied into $merged[0..k]$.
- The tail end of $A[]$, namely $A[k + 1..to]$ is in place and only $merged[0..k]$ needs to be copied into its right place at $A[from..from + k]$.

Merging— i runs out first, i.e. $i > p$

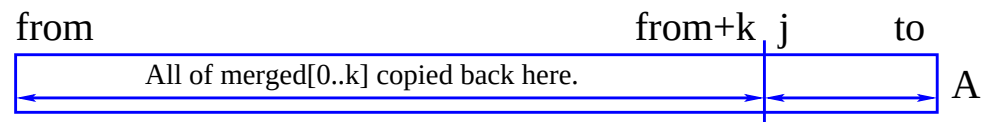
- At some stage either: $i > p$, or $j > to$.

Assume $i > p$,

then all of $A[from..p]$ and only some of $A[p+1..to]$ has been copied into $merged[0..k]$.



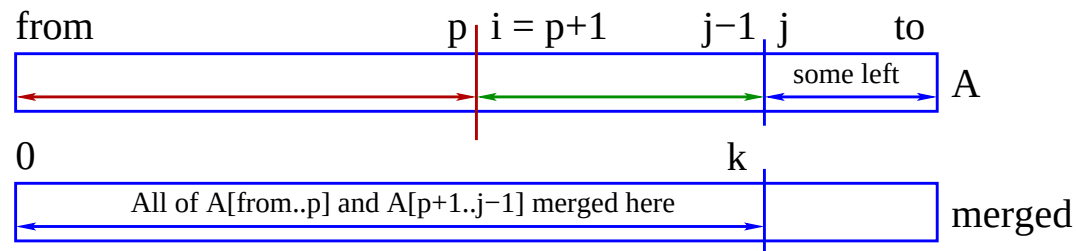
- Notice that, here, j coincides with $from + k + 1$.
- To complete the sorting of $A[from..to]$, $merged[0..k]$ must be copied back into $A[from..j-1]$ since $A[j..to]$ is already in place.



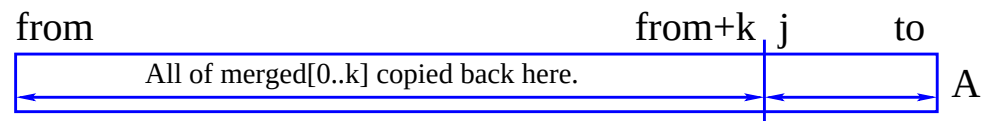
The entire $A[from..to]$ is now in sorted order.

Merging—the code when i runs out first

- We will only copy the contents of $merged[0..k]$ back into $A[from..j-1]$ since $A[j..to]$ is already in place.



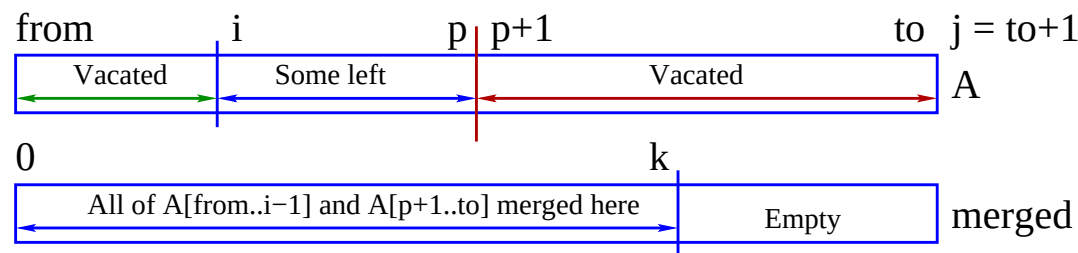
```
1 // Copy merged[] and A[i..p] or A[j..to] back to A[]
2 if (i == p+1) { // rest of A[j..to] already in place
3     r = from; s = 0;
4     while (r < j) // OR while (s <= k)
5         A[r++] = merged[s++]; }
6 else ...
```



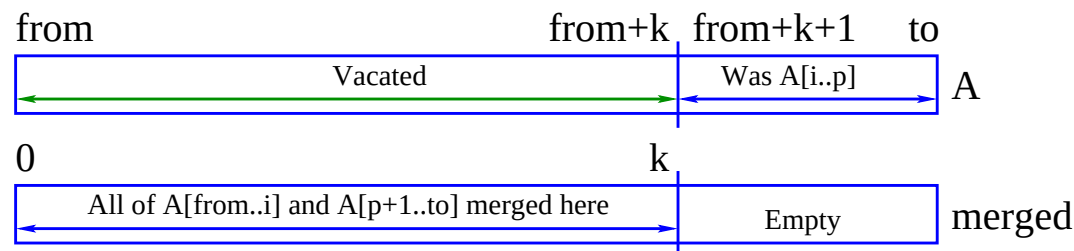
The entire $A[from..to]$ is now in sorted order.

Merging— j runs out first

- Put $i \leftarrow from$, $j \leftarrow to$, and put $k \leftarrow 0$. Compare $A[i]$ with $A[j]$ and store the smallest into $merged[k]$. Update i or j , and k appropriately.
- At some stage $i > p$, or $j > to$. Assume $j > to$, then all of $A[p+1..to]$ and some of $A[from..p]$ has been copied into $merged[0..k]$.



- First copy the contents of $A[i..p]$ into $A[k+1..to]$:

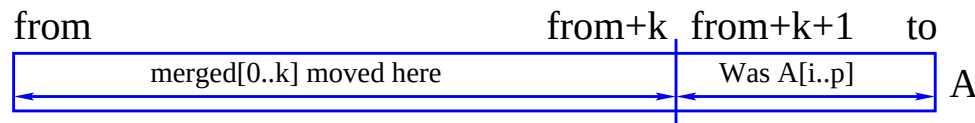


Only $A[from+k+1..to]$ is now in its final position.

Completing the merge when $j > to$

```
1  . . .
2  else { // first park A[i..p] into A[from]k+1..to]
3      r = to; s = p;
4      while (s >= i)
5          A[r--] = A[s--];
6      // move merged[0..k] to A[from..from+k]
7      r = from; s = 0;
8      while (s <= k)
9          A[r++] = merged[s++]; }
```

- To complete the sorting of $A[from..to]$ the contents of $merged[0..k]$ must be copied back into $A[from..from + k]$ since $A[from + k + 1..to]$ is already in place.



The entire $A[from..to]$ is now in sorted order.

mergeSort—the code

```
1 public void mergeSort(int A[], int from, int to) {  
2   // in : A[from..to] array of elements in any order  
3   // int from, and int in, within bounds  
4   //  
5   // out: Array[from..to] in sorted order  
6   int p;  
7   if (from + 1 >= to) {  
8     if (A[from] <= A[to])  
9       return;  
10    swap(A, from, to);  
11    return; }  
12   else {  
13     p = (from+to)/2;  
14     mergeSort (A, from, p);  
15     mergeSort (A, p+1, to);  
16     merge (A, from, p, to);  
17     return; } }
```

Code for merge

```
1 public void merge(int A[], int from, int p, int to) {
2 // in : A[from..p] array of elements in sorted order
3 // A[p+1..to] array of elements in sorted order
4 // int from, int p, and int to, within bounds
5 // out: returns array A[from..to] in sorted order
6 //
7 int merged[] = new int [to -from + 1], k = 0,
8     i = from, j = p+1, r, s;
9
10 while (i <= p && j <= to)
11     if (A[i] < A[j])
12         merged[k++] = A[i++];
13     else
14         merged[k++] = A[j++];
15 // Copy merged[] and A[i..p] or A[j..to] back to A[]
16 if (i == p+1) { // rest of A[j..to] already in place
17     r = from; s = 0;
18     while (s < k)
19         A[r++] = merged[s++]; }
20 else { // first park A[i..p] into A[..to]
21     r = to; s = p;
22     while (s >= i)
23         A[r--] = A[s--];
24     r = from; s = 0;
25     while (s < k)
26         A[r++] = merged[s++]; } }
```

Code for swap

The code for **swap**:

```
1 public void swap (int A[], int x, int y) {  
2   int keep = A[x];  
3   A[x] = A[y];  
4   A[y] = keep; }
```

In hindsight we see that the code required to copy $merged[0..k-1]$ back into $A[from..from+k-1]$ is duplicated. So **merge** is improved as follows by altering the code of the if statement:

```
1   if (i != p+1) { // first park A[i..p] into A[..to]  
2     r = to; s = p;  
3     while (s >= i)  
4       A[r--] = A[s--]; }  
5   r = from; s = 0;  
6   while (s < k)  
7     A[r++] = merged[s++];
```

Note:

- The copying back of the merged array $merged[0..k]$ can be replaced with the copying of a few links if the $A[]$ is stored as a linked list.

Quicksort—another $O(n \log n)$ sort

- Quicksort was published in the Computer Journal in 1962 by Tony Hoare. He invented the **case** statement.
‘Partition’ $A[from..to]$ into two parts: first select $pivot \in A[from..to]$ and save its *index*. Next, move all elements $< pivot$ to the left of $pivot$ and all the others to its right. Finally place $pivot$ at its correct position p in the final list, then sort $A[from..p - 1]$ and $A[p + 1..to]$ similarly.

```
1 public void quickSort(int A[], int from, int to) {  
2 // in : A[from..to] array of elements in any order  
3 int p;  
4     if (to < 0 || from >= to) return; // nothing to do  
5     if (from - to == 1) // A[from..to] == 2  
6         if (A[from] < A[to]) return;  
7         swap(A, from, to);  
8         return; }  
9     else {  
10        p = partition (A, from, to);  
11        quickSort (A, from, p-1);  
12        quickSort (A, p+1, to); } }
```

partition—the $O(n)$ heart of quicksort

‘Partition’ $A[from..to]$ into two parts: first select a $pivot \in A[from..to]$ and save its *index*. Next, move all elements $< pivot$ to the left of $pivot$ and all the others to its right. Finally place $pivot$ at its correct position p in the final list.

```
1 public int partition(int A[], int from, int to) {  
2   int i, j, pivot = A[from], lower=from+1, upper=to;  
3   while (lower < upper) {  
4     while (A[lower] < pivot && lower < upper)  
5       lower++;  
6     while (pivot <= A[upper] && lower < upper)  
7       upper--;  
8     if (lower < upper) swap(A, lower, upper); }  
9   swap(A, from, lower - 1);  
10  return upper; }
```

Quicksort—time complexity, $T(n)$

- If *pivot* always partitions the $A[from..to]$ into two equal parts then $T(1) = 0$, $T(2) = 1$, and $T(n) = 2T(n/2) + O(n)$ implying that $T(n) = n \log n$. Of course this hardly ever happens.

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- Suppose that every $A[i] \in A[from..to]$ is equally likely to be chosen as the pivot, then the running time of quicksort, if i is the i -th smallest element, is $T(n) = n - 1 + T(i - 1) + T(n - i)$.

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- To calculate the average, consider:

$$T(n) = n - 1 + T(1 - 1) + T(n - 1)$$

$$T(n) = n - 1 + T(2 - 1) + T(n - 2)$$

$$T(n) = n - 1 + T(3 - 1) + T(n - 3)$$

...

$$T(n) = n - 1 + T(i - 1) + T(n - i)$$

...

$$T(n) = n - 1 + T(n - 1) + T(n - n)$$

Now sum these values and divide by n .

Quicksort—average running time

- Given that each element was uniform randomly chosen the average running time is:

$$T(n) = n - 1 + T(1 - 1) + T(n - 1)$$

$$T(n) = n - 1 + T(2 - 1) + T(n - 2)$$

$$T(n) = n - 1 + T(3 - 1) + T(n - 3)$$

...

$$T(n) = n - 1 + T(i - 1) + T(n - i)$$

...

$$T(n) = n - 1 + T(n - 1) + T(n - n)$$

$$nT(n) = n(n - 1) + \sum_{i=1}^n (T(i - 1) + T(n - i)), \text{ so that}$$

$$T(n) = n - 1 + \frac{1}{n} \sum_{i=1}^n (T(i - 1) + T(n - i))$$

$$= n - 1 + \frac{1}{n} \sum_{i=1}^n T(i - 1) + \frac{1}{n} \sum_{i=1}^n T(n - i)$$

$$= n - 1 + \frac{2}{n} \sum_{i=0}^{n-1} T(i) \quad \text{A so-called recurrence with full history.}$$

$$= O(n \log n)$$

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Now $T(2) = T(1) + c$, so by telescoping,

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Now consider the recurrence relation with full history

$$T(n) = n - 1 + \frac{2}{n} \sum_{i=0}^{n-1} T(i).$$

Closed form of $T(n) = n - 1 + \frac{2}{n} \sum_{i=0}^{n-1} T(i)$

$$T(n) = n - 1 + \frac{2}{n} \sum_{i=0}^{n-1} T(i), \quad n \geq 2 \quad (1)$$

$$T(n+1) = (n+1) - 1 + \frac{2}{n+1} \sum_{i=0}^n T(i), \quad n \geq 2 \quad (2)$$

Multiply Equation(1) by n and Equation(2) by $n+1$ giving:

$$nT(n) = n(n-1) + 2 \sum_{i=0}^{n-1} T(i), \quad n \geq 2 \quad (3)$$

$$(n+1)T(n+1) = n(n+1) + 2 \sum_{i=0}^n T(i), \quad n \geq 2 \quad (4)$$

Subtract Equation (3) from Equation (4):

Closed form of $T(n) = n - 1 + \frac{2}{n} \sum_{i=0}^{n-1} T(i)$

$$nT(n) = n(n-1) + n \sum_{i=0}^{n-1} T(i), \quad n \geq 2 \quad (3)$$

$$(n+1)T(n+1) = n(n+1) + 2 \sum_{i=1}^n T(i), \quad n \geq 2 \quad (4)$$

Equation (4)–(3):

$$\begin{aligned} (n+1)T(n+1) - nT(n) &= n(n+1) - n(n-1) + 2T(n) \\ &= 2n + 2T(n), \quad n \geq 2 \end{aligned}$$

Giving,

$$\begin{aligned} (n+1)T(n+1) &= (n+2)T(n) + 2n, \\ T(n+1) &= \frac{n+2}{n+1}T(n) + \frac{2n}{n+1}, \quad n \geq 2 \end{aligned}$$

$\frac{2n}{n+1} < 2$, because $\frac{n}{n+1} < 1$,

$$T(n+1) \leq \frac{n+2}{n+1}T(n) + 2, \quad n \geq 2$$

$$\text{and thus } T(n) \leq 2 + \frac{n+1}{n}T(n-1), \quad n \geq 1$$

Closed form of $T(n) = n - 1 + \frac{2}{n} \sum_{i=0}^{n-1} T(i)$

Once again we telescope:

$$\begin{aligned} T(n) &\leq 2 + \frac{n+1}{n} T(n-1), \quad n \geq 1 \\ &= 2 + \frac{n+1}{n} \left[2 + \frac{n}{n-1} \left[2 + \frac{n-1}{n-2} \left[\cdots \frac{4}{3} \right] \right] \right] \\ &= 2 \left[1 + \frac{n+1}{n} + \frac{n+1}{n} \frac{n}{n-1} + \frac{n+1}{n} \frac{n}{n-1} \frac{n-1}{n-2} + \cdots \right. \\ &\quad \left. \cdots + \frac{n+1}{n} \frac{n}{n-1} \frac{n-1}{n-2} \cdots \frac{5}{4} \frac{4}{3} \right] \\ &= 2 \left[1 + \frac{n+1}{n} + \frac{n+1}{n-1} + \frac{n+1}{n-2} + \cdots + \frac{n+1}{3} \right] \\ &= 2(n+1) \left[1 + \frac{1}{n} + \frac{1}{n-1} + \frac{1}{n-2} + \cdots + \frac{1}{3} \right] \end{aligned}$$

Closed form of $T(n) = n - 1 + \frac{2}{n} \sum_{i=0}^{n-1} T(i)$

$$\begin{aligned} T(n) &\leq 2 + \frac{n+1}{n} T(n-1), \quad n \geq 1 \\ &= \dots \\ &= 2 \left[1 + \frac{n+1}{n} + \frac{n+1}{n-1} + \frac{n+1}{n-2} + \dots + \frac{n+1}{3} \right] \\ &= 2(n+1) \left[1 + \frac{1}{n} + \frac{1}{n-1} + \frac{1}{n-2} + \dots + \frac{1}{3} \right] \\ &= 2(n+1) \left[H(n+1) - \frac{3}{2} \right] \end{aligned}$$

Where $H(n) = 1 + 1/2 + 1/3 + \dots 1/n$ is the harmonic series,
 $H(n) = \ln n \gamma + O(1/n)$, where Euler's constant $\gamma = 0.57721566 \dots$
The closed formula for $T(n)$ is

$$T(n) \leq 2(n+1)(\ln n + \gamma - 1.5) = O(n \log n)$$

Pattern matching: $p[0..m-1] \in T[0..n-1]$

- A smaller *pattern*, p , must be found in a larger *text*, T , and its *position*, $i \in T$, must be reported or on failure -1 is returned.
- When programming a matching algorithm, great care must be taken at the boundaries:
 1. If the *pattern* is empty $p = ""$ then $i = 0$ should be returned

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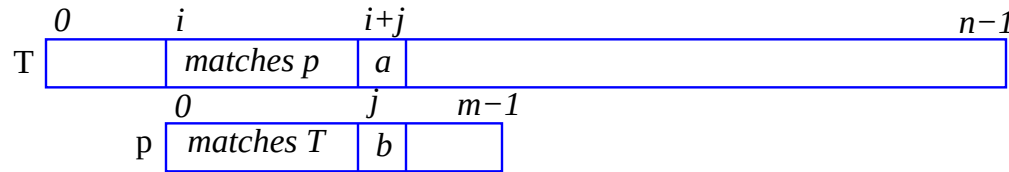
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 4. Test that a pattern can match $T[0..m-1]$.
 5. Test that a pattern can match $T[n-m..n-1]$.
 6. Test that the pattern can match somewhere in $T[1..n-2]$, i.e. it does not fall on a boundary.

Naive, greedy or Brute Force (BF) pattern matching

- Find the *pattern*, $\mathbf{p}$, in the *text*, $\mathbf{T}$, and return its *position* or the return -1 on failure.

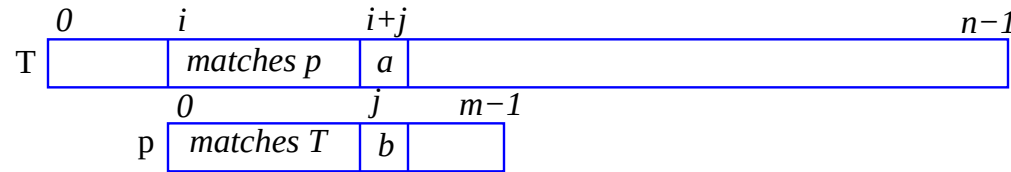
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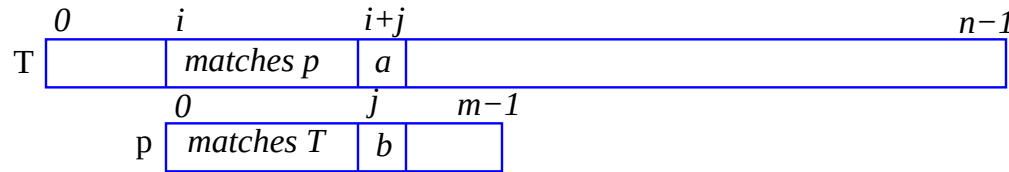
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Here $\mathbf{p}[0 \dots j-1]$ has matched $\mathbf{T}[i \dots i+j-1]$ but $\mathbf{p}[j]$ and $\mathbf{T}[i+j]$ do not match.

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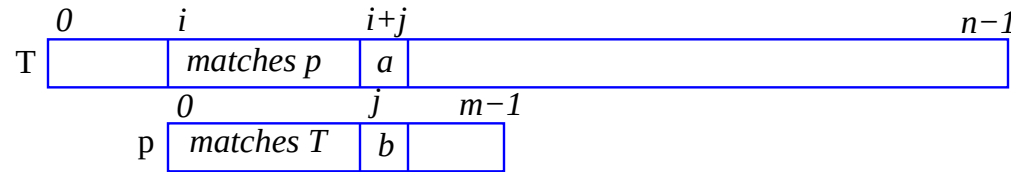


Here $p[0..j-1]$ has matched $T[i..i+j-1]$ but $p[j]$ and $T[i+j]$ do not match.

- Next: $i++$; $j = 0$; and try to match the pattern $p[0..m-1]$ with the substring $T[i..i+m-1]$ from the text, until a match is found.

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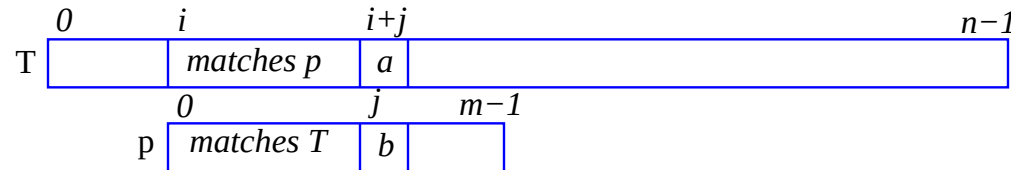


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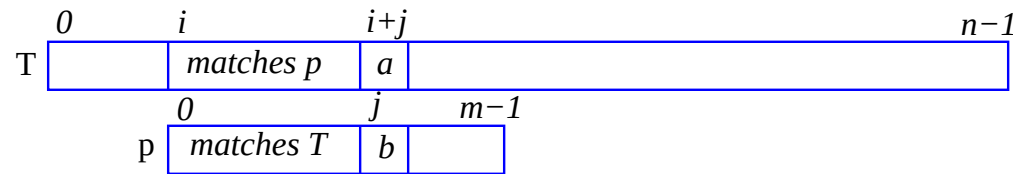
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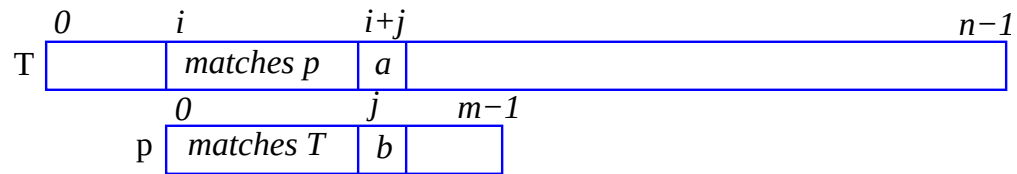
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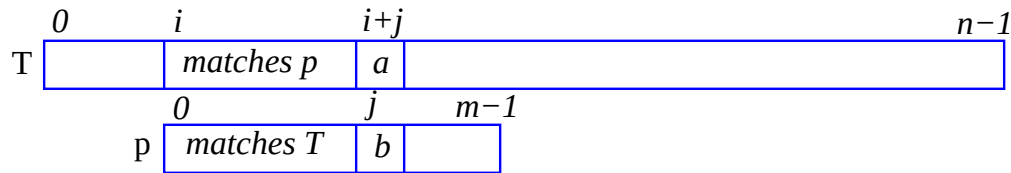
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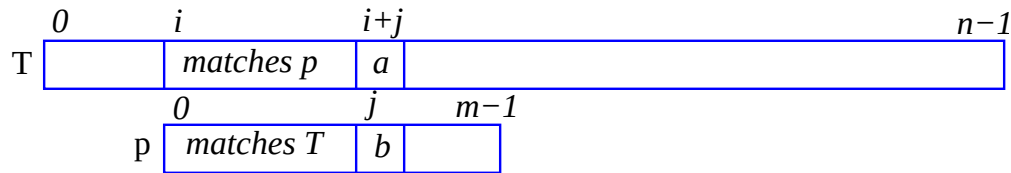
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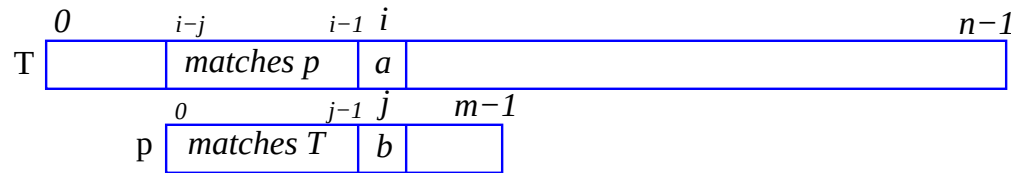
but $p[j] \neq T[i+j]$.

Try matching $p[0]$ and $T[i+1]$ next.

```
1 int matchBF (String p, String T) {
2   int m = p.length(); if (m == 0) return 0;
3   int n = T.length();
4   int i = 0, found = -1;
5
6   while (found < 0 && i <= n-m) {
7     int j = 0;
8     while (j < m && p[j] == T[i+j])
9       j++;
10    if (j == m)
11      found = i;
12    i++; }
13   return found; }
```

Naive, greedy or Brute Force (BF) pattern matching

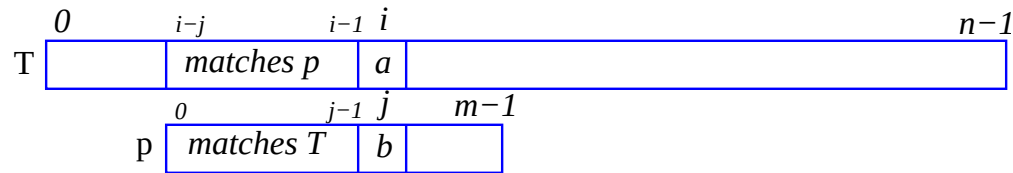
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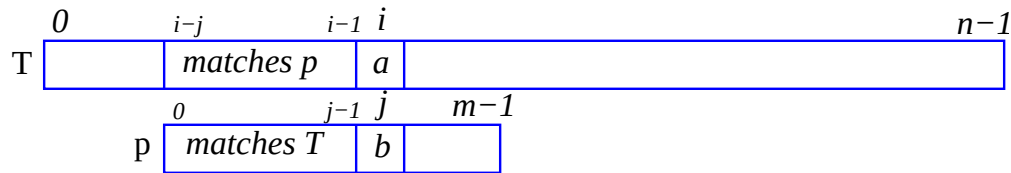


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4   int i = 0, j = 0;
5   do {
6     if (p[j] == T[i])
7       if (j == m - 1) return i - j;
8     else { i++; j++; }
9   } else {
10    i = i - j + 1;
11    j = 0; }
12  } while (i < n - m);
13  return -1; }
```

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RK, BM and KMP matching—briefly

Three methods that improve the time complexity:

Rabin-Karp-matching compares $\text{hash}(p) :: \text{hash}(T[i..i+m-1])$.
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Rabin-Karp matching (RK)

- $|p[0..m-1]| = m$ and $|T[i..i+m-1]| = m$.
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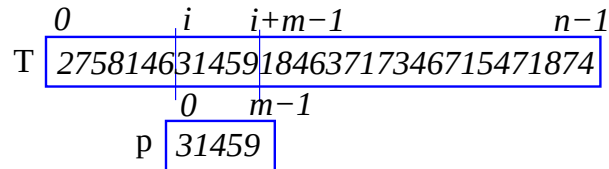
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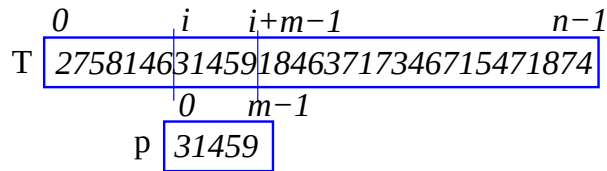


$T = 27581463145918463717346715471874$ First get $pHash = 31459$, then one-by-one match it with each $THash_i$:

$27581_0, 75814_1, 58146_2, 81463_3, 14631_4, 46314_5, 63145_6, 31459_7$.

Since $pHash$ matches with $THash_7$ the position 7 is returned.

Rabin-Karp—rolling hash for decimal digits



$T = 27581463145918463717346715471874$

$$\begin{aligned} pHash &= 31459 \\ THash_0 &= 27581 \\ THash_1 &= 75814 = (THash_0 - 20000) \times 10 + 4 \\ THash_2 &= 58146 = (THash_1 - 70000) \times 10 + 6 \\ THash_3 &= 81463 = (THash_2 - 50000) \times 10 + 3 \\ THash_4 &= 14631 = (THash_3 - 80000) \times 10 + 1 \\ THash_5 &= 46314 = (THash_4 - 10000) \times 10 + 4 \\ THash_6 &= 63145 = (THash_5 - 40000) \times 10 + 5 \\ THash_7 &= 31459 = (THash_6 - 60000) \times 10 + 9 \\ THash_8 &= 14591 = (THash_7 - 30000) \times 10 + 1 \end{aligned}$$

In general

$$THash_{i+1} = (THash_i - \lfloor THash_i / 10000 \rfloor 10000) \times 10 + (int)T[i + m]$$

and in Java this could be coded as

```
1 THash = (THash - (THash/10000)*10000) * 10
2   + (int)T.charAt(i+m)-(int)'0';
```

Rabin-Karp—rolling hash for characters

Let $|T| = m$, and $h_i \equiv \text{hash}(T[i..i + m - 1])$. Given h_i calculate h_{i+1} .
Design of a hashing function for Unicode characters is based on the code for digits

1. The contribution of the leftmost character to h_i :

$$h_{\text{leftmost digit}} = \lfloor h_i / 10^{m-1} \rfloor \times 10^{m-1}$$

2. Deduct it from h_i

$$\text{hash}(T[i + 1..i + m - 1]) = h_i - h_{\text{leftmost digit}}$$

3. Shift this one character left

$$\text{hash}(T[i + 1..i + m - 1]) \times 10$$

4. Add the character $T[i + m]$.

$$\begin{aligned} h_{i+1} &= \text{hash}(T[i + 1..i + m]) \\ &= \text{hash}(T[i + 1..i + m - 1]) \times 10 + T[i + m] \end{aligned}$$

Rabin-Karp—rolling hash for characters

Let $|T| = m$, and $h_i \equiv \text{hash}(T[i..i + m - 1])$. Given h_i calculate h_{i+1} .
The hash function for Unicode characters is based on the code for digits with the base changed from 10 to b .

1. The contribution of the leftmost character to h_i :

$$h_{\text{leftmost digit}} = \lfloor h_i / b^{m-1} \rfloor \times b^{m-1}$$

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3. Shift this one character left

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The biggest Unicode character has the value $2^{16} - 1$.

Using base $b = 2^{16}$ makes the above method work. *There is a snag.*

Rabin-Karp—rolling hash for characters

Basing the hash function for Unicode characters by using the base $b = 2^{16}$ has some snags:

1. The contribution of the leftmost character to h_i :

$$h_{\text{leftmost digit}} = \lfloor h_i / b^{m-1} \rfloor \times b^{m-1}$$

The value of $b^{m-1} = 2^{16(m-1)}$ uses ... bits. So when $m > \dots$ it will overflow.

This is remedied by using modular arithmetic. Every arithmetic operation must be calculated modulo some large prime that is less than 2^{16} .

2. Deduct it from h_i

$$\text{hash}(T[i + 1..i + m - 1]) = h_i - h_{\text{leftmost digit}}$$

There is a new snag.

The result may become negative. It must first be made positive.

Rabin-Karp—rolling hash for characters

There is a new snag.

The result may become negative. It must first be made positive. This is done by adding **modHash** to the number as many times as needed to make it positive. Suppose that $h < 0$, then

3. Shift this one character left

$$\text{hash}(T[i + 1..i + m - 1]) \times b$$

4. Add the character $T[i + m]$.

$$\begin{aligned} h_{i+1} &= \text{hash}(T[i + 1..i + m]) \\ &= \text{hash}(T[i + 1..i + m - 1]) \times b + T[i + m] \end{aligned}$$

The biggest Unicode character has the value $2^{16} - 1$.
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Rabin-Karp—code

```
1 public static int modHash = 65521;
2 public static final int base
3     = (Character.MAX_VALUE + 1) % modHash;
4
5 static int firstHash(String s) {
6     int m = s.length();
7     int j, h = 0;
8     for(j = 0; j < m; j++){
9         h = mod(h * base);
10        h = mod(h + s.charAt(j));
11    }
12    return h;
13 }
```

Rabin-Karp—code

```
1 static int hash(String s) {  
2     return firstHash(s);  
3 }  
4  
5 static int baseLeftmost (String T) {  
6     int baseLeft = 1;  
7     int i;  
8  
9     for (i = 1; i < T.length(); i++)  
10         baseLeft = mod(baseLeft*base);  
11     return baseLeft;  
12 }  
13 }
```

Rabin-Karp—code

```
1 static int matchRK(String p, String T){
2   int m = p.length();
3   int n = T.length();
4   if (m > n)
5     return -1;
6   int pHash = hash(p);
7   int THash = hash((String)T.subSequence(0,m));
8   int baseLeft = baseLeftmost(p);
9   int i;
```

Rabin-Karp—code

```
1 // Get the subsequence=m starting at 0
2
3 for (i = 0; i + m <= n; i++) {
4     if (THash == pHash) {
5         if (stringCompare(
6             (String)T.subSequence(i, i+m), p))
7             return i;
8     }
9     if (i == n-m)
10        return -1;
11    // deduct contribution of left character
12    THash = mod(THash
13        - mod(baseLeft * T.charAt(i)));
14    // Shift left
15    // add character at right
16    THash = mod(mod(THash*base)
17        + T.charAt(i+m)) ;
18 }
19 return -1;
20 }
```

Rabin-Karp—code

```
1 static boolean stringCompare(String p,  
2                               String T) {  
3     int m = p.length();  
4     int n = T.length();  
5  
6     int i;  
7  
8     if (m != n)  
9         return false;  
10    for (i = 0; i < m; i++)  
11        if (p.charAt(i) != T.charAt(i))  
12            return false;  
13    return true;  
14 }
```

Rabin-Karp—code

```
1 static int mod (int top) {  
2   int modulo = top % modHash;  
3  
4   if (modulo < 0)  
5     return modulo  
6       + modHash * (1 + (-modulo/modHash));  
7   else  
8     return modulo;  
9  
10 }
```

Rabin-Karp matching—rolling hash

- Normally the patterns may get too big to be stored as a unique number. Apply modulo arithmetic when hashing. Use a large prime number `modHash` to `mod` every arithmetic result.
- Calculate the hash roughly as follows:

```
1 int firstHash(String s) {
2   int m = s.length();
3   int j, h=0;
4   for(j = 0, j < m, j++)
5     h = (h+mod(s.charAt(j)*base));
6   return h;
7 }
8
9 int nextHash(char left, char right,
10              int THash) {
11   THash = mod(THash - mod(left*base^(m-1)));
12   return mod(THash*base + right);
13 }
```

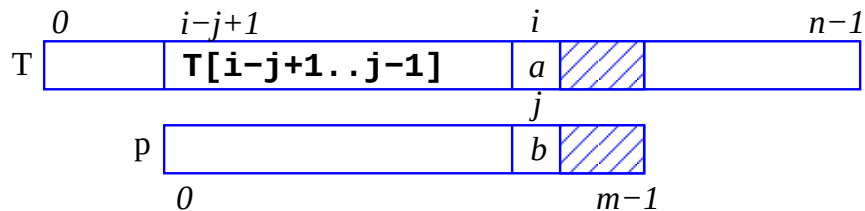
Rabin-Karp matching—pseudo code

```
1 int rabinKarpMatch(String p, String T){
2   int m = p.length();
3   int n = T.length();
4   int pHash = hash(p[0..m-1]);
5   int THash = hash(T[0..m-1]);
6
7   for (i = 0; i < n; i++) {
8     if (THash == pHash)
9       // verify equality of patterns
10      if (T[i..i+m-1] == p)
11        return i;
12     THash = hash(T[i+1..i+m]);
13   }
14   return -1;
15 }
```


Rabin-Karp matching (RK)—complexity

- Since every character in the *text* file must be used in the hashing formula, at worst all characters need to be scanned, taking $O(n)$.
- The pattern must be hashed, $O(m)$.
- The worst case time is thus $O(m + n)$.
- This is at least m times faster than the BF method.

Boyer-Moore matching (BM)



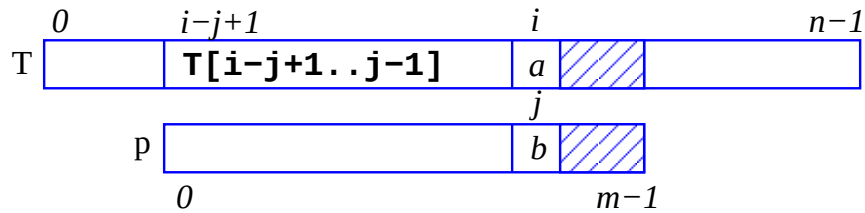
If the letter a never appears in p , the pattern may be moved on, placing $p[0]$ over $T[i+1]$ and $p[m-1]$ over $T[i+m]$ —*ignoring* the characters in $T[i-j+1..i-1]$.

Boyer-Moore-matching preprocesses p to find the last occurrence of each character. It matches from $p[m-1]$ backwards through to $p[0]$. On non-matching $p[j]$ and $T[i]$, if $T[i]$ does not occur in p , then $p[0]$ is moved to next match $T[i+1]$ —ignoring characters in $T[i-j+1..i-1]$ —*otherwise* p is moved putting $p[m-1]$ over $T[i']$ where

$$i' = i + m - \min(j, 1 + \text{lastInP}(T[i]));$$

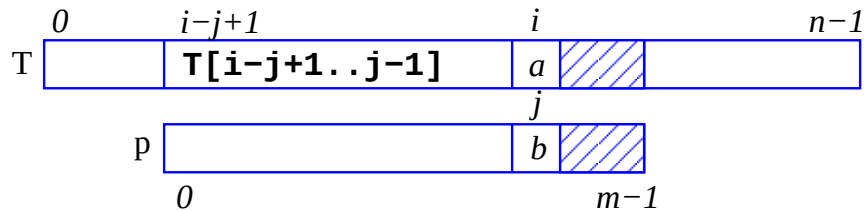
next, match $p[m-1]::T[i']$. In the worst case *BM* can degenerate to $O(mn + |\Sigma|)$ but it is likely to scan less than n characters in T .

Boyer-Moore matching (BM)



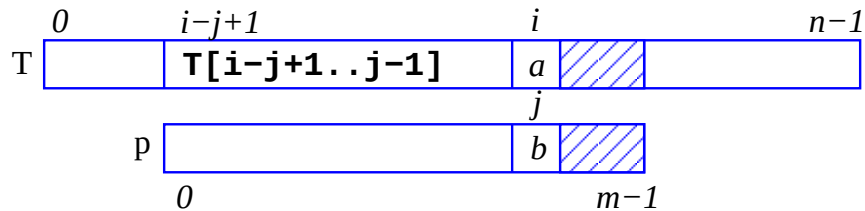
- For p , calculate the table `lastInP` and place $p[0]$ over $T[0]$, i.e. set $i = m-1$ and $j = m-1$.

Boyer-Moore matching (BM)



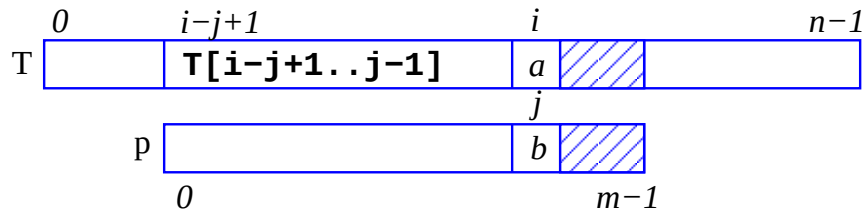
- For p , calculate the table `lastInP` and place $p[0]$ over $T[0]$, i.e. set $i = m-1$ and $j = m-1$.
- Compare $p[j]::T[i]$, i.e. compare $p[m-1]::T[m-1]$. If equal, then $j--$ and $i--$, until a match of p at $j == 0$.

Boyer-Moore matching (BM)



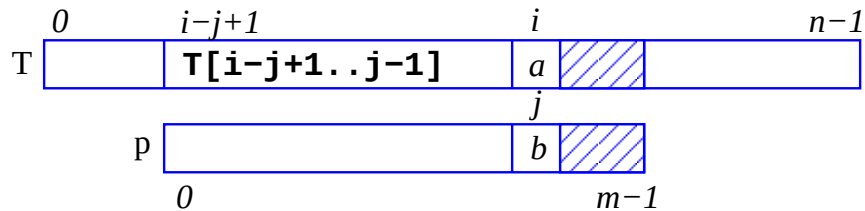
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- Compare $p[j]::T[i]$, i.e. compare $p[m-1]::T[m-1]$. If equal, then $j--$ and $i--$, until a match of p at $j == 0$.
- If $p[i]::T[i]$ are not equal: Look up the entry in `lastInP(T[i])`. It is `-1` if $T[i] \notin p$.

Boyer-Moore matching (BM)



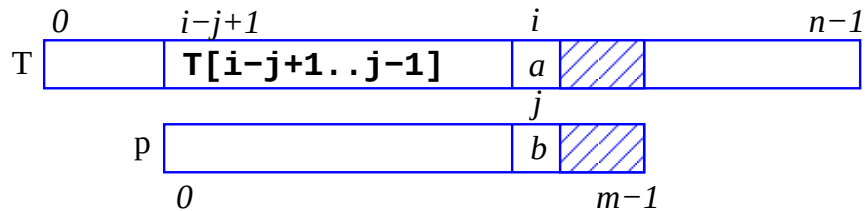
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- If $p[i]::T[i]$ are not equal: Look up the entry in `lastInP(T[i])`. It is -1 if $T[i] \notin p$.
- If $T[i]$ does not appear in the pattern p then move i forward: $i' = i + m$.

Boyer-Moore matching (BM)



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- Compare $p[j]::T[i]$, i.e. compare $p[m-1]::T[m-1]$. If equal, then $j--$ and $i--$, until a match of p at $j == 0$.
- If $p[i]::T[i]$ are not equal: Look up the entry in `lastInP(T[i])`. It is -1 if $T[i] \notin p$.
- If $T[i]$ does not appear in the pattern p then move i forward: $i' = i + m$.
- If the last appearance of $T[i] \in p$ is at k then move i forward: $i' = i + m - \min(j, 1+k)$.

Boyer-Moore matching (BM)



- For p , calculate the table `lastInP` and place $p[0]$ over $T[0]$, i.e. set $i = m-1$ and $j = m-1$.
- Compare $p[j]::T[i]$, i.e. compare $p[m-1]::T[m-1]$. If equal, then $j--$ and $i--$, until a match of p at $j == 0$.
- If $p[i]::T[i]$ are not equal: Look up the entry in `lastInP(T[i])`. It is -1 if $T[i] \notin p$.
- If $T[i]$ does not appear in the pattern p then move i forward: $i' = i + m$.
- If the last appearance of $T[i] \in p$ is at k then move i forward: $i' = i + m - \min(j, 1+k)$.
- Repeat `while (i < n)`.

Boyer-Moore—getLast(p, lastInP)

```
1 static void getLast(String p, int lastInP[]) {
2 // pre: p[0..m-1] pattern for which
3 // lastInP must be created
4 // post: Boyer-Moore's_lastInP_vector
5 //
6
7 int _j;
8 int _s=_256;
9 int _m=_p.length();
10
11 for_(j=0;_j<s;_j++)
12     lastInP[j]=_1;
13 for_(j=0;_j<m;_j++)
14     lastInP[(int)(p.charAt(j))]=_j;
15 }
```

Boyer-Moore—matchBM(p, T)

Start with the usual tests on the bounds `m` and `n` and set up `lastInP`.

```
1 static int matchBM(String p, String T) {
2 // pre: p[0..m-1] pattern to match
3 // T[0..n-1] text in which to find pattern
4 // post: returns index of first matched position
5
6 int m = p.length();
7 int n = T.length();
8
9     if (m == 0) return 0;
10    else if (n == 0) return -1;
11    else if (m > n) return -1;
12
13 int s = 256;
14 int last[] = new int [s];
15 int i; int j;
16
17    getLast(p, last);
```

Boyer-Moore—matchBM(p, T)

The code puts i , so that $p[0]$ lies at $T[i-m+1]$.

```
1  i = j = m - 1;
2  do {
3      if (p.charAt(j)==T.charAt(i))
4          if (j == 0 )
5              return i; // first match
6          else {
7              i--;
8              j--;
9          }
10     else {
11         i = i + m // jump step
12             - Math.min(j, 1+last[T.charAt(i)]);
13         j = m - 1;
14     }
15 } while (i < n );
```

Boyer-Moore matching—complexity

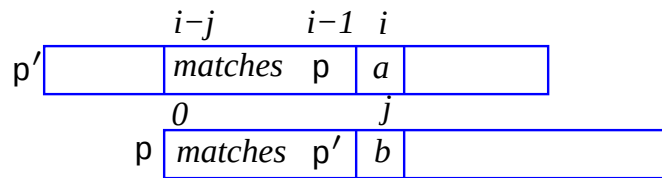
- BM must set up `lastInP[]`.
This costs $|\text{lastInP}[]| = |\Sigma|$.
Then it starts comparing from the back of `p`.
Unless `p` is covering itself in `T`,
or if the $|\mathbf{p}| \ll |\mathbf{T}|$ or $|\mathbf{p}| \ll |\Sigma|$,
most comparisons are going to be unequal.

So, on *average*, the complexity is $O(n + \Sigma) = O(n)$ since Σ is a constant.

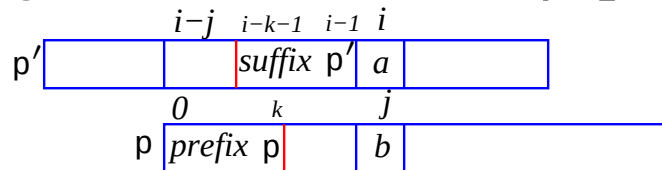
Boyer-Moore matching—complexity

- Many jumps might be m long, so the complexity could become $O(n/m)$. e.g. $\text{p} = \text{"yyyyyyy"}^m$, i.e. m ‘y’s, and no ‘x’s
 $\text{T} = \text{"xxxxxxxxxxxxxxxxxxxxxyyyyyyy"}^m$, $|\text{T}| = n$.
- The *worst case* for BM appears when
 $\text{p} = \text{"yx"}^{m-1}$, i.e. a ‘y’ and $m - 1$ ‘x’s and $\text{T} = \text{"xxxxxxxxxxxxxxxxxyxxxxxxxxxxx"}^m$, $|\text{T}| = n$.
BM degenerates to $O(mn + |\Sigma|)$
- Generally, BM is likely to scan less than n characters in T .

Knuth-Morris-Pratt matching—preview



Knuth-Morris-Pratt-matching (KMP) preprocesses p by first matching p with itself storing information about where to move $p[0]$ next when it gets a mismatch at $p'[i]$: We know $p[0..j-1] == p'[i-j..j-1]$.



In the matching piece KMP finds a *prefix* of $p \equiv p[0..k]$ that matches a *suffix* of $p' \equiv p'[i-k-1..i-1]$.

- When there *is a match*, KMP always sets the next value of j to $j++$ and steps i to $i++$.
- When there *is no match*, j becomes $\text{next}[j] \equiv k$ and if $j \geq 0$ then i stays the same and KMP avoids redundant comparisons in $p'[i-j..i-1]$, otherwise when $j < 0$ then $j = 0$ and $i++$.

KMP matching—examples

Let i run in T and j run through p , comparing $p[j]$ with $T[i]$ and putting $i++$; $j++$; when they are equal.

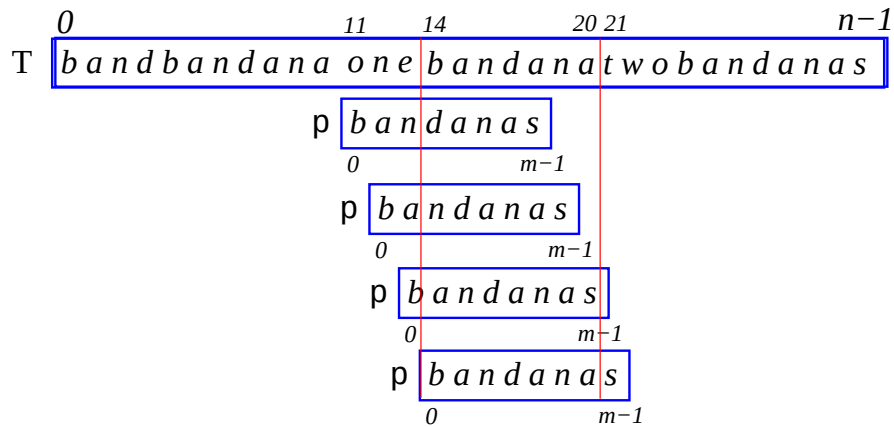
T $b a n d b a n d a n a o n e b a n d a n a t w o b a n d a n a s$
 $0 \qquad i \qquad n-1$
 p $b a n d a n a s$
 $0 \qquad j \qquad m-1$

$p[0..3] \equiv T[0..3]$ but $p[j=4] \equiv 'a' \neq T[i=4] \equiv 'b'$, Since $p[0..3] \equiv 'band'$ does not match with any substring of itself, next put $j=0$ — i stays the same—comparing $p[0..] :: T[4..]$. KPM ‘knows’ $p[0] = 'band'$ and safely jumps over it.

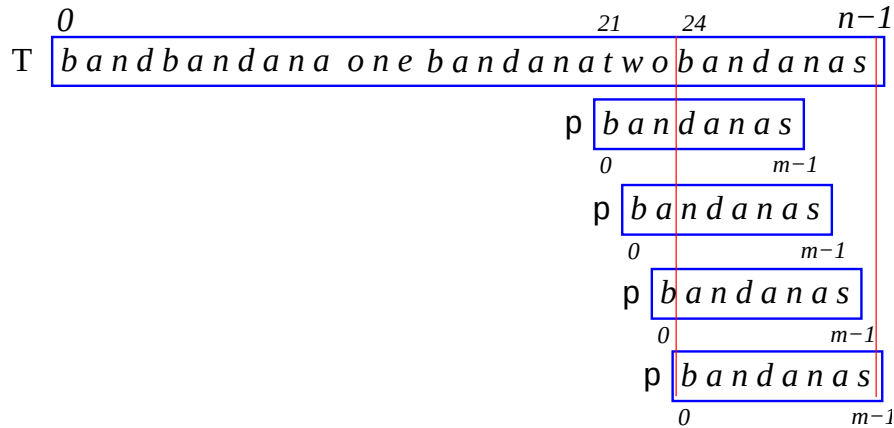
T $b a n d b a n d a n a o n e b a n d a n a t w o b a n d a n a s$
 $0 \qquad 4 \qquad 10 \quad i \quad 12 \qquad n-1$
 p $b a n d a n a s$
 $0 \qquad j \qquad m-1$

Now $p[0..6] \equiv 'bandana'$ matches $T[4..10]$ and j becomes 0 putting $p[0..]$ over $T[11..]$.

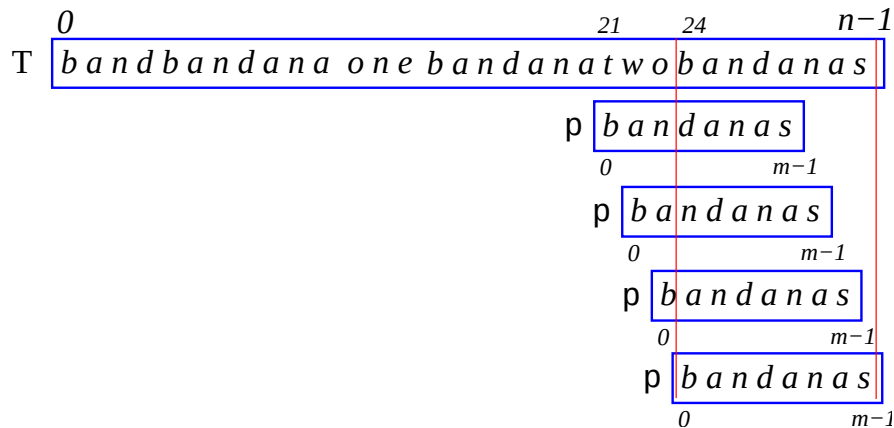
Knuth-Morris-Pratt matching (KMP)



Subsequently j hobbles forward one-by-one until **bandana** is matched at $T[14..20]$. The mismatch at $p[7]='s'$ with $T[21]='t'$ causes p to jump forward to try matching $p[0..]$ over $T[21..]$



Knuth-Morris-Pratt matching (KMP)



When $p[0]$ mismatches both i and j are incremented:

$p[j=0] \equiv 'b'$ mismatches $T[i=21] \equiv 't'$ and then

$p[j=0] \equiv 'b'$ mismatches $T[i=22] \equiv 'w'$ and then

$p[j=0] \equiv 'b'$ mismatches $T[i=23] \equiv 'o'$ and then

$p[j=0] \equiv 'b'$ matches $T[i=24] \equiv 'b'$, and eventually the pattern

$p[0..7] \equiv \text{'bandanas'}$ matches $T[24..31]$.

The value of $\text{next}[0] = -1$ and all the other values of $\text{next}[j]$ here are 0 because 'bandanas' has no suffixes that match prefixes of itself.

The following example has substrings $p[0..k]$, $k > 0$, which have prefixes that match their own suffixes.

Consider the pattern **ababbababaa**.

Knuth-Morris-Pratt matching (KMP)

The pattern **ababbababaa** contains substrings $p[0..k]$, $k > 0$, with prefixes that match their own suffixes.

j	next[j]
0	a -1 -
1	b 0 a-
2	a 0 ab-
3	b 1 aba-: $p[0..0] \equiv a \equiv p[2..2]$
4	b 2 abab-: $p[0..1] \equiv ab \equiv p[2..3]$
5	a 0 ababb-
6	b 1 ababba-: $p[0..0] \equiv a \equiv p[5..5]$
7	a 2 ababbab-: $p[0..1] \equiv ab \equiv p[5..6]$
8	b 3 ababbaba-: $p[0..2] \equiv aba \equiv p[5..7]$
9	a 4 ababbabab-: $p[0..3] \equiv abab \equiv p[5..8]$
10	a 3 ababbababa-: $p[0..2] \equiv aba \equiv p[7..9]$

If the pattern **p=ababbababaa** mismatches the text

T=ababbababababababababababababab at p[j=10]≡‘a’
 != T[i=10]≡‘b’, the next[j] value is 2 and j must be set to j =
 next[j]+1≡3, leaving i the same.

Knuth-Morris-Pratt matching (KMP)

The pattern **abababababa** contains substrings $p[0..k]$, $k > 0$, with prefixes that match their own suffixes. next

0	a	-2	-
1	b	-1	a-
2	a	-1	ab-
3	b	0	aba-: $p[0..0] \equiv a \equiv p[2..2]$
4	a	1	abab-: $p[0..1] \equiv ab \equiv p[2..3]$
5	b	2	ababa-: $p[0..2] \equiv aba \equiv p[2..4]$
6	a	3	ababab-: $p[0..3] \equiv abab \equiv p[2..5]$
7	b	4	abababa-: $p[0..4] \equiv ababa \equiv p[2..6]$
8	a	5	abababab-: $p[0..5] \equiv ababab \equiv p[2..7]$
9	b	6	ababababa-: $p[0..6] \equiv abababa \equiv p[2..8]$
10	a	7	ababababab-: $p[0..7] \equiv abababab \equiv p[2..9]$

If the pattern **p=abababababa** mismatches the text

T=abababababbababababababababababab at p[j=10]≡‘a’
 != T[i=10]≡‘b’, the next[j] value is 7 and j must be set to j =
 next[j]+1≡8, leaving i the same.

Knuth-Morris-Pratt matching (KMP)

- In KMP the text T is processed forward.
- Since all of $p[0 \dots j-1]$ has been matched, none of the corresponding characters in $T[i-j \dots i-1]$ need ever be matched again.
- Sometimes the character at $T[i]$ which mismatched with character $p[j]$ may be repeatedly matched with different characters of the pattern p .
- The next placement of p over T depends entirely on what has already been matched in p .
- This information is applied to decide which character of p must next be matched with $T[i]$.

Knuth-Morris-Pratt matching—recap

- This information is applied to decide which character of p must next be matched with $T[i]$.
- k is calculated to place $p[k+1]$ over $T[i]$ where $p[0..k]$ is the *greatest* prefix in $p[0..j-1]$ that matches a suffix $p[j-k-1..j-1]$. Note that length $|p[0..k]| = k + 1 = |p[j-k-1..j-1]|$.
- We call the table of k 's `next[]`.

Knuth-Morris-Pratt matching—recap

- The longest prefix-suffix pair is placed over one another first.
- If there is a match: $i++;$ $j++;$, else
- If there is a mismatch a next j that is always smaller than the previous one is calculated using $j = \text{next}[j] + 1$.
- Note that i is only advanced when there is a match or when j becomes 0 .
- We next construct the `next[]` table.
- When there is a mismatch of $p[j]$ with $T[i]$ the value of `next[j]` tells us to advance j to `next[j]+1`.
- Usually i stays the same but can advance by 1 .

KMP—constructing `next[]`

- When `p[j]` and `T[i]` match, simply increment both `i` and `j`, using `i++; j++;`.
- We will now consider mismatches at various values of `j ∈ [0..m-1]`.
- At a mismatch of `p[0]`, simply put `i++; j++;`.
- We will construct `next[]` for Manber's example from Udi Manber [Section 6.7, pp. 148–155, *Introduction to Algorithms A Creative Approach*, Addison-Wesley Publishing Co., Reading, MA., 1989]

Note that the **ababbababaa** example is this example in slightly disguised form.

KMP—constructing next[]

Consider a mismatch at $p[0]$. Since no information has been gained, simply $i++$; $j++$;

$\begin{array}{c} 0 \qquad \qquad m-1 \\ \text{T } \boxed{x \mid y \ x \ y \ y \ x \ y \ x \ y \ x \ x} \\ \text{p } \boxed{_ \mid y \ x \ y \ y \ x \ y \ x \ y \ x \ x} \end{array}$

$\begin{array}{c} 1 \qquad \qquad m-1 \\ \text{T } \boxed{x \mid y \ x \ y \ y \ x \ y \ x \ y \ x \ x} \\ \text{p } \boxed{x \mid _ \ x \ y \ y \ x \ y \ x \ y \ x \ x} \end{array}$

On mismatching $p[1]$ —above right— $p[0..0]$ is known, but it does not help so $j=0$ and i remains the same.

After a mismatch at $p[2]$ —below left—the substring $p[0..1]$ is known, but no prefix equals any suffix so again $j=0$ and i remains the same.

$\begin{array}{c} 2 \qquad \qquad m-1 \\ \text{T } \boxed{x \ y \mid x \ y \ y \ x \ y \ x \ y \ x \ x} \\ \text{p } \boxed{x \ y \mid _ \ y \ y \ x \ y \ x \ y \ x \ x} \end{array}$

$\begin{array}{c} 3 \qquad \qquad m-1 \\ \text{T } \boxed{x \ y \ x \mid y \ y \ x \ y \ x \ y \ x \ x} \\ \text{p } \boxed{x \ y \ x \mid _ \ y \ x \ y \ x \ y \ x \ x} \end{array}$

The mismatch at $p[3]$ means that we have covered $p[0..2] \equiv "xyx"$.

This has a prefix x - that equals a suffix $-x$ so now $j=1$ and i remains the same.

j	0	1	2	3	4	5	6	7	8	9	10
p	x	y	x	y	y	x	y	x	y	x	x
next[j]	-1	0	0	1	.	.	.	.	.	.	.

KMP—constructing next[]

When mismatching at $p[4]$, $p[0..3] \equiv \text{"xyxy"}$ It has a prefix xy- that equals a suffix -xy , so now $j=2$ and i remains the same.

T $\boxed{\text{x y x y} \mid \text{y x y x y x x}}^{4 \quad m-1}$
 p $\boxed{\text{x y x y} \mid \text{- x y x y x x}}$

T $\boxed{\text{x y x y y} \mid \text{x y x y x x}}^{5 \quad m-1}$
 p $\boxed{\text{x y x y y} \mid \text{x y x y x x}}$

On mismatching $p[5]$ —above right— $p[0..4] \equiv \text{"xyxyy"}$, has no prefix-suffix pair, so $j=0$ and i static.

After a mismatch at $p[6]$ —below left—the substring $p[0..5] \equiv \text{"xyxyyx"}$ has the prefix-suffix pair x- , -x and $j=1$ and i is static.

T $\boxed{\text{x y x y y x} \mid \text{y x y x x}}^{6 \quad m-1}$
 p $\boxed{\text{x y x y y x} \mid \text{- x y x x}}$

T $\boxed{\text{x y x y y x y} \mid \text{x y x x}}^{7 \quad m-1}$
 p $\boxed{\text{x y x y y x y} \mid \text{- y x x}}$

The mismatch at $p[7]$ means that we have covered $p[0..6] \equiv \text{"xyxyxyx"}$. This has a prefix xy- that equals a suffix -xy so now $j=2$ and i remains the same.

j	0	1	2	3	4	5	6	7	8	9	10
p	x	y	x	y	y	x	y	x	y	x	x
next[j]	-1	0	0	1	2	0	1	2	.	.	.

KMP—constructing next[]

When mismatching at $p[8]$, $p[0..7] \equiv \text{"xyxyxyxy"}$ It has a prefix xyx- that equals a suffix -xyx , so now $j=3$ and i remains the same.

T x y x y y x y x y x x
p x y x y y x y x _ x x

T x y x y y x y x y x x
p x y x y y x y x y _ x

If $p[9] \neq T[9]$ —above right— $p[0..8] \equiv \text{"xyxyxyxyxy"}$, This has a prefix xyxy- that equals a suffix -xyxy , so now $j=4$ and i remains the same.

After the mismatch at $p[10]$ —below—the substring $p[0..9] \equiv \text{"xyxyxyxyxyx"}$ has the prefix-suffix pair xyx- , -xyx , so $j=3$ and i is static.

T x y x y y x y x y x x
p x y x y y x y x y x _

The table has now been completed.

j	0	1	2	3	4	5	6	7	8	9	10
p	x	y	x	y	y	x	y	x	y	x	x
next[j]	-1	0	0	1	2	0	1	2	3	4	3

KMP—buildNext

Pseudocode for buildNext.

```
1 void buildNext(String p, int [] next) {  
2   int m = p.length();  
3   int i, j;  
4  
5   next[0] = -2;  
6   next[1] = -1;  
7   for (i = 2; i < m; i++) {  
8     j = next[i-1]+1;  
9     while (j >= 0 && p[i-1] != p[j]){  
10      j = next[j]+1; }  
11     next[i] = j; }  
12   for (j = 0; j < m; j++)  
13     next[j]++; }
```

Pseudocode for buildNext

```
1 int matchKMP(String p, String T) {  
2     int m = p.length();  
3     int n = T.length();  
4  
5     if (m == 0) return 0;  
6     else if (n == 0) return -1;  
7     else if (m > n) return -1;  
8  
9     int i, j, next[] = new int [m+1];  
10  
11     buildNext(p, next);  
12     ...
```

KMP—KMPmatch

```
1   ...
2   buildNext(p, next);
3   i = 0;
4   j = 0;
5   while (i < n) {
6       if (p[j] == T[i]){
7           i++;
8           j++; }
9       else {
10          j = next[j];
11          if (j < 0) {
12              j = 0;
13              i++; } }
14       if (j == m)
15           return i - m; }
16   return -1; }
```

- Calculating `buildNext[]` takes $O(m)$. Matching the pattern using the `next[]` table takes $O(n)$. KMP runs in $O(m + n)$.
- One character in `T[i]` may be matched with many characters from the pattern `p`. How many?
- When there is a mismatch with `T[i]` another character from `p` is matched if `next[j] >= 0`.
- Suppose the first mismatch involved `p[j]`, since each evaluation of `next[j]` leads us to a smaller index $j \in [0..j-1]$, so we can only j times. However, to reach `p[j]` we must have gone forward j times without any backtracking.
- If we assign the costs of backtracking to the forward moves then we at most double the number of forward moves.
- There are only n forward moves so the number of comparisons is $O(n)$.

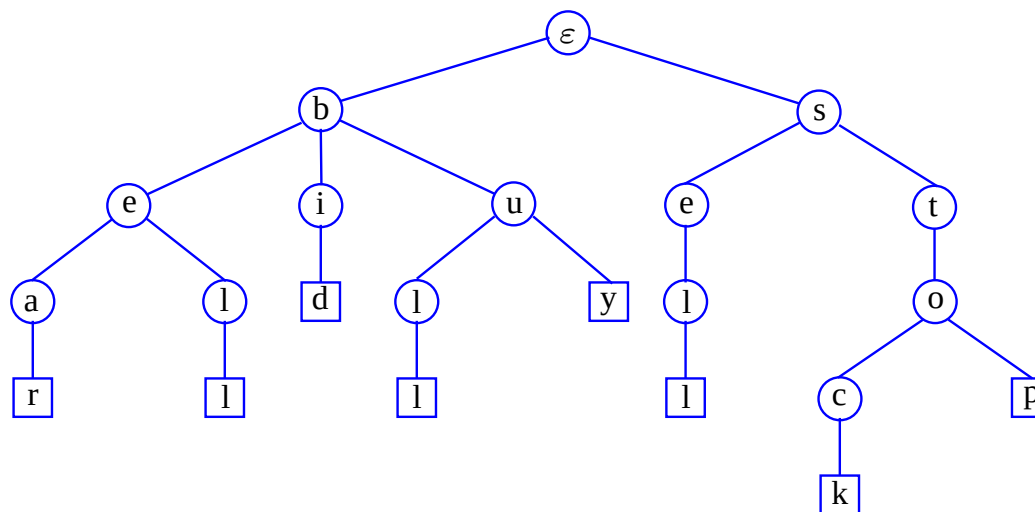
Preprocess the *text* T —not the *pattern* p

- In Rabin-Karp, Boyer-Moore, and Knuth-Morris-Pratt pattern matching the pattern p is *preprocessed* to speed up searching for its occurrence in a text T .
- If many searches $p \in T$ are done, such as searching for the presence of a word p in a dictionary T , then *preprocessing* the text T may help to speed up queries, especially if T is large or when many different queries p are made.
- A *trie* is a very fast, dense structure that allows searches in time proportion to the *pattern* size, i.e. $O(m)$.

Standard tries

The standard trie for a set of strings T is an ordered tree such that:

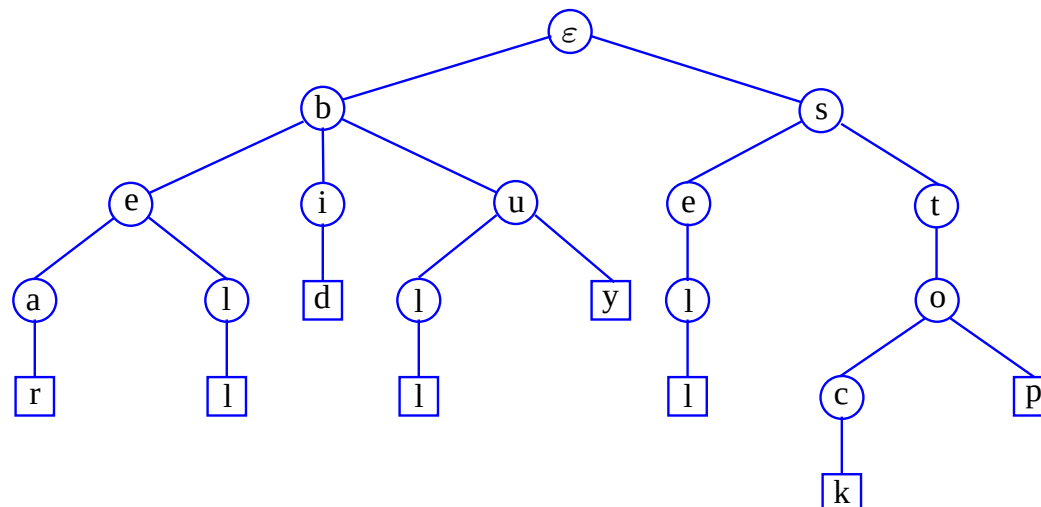
- Each node but the root is labelled with a character.
- The children of a node are alphabetically ordered.
- The paths from the leaves to the root yield the strings of T .
- e.g. standard trie for the set of strings
 $T = \{bear, bell, bid, bull, buy, sell, stock, stop\}$



Analysis of standard tries

A standard trie uses $O(n)$ space and supports searches, insertions and deletions in time $O(md)$, where:

- n total size of the strings in T , and m size of the string parameter of the operation, while $d = |\Sigma|$ size of the alphabet.
- Using binary search at each node for the next letter, searches, insertions and deletions may be done in $O(m \log d)$ time.
- By using hashing or direct lookup this is further reduced to $O(m)$.



Word matching with a trie

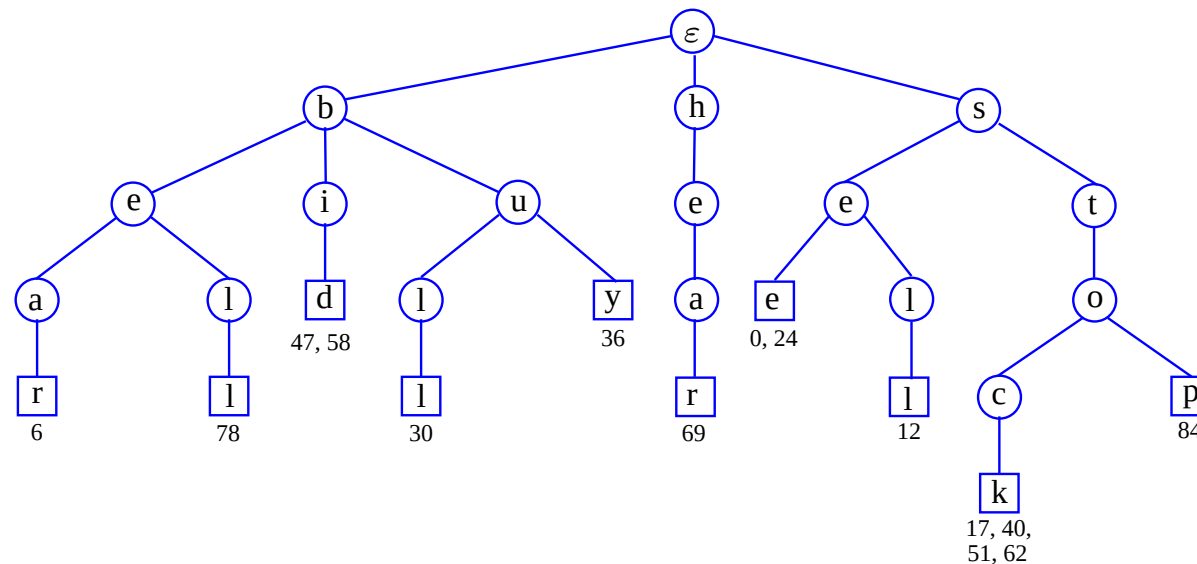
s	e	e		a		b	e	a	r	?		s	e	l	l		s	t	o	c	k	!		
0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	

s	e	e		a		b	u	l	l	?		b	u	y		s	t	o	c	k	!		
24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45	46	

b	i	d		s	t	o	c	k	!		b	i	d		s	t	o	c	k	!			
47	48	49	50	51	52	53	54	55	56	57	58	59	60	61	62	63	64	65	66	67	68		

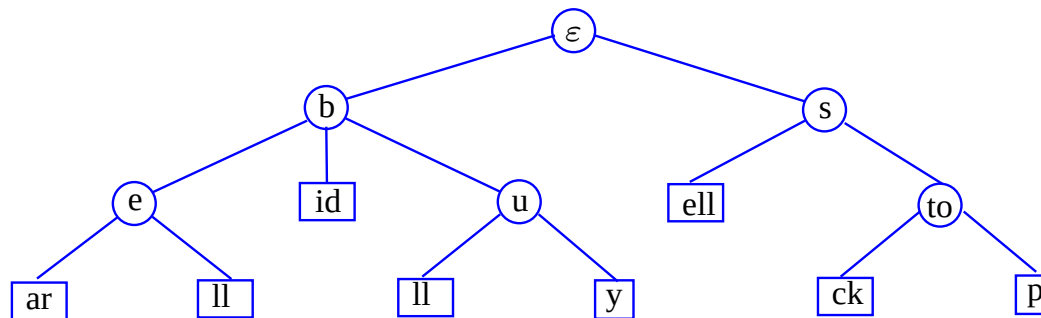
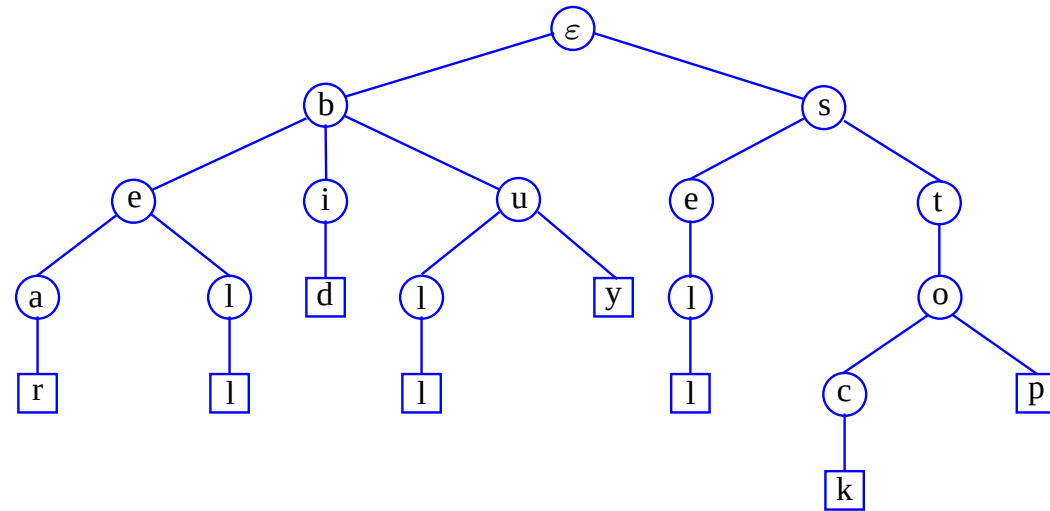
h	e	a	r		t	h	e		b	e	l	l	?		s	t	o	p	!			
69	70	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85	86	87	88			

- We insert some of the words of the text into a trie.
- Each leaf stores the occurrences of the associated word in the text.



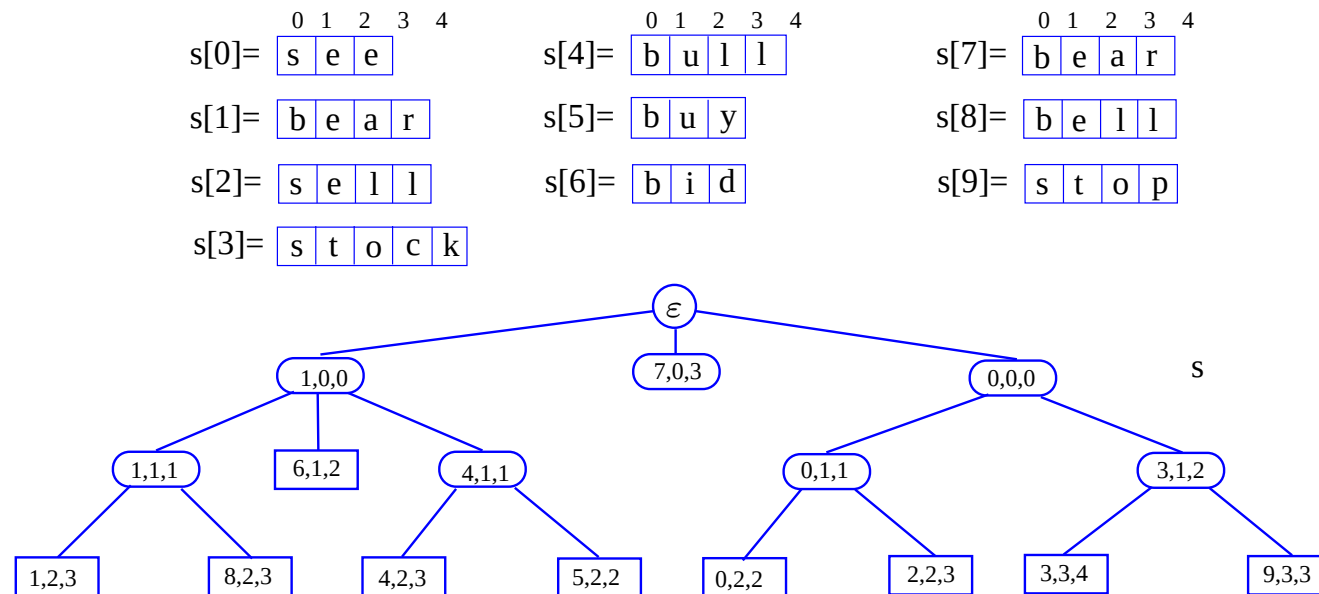
Compressed tries

- A compressed trie has internal nodes of degree at least two.
- It is obtained from a standard trie by compressing chains of *redundant* nodes.



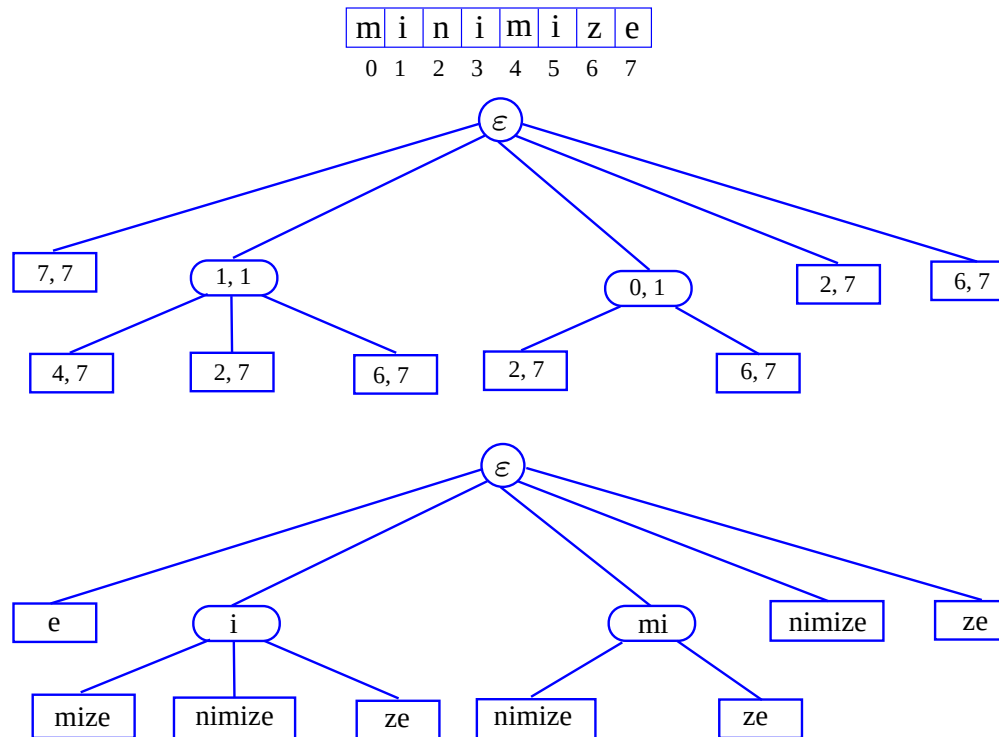
Compact Representation

- Compact representation of a compressed trie for an array of strings:
- Stores at the nodes ranges of indices instead of substrings
- Uses $O(s)$ space, where s is the number of strings in the array
- Serves as an auxiliary index structure



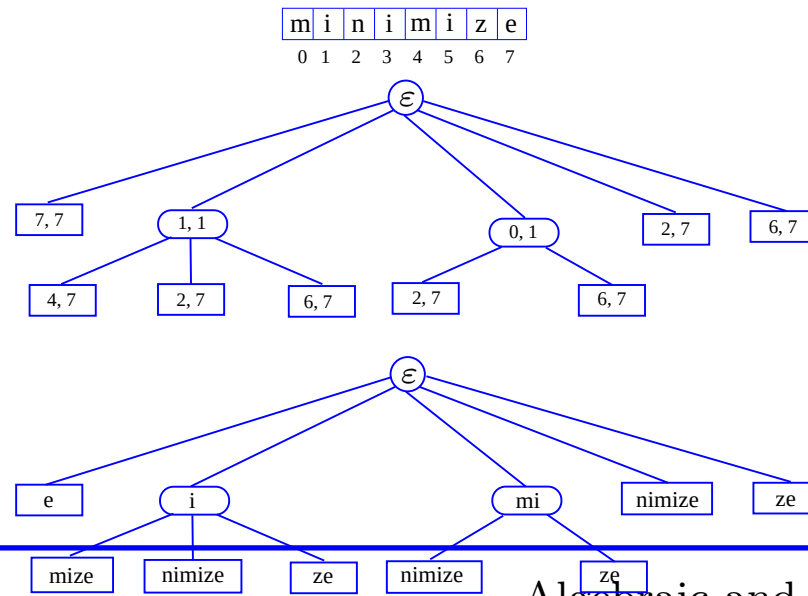
Suffix trie

- The suffix trie of a string T is the compressed trie of all the suffixes of T .



Analysis of suffix tries

- Compact representation of the suffix trie for a string T of size n from an alphabet of size d .
- Uses $O(n)$ space.
- Supports arbitrary pattern matching queries in T in $O(dm)$ time, where m is the size of the pattern.
- Can be constructed in $O(n)$ time.



Text compression with Huffman's code

- Given a text T reproduce it in a lossless way using less memory.

T is a piece of text in which we count the characters to find their frequencies.

- Huffman scheme counts the number of times characters appear in T .
- Pseudo code for getting the frequencies of the characters T in follows:

Text compression with Huffman's code

Ch		a	b	c	d	e	f	h	i	k	n	o	r	s	t	u	v
F	9	5	1	3	7	3	1	1	1	1	4	1	5	1	2	1	1

- By combining pairs of these frequencies, with the lowest first, and then adding their combinations to the frequency list, and removing the originals a binary tree is produced from which an optimal code can be written down.
- The compressed code may then rapidly be produced from the binary tree.

Huffman code—adapted from GT page 566.

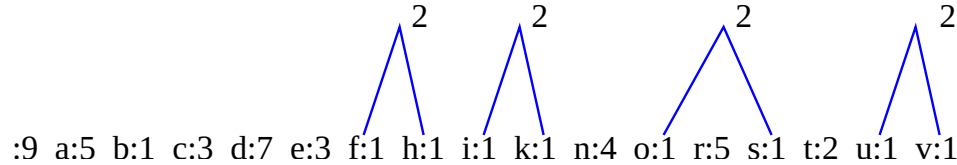
Note that the text I have used differs slightly from that of GT page 566.

Char.		a	b	c	d	e	f	h	i	k	n	o	r	s	t	u	v
Freq.	9	5	1	3	7	3	1	1	1	1	4	1	5	1	2	1	1

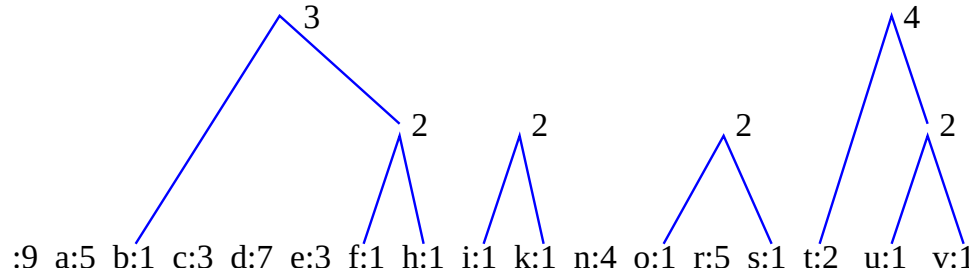
- In order to find the lowest frequency characters first, an obvious method is to: sort the characters by frequency; combine the top pair; remove the top two characters but insert their combination back into the list at its correct position.
- The very first pair with the lowest frequency could appear at the lowest node in the final tree.
- Subsequent combinations fill the tree from the lowest level.
- The last combination is hooked into the root of the tree.

Huffman coding—combining the lower nodes

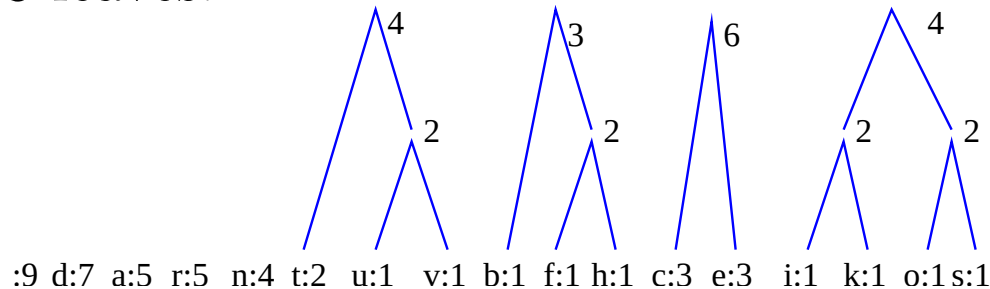
First, combine nodes that are pairs of 1s into 2-nodes.



Then combine the last 1-node with a 2 node, and start combining pairs of 2 nodes.

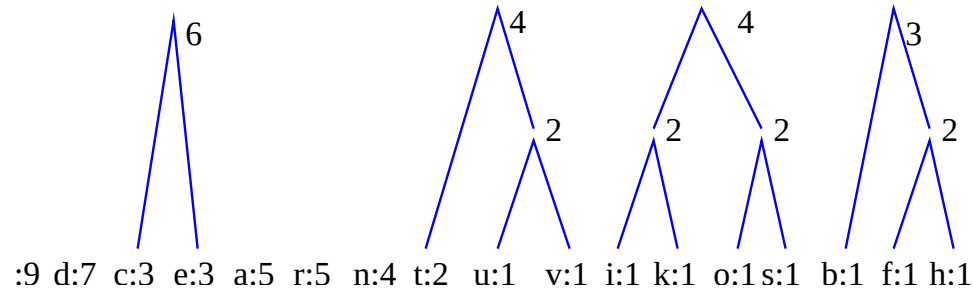


Reorder the list and combine the remaining 2 nodes. Combine the pair of 3-leaves.

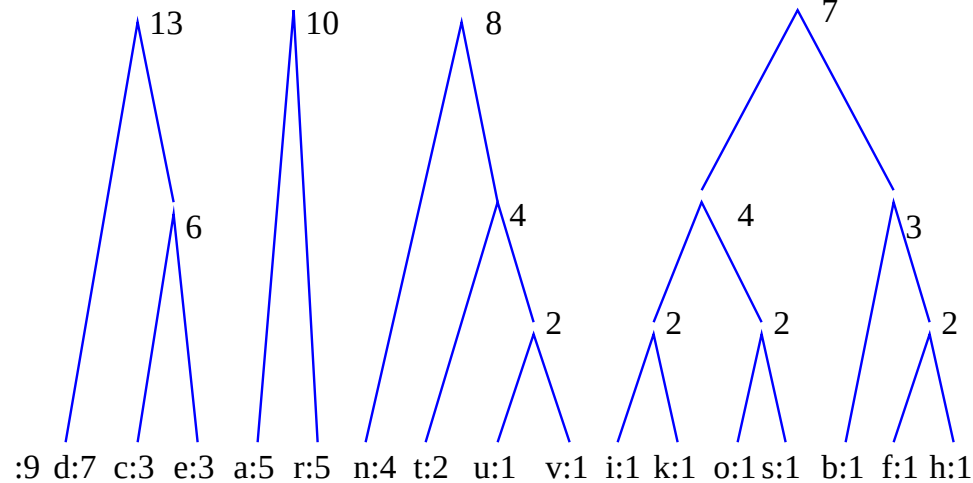


Huffman coding—reorder and combine

Simply reorder the list.



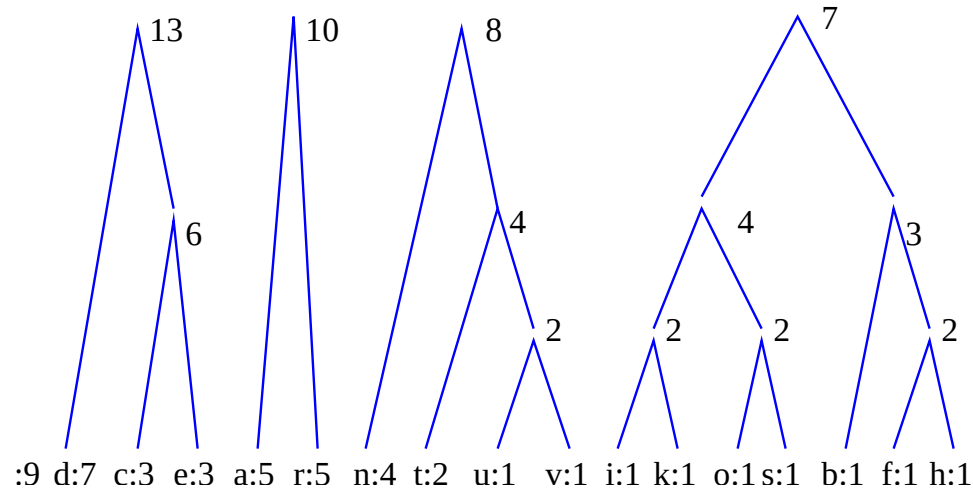
Combine lowest nodes to the next level.



Note that there are only five nodes left to combine.

Huffman coding—combine the 1s

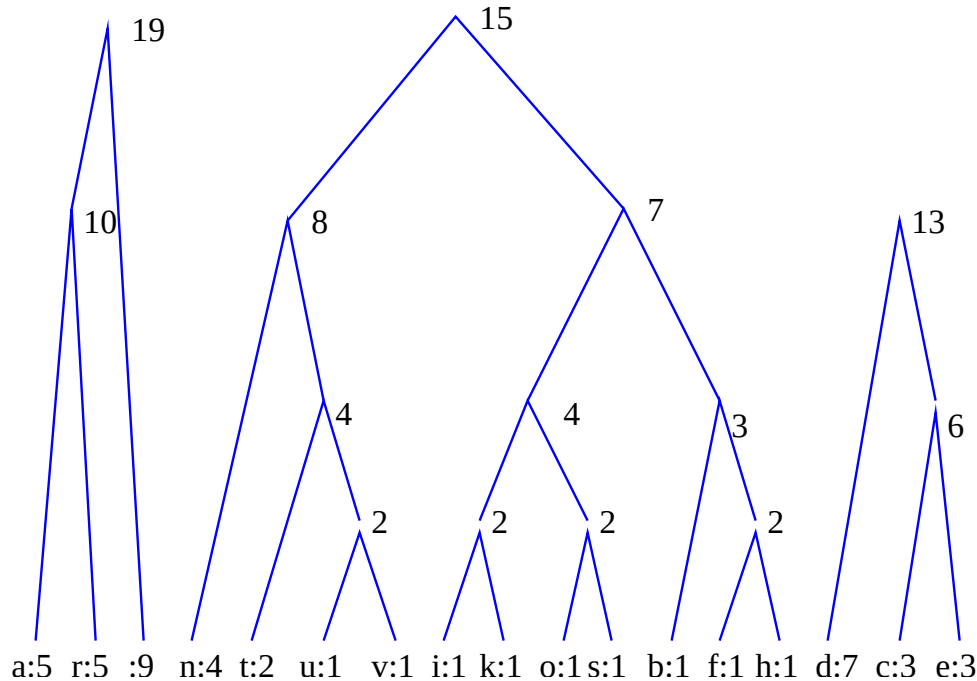
After combining 7 + 8 and 9 + 10, we get:



Next we will rearrange the nodes—high frequencies to the left.

Huffman coding—rearrange the forest of trees from highest to lowest

The nodes rearranged—high counts to the left, lowering to the right.

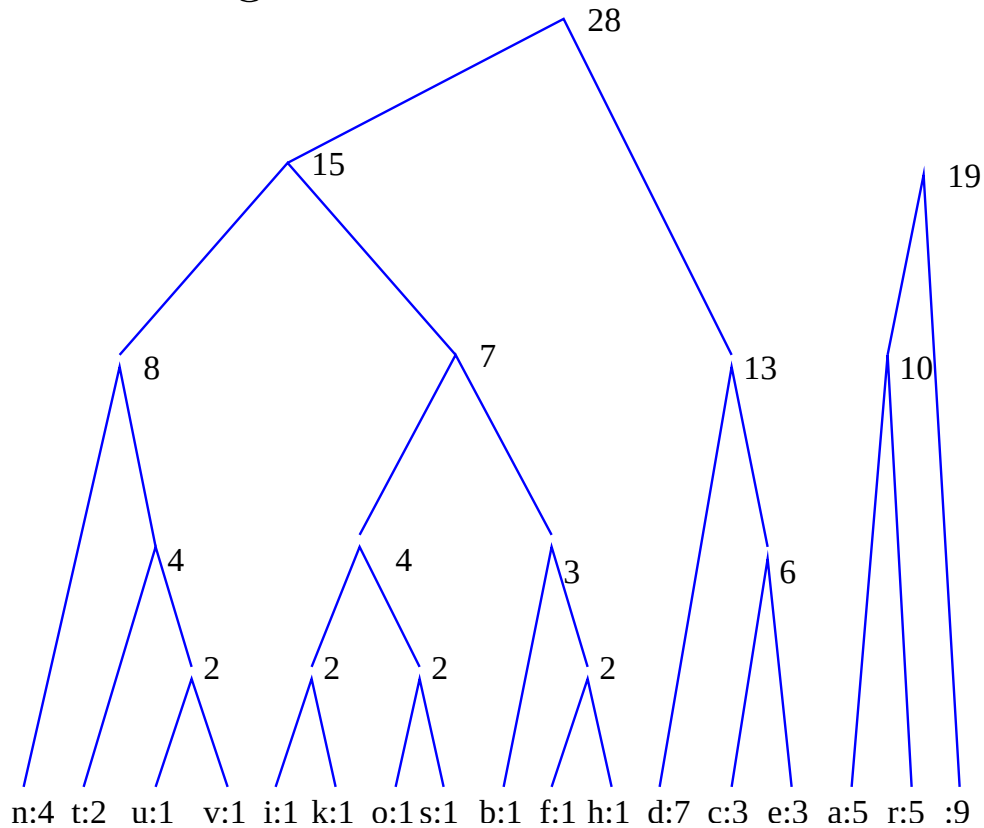


There are only three nodes left.

We will combine the lowest two, resort and again combine the lowest two to create a single rooted tree from the forest.

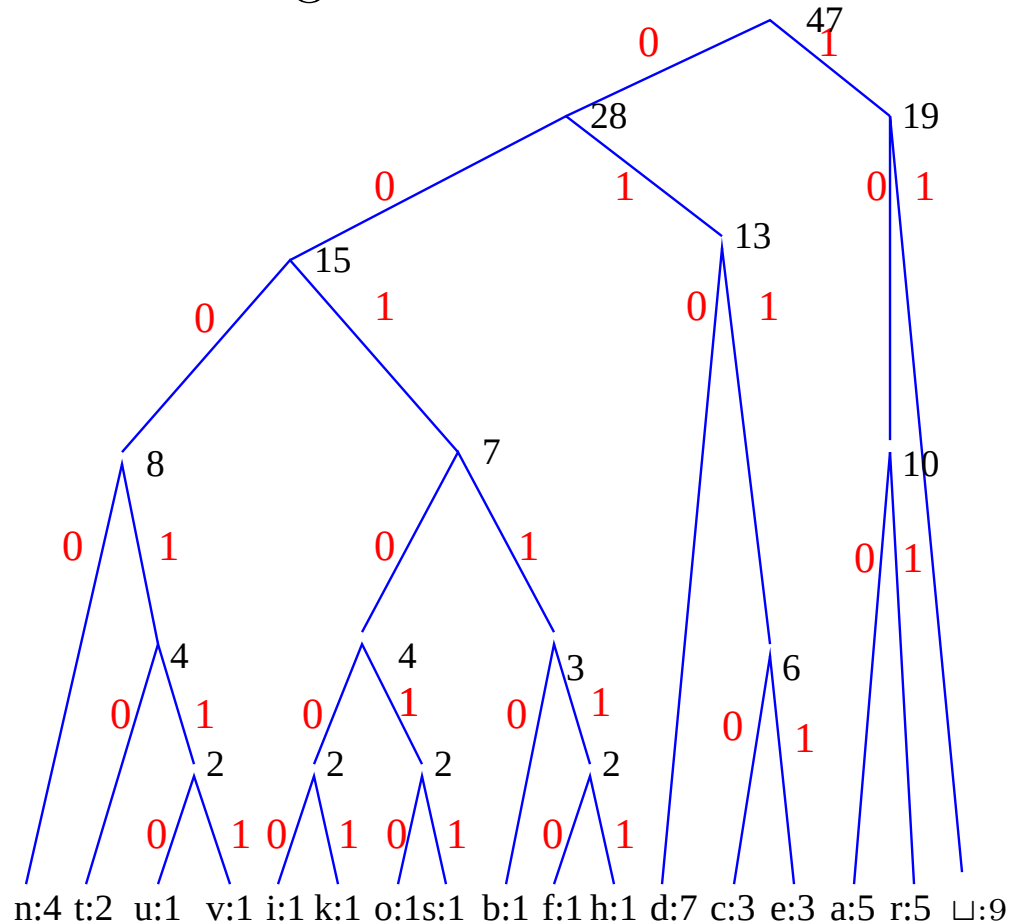
Huffman coding—combine and sort

Now combine the 13-tree and the 15-tree into a 28-tree, and resort the remaining forest of two trees.



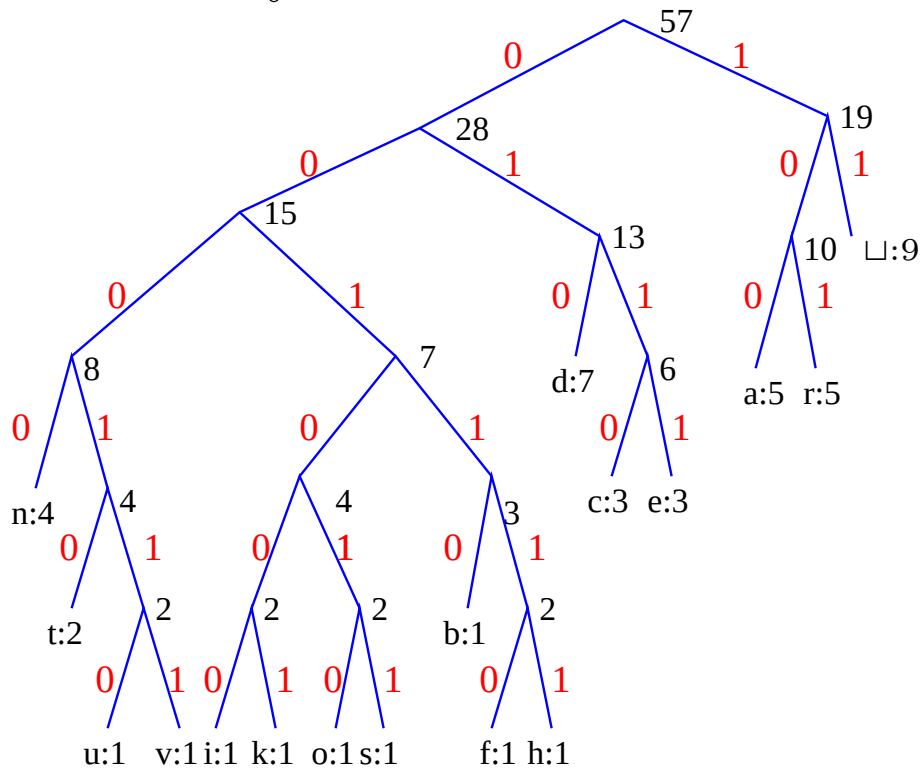
Huffman code—finally combine and label

Combine the last two trees and label each left branch of the tree with a 0 and each right branch with a 1.



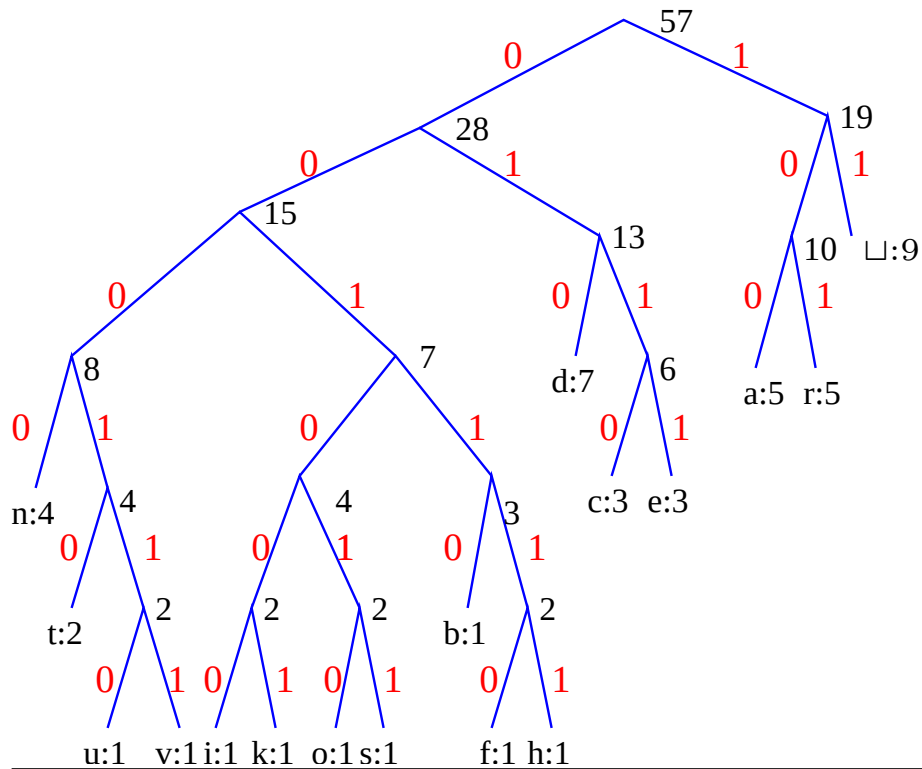
Huffman code—finally combine and label

We rearrange the tree such that the levels of each character can be seen more easily.



From the figure it follows that the number of digits needed to represent each character is the same as its depth in the Huffman tree.

Huffman code in tabular form



Char.		a	b	c	d	e	f	h	i
Freq.	9	5	1	3	7	3	1	1	1
Code	11	100	00110	0110	010	0111	001110	001111	001000

Char.	k	n	o	r	s	t	u	v
Freq.	1	4	1	5	1	2	1	1
Code	001001	0000	001010	101	001011	00010	000110	000111

"a fast runner need never be afraid of the dark" coded is:

1001100111 0100001011 0001011101 0001100000 0000011110
 1110000011 1011101011 0000011100 0111011110 1110011001
 1111100001 1101011000 0100001011 0010100011 1011000100

0111101111 1010100101 001001

Huffman code for T

Char.	□	a	b	c	d	e	f	h	i
Freq.	9	5	1	3	7	3	1	1	1
Code	11	100	00110	0110	010	0111	001110	001111	001000

Char.	k	n	o	r	s	t	u	v
Freq.	1	4	1	5	1	2	1	1
Code	001001	0000	001010	101	001011	00010	000110	000111

"a fast runner need never be afraid of the dark" coded is:

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1111100001 1101011000 0100001011 0010100011 1011000100
```

```
0111101111 1010100101 001001
```

The Huffman code has 176 bits. Since $|T| = 46$ and it has an alphabet of 16 characters a fixed length code of 4 bits per character is needed to encode this string, i.e. $46 \times 4 = 184$.

This is *not* the optimal Huffman code for this string. Why not?

Huffman code for T

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Code	11	100	00110	0110	010	0111	001110	001111	001000

Char.	k	n	o	r	s	t	u	v
Freq.	1	4	1	5	1	2	1	1
Code	001001	0000	001010	101	001011	00010	000110	000111

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```

```
0111101111 1010100101 001001
```

The Huffman code has 176 bits. Since $|T| = 46$ and it has an alphabet of less than 16 characters a fixed length code of 4 bits per character is needed to encode this string, i.e. $46 \times 4 = 184$.

This is *not* the optimal Huffman code for this string. Why not?

Since our code caters for some of characters that have different counts from T, e.g. **b**, **c**:3, **d**:7, **e**:3, and **f**:1, than this string has, the code cannot be optimal.

H.W. Calculate the optimal Huffman code for T.

Huffman with sorting and insertion into the sorted list

- Sorting the m characters of the text according to their frequencies may be done in $O(m \log m)$ time.
- The position of each insertion is found in $O(\log m)$ time. How? But there are this does not really help because insertions entail moving the rest of the list down—on average $m/2$ such items are shifted. So this part takes at worst $O(m)$ time for each of m entries, i.e. it takes $O(m^2)$ time.
- So the Huffman tree is built in $O(m \log m) + m^2$ which yields $O(m^2)$ time using the sort-and-insert method.
- This time can be improved.

Huffman code using a heap

- Instead of repeated sorting and insertion into the sorted list, a heap should rather be used.
- The $\langle \text{character}(s), \text{frequency} \rangle$ entries are placed into a heap with the entries with the lowest frequency appearing on the top of the heap.
- The top two entries are removed from the heap and combined into a new entry.
- The new entry is then inserted into the heap at its appropriate position.
- Building the heap *bottom-up* takes $O(m)$ time. Doing m insertions into a heap takes $O(m \log m)$ time.
- The total cost of constructing the Huffman tree is thus $m \log m$.

Huffman code explained

Consider the set {a, b, c, d, e, f, r} with the frequencies and probabilities for each letter:

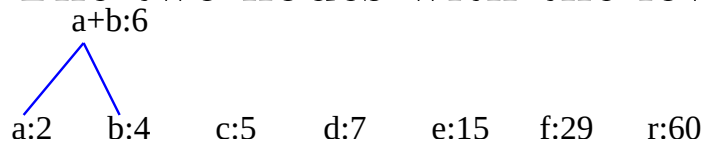
Character	a	b	c	d	e	f	r	All
Frequency	2	4	5	7	15	29	60	122
Probability	0.016	0.033	0.041	0.057	0.123	0.238	0.492	1.000

The total of the frequencies is 122.

The letters are sorted according to counts or probability—smallest first.

The probability is frequency divided by the total frequency.

The two nodes with the lowest counts are combined:



Insert the combined node [a+b : 6] after node [c : 5] and before node [d : 7].

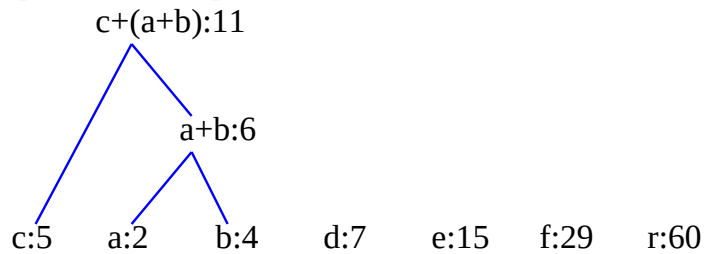


Huffman coding

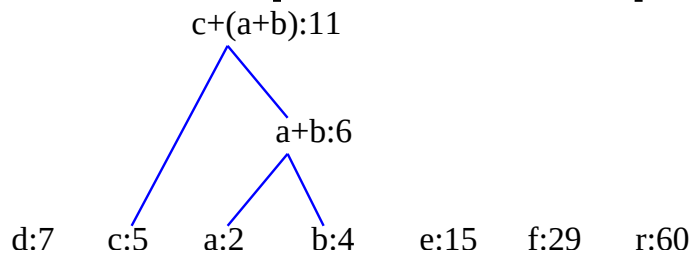
Starting with



form node $[c+(a+b) : 11]$ by combining combine nodes $[c : 5]$ and node $[a+b : 6]$:

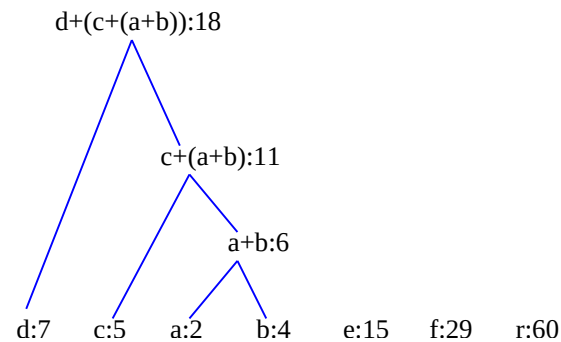


Put node $[c+(a+b) : 11]$ in its correct place:

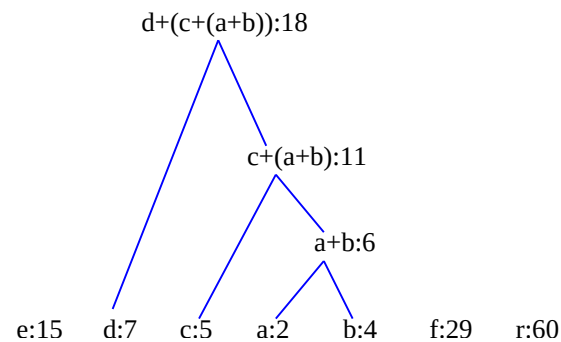


Huffman coding

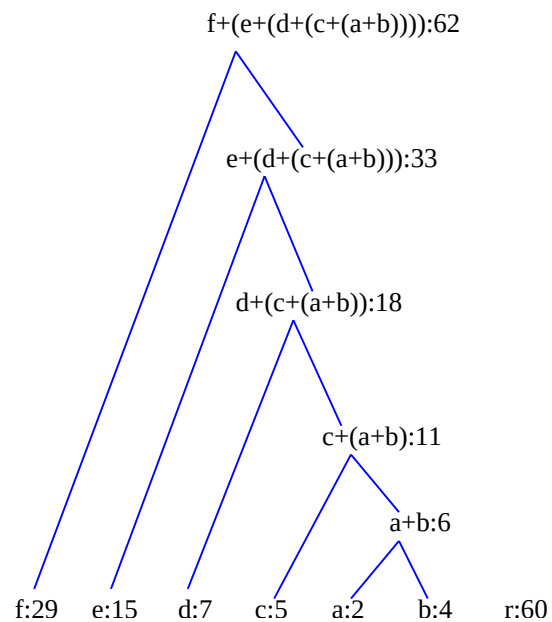
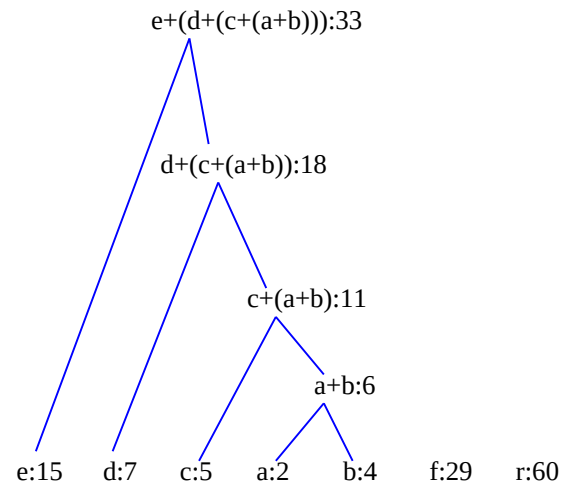
Form node $[d+(c+(a+b)) : 18]$ by combining combine nodes $[d : 7]$ and node $[c+(a+b) : 11]$:



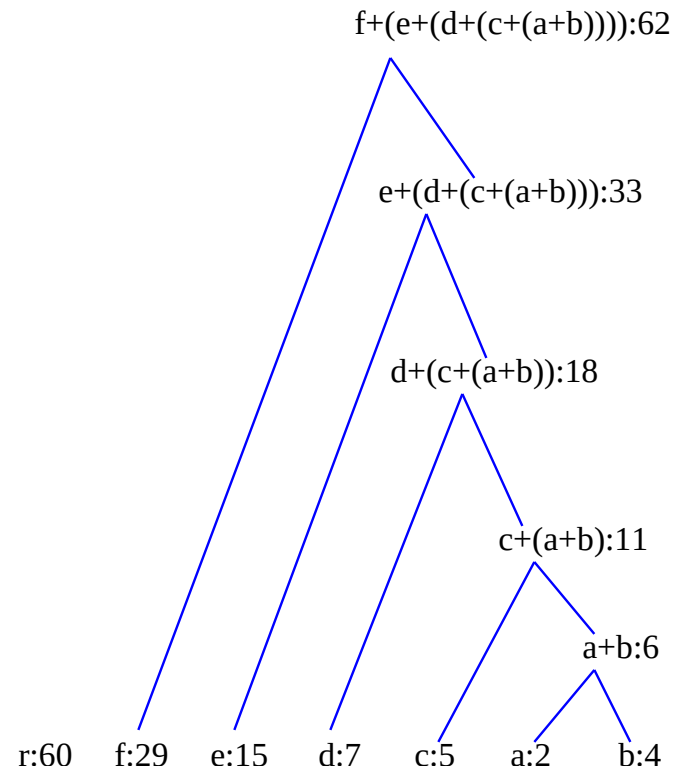
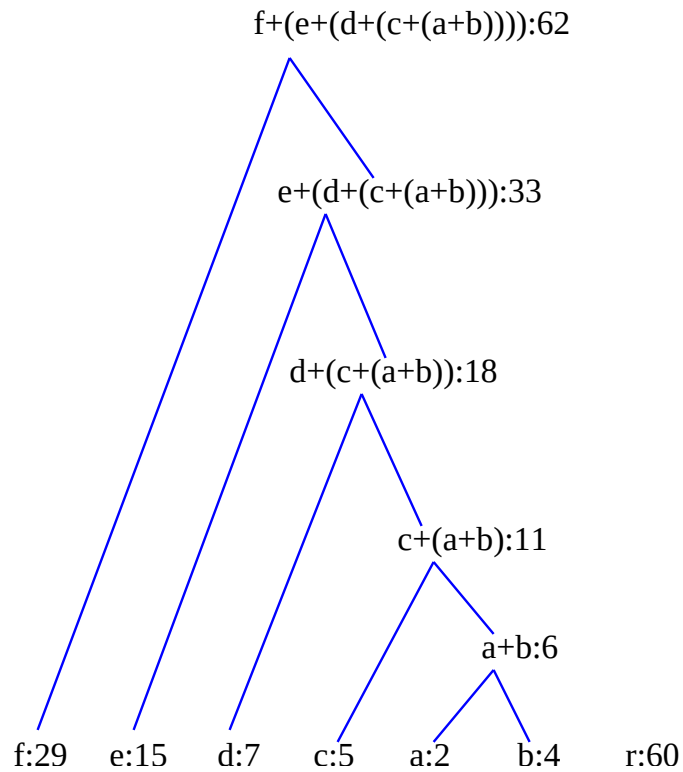
Put node $[d+(c+(a+b)) : 18]$ in its correct place:



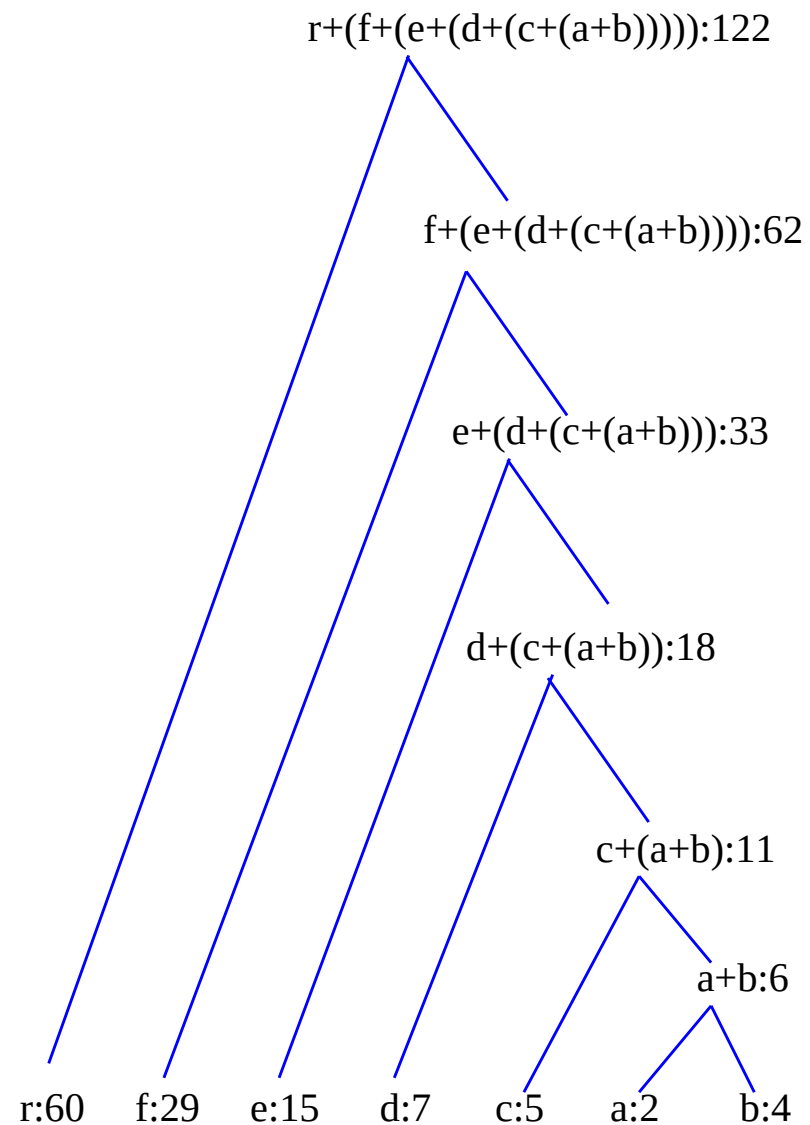
Huffman coding—repeat steps



Huffman coding—repeat steps

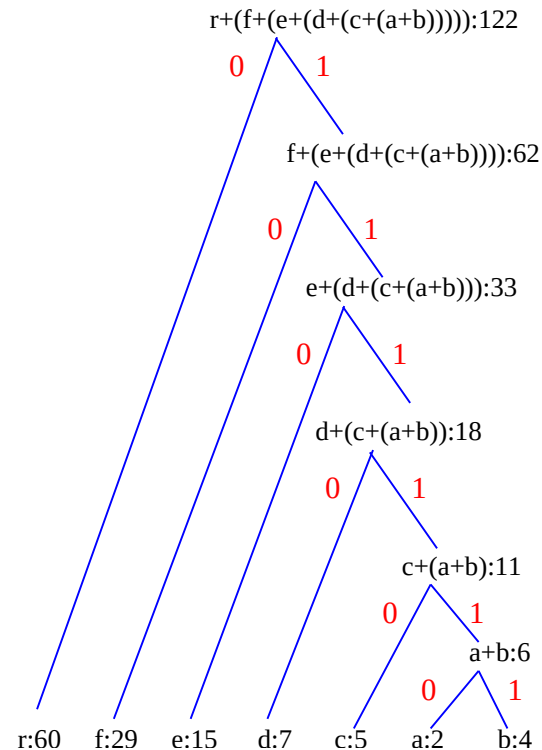


Huffman coding—last combination



Huffman coding—enter coding

Tag each left leg of the tree with a **0** and each right leg with a **1**.



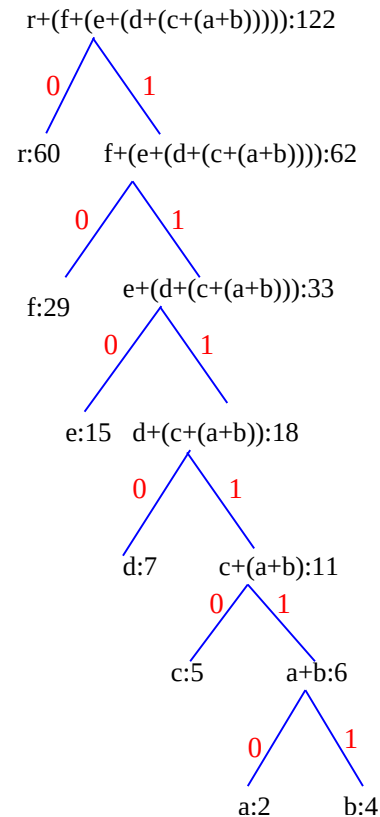
Read off the codes for each letter in the tree:

Character	a	b	c	d	e	f	r
Frequency	2	4	5	7	15	29	60
Probability	0.016	0.033	0.041	0.057	0.123	0.238	0.492
Code	111110	111111	11110	1110	110	10	0

What does **1111110111110111101101110** represent?

Huffman coding—enter coding

A Huffman tree from which the levels of each character may be seen.



The code in tabular form.

Character	a	b	c	d	e	f	r
Frequency	2	4	5	7	15	29	60
Probability	0.016	0.033	0.041	0.057	0.123	0.238	0.492
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Divide-and-conquer methods

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7         return combine(divide_and_conquer( $P_1$ ),
8                         divide_and_conquer( $P_2$ ),
9                         ...,
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Divide-and-conquer methods

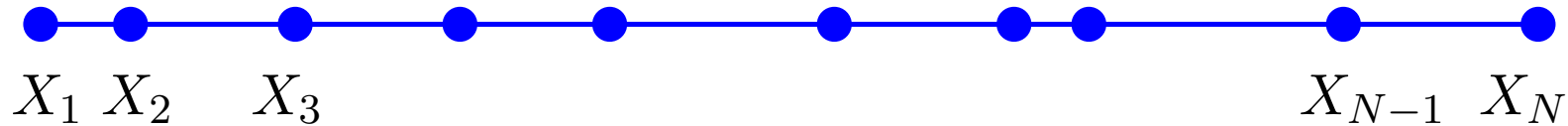
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- The resulting timing depends largely on the time taken by **combine**.
- If $T(\text{combine}) = O(N)$,
$$T(N) = kT(N/k) + O(N) = O(N \log_k N)$$

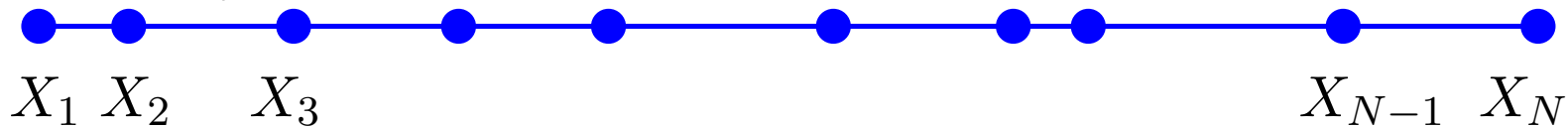
An $O(N^2)$ method for 1-D closest pair

Given a set of N points in a straight line—use the X -axis for simplicity—find the closest pair of points.



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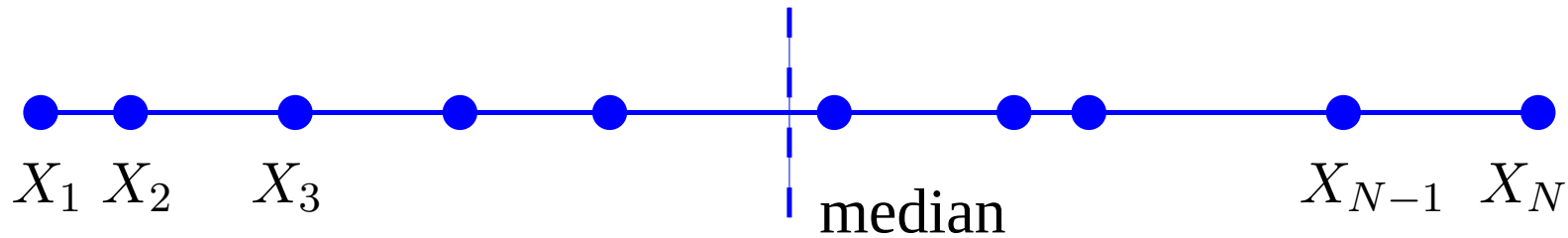


- The problem can be solved brute force by running through all the pairs in $O(N^2)$ time as follows

```
1  shortestDistance = |X[1] - X[2]|
2  indexJ = 1
3  indexK = 2
4  for j in [1..N-1]:
5      for k in [j+1..N]:
6          difference = |X[j] - X[k]|
7          if difference < shortestDistance:
8              shortestDistance = difference
9              indexJ = j
10             indexK = k
```

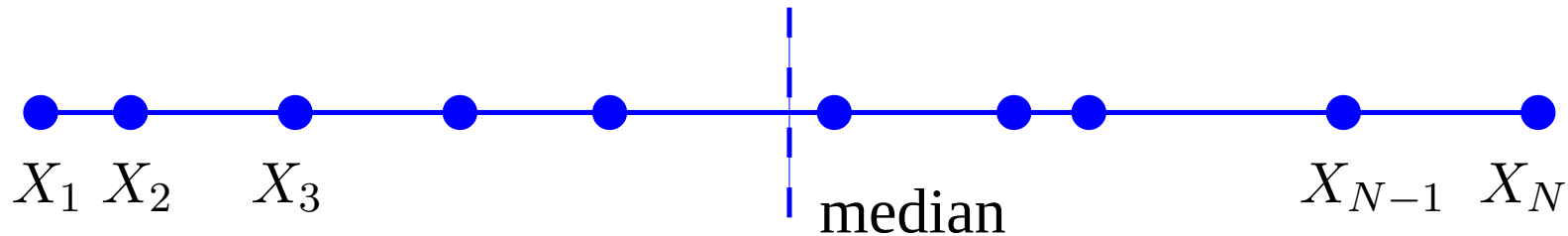
An $O(N \log N)$ divide-and-conquer method for 1-D closest pair

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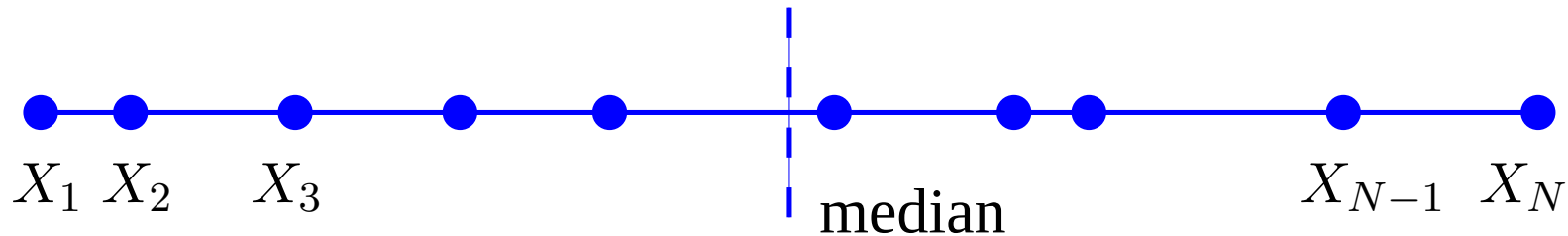


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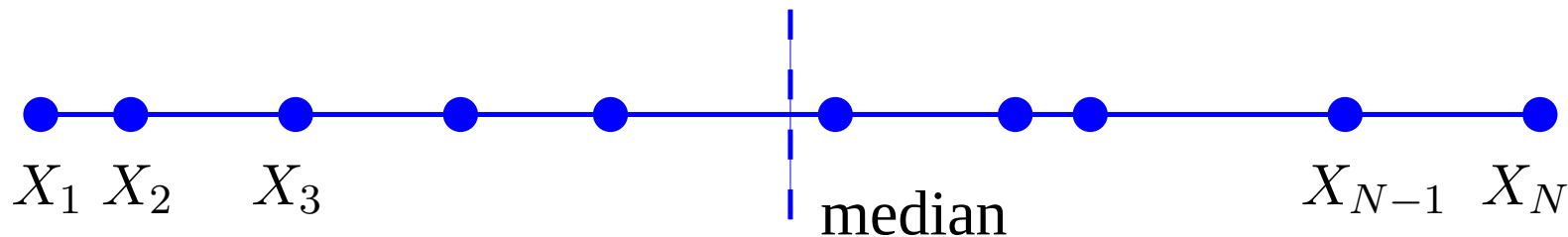


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```
1 def closestPairs(X, low, high):  
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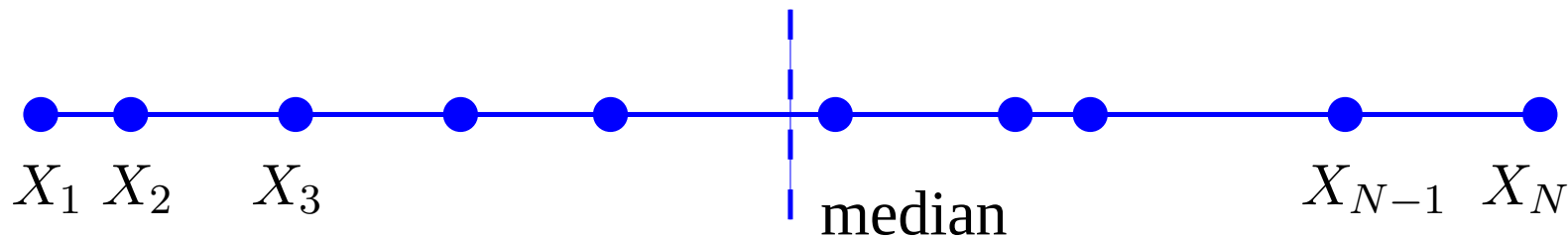


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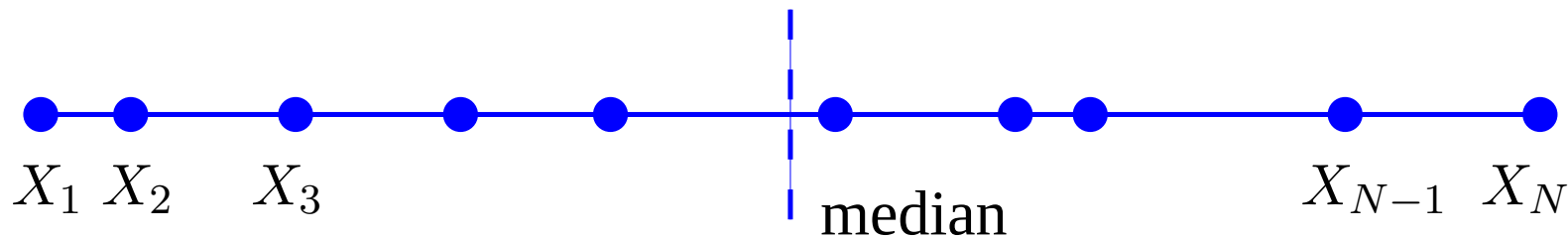


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9         straddle = minimum (X, m- $\delta$ , m+ $\delta$ )
```

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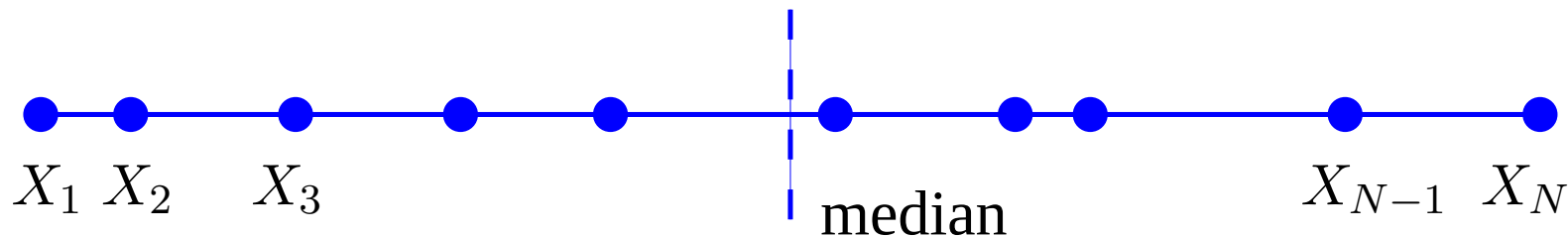


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```

It takes time $T(N) = 2T(N/2) + O(N)$, i.e.,
 $T(N) = O(N \log N)$.

The longest maximum subvector problem

In a list or vector X of N integers find the maximum sum of any contiguous subvector.¹³

-66	-15	87	91	32	5	-47	47	4	-77
0	1	2	3	4	5	6	7	8	9

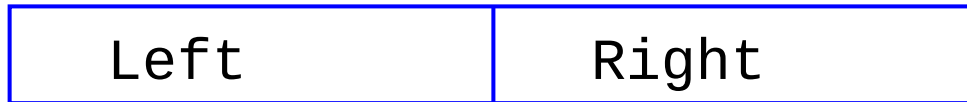
In the list X above the subvector $X[2..8]$ has the maximum sum 219. Clearly if all the numbers are negative then the maximum contiguous subvector (MCS) is simply the largest number. In this case we define the MCS as 0.

We know how to solve this problem in $O(N)$ time but we are now interested in using divide-and-conquer to solve it in the slower time of $O(N \log N)$

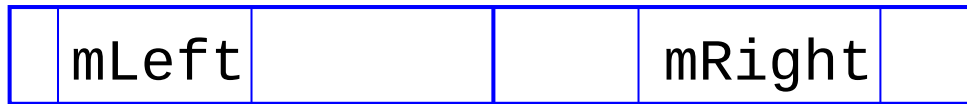
¹³This example comes from pages 77–81, “Programming Pearls”, Jon Bentley, ACM Press and Addison-Wesley, New York, 2000.

A divide-and-conquer method for MCS

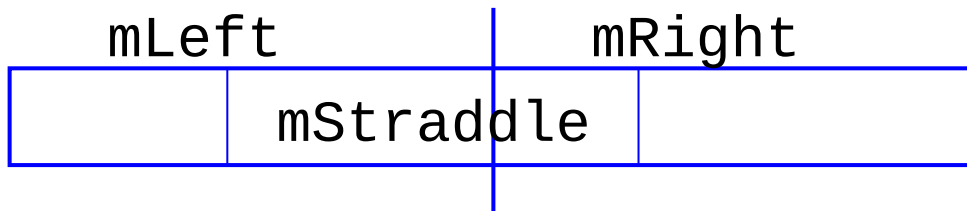
First, we divide the given array $X[]$ into two parts, **Left** and **Right**:



and then find the MCS **mLeft** in **Left** and the MCS **mRight** in **Right**.



There is still a snag—the MCS might straddle both **Left** and **Right**.



The MCS is the subvector that gives the biggest of these three values.

The final algorithm is (1) $\text{middle} = N/2$,

(2) Get MCS, **mLeft** of $X[\text{low}..\text{middle}]$, using

$\text{mLeft} = \text{MCS}(X, \text{low}, \text{middle})$, and get the MCS, **mRight** of $X[\text{middle}+1..N]$, using $\text{mRight} = \text{MCS}(X, \text{middle}+1, N)$.

More difficult is getting the straddling MCS, **mStraddle**.

Calculating the straddling MCS

The idea of the straddling MCS is first to calculate the MCS, called ‘maxsofarLeft’ that ends at $X[\text{middle}]$ but starts somewhere in **Left**—running backwards, and also calculating the MCS called ‘maxsofarRight’ that starts at $X[\text{middle}+1]$ and ends somewhere in **Right**. The sum of these two values is **mStraddle**.

```
1 int maxStraddle(low, high)
2     middle = (low + high) / 2
3     sum = 0; maxsofarLeft = 0
4     for i in [low..middle]
5         sum += X[middle - i + 1]
6         if sum > maxsofarLeft
7             maxsofarLeft = sum
8     sum = 0; maxsofarRight = 0
9     for i in [middle+1..high]
10        sum += X[i]
11        if sum > maxsofarRight
12            maxsofarRight = sum
13    return maxsofarLeft + maxsofarRight
```

A divide-and-conquer method for MCS

```
1 int MCS(low, high)
2     if low > high
3         return 0
4     if low == high
5         return max(0, X[low])
6     middle = (low + high) / 2
7     mLeft = MCS(X, low, middle)
8     mRight = MCS(X, middle+1, high)
9     mStraddle = maxStraddle (X, low, high)
10    return max (mLeft, mRight, mStraddle)
```

Suppose ‘MCS’ takes time $T(N)$ for an input of N . Then ‘mLeft’ takes time $T(N/2)$ and ‘mRight’ takes time $T(N/2)$. ‘maxStraddle()’ takes $O(N)$ because, at worst, it scans the entire input once. We conclude that

$$\begin{aligned} T(N) &= 2T(N/2) + O(N) \\ &= O(N \log N) \end{aligned}$$

Divide-and-conquer methods— $O(N \log N)$

Suppose the divide-and-conquer method has time complexity $T(N)$ when it has an input of size N .

Divide-and-conquer methods entail:

- (1) *dividing* the problem into equal parts,
- (2) *solving each part* of the problem separately, and
- (3) *joining* the results somehow.

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4. In MCS straddle passes through all N elements once, giving the same formula.

Divide-and-conquer methods— $O(N \log N)$ —I

Solving the recursive equation for merge sort and MCS

$$T(N) = 2T(N/2) + cN.$$

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Continue until $N/2^k = 1$.

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Since there is nothing to do when the input is of size 1, it follows that $T(1) = 0$, so we set $N/2^k = 1$ giving $N = 2^k$.

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Continue until $N/2^k = 1$.

Since there is nothing to do when the input is of size 1, it follows that $T(1) = 0$, so we set $N/2^k = 1$ giving $N = 2^k$. This happens when $k = \log_2 N$ and

$$\begin{aligned} T(N) &= T(1) + kcN \\ &= kcN \\ &= cN \log_2 N \\ &= O(N \log N) \end{aligned}$$

An $O(n)$ method for MCS(X , low, high)

This method starts at $X[\text{low}]$ and scans through all the elements up to $X[\text{high}]$ keeping book of the maximum contiguous subvector found so far.



The maximum sum in the first i elements is either the maximum in the first $i - 1$ elements, called ‘maxSoFar’ or that of the subvector that ends in $X[i]$, called ‘maxToHere’.

‘maxToHere’ contains the biggest sum *ending* at $X[i - 1]$. Considering item $X[i]$ take the biggest of $\text{maxToHere} + X[i]$ and 0.

```
1 int MCS(low, high)
2     maxSoFar = 0.0
3     maxToHere = 0.0
4     for i in [1, N)
5         maxToHere = max(maxToHere + X[i], 0.0)
6         maxSoFar = max(maxSoFar, maxToHere)
7     return maxSoFar
```

The 0–1 Knapsack Problem

Given any set S of n items where item i has a weight, $w_i > 0$, and has a benefit, $b_i > 0$, such that the total benefit is

$$B = \sum_{i \in S} b_i \text{ while } \sum_{i \in T} w_i < C,$$

where C is the maximum weight that is called the constraint or the capacity of the container.

If the number of objects is always an integer, this is called the 0–1 knapsack problem.

Let T denote the subset of items selected then the objective is to

$$\text{maximize } \sum_{i \in T} b_i \text{ while } \sum_{i \in T} w_i < C.$$

The 0–1 knapsack problem: example

Given a set S of n items where item i has a weight, $w_i > 0$, and has a benefit, $b_i > 0$, then select some items that maximize the total benefit B , with weight of at most $C = 9$.

	1	2	3	4	5
weight	4	2	2	6	2
benefit	R20	R3	R6	R25	R80

The solution is

$T = \{S_1, S_3, S_5\}$, with weight $C = 8$.

0–1 Knapsack algorithm: Try I

Let $S = \{S_k\}_{k=1}^n$ be a set of items and let
 $B[k] = \{\text{best selection from } S_k\}$.

This problem does not have *subproblem optimality*, for example the set

$$S = \{(3, 2), (5, 4), (8, 5), (4, 3), (10, 9)\}$$

of (benefit, weight) pairs with weight constrained by $C \leq 20$

Best selection from S_4 :

$$B[4] = \{(3, 2), (5, 4), (8, 5), (4, 3)\}$$

Best selection from S_5 :

$$B[5] = \{(3, 2), (5, 4), (8, 5), (10, 9)\}$$

Since we cannot derive $B[5]$ from $B[4]$ the problem does not possess *subproblem optimality*.

0–1 Knapsack algorithm: Try II

Let $S = \{S_k\}_{k=1}^n$ be a set of items and let

$B[k, C]$ = best selection from S_k with weight at most C .

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This problem *has subproblem optimality*.

$$B[k, C] = \begin{cases} B[k-1, C] & \text{if } w_k > C \\ \max\{B[k-1, C], B[k-1, C - w_k] + b_k\} & \text{otherwise} \end{cases}$$

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i.e., the best subset of S_k with weight at most C is

either the best subset of S_{k-1} with weight at most C

or the best subset of S_{k-1} with weight at most $C - w_k$ plus item k .

0–1 Knapsack problem: algorithm

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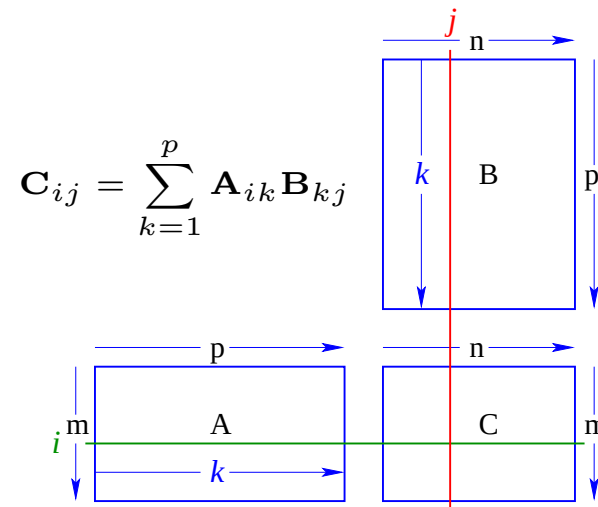
- Recall the definition of $B[k, C]$
 - Since $B[k, C]$ is defined in terms of $[k-1, *]$, we can use two arrays instead of a matrix
 - Running time: $O(nC)$.
 - Not a polynomial-time algorithm since C may be large
 - This is a *pseudo-polynomial* time algorithm
- b_i —a positive “benefit”
 w_i —a positive “weight”

0–1 Knapsack problem: algorithm

Listing 1: 0–1 Knapsack problem: algorithm

```
1 Algorithm 01Knapsack(S, C):  
2 In: set S of n items with benefit b_i  
3   and weight w_i; maximum weight C  
4 Out: benefit of best subset of S with  
5   weight at most C  
6 let A and B be arrays of length C + 1  
7 for w = 0 to C do  
8   B[w] = 0  
9 for k = 1 to n do  
10  copy array B into array A  
11  for w = w_k to C do  
12    if A[w-w_k] + b_k > A[w] then  
13      B[w] = A[w-w_k] + b_k  
14 return B[C]
```

Matrix multiplication

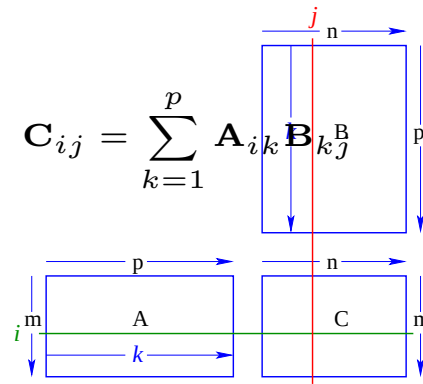


The ij th element of the $m \times n$ matrix $\mathbf{C}$, the product of the $m \times p$ matrix $\mathbf{A}$ and the $p \times n$ matrix $\mathbf{B}$ is:

$$\begin{aligned} C_{ij} &= \mathbf{A}_{i1}\mathbf{B}_{1j} + \mathbf{A}_{i2}\mathbf{B}_{2j} + \dots + \mathbf{A}_{ip}\mathbf{B}_{pj} \\ &= \sum_{k=1}^p \mathbf{A}_{ik}\mathbf{B}_{kj} \end{aligned}$$

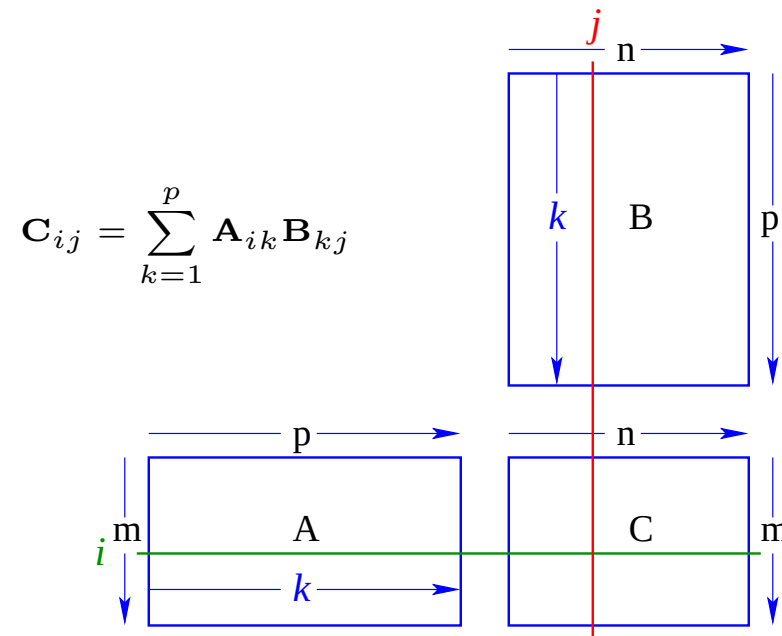
$\mathbf{C}$ is calculated by running through all the possible combinations of i and j using two nested **for** loops.

Matrix multiplication



```
1 void matMultiply (double [][] A, double [][] B,  
2                   double [][] C, int m, p, n) {  
3   int i, k, j; double Cij;  
4  
5   for (i=1; i<=m; i++)  
6     for (j=1; j<=n; j++){  
7       Cij = A[i][1]*B[1][j];  
8       for (k=2; k<=p; k++)  
9         Cij += A[i][k]*B[k][j];  
10      C[i][j] = Cij; } }
```

Matrix multiplication



Suppose $\{\mathbf{A}_i\}_{i=1}^n$ is a chain of compatible matrices with dimensions: $d_i \times d_{i+1}$. We want to compute:

$$\mathbf{C} = \mathbf{A}_1 \times \mathbf{A}_2 \times \dots \times \mathbf{A}_n$$

This is known as a chain product.

The dimension of the final product $\mathbf{C}$ is $d_1 \times d_{n+1}$.

Matrix Chain Products

Let $\{\mathbf{A}_\alpha\}_{\alpha=1}^n$ be a chain of compatible matrices with dimensions: $d_\alpha \times d_{\alpha+1}$. We want to compute:

$$\mathbf{C} = \mathbf{A}_1 \times \mathbf{A}_2 \times \dots \times \mathbf{A}_n$$

The products are done pairwise: e.g. $\mathbf{P} = \mathbf{A}_\alpha \times \mathbf{A}_{\alpha+1}$ where the elements of $\mathbf{P}$ are

$$\mathbf{P}_{ij} = \sum_{k=1}^{d_{\alpha+1}} \mathbf{A}_{\alpha ik} \mathbf{A}_{\alpha+1 kj}$$

The final dimensions of $\mathbf{P}$ are $d_\alpha \times d_{\alpha+2}$.

The problem is: Given the dimensions $\{d_\alpha\}$, how should the corresponding matrices be paired?

$\mathbf{B}$ is 3×100 , $\mathbf{C}$ is 100×5 , $\mathbf{D}$ is 5×5 .

$(\mathbf{B} * \mathbf{C}) * \mathbf{D}$ takes $1500 + 75 = 1575$ ops

$\mathbf{B} * (\mathbf{C} * \mathbf{D})$ takes $1500 + 2500 = 4000$ ops

The Enumeration Approach is $O(4^n)$

Matrix Chain-Product Algorithm:

- Try all possible ways to parenthesize

$$\mathbf{A}_1 \times \mathbf{A}_2 \times \dots \times \mathbf{A}_n$$

- Calculate number of operations for each one
- Pick the one that is best
- Running time:
- The number of parenthesizations is equal to the number of binary trees with n nodes
- This is exponential.
- This is the *Catalan* number, which is almost 4^n .
- This algorithm takes too long.

A Greedy Approach

- Idea α : repeatedly select the product that uses (up) the most operations.
- Counter example:

A is 10×5 , **B** is 5×10 , **C** is 10×5 , **D** is 5×10 .

Greedy idea α gives

$(\mathbf{A} \times \mathbf{B}) \times (\mathbf{C} \times \mathbf{D})$,

which takes

$500 + 1000 + 500 = 2000$ operations.

But the alternative bracketing yields:

$\mathbf{A} \times ((\mathbf{B} \times \mathbf{C}) \times \mathbf{D})$ takes

$500 + 250 + 250 = 1000$ operations.

Another Greedy Approach

- Idea β : repeatedly select the product that uses the fewest operations.
- Counter example

A is 101×11

B is 11×9

C is 9×100

D is 100×99

- Greedy idea β gives
 $\mathbf{A} \times ((\mathbf{B} \times \mathbf{C}) \times \mathbf{D})$, which takes
 $109989 + 9900 + 108900 = 228789$ ops., but
 $(\mathbf{A} \times \mathbf{B}) \times (\mathbf{C} \times \mathbf{D})$ takes
 $9999 + 89991 + 89100 = 189090$ ops.
- The greedy approach β also does not yield the optimal value.

A Recursive Approach

Define subproblems: Find the *best* parenthesization of $\mathbf{A}_i \times \mathbf{A}_{i+1} \times \dots \times \mathbf{A}_j$

Let $N_{i,j}$ denote the number of operations done by this subproblem.

The optimal solution for the whole problem is $N_{1,n}$.

Subproblem optimality: The optimal solution can be defined in terms of optimal subproblems.

- There has to be a final multiplication, i.e. the root of the expression tree, for the optimal solution. Suppose the final multiplication is at index i :

$$(\mathbf{A}_1 \times \dots \times \mathbf{A}_i) \times (\mathbf{A}_{i+1} \dots \times \mathbf{A}_n)$$

- Then the optimal solution $N_{1,n}$ is the sum of two optimal subproblems, $N_{1,i}$ and $N_{i+1,n}$ *plus* the time for the final multiplication.
- If the global optimum did not have these optimal subproblems, we could define an even better *optimal* solution.

A Characterizing Equation

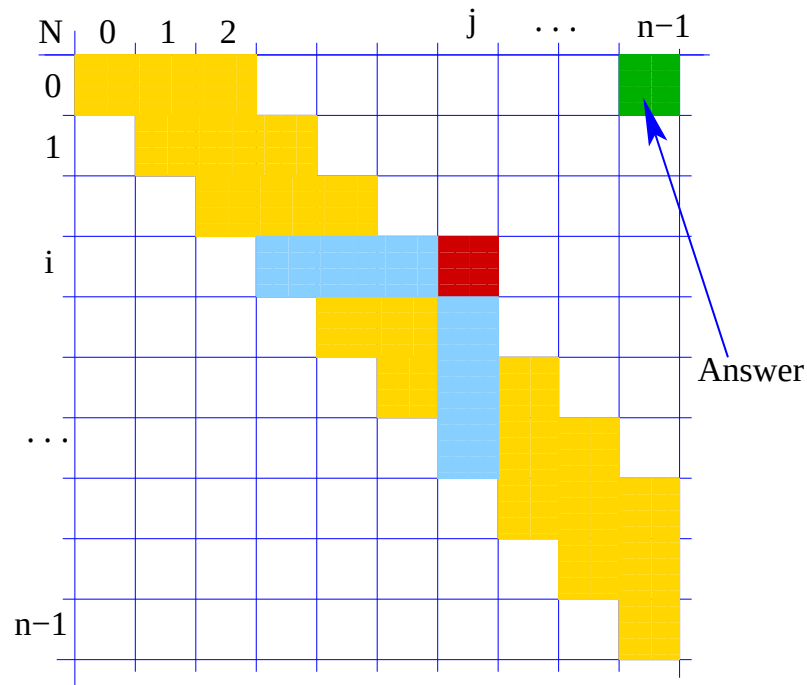
- Then the optimal solution $N_{1,n}$ is the sum of two optimal subproblems, $N_{1,i}$ and $N_{i+1,n}$ plus the time for the final multiplication.
- The global optimal has to be defined in terms of optimal subproblems, depending on where the final multiplication appears.
- Let us consider all possible places for the final multiplication: Recall that $\mathbf{A}_i$ is a $d_i \times d_{i+1}$ dimensional matrix. So, a characterizing equation for $N_{i,j}$ is the following:

$$N_{i,j} = \min_{i \leq k \leq j} [N_{i,k} + N_{k+1,j} + d_i d_{k+1} d_{j+1}]$$

Note that subproblems are not independent—the subproblems overlap.

Visualizing dynamic programming

$$N_{i,j} = \min_{i \leq k \leq j} [N_{i,k} + N_{k+1,j} + d_i d_{k+1} d_{j+1}]$$



First, fill in the $N_{i,i} = 0, \forall i \in [1..n - 1]$.

Get $N_{i,j}$ from previous entries in row i and column j .

Filling in each entry in the N table takes $O(n)$ time.

Getting actual parenthesization can be done by remembering k for each N entry.

Total run time: $O(n^3)$. Space requirements n^2 .

A Dynamic Programming Algorithm

Since subproblems overlap, we can't use recursion. Instead, we construct optimal subproblems *bottom-up*. Start with $N_{i,i} = 0, \forall i \in [1..n]$; then do length 2,3, ... subproblems. Running time: $O(n^3)$.

```
1 Algorithm matrixChain(S):
2   pre: sequence S of n matrices to be multiplied
3   post: number of operations in an optimal
4         parenthesization of S
5
6   for i = 1 to n-1 do
7     N[i,i] = 0
8   for b = 1 to n-1 do
9     for i = 0 to n-b-1 do
10      j = i+b
11      N[i,j] = +infinity
12      for k = i to j-1 do
13        N[i,j] = min(N[i,j], N[i,k] + N[k+1,j] +
14                     d[i]*d[k+1]*d[j+1])
```

Dynamic programming in general

- Applies to a problem that appears to need a lot of time, possibly exponential, provided we have:
- *Simple subproblems*: the subproblems can be defined in terms of a few variables, such as j , k , l , m , and so on.
- *Subproblem optimality*: the global optimum value can be defined in terms of optimal subproblems
- *Subproblem overlap*: the subproblems are not independent, but instead they overlap: hence they should be constructed bottom-up.

Substrings

- A substring of a character string $x_0x_1x_2 \dots x_{i_{n-1}}$ is a string of the form $x_{i_0}x_{i_1}x_{i_2} \dots x_{i_k}$, where $i_{j+1} = i_j + 1$ and $i_0 \geq 0$ and $i_k \leq i_{n-1}$.
- Example String: "ABCDEFGH IJK"
- Substring: "ABCDEFGH IJK"
- Substring: "DEFGH"
- Substring: ""
- Not substring: "DCEFGH IJ"

Subsequences—Section 11.5.1

- A subsequence of a character string $x_0x_1x_2 \dots x_{i_n-1}$ is a string of the form $x_{i_0}x_{i_1}x_{i_2} \dots x_{i_k}$, where $i_j < i_{j+1}$.
- Example String: "ABCDEFGHGIJK"
- Subsequence: "ACEGJIK"
- Subsequence: "DFGHK"
- Not subsequence: "DAGH"
- Subsequence, but also a substring: "DEFGHI"
- A subsequence is *not the same* as substring.

The Longest Common Subsequence (LCS)

- Given two strings X and Y , the longest common subsequence (LCS) problem is to find a longest subsequence common to both X and Y
- "ABCDEFGH" and "XZACKDFWGH" have "ACDFG" as a longest common subsequence
- Note that a *subsequence* is not a *substring*:
"ZACK" is not only a *substring* but it is also a *subsequence* of "XZACKDFWGH" a subsequence.
- Every *substring* of T is a *subsequence* of T .

The Longest Common Subsequence (LCS)—applications

- LCS has applications to DNA similarity testing—alphabet is $\{\text{A, C, G, T}\}$
- LCS is used by spelling checkers to find the best word to fit your spelling mistake.
- LCS is used by `diff` in Linux/Unix to find differences between two files.

A Poor Approach to the LCS Problem

A Brute-force solution:

- Enumerate all subsequences of X
- Test which ones are also subsequences of Y
- Pick the longest one.

Analysis:

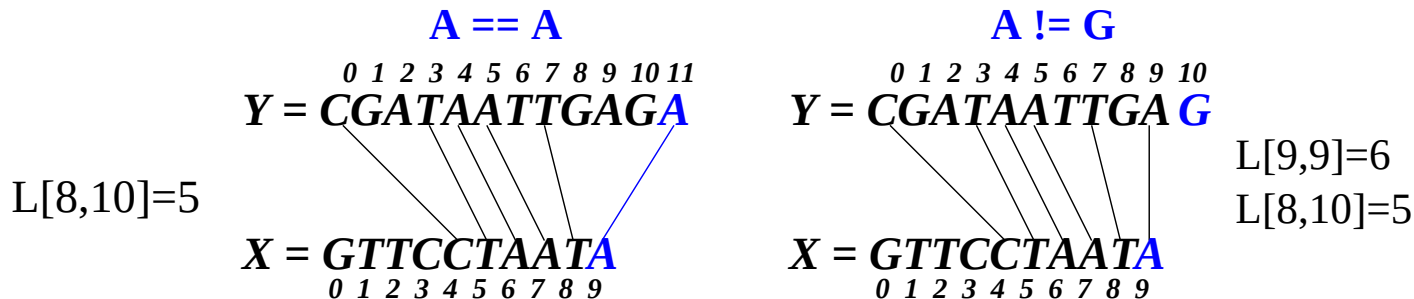
- If X is of length n , then it has 2^n subsequences
- This is an exponential-time algorithm

LCS solved with Dynamic Programming

- Define $L[i, j]$ to be the length of the longest common subsequence of $X[0..i]$ and $Y[0..j]$.
- Allow for -1 as an index, so $L[-1, k] = 0$ and $L[k, -1] = 0$, to indicate that the null part of X or Y has no match with the other.

Then we can define $L[i, j]$ in the general case as follows:

- if $x_i = y_j$, then \ \ add this match
$$L[i, j] = L[i - 1, j - 1] + 1$$
- if $x_i \neq y_j$, then \ \ no match here
$$L[i, j] = \max\{L[i - 1, j], L[i, j - 1]\}$$



An LCS Algorithm—in pseudocode

$$L_{i,j} = \begin{cases} \text{for } x_i = y_j & L_{i-1,j-1} + 1 \\ \text{otherwise} & \max\{L_{i-1,j}, L_{i,j-1}\} \end{cases}$$

L	-1	0	1	2	3			j	...	n-1	
-1	0	0	0	0	0	0	0	0	0	0	0
0	0										
1	0										
2	0										
3	0										
	0										
i	0										
	0										
...	0										
n-1	0										

LCS Algorithm—closer to Java

Listing 2: LCS Java-like algorithm— $O(N^2)$

```
1 char [] void LCS(char X[], int n, char Y[], int m){
2 //pre: Strings X of length n, and Y of length m.
3 //post: For i in [1..n-1], j in [1..m-1]:
4 // L[i,j] = length of LCS of X[0..i] and Y[0..j]
5 int L[][] = new int [n][m];
6     for (i=1; i < n; i++)
7         L[i][0] = 0;
8     for (j=0; j < m; j++)
9         L[0][j] = 0;
10    for (i=1; i < n; i++)
11        for (j=1; j < m; j++)
12            if (x[i] == y[j])
13                L[i][j] = L[i-1][j-1] + 1;
14            else
15                L[i][j] = max(L[i-1][j], L[i][j-1]);
```

LCS Algorithm—backtracking

Listing 3: LCS—backtracking)

```
1 // Backtrack for longest common subsequence
2 int length = L[n-1][m-1];
3 char lcs[] = new char[length+1];
4   for (i = n-1, j = m-1; i > 0 && j > 0 && length > 0; )
5     if (x[i] == y[j]) {
6       lcs[length--] = x[i--];
7       j--; }
8   else if (L[i-1][j] > L[i][j-1])
9     i--;
10  else
11    j--;
12
13  return lcs; }
```


Visualizing the LCS Algorithm

Analysis of LCS Algorithm

- The simple loops take $O(m)$ and $O(n)$ each.
- Outer nested loop takes $O(n)$ and repeats the inner loop $O(m)$ times.
- The body of the inner loop takes $O(1)$, i.e. it is independent of n or m .
- The backtracking is $O(1)$ per step and is at most $\max(O(n), O(m))$, therefore it can be ignored.
- So, the total running time is $O(nm)$
- The result is in $L[n, m]$ from which the subsequence may be recovered.

Longest maximum subvector problem solved in 5 ways

In a list or vector X of N integers find the maximum sum of any contiguous subvector.¹⁴

-66	-15	87	91	32	5	-47	47	4	-77
0	1	2	3	4	5	6	7	8	9

In the list X above the subvector $X[2..8]$ has the maximum sum 219. Clearly if all the numbers are negative then the maximum contiguous subvector (MCS) is simply the largest number. In this case we define the MCS as 0.

¹⁴This example comes from pages 77–81, “Programming Pearls”, Jon Bentley, ACM Press and Addison-Wesley, New York, 2000.

An $O(n^3)$ method for MCS

In our first attempt we simply calculate *all* possible subvectors $X[\text{low}..\text{high}]$ and find the greatest.

```
1 int max_n^3 (x)
2     n = len(x)
3     maxsofar = 0
4     for low in [0,n)
5         for high in [low,n)
6             sum = 0
7             for r in [low,high)
8                 sum += x[r]
9                 if (sum > maxsofar)
10                     maxsofar = sum
11     return maxsofar
```

Clearly three nested loops yield an $O(N^3)$ method.

There is a simple way to reduce this to only two loops by preserving partial sums.

An $O(n^2)$ method for MCS—1

In our second attempt again calculate all the subvectors $X[\text{low}..\text{high}]$ but use partial sums to determine them and find the greatest.

```
1 int max_n^2(x)
2     n = len(x)
3     maxsofar = 0
4     for low in [0,n)
5         sum = 0
6         for high in [low,n)
7             sum += x[r]
8             if (sum > maxsofar)
9                 maxsofar = sum
10    return maxsofar
```

There are now only two nested loops clearly making it an $O(N^2)$ method. There is another way of using only two loops.

Another $O(n^2)$ method for MCS—2

In our third attempt we first calculate the cumulative sums for for each possible range of numbers subvectors $X[0..i]$ where where $i \in [0..N]$

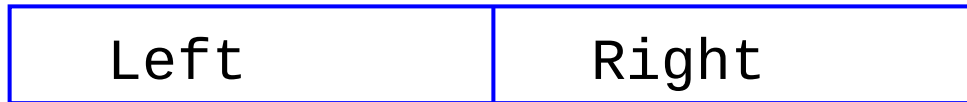
```
1 int max_n^2(x)
2     n = len(x)
3     sumTo[0] = 0
4     for i in [0,n)
5         sumTo[i] = sumTo[i-1] + X[i]
6     maxsofar = 0
7     for low in [0,n)
8         for high in [low,n)
9             sum = sumTo[high] - sumTo[low-1]
10            // sum of all elements in X[low..high)
11            if (sum > maxsofar)
12                maxsofar = sum
13     return maxsofar
```

With two nested loops, this is still $O(N^2)$.

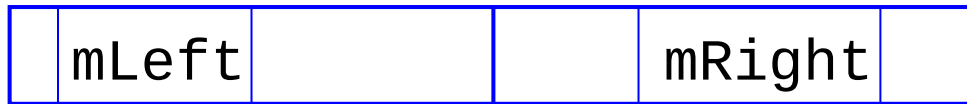
The next method is subquadratic.

A divide-and-conquer method for MCS

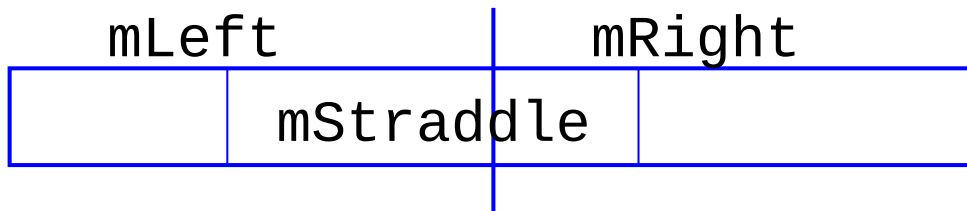
First, we divide the given array $X[]$ into two parts, **Left** and **Right**:



and then find the MCS **mLeft** in **Left** and the MCS **mRight** in **Right**.



There is still a snag—the MCS might straddle both **Left** and **Right**.



The MCS is the subvector that gives the biggest of these three values.

The final algorithm is (1) $\text{middle} = N/2$,

(2) Get MCS, **mLeft** of $X[\text{low}..\text{middle}]$, using

$\text{mLeft} = \text{MCS}(X, \text{low}, \text{middle})$, and get the MCS, **mRight** of $X[\text{middle}+1..N]$, using $\text{mRight} = \text{MCS}(X, \text{middle}+1, N)$.

More difficult is getting the straddling MCS, **mStraddle**.

Calculating the straddling MCS

The idea of the straddling MCS is first to calculate the MCS, called ‘maxsofarLeft’ that ends at $X[\text{middle}]$ but starts somewhere in **Left**—running backwards, and also calculating the MCS called ‘maxsofarRight’ that starts at $X[\text{middle}+1]$ and ends somewhere in **Right**. The sum of these two values is **mStraddle**.

```
1 int maxStraddle(low, high)
2     middle = (low + high) / 2
3     sum = 0; maxsofarLeft = 0
4     for i in [low..middle]
5         sum += X[middle - i + 1]
6         if sum > maxsofarLeft
7             maxsofarLeft = sum
8     sum = 0; maxsofarRight = 0
9     for i in [middle+1..high]
10        sum += X[i]
11        if sum > maxsofarRight
12            maxsofarRight = sum
13    return maxsofarLeft + maxsofarRight
```

A divide-and-conquer method for MCS

```
1 int MCS(low, high)
2     if low > high
3         return 0
4     if low == high
5         return max(0, X[low])
6     middle = (low + high) / 2
7     mLeft = MCS(X, low, middle)
8     mRight = MCS(X, middle+1, high)
9     mStraddle = maxStraddle (X, low, high)
10    return max (mLeft, mRight, mStraddle)
```

Suppose ‘MCS’ takes time $T(N)$ for an input of N . Then ‘mLeft’ takes time $T(N/2)$ and ‘mRight’ takes time $T(N/2)$. ‘maxStraddle()’ takes $O(N)$ because, at worst, it scans the entire input once. We conclude that

$$\begin{aligned} T(N) &= 2T(N/2) + O(N) \\ &= O(N \log N) \end{aligned}$$

Divide-and-conquer methods— $O(N \log N)$

Suppose the divide-and-conquer method has time complexity $T(N)$ when it has an input of size N .

Divide-and-conquer methods entail:

- (1) *dividing* the problem into equal parts,
- (2) *solving each part* of the problem separately, and
- (3) *joining* the results somehow.

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1. *Dividing* the problem in two, has the low cost of $O(1)$. All that needs to be done is $N = N/2$.

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2. *Solving each half* of the problem separately costs $2T(N/2)$.

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3. *Joining the two results* is the critical part. Try to make this depend *linearly* on N . If it is sublinear all the better. Usually this part entails looking at every input once. In mergesort the two sorted halves are merged in time cN , making the cost of merge sort, $T(N) = 2T(N/2) + cN$.

Divide-and-conquer methods— $O(N \log N)$

Suppose the divide-and-conquer method has time complexity $T(N)$ when it has an input of size N .

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4. In MCS straddle passes through all N elements once, giving the same formula.

Divide-and-conquer methods— $O(N \log N)$ —I

Solving the recursive equation for merge sort and MCS

$$T(N) = 2T(N/2) + cN.$$

is quite easy.

$$T(N) = 2T(N/2) + cN$$

Divide-and-conquer methods— $O(N \log N)$ —I

Solving the recursive equation for merge sort and MCS

$$T(N) = 2T(N/2) + cN.$$

is quite easy.

$$\begin{aligned} T(N) &= 2T(N/2) + cN \\ &= 2[2T(N/4) + cN/2] + cN \\ &= 2^2T(N/2^2) + cN + cN \end{aligned}$$

Divide-and-conquer methods— $O(N \log N)$ —I

Solving the recursive equation for merge sort and MCS

$$T(N) = 2T(N/2) + cN.$$

is quite easy.

$$\begin{aligned} T(N) &= 2T(N/2) + cN \\ &= 2[2T(N/4) + cN/2] + cN \\ &= 2^2T(N/2^2) + cN + cN \\ &= 2^2[2T(N/2^3) + cN/2^2] + cN + cN \end{aligned}$$

Divide-and-conquer methods— $O(N \log N)$ —I

Solving the recursive equation for merge sort and MCS

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$$\begin{aligned} T(N) &= 2T(N/2) + cN \\ &= 2[2T(N/4) + cN/2] + cN \\ &= 2^2T(N/2^2) + cN + cN \\ &= 2^2[2T(N/2^3) + cN/2^2] + cN + cN \\ &= 2^3T(N/2^3) + cN + cN + cN \\ &= 2^3T(N/2^3) + 3cN \end{aligned}$$

Divide-and-conquer methods— $O(N \log N)$ —I

Solving the recursive equation for merge sort and MCS

$$T(N) = 2T(N/2) + cN.$$

is quite easy.

$$\begin{aligned} T(N) &= 2T(N/2) + cN \\ &= 2[2T(N/4) + cN/2] + cN \\ &= 2^2T(N/2^2) + cN + cN \\ &= 2^2[2T(N/2^3) + cN/2^2] + cN + cN \\ &= 2^3T(N/2^3) + cN + cN + cN \\ &= 2^3T(N/2^3) + 3cN \\ &\vdots \\ &= 2^kT(N/2^k) + kcN \end{aligned}$$

Divide-and-conquer methods— $O(N \log N)$ —I

Solving the recursive equation for merge sort and MCS

$$T(N) = 2T(N/2) + cN.$$

is quite easy.

$$\begin{aligned} T(N) &= 2T(N/2) + cN \\ &= 2[2T(N/4) + cN/2] + cN \\ &= 2^2T(N/2^2) + cN + cN \\ &= 2^2[2T(N/2^3) + cN/2^2] + cN + cN \\ &= 2^3T(N/2^3) + cN + cN + cN \\ &= 2^3T(N/2^3) + 3cN \\ &\vdots \\ &= 2^kT(N/2^k) + kcN \end{aligned}$$

Continue until $N/2^k = 1$.

Divide-and-conquer methods— $O(N \log N)$ —II

$$\begin{aligned} T(N) &= 2T(N/2) + cN \\ &\vdots \\ &= 2^k T(N/2^k) + kcN \end{aligned}$$

Continue until $N/2^k = 1$.

$$\begin{aligned} T(N) &= 2T(N/2) + cN \\ &\vdots \\ &= 2^k T(N/2^k) + kcN \end{aligned}$$

Continue until $N/2^k = 1$.

Since there is nothing to do when the input is of size 1, it follows that $T(1) = 0$, so we set $N/2^k = 1$ giving $N = 2^k$.

Divide-and-conquer methods— $O(N \log N)$ —II

$$\begin{aligned} T(N) &= 2T(N/2) + cN \\ &\vdots \\ &= 2^k T(N/2^k) + kcN \end{aligned}$$

Continue until $N/2^k = 1$.

Since there is nothing to do when the input is of size 1, it follows that $T(1) = 0$, so we set $N/2^k = 1$ giving $N = 2^k$. This happens when $k = \log_2 N$ and

$$\begin{aligned} T(N) &= T(1) + kcN \\ &= kcN \\ &= cN \log_2 N \\ &= O(N \log N) \end{aligned}$$

An $O(n)$ method for MCS(X , low, high)

This method starts at $X[\text{low}]$ and scans through all the elements up to $X[\text{high}]$ keeping book of the maximum contiguous subvector found so far.



The maximum sum in the first i elements is either the maximum in the first $i - 1$ elements, called ‘maxSoFar’ or that of the subvector that ends in $X[i]$, called ‘maxToHere’.

‘maxToHere’ contains the biggest sum *ending* at $X[i - 1]$. Considering item $X[i]$ take the biggest of $\text{maxToHere} + X[i]$ and 0.

```
1 int MCS(low, high)
2     maxSoFar = 0.0
3     maxToHere = 0.0
4     for i in [1, N)
5         maxToHere = max(maxToHere + X[i], 0.0)
6         maxSoFar = max(maxSoFar, maxToHere)
7     return maxSoFar
```

```
String T = "The cat sat on the mat.";
int n = T.length(), i;

for(i=0; i <=n-5; i++) {
    // get T[0..0+i-1], length = i.
    System.out.println(T.subSequence(0,i));
    // get T[i..i+5], length 6.
    System.out.println(T.substring(i,i+5));
}
```


Java's `toString()` method

- The `String` representation of `object` is a convenient way to print it.
- Java provides for a `toString` method in any `class`.
- Suppose that your `Node` consists of three fields defined by

```
1 long key;  
2 String data;  
3 Node next;
```

This is typical of a node in a linked list.

- A simple node `Node` defined by

`Node N;`

may be instantiated by the statement

```
N = new Node(123L, "one_two_three");
```

```
String toString()
```



COS 211 Examination

Thursday 23rd May at 8h30
in Prefab A

Note: I do not take any responsibility for informing you of venue or timetable changes.

Useful mathematical facts—Logarithms

We write $\log$ for $\log_2$, and $\ln$ for $\log_e$ —some authors use $\lg$ for $\log_2$

If $x = b^n$, then $\log_b x =$

Useful mathematical facts—Logarithms

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Useful mathematical facts—Logarithms

We write $\log$ for $\log_2$, and $\ln$ for $\log_e$ —some authors use $\lg$ for $\log_2$

If $x = b^n$, then $\log_b x = n$.

$\log_b 1 = 0$, $\log_b b = 1$.

$\log x \times y =$

Useful mathematical facts—Logarithms

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Summing logarithms

$$\begin{aligned}\sum_{r=1}^n \lfloor \log_2 r \rfloor &= (n+1) \lfloor \log_2 n \rfloor - 2^{\lfloor \log_2 n \rfloor + 1} + 2 \\ &= \Theta(n \log n).\end{aligned}$$

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Sum of functions bounded by integral

If $f(x)$ is a monotonically increasing continuous function,

$$\sum_{r=1}^n f(r) \leq \int_1^{n+1} f(x) dx.$$

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The formula, $\log_2 x =$

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The formula $\log_{10} x \approx 0.3 \times \log_2 x$ shows us how to get the number of digits represented by a given number of bits— $\log_2 x$ —of a number x .

e.g. The largest unsigned 24-bit number is $2^{24} - 1$,

$$0.3 \log_2(2^{24} - 1) \approx 0.3 \times 24 = 7.2 \text{ digits.}$$

So 24 bits accurately represent 7 digits.

On the other hand, by the same argument a 23 bit number can only represent numbers of $\lfloor 6.9 \rfloor = 6$ digits accurately.

This is where the *hidden 1 bit* of the 23-bit mantissa of the IEEE 754 Floating Point Standard is used to turn a 23 mantissa into a 24 bit mantissa by adding a hidden bit that is always 1 excepting when the number is zero.

Useful mathematical facts—Asymptotic analysis

Since we are making approximations of the time that algorithms take, we consider non-negative increasing functions.

Big- O notation— $T(n) = O(g(n))$ —“ $T(n)$ is of order $g(n)$ ”
 $\iff$ there is a constant c such that $T(n) \leq cg(n)$, when n is big enough.

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$\iff$ both $T(n) = O(g(n))$ and $T(n) = \Omega(g(n))$ hold.
 $T(n)$ is asymptotically *tightly* bounded by $g(n)$.

Useful mathematical facts—Asymptotic analysis, examples

e.g. Let $T(n) = 64n^3 + 32n^2 + 16n + 8$

$T(n)$ is $O(n^3)$, $O(n^4)$, $\dots O(n^{100})$, $\dots$, $\Omega(n^3)$, $\Omega(n^2)$, $\Omega(n)$, and $\Theta(n^3)$.

$T(n)$ is not $O(n)$, $\Omega(n^4)$, $\Theta(n)$, $\Theta(n^2)$.

Although $T(n)$ is $O(n^4)$ it is not $\Omega(n^4)$. So it is not $\Theta(n^4)$.

Small o notation

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This implies that if $g(n) = o(T(n))$ then $O(T(n) + g(n)) = O(T(n))$,

e.g. $O(n^n + n^2 + n \log n + n + 1000000000) = O(n^n)$.

Asymptotically equal.

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L’Hospital’s Rule (1696)—was discovered by **Jacob Bernoulli** (1654–1705) but named after **Guilliaume L’Hospital** (1661–1704).

For functions f and g ,
if both $\lim_{x \rightarrow a} f(x) = 0$ and $\lim_{x \rightarrow a} g(x) = 0$,
or both $\lim_{x \rightarrow a} f(x) = \pm\infty$ and $\lim_{x \rightarrow a} g(x) = \pm\infty$,
and $\lim_{x \rightarrow a} \frac{f'(x)}{g'(x)}$ exists,
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R.W. Gosper, in 1975, gives a better approximation of $n!$ which yields reasonable values of $n!$ for all n with a small relative error,

$$\sqrt{2n + \frac{1}{3}\pi} \left(\frac{n}{e}\right)^n$$

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- Be prepared for repeat questions out of old exam papers and class test papers—please note that I do not supply these. Beware that when old questions are repeated they invariably have changes.

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Do you understand what is being asked?

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- Write code to declare a BST class.

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- Find k th smallest element using a heap—explain how to do this without a heap. How much time does each of your methods take?
- Be able to code all of the heap methods and code a **heap** class.

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- Heap sort. Build the heap bottom up in time $O(N)$, then swap elements $A[1]$ and $A[n]$, reduce size of tree by one and `heapify(1)` until there is only one element left.
- Explain radix sort. It is so fast it should always be used—why is it not always used?

Memo for examination

- Boyer-Moore, Knuth-Morris-Pratt, Rabin-Karp versus brute force. Know all their timings. Know details of algorithms. Be able to fill in gaps in given programs.
- Huffman encoding and decoding. How does Wayner encryption—based on Huffman coding—work.
- Speed of Huffman using heap versus not using a heap.
- Be able to describe efficient spelling checkers given constraints such as: “Using a preprocessed dictionary the checking algorithm must independent of the size of the dictionary.” or “The program may not use a method that relies on hashing.” or “The dictionary and data must be processed at the same time.”

Memo for examination

- Longest common subsequence. What is dynamic programming?
What other methods have you seen that use dynamic programming.
- Graphs: definitions. Depth-first search (DFS) for trees. Breadth-first search (BFS) for trees. DFS and BFS for graph traversal. Dijkstra's shortest path algorithm,
- For graphs also be able to explain the datastructures to store them: edge list, node list, adjacency list, etc.
- Topological sorting.
- Be able to demonstrate all algorithms, given a small set of data.

Memo: divide-and-conquer algorithms

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- Be able to code and explain the nearest point algorithm in 1-D
- Find the maximum element in an array using a divide-and-conquer algorithm.

Turning exponential algorithms into polynomial time algorithms

- Consider the code to calculate the n-th fibonacci number

```
1 def F(n):  
2     if n == 0: return 0  
3     if n == 1: return 1  
4     return F(n-1) + F(n-2)
```

- Explain why it is exponential.
- Alter the code to speed it up.
- Give code to calculate the n-th Fibonacci number without using a large array.
- Explain why this recursive code for factorial does not work for large values of n.

```
1 def factorial(n):  
2     if n < 2: return 1  
3     return n * factorial(n-1)
```

- Alter the code so that it does not show this behaviour.

Memo: exponential algorithms

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Some other algorithms

- Multiply matrices
- Do x^n by turning the n into a binary number.
- Find sums and products of polynomials.

Another approach to programming

Dynamic programming

- Matrix chain products
- The knapsack algorithm.
- Note we will look at the algorithm for the longest common subsequence problem next semester but you must understand the idea subproblem optimality.